



SM484

System Dynamics Simulation



SM484 SYSTEM DYNAMICS SIMULATION

ABET eCOURSEBOOK

AY2026

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UNITED STATES MILITARY ACADEMY
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SECTION I

COURSE ASSESSMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A

AY2026 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

15 DEC 2025

MEMORANDUM FOR COL Julia Coxen, Professor and Department Head, Department of
Systems Engineering, United States Military Academy, West Point, New York 10996.

SUBJECT: Academic Term (AT) 26-1 Course Assessment –SM484 System Dynamics
Simulation

1. The attached report summarizes the SM484 assessment for AT 2026-1
2. POC for this action is MAJ Jacob Lueder at jacob.lueders@westpoint.edu.

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Graded Events
5. Cadet Course End Feedback (TBP)

JACOB T. LUEDERS
MAJ, EN/FA49
SM484 Course Director

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY21			AY22		AY23		AY24		AY25		AY26	
21-1	21-2	21-3	22-1	21-2	23-1	23-2	24-1	24-2	25-1	25-2	26-1	26-2
X		X	X		X		X		X		X	

Table 1: Historical Course Offerings

b. Enrollments.

Academic Term	Enrollment
21-1	46
22-1	52
23-1	62
24-1	59
25-1	65
26-1	59

Table 2: Historical Enrollment

c. Historical Course Final Grade Averages.

Academic Term	Course Average
21-1	87.1%
22-1	84.0%
23-1	86.0%
24-1	84.1%
25-1	86.2%
26-1	83.0%

Table 3: Historical Course Average

2. Course Strategy.

- a. This course is a simulation elective for the Engineering Management, Systems Engineering, Operations Research, and Systems and Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The

techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive.

- b. Cadets will use the properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, implement these models, and perform an analysis of the results. Topics include applications of systems dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting design results in written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers.

Section II – Course Assessment

1. Assessment of Course Goals.

a. Course Objectives

Event	Title	Type	Issued	Lesson	Due	Lesson	Points	Apply system dynamics methodology to the solution of large scale, complex problems Create models of dynamic systems, operations, and processes Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes Develop policies and controls to improve systems, operations, and processes Interpret the output from continuous time simulations of system dynamics models as an aid for decision making Prepare and present the results of system dynamics analyses in oral and written form Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results						
								1	2	3	4	5	6	7
1	Homework 1-Prob Def and Feedback Structure (22 UN Case Studies)	Individual	20-Aug	2	10-Sep	8	75	X	X			X	X	
2	Homework 2-Modeling a Pandemic (Vensim)	Individual	5-Sep	6	19-Sep	11	75	X	X	X			X	
3	Homework 3-Stocks and Flows	Individual	29-Sep	13	27-Oct	23	75		X			X	X	
4	Homework 4-Fishbanks Policy Analysis	Individual	29-Oct	24	19-Nov	30	75				X	X	X	X
5	Quiz 1	Individual	15-Sep	9	15-Sep	9	40	X						
6	Quiz 2	Individual	24-Oct	22	24-Oct	22	40	X				X		
7	VPR 1	Individual	22-Sep	12	22-Sep	12	150	X	X			X		
8	VPR 2	Individual	3-Nov	25	3-Nov	25	150	X			X	X		
9	Project Part I	Group	8-Sep	7	1-Oct	14	50	X						
10	Project Part II	Group	8-Sep	7	20-Oct	20	75	X					X	
11	Project Part III	Group	8-Sep	7	7-Nov	27	50		X	X			X	
12	Project Presentation	Group	8-Sep	7	26-Nov	32	50	X		X	X	X	X	X
13	Project Report	Group	8-Sep	7	10-Dec	35	75	X		X	X	X	X	X
14	Instructor Points	Individual	18-Aug	1	10-Dec	35	20	X						
Individual							700							
Group							300							
Total							1000							

- b. Results of Course End Feedback. To be published See enclosure 5.

2. **Assessment of Course Content.**

- a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge through the identification and application of knowledge, skills and behaviors. This course teaches the tools necessary to understand System Dynamics concepts, tools, and models. Block I focused on an introduction to System Dynamics. Block II focused on System Dynamics modeling, Block IV focused on model validation and policy analysis. These blocks adequately covered the topics and course objectives.
- b. **Quality of the Assessment Mechanisms.** The end of course survey and the department's major-graded-event content approval process provides an adequate assessment tool for an impartial and effective assessment of the course objectives.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Assessment of Changes to Current Academic Term.** Based on previous course assessments.

- a. **Course Topics.** Removed climate change references from lessons and homework IAW executive orders.
- b. **Textbook.** No change.
 - i. Sterman, John D. Business Dynamics: Thinking and Modeling for a Complex World, Boston: Irwin McGraw-Hill, 2000.
- c. **Curriculum.** Changed the delivery order of lessons to build fundamental SD knowledge before building in Vensim. These changes combined blocks, so the structure now had only three blocks of instruction. This improved cadet performance on WPRs and enhanced their ability to understand the material. Added a DROP the week of Thanksgiving, and required groups to meet with instructors outside of class time to work on their model and explain how they will meet the end of term requirements. This time was well spent and recommend continuing.
- d. **Graded Requirements.**
 - i. Homework: The order of homework was changed to match the lesson changes to the curriculum. There was no noticeable difference in the performance of the homework with this change, as they still had similar material and time associated.
 - ii. Course Project: Added new Part 1 to the project to have an initial assessment of the Causal Loop Diagram that cadets build their projects on throughout the term. This additional requirement also lowered some of the point values for later group project components to maintain a total of 300 points. Both instructors felt that this

resulted in significantly improved models in the group project. Additionally, the project scopes were limited to three options related to the military, which resulted in better instructor assessments of projects than previous terms.

- iii. **WPRs:** Though challenging for many parts of the WPR as testing on definitions and processes will be similar, the parts where possible should be completely rewritten as past WPRs are used to study for current WPRs.
- e. **Course Administration.** Continued the use of Gradescope for WPRs, and expanded to include Gradescope for homework when applicable, recommend sustaining this practice.

2. Proposed Change for Next Academic Term.

- a. **Course Topics.** None.
- b. **Textbook.** Not necessarily a textbook concern, but the Vensim Tutorial needs to be significantly updated with the new version of Vensim that was released this fall. The user interface in the new version of Vensim is significantly different from the previous version and there is some additional functionality in the new version. Although this tutorial was updated for 26-1, an additional update occurred, necessitating further changes. Instead of a separate Technical Writing in SM484 handout, include it in the same PDF as the project prompt and ensure they are synced.
- c. **Curriculum.** Look at modifying some lessons to include additional modeling techniques and topics. Specifically incorporate the input/output objects and custom tables and graphs in Vensim into the curriculum. This is done informally, but should be added as a formal lesson objective. Additionally, look at incorporating co-flows into the topics that are covered in the lesson; consider putting this topic with aging chains as they generally work together. A larger task is restructuring the way examples are given throughout the course. Having a single example built similar to how the project builds would be useful to guide the teams a little more deliberately as they work on their project. Expand the rate-equation lesson to more deliberately cover the types of rate equations cadets need to use in their projects, potentially via a rate equation workshop. Consider the use of the People Express Flight Simulator (or similar) instead of the beer game.
- d. **Graded Requirements.** The timing of graded events needs some work. Right now, HW3 is due after WPR1, but it reinforces material from WPR1 (graphical integration & differentiation). Could just shift the timing of WPR and homework assignments, or consider major adjustments to have a homework, a quiz, and a WPR for each block. The first block could likely support two separate homework assignments, as significant in-class workshop time is dedicated to them. For the project, consider just a single prompt to facilitate better, and have available data for that topic. Given 3 options, 14/15 teams chose the same project this term (Army Retention), but then scoped them differently. Could run similarly to EM481, where after a submission, the cadets are then provided with a solution to better enable the next phase.

MADN-SE
AT 26-1 SM484 End-of-Course Assessment

Enclosure 2: Instructional Memorandum



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

04 August 2025

INSTRUCTIONAL MEMORANDUM NO. 484-26-1

SM484: SYSTEM DYNAMICS SIMULATION

1. PURPOSE. The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete SM484: System Dynamics Simulation.

2. REFERENCES.

- a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 07 August 2025.
- b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 07 August 2025.
- c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 07 August 2025.
- d. USMA publication. "Documentation and Acknowledgement of Academic Work," Office of the Dean, June 2025.

3. GENERAL.

- a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, practice problem guidance, and criteria for passing the course.
- b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

- a. Course Scope. This course is a simulation elective for the Systems Engineering, Engineering Management, Operations Research, and System Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes and evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part

of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, Complex Adaptive Systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive. Cadets use properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, produce general estimates of model parameters, and perform an analysis of the results to inform decision-makers and policy outcomes. Topics include applications of System Dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting their design results in both written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers.

b. Course Strategy. This course consists of three major blocks.

- 1) Block 1, ***Introduction to System Dynamics***, consists of 12 lessons introducing the basic concepts of modeling dynamic systems and the Vensim PLE software. After the base knowledge, lessons focus on extending the cadets' abilities to understand and model dynamic systems using causal loop diagrams and stock and flow models. This block contains two graded homework sets that introduce students to extracting pertinent system dynamics concepts from real-world problems and then linking those concepts to the underlying equations in the models. This block uses two workshops to reinforce lesson objectives regarding causal loop diagrams and stock and flow diagrams. This block concludes with a WPR.
- 2) Block 2, ***Basic Modeling***, consists of 13 lessons focusing on the development and analysis of continuous time models. The block starts the course project, where each cadet communicates a system dynamics problem and a dynamic hypothesis for a topic chosen from a list of options related to current military challenges. This writing will form the initial pool of problems that the cadets will fully model by the end of the course. Cadets will practice their modeling skills in both the classroom and through technical writing and initial model development for their final projects. Cadets will have multiple workshop sessions, culminating in the development of a full stock and flow model in support of their semester project. This block concludes with a WPR.
- 3) Block 3, ***Model Validation & Policy Analysis***, consists of 10 lessons focusing on model testing (verification and validation) and policy analysis. The Promotion simulation exercise will be used as a lab to help cadets better understand how to formally test, describe, and explain policy alternatives with a system dynamics model. Cadets will complete and present their semester project during this block.

c. Course Objectives:

- 1) Apply system dynamics methodology to the solution of large scale, complex problems.
- 2) Create models of dynamic systems, operations, and processes
- 3) Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes.
- 4) Develop policies and controls to improve systems, operations, and processes.
- 5) Interpret the output from continuous time simulations of system dynamics models as an aid for decision making.
- 6) Prepare and present the results of system dynamics analysis in oral and written form.
- 7) Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results.

d. Texts:

- 1) Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000.

e. Additional References.

- 1) Vensim User Guide: http://www.vensim.com/documentation/users_guide.html

f. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and practice problems.

- 1) The course consists of 1000 points.
 - a) Written Partial Review. (Two at **150 points**, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
 - b) Graded Homework. (Four at **75 points**, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
 - c) Quizzes. (Two at **40 points**, Individual Work). The purpose of these Quizzes is to provide additional assessment opportunity of cadets' understanding and ability to apply the concepts taught, but under more time constraints than a full homework assignment.

- d) Project. (Separate submissions for a total of **300 Points**, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.
- e) Instructor Points. (**20 points**, Individual Work). These instructor points will be used at the discretion of the individual instructor and can factor in class participation and checks-on-learning. Instructors will inform cadets NLT lesson 6 on how instructor points will be assigned.

2) Student performance assessment summary:

Title	Type	Issued	Lesson	Due	Lesson	Points
Homework 1-Prob Def and Feedback Structure (22 UN Case Studies)	Individual	20-Aug	2	10-Sep	8	75
Quiz 1	Individual	15-Sep	9	15-Sep	9	40
Homework 2-Modeling a Pandemic (Vensim)	Individual	5-Sep	6	19-Sep	11	75
WPR 1	Individual	22-Sep	12	22-Sep	12	150
Project Part I	Group	8-Sep	7	1-Oct	14	50
Project Part II	Group	8-Sep	7	20-Oct	20	75
Quiz 2	Individual	24-Oct	22	24-Oct	22	40
Homework 3-Stocks and Flows	Individual	29-Sep	13	27-Oct	23	75
WPR 2	Individual	3-Nov	25	3-Nov	25	150
Project Part III	Group	8-Sep	7	7-Nov	27	50
Homework 4-Fishbanks Policy Analysis	Individual	29-Oct	24	19-Nov	30	75
Project Presentation	Group	8-Sep	7	26-Nov	32	50
Project Report	Group	8-Sep	7	10-Dec	35	75
Instructor Points	Individual	18-Aug	1	10-Dec	35	20
Individual						700
Group						300
Total						1000

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Complete the course with at least 65% of the point total and 50% on the Course Project.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take the WPRs, complete all exercises and projects, and participate in all briefs. Cadets who will miss an event must contact the instructor as soon they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed. Graded Events will be administered IAW USMA and DSE policy—Cadets will report to the designated room for testing.

- c. Late Submissions. Failure to submit a graded event on time will result in a reduced grade (5% per 24-hour period IAW reference 2.a Department of Systems Engineering Admin Memo 1-1) and may result in disciplinary action. If you incorrectly/fail to include a Cover Page with an Acknowledgement Statement [with each submitted graded event], your assignment will be considered late until proper documentation is submitted.
- d. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).
- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2a through 2d. Generative AI is welcome for use, but assistance must be documented according to the DAAW, and remember you are responsible for checking the correctness of generated material. Generally acceptable use of AI is brainstorming, using it to clarify a specific unclear point, help with generating ideas for writing. Generally unacceptable use of AI is using it to either write from scratch, or completely edit large portions of your paper. This use must still be documented, but will receive the same deduction as if you had a person outside your group do that same work.
- f. Collaboration.
 - 1) ALL collaboration must be documented. Any discussion of assignments with anyone other than an SM484 instructor requires documentation. Documentation must be specific and detail topics discussed and actions taken.
 - 2) Sharing of electronic files via email or any other electronic means is strongly discouraged.
 - 3) Collaboration will result in a minor deduction of points in accordance with the following constructed scale:
 - a) I did this problem myself with help only from web-based resources, resources provided in class, or through conceptual help from another person (i.e. help understanding concepts or similar problems, but not help on the specific problem). **(100% credit)**
 - b) I received verbal help from another person beyond conceptual that did not include access to or copying their work. I did all typing myself. **(95% credit)**
 - c) I received help from another person that included looking at and copying their work. I did all the typing myself. **(85% credit)**
 - d) I received help from another person that included electronic access to their work. I did all the typing myself. **(75% credit)**
 - e) I received help from another person that included copying portions of their solution into my work. **(50% credit)**
 - 4) You must be very specific (which problem, etc.) when explaining any assistance used in your documentation or you will be deducted at a higher

penalty. This specific verbiage should be included within your e-Acknowledgement Statement depending on the type of assistance you received on each problem.

- g. Additional Instruction. AI is available at mutually agreed upon times between a cadet and the instructor. Cadets are expected to have specific questions for any AI session. General review sessions for groups of cadets prior to the WPRs will not be provided outside of regularly scheduled class periods. **AI closes 24 hours prior to homework submission deadlines**, so be proactive with your studies and don't wait until the last minute to ask for help. Be prepared to show your Challenge Problems at any AI session.
- h. Challenge Problems. Challenge Problems are assigned along with select reading assignments. These problems are intended to enhance understanding of text material. Although these problems will not always be collected and graded, you are expected to work them, particularly if you are struggling with the material.
- i. Class. You are expected to attend all synchronous lessons. If you miss class, you must let the instructor know before the lesson, and you are responsible for coordinating with peers and the instructor to ensure you receive all the missed content.
- j. Canvas: All course material, including the syllabus, graded events, and lesson material, will be posted in your course specific Canvas site. Please note that all sections use the same Canvas site, however, certain assignments and materials may only be viewable to a specific section if your instructor chooses to share specific additional material. Your instructor may choose to post supplemental materials under their specific folder. It is your responsibility to routinely check Canvas for announcements and course material.

FOR THE HEAD OF THE DEPARTMENT:

JACOB T LUEDERS
MAJ, EN (FA49)
Course Director

JOHN D CADDELL
LTC, IN (FA47)
SDS Program Director

DISTRIBUTION:

1 each instructor, SM484
1 each cadet, SM484

Enclosures:

SM484 Syllabus

Day	Lsn	Date	Day	Title	Assignment (Complete Prior to Class)	Graded Event	Lesson Objectives
1	1	18-Aug	Mon	Introduction to System Dynamics	Read: Business Dynamics (BD) Sections 1.1 - 1.3 (p 3 - 20) Do: Challenge Problem (p 14)		Define and describe the field of System Dynamics Describe the applications of System Dynamics Articulate a System Dynamics Problem
2	2	20-Aug	Wed	Causal Loop Diagrams (CLD)	Read: BD Chapter 5 (p 137-158) Do: Challenge Problem (p 147)	HW 1 Issued	Create a causal loop diagram Describe the cause and effect relationship in complex systems Describe the difference between causation and correlation Identify link and loop polarity
3	3	25-Aug	Mon	Structure and Behavior of Dynamic Systems	Read: BD Chapter 4 (p 107-133)		List and describe the fundamental modes of behavior and describe the feedback structure Define and Develop Reference Modes Formulate a Dynamic Hypothesis
4	4	27-Aug	Wed	Causal Loop Diagram Workshop	Read: BD Chapter 5 (p 159-176) Do: Challenge Problem 2 (p 168)		Create a causal loop diagram Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
5	X-Hour	29-Aug	Fri				
6	5	3-Sep	Wed	Stocks and Flows	Read: BD Chapter 6 (p 191-210) Do: Challenge Problem (p 211 - pick one 1, 2, or 3)		Define a Stock Define a Flow Classify an object as a stock or flow Apply the principles of stock and flow diagramming to represent a dynamic system Convert a causal loop diagram to a stock and flow diagram
7	6	5-Sep	Fri	Introduction to Vensim	Read: Vensim Guide (VG) Chapter 2: The Vensim User Interface (https://www.vensim.com/documentation/ustr02.html)	HW 2 Issued	Create a stock and flow diagram in Vensim Define and Develop Reference Modes Design and evaluate policy based on System Dynamics Model
8	7	8-Sep	Mon	The SD Modeling Process	Read: BD Section 2.5 (p 79-81) and Ch 3 (p 83 - 105) Read: Project Guidance	Project Issued	List and describe the 5 steps of the System Dynamics Modeling Process Articulate a System Dynamics Problem Formulate a Dynamic Hypothesis Understand how to utilize System Dynamics within the SDP
9	8	10-Sep	Wed	Stocks and Flows Workshop	Read: https://climate.nasa.gov/news/2365/seven-case-studies-in-carbon-and-climate/ Read: BD Chapter 6 (p 210-230)	HW1 Due by 2359	Convert a causal loop diagram to a stock and flow diagram Create a stock and flow diagram in Vensim
10	9	15-Sep	Mon	Dynamics of Stocks and Flows I	Read: BD Chapter 7 (p 231-241) Do: Challenge Problem (p 239)	Quiz 1	Understand the primary elements of Dynamic Complexity and how it arises Apply graphical integration and differentiation Determine Net Flow
11	10	17-Sep	Wed	Dynamics of Stocks and Flows II	Review: VG Chapters 2 (p 11-22), 4 (p 41-62) , & 5 (p 63 - 74) Read: HW 3 Guidance		Describe the cause and effect relationship in complex systems Apply graphical integration and differentiation Determine Net Flow Articulate a System Dynamics Problem
12	11	19-Sep	Fri	Dynamic Complexity	Read: BD Sections 1.3.1 - 1.5 (p 21 - 39)	HW2 Due by 2359	Articulate a System Dynamics Problem Understand the primary elements of Dynamic Complexity and how it arises Recognize sources of Policy Resistance Design and evaluate policy based on System Dynamics Model
13	12	22-Sep	Mon	WPR 1	Review: Lessons 1 - 11	WPR 1	
14	X-Hour	24-Sep	Wed				
15	13	29-Sep	Mon	Dynamics of Simple Structures: Intro	Read: BD Chapter 8 (p 263-280) Do: Challenge Problem (p 281 - part 1)	HW3 Issued	Understand the characteristics of a first-order, linear feedback system Understand the relationship between a first-order, positive feedback loop and exponential growth Understand the relationship between a first-order, negative feedback loop and goal-seeking behavior
16	14	1-Oct	Wed	Dynamics of Simple Structures: S-shaped Growth	Read: BD Chapter 9 (p 295 - 309, 323 - 334)	Project Part I due by 2359	Understand the dynamics and behavior of systems that experience S-Shaped growth Understand how the Logistic Growth model is used to model S-shaped growth Understand the SIR Model Understand the relationship between a nonlinear, first-order system and S-shaped growth
17	15	3-Oct	Fri	Dynamics Lab: Beer Game	Read: Skim BD Chapter 2 - Focus on one example		Articulate a System Dynamics Problem Model the behavior of the system using a management flight simulator Evaluate the effectiveness of your policies
18	16	6-Oct	Mon	Delays - Material	Read: BD Chapter 11 (p 409-425) Do: Challenge (p 425 - part 1)		Define a Delay Describe a material delay Describe a pipeline delay Determine the order of a delay Understand the structure and behavior of material delays Model a Material Delay
19	17	8-Oct	Wed	Delays - Information	Read: BD Chapter 11 (p 426-448) Skim: Section 11.6; p. 449-467		Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delays
20	18	14-Oct	Tue	Dynamics of Growth: Path Dependence	Read: BD Chapter 10 (p 349-406) Do: Challenge Problem (p 353)		Understand the characteristics and behaviors of multiple-loop systems Understand the concept of loop dominance and how it impacts behavior in multiple-loop systems Understand the dynamics and behaviors that experience path dependence Understand how positive feedback drives path dependence
21	19	16-Oct	Thu	Project Working Session	Review: Project Guidance		Create a System Dynamics Model
22	20	20-Oct	Mon	Dynamics of Growth: Aging Chains	Read: BD Chapter 12 (p 469 - 497) Do: Challenge (p 495 - part 3)	Project Part II Due by 2359	Describe an aging chain Understand and structure and behavior of an aging chain Model an Aging Chain
23	21	22-Oct	Wed	Modeling Decision Making	Read: BD Chapter 13 (p 513-550)		Understand the principles for modeling decision making Be familiar with common formulations for rate equations Understand the common pitfalls of modeling decision rules that should be avoided
24	22	24-Oct	Fri	Formulating Nonlinear Relationships	Read: BD Chapter 14 (p 551-595) Do: Challenge (p 566)	Quiz 2	Understand the guidelines for formulating table functions to capture nonlinear relationships Understand the common pitfalls in the specification and use of nonlinear functions
25	23	27-Oct	Mon	Project Working Session	Review: Project Guidance	HW 3 Due by 2359	Create a System Dynamics Model
26	24	29-Oct	Wed	Modeling Human Behavior	Read: BD Chapter 15 (pp 597-630)	HW4 Issued	Understand the cognitive limitations of decision makers Incorporate human behavior into a model
27	25	3-Nov	Mon	WPR 2	Review Lessons 13 - 24	WPR 2	
28	26	5-Nov	Wed	Model Testing	Read: BD Chapter 21 (p 845 - 891) Do: Challenge (p 871)		List the types of assessment tests for dynamic models Produce a calibrated model to given data Conduct goodness of fit test using Vensim
29	27	7-Nov	Fri	Policy Design and Evaluation Workshop	Review: BD Section 3.5.5 (p 103 - 104) Read: HW 4 Guidance	Project Part III Due by 2359	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
30	28	13-Nov	Thu	Promotion Lab - Part I	Read: BD Chapter 12 (p 469-496) Focus: BD p. 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
31	29	17-Nov	Mon	Promotion Lab - Part II	Review: BD Chapter 12 (p 469-496)		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
32	30	19-Nov	Wed	Challenges in System Dynamics	Read: System Dynamics and Decision Making in the Context of Dynamically Complex Multi-Dimensional Societal Issues	HW4 Due by 2359	Understand the challenges faced by systems dynamics modelers Understand the complexity of communicating a systems dynamics model
33	31	24-Nov	Mon	Project Working Session	Review: Project Guidance		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
34	32	26-Nov	Wed	Project Working Session	Review: Project Guidance	Project Presentations Due by 2359	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
35	33	2-Dec	Tue	Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
36	X-Hour	4-Dec	Thu				
37	34	8-Dec	Mon	Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
38	35	10-Dec	Wed	Course Wrap-up / Feedback	Complete end of course survey online	Project Report Due by 1600	Create a System Dynamics Model Describe the applications of System Dynamics Articulate a System Dynamics Problem Effectively communicate analysis and findings to a decision maker through written reports
39	X-Hour	12-Dec	Fri				
40	X-Hour	15-Dec	Mon				

MADN-SE
AT 26-1 SM484 End-of-Course Assessment

Enclosure 3: AT 26-1 SM484 Syllabus

Day	Lsn	Date	Day	Title	Assignment (Complete Prior to Class)	Graded Event	Lesson Objectives
1	1	18-Aug	Mon	Introduction to System Dynamics	Read: Business Dynamics (BD) Sections 1.1 - 1.3 (p 3 - 20) Do: Challenge Problem (p 14)		Define and describe the field of System Dynamics Describe the applications of System Dynamics Articulate a System Dynamics Problem
2	2	20-Aug	Wed	Causal Loop Diagrams (CLD)	Read: BD Chapter 5 (p 137-158) Do: Challenge Problem (p 147)	HW 1 Issued	Create a causal loop diagram Describe the cause and effect relationship in complex systems Describe the difference between causation and correlation Identify link and loop polarity
3	3	25-Aug	Mon	Structure and Behavior of Dynamic Systems	Read: BD Chapter 4 (p 107-133)		List and describe the fundamental modes of behavior and describe the feedback structure Define and Develop Reference Modes Formulate a Dynamic Hypothesis
4	4	27-Aug	Wed	Causal Loop Diagram Workshop	Read: BD Chapter 5 (p 159-176) Do: Challenge Problem 2 (p 168)		Create a causal loop diagram Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
5	X-Hour	29-Aug	Fri				
6	5	3-Sep	Wed	Stocks and Flows	Read: BD Chapter 6 (p 191-210) Do: Challenge Problem (p 211 - pick one 1, 2, or 3)		Define a Stock Define a Flow Classify an object as a stock or flow Apply the principles of stock and flow diagramming to represent a dynamic system Convert a causal loop diagram to a stock and flow diagram
7	6	5-Sep	Fri	Introduction to Vensim	Read: Vensim Guide (VG) Chapter 2: The Vensim User Interface (https://www.vensim.com/documentation/ustr02.html)	HW 2 Issued	Create a stock and flow diagram in Vensim Define and Develop Reference Modes Design and evaluate policy based on System Dynamics Model
8	7	8-Sep	Mon	The SD Modeling Process	Read: BD Section 2.5 (p 79-81) and Ch 3 (p 83 - 105) Read: Project Guidance	Project Issued	List and describe the 5 steps of the System Dynamics Modeling Process Articulate a System Dynamics Problem Formulate a Dynamic Hypothesis Understand how to utilize System Dynamics within the SDP
9	8	10-Sep	Wed	Stocks and Flows Workshop	Read: https://climate.nasa.gov/news/2365/seven-case-studies-in-carbon-and-climate/ Read: BD Chapter 6 (p 210-230)	HW1 Due by 2359	Convert a causal loop diagram to a stock and flow diagram Create a stock and flow diagram in Vensim
10	9	15-Sep	Mon	Dynamics of Stocks and Flows I	Read: BD Chapter 7 (p 231-241) Do: Challenge Problem (p 239)	Quiz 1	Understand the primary elements of Dynamic Complexity and how it arises Apply graphical integration and differentiation Determine Net Flow
11	10	17-Sep	Wed	Dynamics of Stocks and Flows II	Review: VG Chapters 2 (p 11-22), 4 (p 41-62) , & 5 (p 63 - 74) Read: HW 3 Guidance		Describe the cause and effect relationship in complex systems Apply graphical integration and differentiation Determine Net Flow Articulate a System Dynamics Problem
12	11	19-Sep	Fri	Dynamic Complexity	Read: BD Sections 1.3.1 - 1.5 (p 21 - 39)	HW2 Due by 2359	Articulate a System Dynamics Problem Understand the primary elements of Dynamic Complexity and how it arises Recognize sources of Policy Resistance Design and evaluate policy based on System Dynamics Model
13	12	22-Sep	Mon	WPR 1	Review: Lessons 1 - 11	WPR 1	
14	X-Hour	24-Sep	Wed				
15	13	29-Sep	Mon	Dynamics of Simple Structures: Intro	Read: BD Chapter 8 (p 263-280) Do: Challenge Problem (p 281 - part 1)	HW3 Issued	Understand the characteristics of a first-order, linear feedback system Understand the relationship between a first-order, positive feedback loop and exponential growth Understand the relationship between a first-order, negative feedback loop and goal-seeking behavior
16	14	1-Oct	Wed	Dynamics of Simple Structures: S-shaped Growth	Read: BD Chapter 9 (p 295 - 309, 323 - 334)	Project Part I due by 2359	Understand the dynamics and behavior of systems that experience S-Shaped growth Understand how the Logistic Growth model is used to model S-shaped growth Understand the SIR Model Understand the relationship between a nonlinear, first-order system and S-shaped growth
17	15	3-Oct	Fri	Dynamics Lab: Beer Game	Read: Skim BD Chapter 2 - Focus on one example		Articulate a System Dynamics Problem Model the behavior of the system using a management flight simulator Evaluate the effectiveness of your policies
18	16	6-Oct	Mon	Delays - Material	Read: BD Chapter 11 (p 409-425) Do: Challenge (p 425 - part 1)		Define a Delay Describe a material delay Describe a pipeline delay Determine the order of a delay Understand the structure and behavior of material delays Model a Material Delay
19	17	8-Oct	Wed	Delays - Information	Read: BD Chapter 11 (p 426-448) Skim: Section 11.6; p. 449-467		Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delays
20	18	14-Oct	Tue	Dynamics of Growth: Path Dependence	Read: BD Chapter 10 (p 349-406) Do: Challenge Problem (p 353)		Understand the characteristics and behaviors of multiple-loop systems Understand the concept of loop dominance and how it impacts behavior in multiple-loop systems Understand the dynamics and behaviors that experience path dependence Understand how positive feedback drives path dependence
21	19	16-Oct	Thu	Project Working Session	Review: Project Guidance		Create a System Dynamics Model
22	20	20-Oct	Mon	Dynamics of Growth: Aging Chains	Read: BD Chapter 12 (p 469 - 497) Do: Challenge (p 495 - part 3)	Project Part II Due by 2359	Describe an aging chain Understand and structure and behavior of an aging chain Model an Aging Chain
23	21	22-Oct	Wed	Modeling Decision Making	Read: BD Chapter 13 (p 513-550)		Understand the principles for modeling decision making Be familiar with common formulations for rate equations Understand the common pitfalls of modeling decision rules that should be avoided
24	22	24-Oct	Fri	Formulating Nonlinear Relationships	Read: BD Chapter 14 (p 551-595) Do: Challenge (p 566)	Quiz 2	Understand the guidelines for formulating table functions to capture nonlinear relationships Understand the common pitfalls in the specification and use of nonlinear functions
25	23	27-Oct	Mon	Project Working Session	Review: Project Guidance	HW 3 Due by 2359	Create a System Dynamics Model
26	24	29-Oct	Wed	Modeling Human Behavior	Read: BD Chapter 15 (pp 597-630)	HW4 Issued	Understand the cognitive limitations of decision makers Incorporate human behavior into a model
27	25	3-Nov	Mon	WPR 2	Review Lessons 13 - 24	WPR 2	
28	26	5-Nov	Wed	Model Testing	Read: BD Chapter 21 (p 845 - 891) Do: Challenge (p 871)		List the types of assessment tests for dynamic models Produce a calibrated model to given data Conduct goodness of fit test using Vensim
29	27	7-Nov	Fri	Policy Design and Evaluation Workshop	Review: BD Section 3.5.5 (p 103 - 104) Read: HW 4 Guidance	Project Part III Due by 2359	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
30	28	13-Nov	Thu	Promotion Lab - Part I	Read: BD Chapter 12 (p 469-496) Focus: BD p. 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
31	29	17-Nov	Mon	Promotion Lab - Part II	Review: BD Chapter 12 (p 469-496)		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
32	30	19-Nov	Wed	Challenges in System Dynamics	Read: System Dynamics and Decision Making in the Context of Dynamically Complex Multi-Dimensional Societal Issues	HW4 Due by 2359	Understand the challenges faced by systems dynamics modelers Understand the complexity of communicating a systems dynamics model
33	31	24-Nov	Mon	Project Working Session	Review: Project Guidance		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
34	32	26-Nov	Wed	Project Working Session	Review: Project Guidance	Project Presentations Due by 2359	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
35	33	2-Dec	Tue	Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
36	X-Hour	4-Dec	Thu				
37	34	8-Dec	Mon	Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
38	35	10-Dec	Wed	Course Wrap-up / Feedback	Complete end of course survey online	Project Report Due by 1600	Create a System Dynamics Model Describe the applications of System Dynamics Articulate a System Dynamics Problem Effectively communicate analysis and findings to a decision maker through written reports
39	X-Hour	12-Dec	Fri				
40	X-Hour	15-Dec	Mon				

Enclosure 4: Graded Events

Graded Events

Title	Type	Issued	Lesson	Due	Lesson	Points
Homework 1-Prob Def and Feedback Structure (22 UN Case Studies)	Individual	20-Aug	2	10-Sep	8	75
Quiz 1	Individual	15-Sep	9	15-Sep	9	40
Homework 2-Modeling a Pandemic (Vensim)	Individual	5-Sep	6	19-Sep	11	75
WPR 1	Individual	22-Sep	12	22-Sep	12	150
Project Part I	Group	8-Sep	7	1-Oct	14	50
Project Part II	Group	8-Sep	7	20-Oct	20	75
Quiz 2	Individual	24-Oct	22	24-Oct	22	40
Homework 3-Stocks and Flows	Individual	29-Sep	13	27-Oct	23	75
WPR 2	Individual	3-Nov	25	3-Nov	25	150
Project Part III	Group	8-Sep	7	7-Nov	27	50
Homework 4-Fishbanks Policy Analysis	Individual	29-Oct	24	19-Nov	30	75
Project Presentation	Group	8-Sep	7	26-Nov	32	50
Project Report	Group	8-Sep	7	10-Dec	35	75
Instructor Points	Individual	18-Aug	1	10-Dec	35	20
Individual						700
Group						300
Total						1000

MADN-SE
AT 26-1 SM484 End-of-Course Assessment

Enclosure 5: Course End Feedback

TBD Upon Receipt.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB B

AY2025 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

18 DEC 2024

MEMORANDUM FOR COL Julia Coxen, Professor and Department Head, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

SUBJECT: Academic Term (AT) 25-1 Course Assessment –SM484 System Dynamics Simulation

1. The attached report summarizes the SM484 assessment for AT 2025-1
2. POC for this action is COL James Enos at james.enos@westpoint.edu.

JAMES R. ENOS
COL, IN/FA47
SM484 Course Director

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Graded Events
5. Cadet Course End Feedback (TBP)

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY20		AY21			AY22		AY23		AY24		AY25	
20-1	20-2	21-1	21-2	21-3	22-1	21-2	23-1	23-2	24-1	24-2	25-1	25-2
X		X		X	X		X		X		X	

Table 1: Historical Course Offerings

b. Enrollments.

Academic Term	Enrollment
20-1	67
20-3	20
21-1	46
22-1	52
23-1	62
24-1	59
25-1	65

Table 2: Historical Enrollment

c. Historical Course Final Grade Averages.

Academic Term	Course Average
20-1	86.3%
20-3	91.5%
21-1	87.1%
22-1	84.0%
23-1	86.0%
24-1	84.1%
25-1	86.2%

Table 3: Historical Course Average

2. Course Strategy.

- This course is a simulation elective for the Engineering Management, Systems Engineering, Operations Research, and Systems and Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The

techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive.

- b. Cadets will use the properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, implement these models, and perform an analysis of the results. Topics include applications of systems dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting design results in written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers. Cadets will spend several lessons in a computer lab environment.

Section II – Course Assessment

1. Assessment of Course Goals.

a. Course Objectives

Event	Title	Type	Issued	Due	Lesson	Points	Apply system dynamics methodology to the solution of large scale, complex problems	Create models of dynamic systems, operations, and processes	Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes	Develop policies and controls to improve systems, operations, and processes	Interpret the output from continuous time simulations of system dynamics models as an aid for decision making	Prepare and present the results of system dynamics analysis in oral and written form	Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results	Average
							1	2	3	4	5	6	7	
1	Homework 1	Individual	28-Aug	9-Sep	8	75								89.8%
2	Homework 2	Individual	11-Sep	25-Sep	14	75								89.3%
3	Homework 3	Individual	2-Oct	21-Oct	23	75								85.9%
4	Homework 4	Individual	4-Nov	25-Nov	35	75								86.3%
5	Quiz 1	Individual	23-Sep	23-Sep	13	25								85.1%
6	Quiz 2	Individual	30-Oct	30-Oct	26	25								81.7%
7	WPR 1	Individual	30-Sep	30-Sep	16	150								82.2%
8	WPR 2	Individual	6-Nov	6-Nov	29	150								91.0%
9	Project Part I	Group	18-Sep	23-Oct	24	50								86.2%
10	Project Part II	Group	18-Sep	14-Nov	31	50								73.4%
11	Project Presentation	Group	18-Sep	5-Dec	37	100								91.2%
12	Project Report	Group	18-Sep	5-Dec	37	100								88.1%
13	Instructor Points	Individual	18-Sep	13-Dec	40	50								84.2%
							86.9%	84.6%	85.6%	89.1%	87.0%	86.3%	88.5%	
Direct Assessment:							3.5	3.4	3.4	3.6	3.5	3.5	3.5	

- b. **Results of Course End Feedback.** To be published See enclosure 5.

2. **Assessment of Course Content.**

- a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge through the identification and application of knowledge, skills and behaviors. This course teaches the tools necessary to understand System Dynamics concepts, tools, and models. Block I focused on an introduction to System Dynamics. Block II focused on System Dynamics Tools. Block III focused on System Dynamics modeling, Block IV focused on policy analysis. These blocks adequately covered the topics and course objectives. The cadets enjoyed the applications of System Dynamics block as it provided cadets with a means to gain a deeper understanding of System Dynamics as they see potential applications.
- b. **Quality of the Assessment Mechanisms.** The end of course survey and the department's major-graded-event content approval process provides an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. **Assessment of Course Processes.**

- a. **Textbooks.** No Change
 - i. Sterman, John D. Business Dynamics: Thinking and Modeling for a Complex World, Boston: Irwin McGraw-Hill, 2000.
- b. **Course Project.** Text
- c. **Graded Requirements.** No recommended changes.

Section III – Assessment of Recent Changes and Proposed Future Changes.

- 1. **Assessment of Changes to Current Academic Term.** Based on previous course assessments.
 - a. **Course Topics.** No change.
 - b. **Textbook.** No change.
 - c. **Curriculum.** No change.
 - d. **Graded Requirements.** No change.
 - e. **Course Administration.** Fully transitioned to Canvas for the course; uses the quiz function on canvas for two graded events. Additionally, built rubrics for each graded event to ensure consistency between instructors. Implemented gradescope for the

grading of WPR 1 and 2 with great success. This move increased the efficiency and consistency of grading the WPRs.

2. Proposed Change for Next Academic Term.

- a. **Course Topics.** None.
- b. **Textbook.** No necessarily a textbook concern, but the Vensim Tutorial needs to be significantly updated with the new version of Vensim that was released over the summer. The user interface in the new version of Vensim is significantly different from the previous version and there is some additional functionality in the new version.
- c. **Curriculum.** Look at modifying some lessons to include additional modeling techniques and topics. Specifically incorporate the input/output objects and custom tables and graphs in Vensim into the curriculum. Additionally, look at incorporating co-flows into the topics that are covered in the lesson; consider putting this topic with aging chains as they generally work together.
- d. **Graded Requirements.** None.

Enclosure 2: Instructional Memorandum

MADN-SE 19 August 2024

INSTRUCTIONAL MEMORANDUM NO. 484-25-1

SM484: SYSTEM DYNAMICS SIMULATION

1. PURPOSE. The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete SM484: System Dynamics Simulation.

2. REFERENCES.

- a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 16 August 2024.
- b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 16 August 2024.
- c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 16 August 2024.
- d. USMA publication. "Documentation and Acknowledgement of Academic Work," Office of the Dean, June 2023.

3. GENERAL.

- a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, practice problem guidance, and criteria for passing the course.
- b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

- a. **Course Scope.** This course is a simulation elective for the Systems Engineering, Engineering Management, Operations Research, and System Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes and evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Design Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, Complex Adaptive Systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive. Cadets use properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, produce general estimates of model parameters, and perform an analysis of the results to inform decision-makers and policy outcomes. Topics include applications of System

Dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting their design results in both written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers.

b. Course Strategy. This course consists of four major blocks.

- 1) Block 1, ***Introduction to System Dynamics***, consists of 6 lessons introducing the system dynamics methodology and explains the basic concepts of modeling dynamic systems. One graded modeling homework will introduce students to system dynamics concepts and then link those concepts to the underlying equations in the models. In this block, two labs will be used to reinforce lesson objectives. The first is a dynamics lab that will demonstrate the concepts of dynamic complexity. The second will be a simulation lab lesson with a tutorial and will introduce the simulation software used in this course, Vensim PLE.
- 2) Block 2, ***System Dynamics Tools***, consists of 7 lessons that extend cadets' abilities to understand and model dynamic systems. This block focuses on the system dynamics basics of causal loop diagrams, stock and flow models, and an understanding of the basic behaviors of dynamic systems. Cadets will begin to use Vensim PLE to model and analyze complex systems. Two graded homework assignments supplement the in-class learning of model structures and simulation tools. This block concludes with a WPR.
- 3) Block 3, ***Basic Modeling***, consists of 12 lessons focusing on the development and analysis of continuous time models. The block starts the course project part 1, where each cadet communicates a system dynamics problem and a dynamic hypothesis for a topic of his/her choice. This writing will form the initial pool of problems that the cadets will fully model by the end of the course. Cadets will practice their modeling skills in both the classroom and through technical writing and initial model development for their final projects. Cadets will have multiple lab sessions, culminating in the development of a full stock and flow model in support of their semester project. This block concludes with a WPR.
- 4) Block 4, ***Model Validation & Policy Analysis***, consists of 10 lessons focusing on model testing (verification and validation) and policy analysis. The Promotion simulation exercise will be used as a lab to help cadets better understand how to formally test, describe, and explain policy alternatives with a system dynamics model. Cadets will complete and present their semester project during this block.

c. Course Objectives:

- 1) Apply system dynamics methodology to the solution of large scale, complex problems.
- 2) Create models of dynamic systems, operations, and processes

- 3) Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes.
 - 4) Develop policies and controls to improve systems, operations, and processes.
 - 5) Interpret the output from continuous time simulations of system dynamics models as an aid for decision making.
 - 6) Prepare and present the results of system dynamics analysis in oral and written form.
 - 7) Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results.
- d. Texts:
- 1) Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000.
- e. Additional References.
- 1) Vensim User Guide: http://www.vensim.com/documentation/users_guide.html
- f. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and practice problems.
- 1) The course consists of 1000 points.
 - a) Written Partial Review. (Two at **150 points**, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
 - b) Graded Homework. (Four at **75 points**, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
 - c) Quizzes. (Two at **25 points**, Individual Work). The purpose of these Quizzes is to provide additional assessment opportunity of cadets' understanding and ability to apply the concepts taught, but under more time constraints than a full homework assignment.
 - d) Project. (Separate submissions for a total of **300 Points**, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.
 - e) Instructor Points. (**50 points**, Individual Work). These instructor points will be used

at the discretion of the individual instructor and can factor in class participation and checks-on-learning. Instructors will inform cadets NLT lesson 6 on how instructor points will be assigned.

2) Student performance assessment summary:

Event	Title	Type	Issued	Due	Lesson	Points
1	Homework 1	Individual	28-Aug	9-Sep	8	75
2	Homework 2	Individual	11-Sep	25-Sep	14	75
3	Homework 3	Individual	2-Oct	21-Oct	23	75
4	Homework 4	Individual	4-Nov	25-Nov	35	75
5	Quiz 1	Individual	23-Sep	23-Sep	13	25
6	Quiz 2	Individual	30-Oct	30-Oct	26	25
7	WPR 1	Individual	30-Sep	30-Sep	16	150
8	WPR 2	Individual	6-Nov	6-Nov	29	150
9	Project Part I	Group	18-Sep	23-Oct	24	50
10	Project Part II	Group	18-Sep	14-Nov	31	50
11	Project Presentation	Group	18-Sep	5-Dec	37	100
12	Project Report	Group	18-Sep	5-Dec	37	100
13	Instructor Points	Individual	18-Sep	13-Dec	40	50
Individual						700
Group						300
Total						1000

5. ADMINISTRATIVE GUIDANCE.

- a. **Minimum Requirements.** Complete the course with at least 65% of the point total and 50% on the Course Project.
- b. **Graded Events.** Unless there are exceptional circumstances, cadets will take the WPRs, complete all exercises and projects, and participate in all briefs. Cadets who will miss an event must contact the instructor as soon they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed. Graded Events will be administered IAW USMA and DSE policy—Cadets will report to the designated room for testing.
- c. **Late Submissions.** Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period IAW reference 2.a Department of Systems Engineering Admin Memo 1-1) and may result in disciplinary action. If you incorrectly/fail to include a Cover Page with an Acknowledgement Statement [with each submitted graded event], your assignment will be considered late until proper documentation is submitted.
- d. **Final Grades.** Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).
- e. **Documentation.** Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2a through 2d. Generative AI is welcome for use, but

assistance must be documented according to the DAAW, and remember you are responsible for checking the correctness of generated material. Generally acceptable use of AI is brainstorming, using it to clarify a specific unclear point, help with generating ideas for writing. Generally unacceptable use of AI is using it to either write from scratch, or completely edit large portions of your paper. This use must still be documented, but will receive the same deduction as if you had a person outside your group do that same work.

f. Collaboration.

- 1) ALL collaboration must be documented. Any discussion of assignments with anyone other than an SM484 instructor requires documentation. Documentation must be specific and detail topics discussed and actions taken.
- 2) Sharing of electronic files via email or any other electronic means is strongly discouraged.
- 3) Collaboration will result in a minor deduction of points in accordance with the following constructed scale:
 - a) I did this problem myself with help only from web-based resources or resources provided in class. **(100% credit)**
 - b) I received verbal help from another person that did not include access to or copying their work. I did all typing myself. **(95% credit)**
 - c) I received help from another person that included looking at and copying their work. I did all the typing myself. **(85% credit)**
 - d) I received help from another person that included electronic access to their work. I did all the typing myself. **(75% credit)**
 - e) I received help from another person that included copying portions of their solution into my work. **(50% credit)**
- 4) You must be very specific (which problem, etc.) when explaining any assistance used in your documentation or you will be deducted at a higher penalty. This specific verbiage should be included within your e-Acknowledgement Statement depending on the type of assistance you received on each problem.

g. Additional Instruction. AI is available at mutually agreed upon times between a cadet and the instructor. Cadets are expected to have specific questions for any AI session. General review sessions for groups of cadets prior to the WPRs will not be provided outside of regularly scheduled class periods. AI closes 24 hours prior to homework submission deadlines, so be proactive with your studies and don't wait until the last minute to ask for help. Be prepared to show your Challenge Problems at any AI session.

h. Challenge Problems. Challenge Problems are assigned along with select reading assignments. These problems are intended to enhance understanding of text material. Although these problems will not always be collected and graded, you are expected to work them, particularly if you are struggling with the material.

i. Class. You are expected to attend all synchronous lessons. If events arise and teaching must

move to a virtual environment, you will attend class via your Instructor's online chosen online platform (Canvas, Teams, Zoom, etc.). When attending class in a remote status, you must turn on your camera for the duration of the lesson. If you are quarantined due to COVID-19, you will (1) inform your Instructor and (2) attend class via the online platform. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.

- j. Canvas: All course material, including the syllabus, graded events, and lesson material, will be posted in your course specific Canvas site. Please note that all sections use the same Canvas site, however, certain assignments and materials may only be viewable to a specific section if your instructor chooses to share specific additional material. Your instructor may choose to post supplemental materials under their specific folder. It is your responsibility to routinely check Canvas for announcements and course material.

FOR THE HEAD OF THE DEPARTMENT:

JAMES R. ENOS
COL, IN/FA47
SDS Program Director /
Course Director

DISTRIBUTION:

1 each instructor, SM484

1 each cadet, SM484

MADN-SE
AT 25-1 SM484 End-of-Course Assessment

Enclosure 3: AT 25-1 SM484 Syllabus

	Lsn	Date	Title	Assignment (Complete Prior to Class)	Graded Event	Lesson Objectives
Intro to SD	1	19-Aug	No Scheduled Class			
	2	21-Aug	Introduction to System Dynamics	Read: Business Dynamics (BD) Sections 1.1 - 1.3 (p 3 - 20) Do: Challenge Problem (p 14)		Define and describe the field of System Dynamics Describe the applications of System Dynamics Articulate a System Dynamics Problem
	3	23-Aug	Dynamic Complexity	Read: BD Sections 1.3.1 - 1.5 (p 21 - 39)		Articulate a System Dynamics Problem Understand the primary elements of Dynamic Complexity and how it arises Recognize sources of Policy Resistance Design and evaluate policy based on System Dynamics Model
	4	26-Aug	Dynamics Lab: Beer Game	Read: Skim BD Chapter 2 - Focus on one example		Articulate a System Dynamics Problem Model the behavior of the system using a management flight simulator Evaluate the effectiveness of your policies
	5	28-Aug	The SD Modeling Process	Read: BD Section 2.5 (p 79-81) and Ch 3 (p 83 - 105)	HW1 Issued	List and describe the 5 steps of the System Dynamics Modeling Process Articulate a System Dynamics Problem Formulate a Dynamic Hypothesis Understand how to utilize System Dynamics within the SDP
	6	3-Sep	Introduction to Vensim	Read: Vensim Guide (VG) Chapter 2 (p 11-22) Read: Project Guidance		Create a stock and flow diagram in Vensim Define and Develop Reference Modes Design and evaluate policy based on System Dynamics Model
	7	5-Sep	Causal Loop Diagrams (CLD)	Read: BD Chapter 5 (p 137-158) Do: Challenge Problem (p 147)		Create a causal loop diagram Describe the cause and effect relationship in complex systems Describe the difference between causation and correlation Identify link and loop polarity
System Dynamics Tools	8	9-Sep	Structure and Behavior of Dynamic Systems	Read: BD Chapter 4 (p 107-133)	HW1 Due by 2359	List and describe the fundamental modes of behavior and describe the feedback structure Define and Develop Reference Modes Formulate a Dynamic Hypothesis
	9	11-Sep	Causal Loop Diagram Workshop	Read: BD Chapter 5 (p 159-176) Do: Challenge Problem 2 (p 168) Skim: VG Chapter 4 (p 41-62)	HW 2 Issued	Create a causal loop diagram in Vensim Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
	10	13-Sep	No Scheduled Class			
	11	16-Sep	Stocks and Flows	Read: BD Chapter 6 (p 191-210) Read: VG Chapter 5 (p 63 - 74) Do: Challenge Problem (p 211 - pick one 1, 2, or 3)		Define a Stock Define a Flow Classify an object as a stock or flow Apply the principles of stock and flow diagramming to represent a dynamic system Convert a causal loop diagram to a stock and flow diagram
	12	18-Sep	Stocks and Flows Workshop	Read: https://climate.nasa.gov/news/2365/seven-case-studies-in-carbon-and-climate/ Read: BD Chapter 6 (p 210-230) Review: VG Chapter 5 (p 63-74)	Project Issued	Convert a causal loop diagram to a stock and flow diagram Create a stock and flow diagram in Vensim
	13	23-Sep	Dynamics of Stocks and Flows I	Read: BD Chapter 7 (p 231-241) Do: Challenge Problem (p 239)	Quiz 1	Understand the primary elements of Dynamic Complexity and how it arises Apply graphical integration and differentiation Determine Net Flow
	14	25-Sep	Dynamics of Stocks and Flows II	Review: VG Chapters 2 (p 11-22), 4 (p 41-62) , & 5 (p 63 - 74) Read: HW 3 Guidance	HW2 Due by 2359	Describe the cause and effect relationship in complex systems Apply graphical integration and differentiation Determine Net Flow Articulate a System Dynamics Problem
Basic Modeling	15	27-Sep	No Scheduled Class			
	16	30-Sep	WPR1	Review: Lessons 1 - 15	WPR 1	
	17	2-Oct	Dynamics of Simple Structures: Intro	Read: BD Chapter 8 (p 263-280) Do: Challenge Problem (p 281- part 1)	HW 3 Issued	Understand the characteristics of a first-order, linear feedback system Understand the relationship between a first-order, positive feedback loop and exponential growth Understand the relationship between a first-order, negative feedback loop and goal-seeking behavior
	18	7-Oct	Dynamics of Simple Structures: S-shaped Growth	Read: BD Chapter 9 (p 295 - 309, 323 - 334)		Understand the dynamics and behavior of systems that experience S-shaped growth Understand how the Logistic Growth model is used to model S-shaped growth Understand the SIR Model Understand the relationship between a nonlinear, first-order system and S-shaped growth
	19	9-Oct	Delays - Material	Read: BD Chapter 11 (p 409-425) Do: Challenge (p 425 - part 1)		Define a Delay Describe a material delay Describe a pipeline delay Determine the order of a delay Understand the structure and behavior of material delays Model a Material Delay
	20	11-Oct	No Scheduled Class			
	21	15-Oct	Delays - Information	Read: BD Chapter 11 (p 426-448) Skim: Section 11.6; p. 449-467		Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delays
	22	17-Oct	Dynamics of Growth: Path Dependence	Read: BD Chapter 10 (p 349-406) Do: Challenge Problem (p 353)		Understand the characteristics and behaviors of multiple-loop systems Understand the concept of loop dominance and how it impacts behavior in multiple-loop systems Understand the dynamics and behaviors that experience path dependence Understand how positive feedback drives path dependence
	23	21-Oct	Project Working Session	Review: Project Guidance	HW3 Due by 2359	Create a System Dynamics Model
	24	23-Oct	Dynamics of Growth: Aging Chains	Read: BD Chapter 12 (p 469 - 497) Do: Challenge (p 495 - part 3)	Project Part I Due by 2359	Describe an aging chain Understand and structure and behavior of an aging chain Model an Aging Chain
	25	28-Oct	Modeling Decision Making	Read: BD Chapter 13 (p 513-550)		Understand the principles for modeling decision making Be familiar with common formulations for rate equations Understand the common pitfalls of modeling decision rules that should be avoided
	26	30-Oct	Formulating Nonlinear Relationships	Read: BD Chapter 14 (p 551-595) Read: VG Chapter 6 (p 107-120) Do: Challenge (p 566)	Quiz 2	Understand the guidelines for formulating table functions to capture nonlinear relationships Understand the common pitfalls in the specification and use of nonlinear functions
	27	1-Nov	Project Working Session	Review: Project Guidance		Create a System Dynamics Model
Model Validation & Policy Analysis	28	4-Nov	Modeling Human Behavior	Read: BD Chapter 15 (pp 597-630)	HW4 Issued	Understand the cognitive limitations of decision makers Incorporate human behavior into a model
	29	6-Nov	WPR 2	Review Lessons 17 - 28	WPR 2	
	30	12-Nov	Model Testing	Read: BD Chapter 21 (p 845 - 891) Read: VG Chapters 14 and 15 (p 193 - 226) Do: Challenge (p 871)		List the types of assessment tests for dynamic models Produce a calibrated model to given data Conduct goodness of fit test using Vensim
	31	14-Nov	Policy Design and Evaluation Workshop	Review: BD Section 3.5.5 (p 103 - 104) Read: HW 4 Guidance	Project Part II Due by 2359	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	32	18-Nov	Promotion Lab - Part I	Read: BD Chapter 12 (p 469-496) Focus: BD p. 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	33	20-Nov	Promotion Lab - Part II	Review: BD Chapter 12 (p 469-496)		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	34	22-Nov	No Scheduled Class			
	35	25-Nov	Challenges in System Dynamics	Read: System Dynamics and Decision Making in the Context of Dynamically Complex Multi-Dimensional Societal Issues	HW4 Due by 2359	Understand the challenges faced by systems dynamics modelers Understand the complexity of communicating a systems dynamics model
	36	3-Dec	Project Working Session	Review: Project Guidance		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	37	5-Dec	Project Working Session	Review: Project Guidance	Project Presentations Due by 2359	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	38	9-Dec	Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	39	11-Dec	Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	40	13-Dec	Course Wrap-up / Feedback	Complete end of course survey online	Project Report Due by 1600	Create a System Dynamics Model Describe the applications of System Dynamics Articulate a System Dynamics Problem Effectively communicate analysis and findings to a decision maker through written reports

Enclosure 4: Graded Events

Graded Events

Event	Type	Lesson	Date	Points
Homework 1	Individual	8	9SEP24	75
Quiz 1	Individual	13	23SEP24	25
Homework 2	Individual	14	25SEP24	75
WPR 1	Individual	16	30SEP24	150
Homework 3	Individual	23	21OCT24	75
Project Part 1	Group	24	23OCT24	50
Quiz 2	Individual	26	30OCT24	25
WPR 2	Individual	29	6NOV24	150
Project Part 2	Group	31	14NOV24	50
Homework 4	Individual	35	25NOV24	75
Project Presentation	Group	37	5DEC24	100
Project Report	Group	39	12DEC24	100
Instructor Points	Individual	40	13DEC24	50
Total				1000

MADN-SE
AT 25-1 SM484 End-of-Course Assessment

Enclosure 5: Course End Feedback

TBD Upon Receipt.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB C

AY2024 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996



MADN-SE

14 December 2023

MEMORANDUM FOR COL Julia Coxen, Professor and Department Head, Department of Systems Engineering

SUBJECT: AY24-1 Course Assessment and Final Grade Approval – SM484: System Dynamics Simulation

1. The attached report summarizes the SM484 course end assessment and curriculum change recommendations from AY24-1.
2. POC for this action is the undersigned.

JACOB T. LUEDERS
MAJ, EN
Course Director, SM484

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback (TBP)

Enclosure 1: Course Assessment**Section I – Course Information and History****1. Historic Course Offerings and Enrollments.****a. Offered.**

AY19		AY20		AY21			AY22		AY23		AY24	
19-1	19-2	20-1	20-2	21-1	21-2	21-3	22-1	21-2	23-1	23-2	24-1	24-2
X		X		X		X	X		X		X	

Table 1: Historical Course Offerings

b. Enrollments.

Academic Term	Enrollment
19-1	51
20-1	67
20-3	20
21-1	46
22-1	52
23-1	62
24-1	59

Table 2: Historical Enrollment

c. Historical Course Final Grade Averages.

Academic Term	Course Average
19-1	84.7%
20-1	86.3%
20-3	91.5%
21-1	87.1%
22-1	84.0%
23-1	86.0%
24-1	84.1%

Table 3: Historical Course Average

d. **Standard Grade Distribution Table**

Grade	Standard Cut	Recommended Cut	Approved Cut	# CDTs Std (Rec.)[App.]	% of Course (App)
A+	97.50%	94.00%	94.00%	0 (3) [3]	5.1%
A	92.50%	91.50%	91.50%	4 (5) [5]	8.5%
A-	90.00%	90.00%	90.00%	9 (5) [5]	8.5%
B+	87.50%	86.00%	86.00%	7 (11) [11]	18.6%
B	82.50%	82.50%	82.50%	15 (11) [11]	18.6%
B-	80.00%	79.00%	79.00%	7 (10) [10]	16.9%
C+	77.50%	76.00%	76.00%	8 (7) [7]	11.9%
C	72.50%	72.50%	72.50%	6 (4) [4]	6.8%
C-	70.00%	70.00%	70.00%	3 (3) [3]	5.1%
D	65.00%	65.00%	65.00%	0 (0) [0]	0.0%
F	0.00%	0.00%	0.00%	0 (0) [0]	0.0%

Table 4: Standard Grade Distribution Table (PRE APPROVAL)

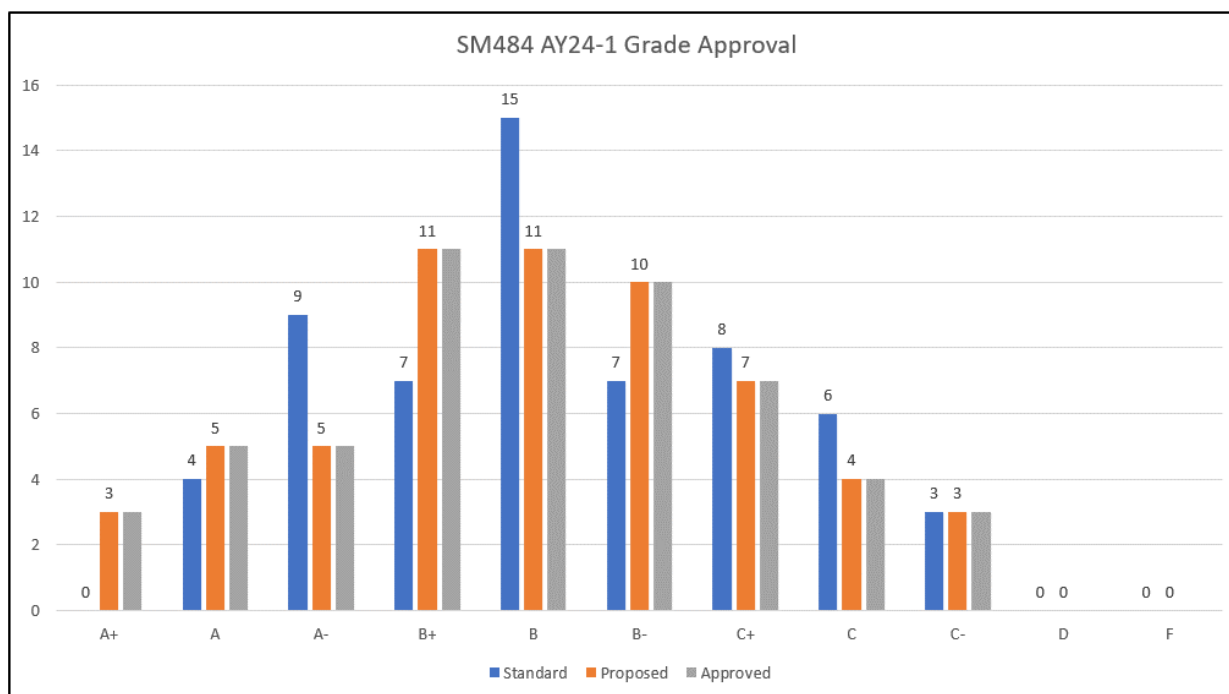


Figure 1: Grade Distribution (PRE APPROVAL)

2. Course Strategy.

a. This course is a simulation elective for the Engineering Management, Systems Engineering, Operations Research, and Systems and Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive.

b. Cadets will use the properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, implement these models, and perform an analysis of the results. Topics include applications of systems dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting design results in written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers. Cadets will spend several lessons in a computer lab environment.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback.** Enclosure 5

2. Course Content.

a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge through the identification and application of knowledge, skills and behaviors. This course teaches the tools necessary to understand System Dynamics concepts, tools, and models. Block I focused on an introduction to System Dynamics. Block II focused on System Dynamics Tools. Block III focused on System Dynamics modeling, Block IV focused on policy analysis. These blocks adequately covered the topics and course objectives. The cadets enjoyed the applications of System Dynamics block as it provided cadets with a means to gain a deeper understanding of System Dynamics as they see potential applications.

b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provides an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. Course Processes.

- a. **Textbook.** Sterman, John D. *Business Dynamics: Thinking and Modeling for a*

Complex World, Boston: Irwin McGraw-Hill, 2000. *Business Dynamics: Thinking and Modeling for a Complex World* by John Sterman is the foremost text book on System Dynamics. Recommend continued use of this textbook for future classes in conjunction with appropriate case studies.

b. **Case Studies.** For this semester, we adjusted the course project, allowing the cadets to pick a project and align into groups of their choosing. Additionally, we used the Fishbanks case study as a means to teach cadets how to develop policies and conduct policy analysis in homework four. The two separate exercises reinforced each other and fostered growth that would allow students to develop models for a case of their choosing for the final project. For WPR questions, graded homework, and in class labs, it is important to seek current case studies through formal and informal channels (e.g. System Dynamics Review and periodicals/news media, respectively) to ensure the cadets understand the relevance of System Dynamics and how it can be applied to analyze and develop solutions to real world problems. Recommended future case studies where instructors vet 3-5 different case studies that are well suited for cadets to build that are applicable to either the military or cadets to analyze (such as Army recruiting/retention).

c. **Graded Requirements.** The graded requirements in SM484 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Based on practices in AY23-1, we maintained four graded homework assignments to assess students learning in a distributed environment. Term 24-1 graded events in the course consisted of the following:

- i. Written Partial Review. (Two at 150 points, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
- ii. Take-Home Quizzes. (Two at 25 points, Individual Work). The purpose of these exercises was to reinforce concepts that proved difficult in the previous year's WPRs.
- iii. Graded Homework. (Four at 75 points, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
- iv. Project. (Separate submissions for a total of 300 Points, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.

The 10 graded events in 24-1 are summarized in Table 1 below:

Table 5: Graded Events

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	2	8	Homework 1 - Vensim Lab	Individual	75	7.5%
2	10	12	Quiz 1	Individual	25	2.5%
3	8	14	Homework 2 - CLD	Individual	75	7.5%
4	14	22	Homework 3 - Stock and Flow	Individual	75	7.5%
5	16	16	WPR 1	Individual	150	15.0%
6	23	25	Quiz 2	Individual	25	2.5%
7	29	29	WPR 2	Individual	150	15.0%
8	28	34	Homework 4 - Fishbanks Policy	Individual	75	7.5%
9	7		Course Project			0.0%
9.1		23	<i>Project Part I - CLD</i>	Group	50	5.0%
9.2		31	<i>Project Part II - Stock and Flow</i>	Group	50	5.0%
9.3		36	<i>Project Presentation</i>	Group	100	10.0%
9.4		40	<i>Project Report</i>	Group	100	10.0%
10	6	40	Instructor Points	Individual	50	5.0%
				TOTAL	1000	

4. Course Pedagogy.

- a. **Feedback Emphasis:** The course is centered on the completion of deliverables in each block with nearly all deliverables building upon previous submissions. This is meant to reinforce learning by allowing students to adjust previous submissions. It also takes some focus off of grades and puts emphasis on growth. As Sterman writes in *Business Dynamics*, "Just as dynamics arise from feedback, so too all learning depends on feedback."
- b. **Group Work:** At the beginning of the semester, cadets were allowed to select project teams. In order to demonstrate the applicability of System Dynamics to a broad range of problems, the course project allowed for cadets to choose the topic. However, the topic had to come from one of 22 case studies found on the United Nations website or an instructor approved topic. In keeping with previous practices, the assignment built on individual work that was completed for Homework 2. I strongly recommend publishing the course project guidance early in the semester to force cadets to start work early.
- c. **Learning Modality.** In AY24-1, SM484 was taught completely in-person over the entire semester.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** None
- b. **Textbook.** None
- c. **Curriculum.** None
- d. **Graded Requirements.** Added two 25pt quizzes prior to WPRs, focusing on questions where cadets struggled in AY23-1. Along with making recommended changes to WPR structure, this had a positive outcome on WPR average results. These 50 points came from a 50 point reduction in instructor points; 50 instructor points seems sufficient to assess student contributions to class. The course project was not restricted to UN case studies as in AY23-1, cadets appreciated some choice in what they could model, but expressed that some choices were more difficult in the end.
- e. **Communications.** All communication occurred over Canvas. It worked well and was easy for cadets to reference throughout the course.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None
- b. **Textbook.** No Change
- c. **Curriculum.** Reorganize the course to build the basics earlier and expand block 1, there was some confusion among cadets when we jumped into some more complex material early and some people struggled to pick it up by the end using this method of instruction. Adjust the project working sessions to bring some forward, and potentially split them up. Cadets requested more practice in Vensim early, so look at incorporating more Vensim desk and challenge problems to build more than just the basics. Look at MIT System Dynamics curriculum and try to incorporate both some of that lesson structure and more current examples. Consider incorporating a singular example that builds throughout the semester so that cadets can see how many of the different learning objectives could combine into their project. Continue to look for interesting new course project topics and case studies. Restructure the Vensim Tutorial with a basic tutorial for in class, and a more in-depth tutorial for out of class. Review assigned readings and focus in on the important pieces versus reading chapters in their entirety.
- d. **Graded Requirements.**
 - i. **Project:** Consider changing the point allocation for the project to a total of 200 from 300 points. Allocate 40 points each to part 1, part 2, and the presentation, while allocating 80 points to the final report. Consider restricting the projects to 2-3 instructor selected options that are well suited for System Dynamics analysis. Many teams did not devote enough time to the project in the initial stages which caused issues later. Look at restructuring the focus from the report to the model building and the process itself. Consider structuring the due-outs as an iterative process so that the groups go through the System Dynamics Modeling Process (SDMP) twice, once with broad strokes and then once to add additional detail. This would also enforce simulating early and often to ensure their model functions as desired. Consider no longer capping deductions for non-participation in group work, cadets need to learn to prioritize tasks and work in groups in their career, and this will foster that learning.
 - ii. **Homework:** Consider adapting all homework to align with a block of the course, and updating the homework examples to better match the types of questions they will see in the project and on WPRs.

- iii. Quizzes: Continue having take-home-quizzes prior to WPRs, students performance on WPRs improved having these mirror questions they could expect to see. Reinforce the importance of taking time and using resources when completing these.
- iv. WPRs: Continue the use of a note sheet on the WPRs, much of System Dynamics modeling is using available references to build structure, so a note sheet helps cadets show that they understand versus straight memorization. Adjust WPRs as mentioned in grade approval memos.
- v. Overall: Consider structuring blocks 1-3 with a 25pt quiz, 75pt homework, and 150pt WPR. Adding this allows for reducing the project by 100 points while also having a very structured approach to the graded assignments for the course. Net is one more WPR and Quiz, and one less Homework (COA seen below in Table 6). Alternatively, a second COA to keep the overall structure but redistribute project points to another homework is shown below in Table 7.

Table 6: Graded Assignment and Block Standardization COA

Block 1	Block 2	Block 3	Block 4
Quiz 1 (25pts)	Quiz 2 (25pts)	Quiz 3 (25pts)	Project Part 1 (40pts)
HW1 (75pts)	HW2 (75pts)	HW3 (75pts)	Project Part 2 (40pts)
WPR1 (150pts)	WPR2 (150pts)	WPR3 (150pts)	Project Presentation (40 pts)
			Project Report (80pts)

Table 7: Add Homework and Quiz COA

Block 1	Block 2	Block 3	Block 4
Quiz 1 (25pts)	Quiz 2 (25pts)	Quiz 3 (25pts)	HW4 (75pts)
HW1.a (75pts)	HW2 (75pts)	HW3 (75pts)	Project Part 1 (40pts)
HW1.b (75pts)	WPR1 (150pts)	WPR2 (150pts)	Project Part 2 (40pts)
			Project Presentation (40 pts)
			Project Report (80pts)

MADN-SE

AY 24-1 Course Assessment SM484 (14 December 2023)

Enclosure 2: Instructional Memorandum



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

11 August 2023

INSTRUCTIONAL MEMORANDUM
NO. 484-24-1

SM484: SYSTEM DYNAMICS
SIMULATION

- 1. PURPOSE.** The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete SM484: System Dynamics Simulation.
- 2. REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 10 August 2023.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 10 August 2023.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 10 August 2023.
 - d. USMA publication. "Documentation and Acknowledgement of Academic Work," Office of the Dean, June 2023.
- 3. GENERAL.**
 - a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, practice problem guidance, and criteria for passing the course.
 - b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
- 4. SPECIFIC.**
 - a. Course Scope. This course is a simulation elective for the Systems Engineering, Engineering Management, Operations Research, and System Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Design Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, Complex

Adaptive Systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive. Cadets use properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, produce general estimates of model parameters, and perform an analysis of the results to inform decision-makers and policy outcomes. Topics include applications of System Dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting their design results in both written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers.

- b. Course Strategy. This course consists of four major blocks.
- 1) Block 1, ***Introduction to System Dynamics***, consists of 6 lessons introducing the system dynamics methodology and explains the basic concepts of modeling dynamic systems. One graded modeling homework will introduce students to system dynamics concepts and then link those concepts to the underlying equations in the models. In this block, two labs will be used to reinforce lesson objectives. The first is a dynamics lab that will demonstrate the concepts of dynamic complexity. The second will be a simulation lab lesson with a tutorial and will introduce the simulation software used in this course, Vensim PLE.
 - 2) Block 2, ***System Dynamics Tools***, consists of 7 lessons that extend cadets' abilities to understand and model dynamic systems. This block focuses on the system dynamics basics of causal loop diagrams, stock and flow models, and an understanding of the basic behaviors of dynamic systems. Cadets will begin to use Vensim PLE to model and analyze complex systems. Two graded homework assignments supplement the in-class learning of model structures and simulation tools. This block concludes with a WPR.
 - 3) Block 3, ***Basic Modeling***, consists of 12 lessons focusing on the development and analysis of continuous time models. The block starts the course project part 1, where each cadet communicates a system dynamics problem and a dynamic hypothesis for a topic of his/her choice. This writing will form the initial pool of problems that the cadets will fully model by the end of the course. Cadets will practice their modeling skills in both the classroom and through technical writing and initial model development for their final

projects. Cadets will have multiple lab sessions, culminating in the development of a full stock and flow model in support of their semester project. This block concludes with a WPR.

- 4) Block 4, **Model Validation & Policy Analysis**, consists of 10 lessons focusing on model testing (verification and validation) and policy analysis. The Promotion simulation exercise will be used as a lab to help cadets better understand how to formally test, describe, and explain policy alternatives with a system dynamics model. Cadets will complete and present their semester project during this block.

c. Course Objectives:

- 1) Apply system dynamics methodology to the solution of large scale, complex problems.
- 2) Create models of dynamic systems, operations, and processes
- 3) Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes.
- 4) Develop policies and controls to improve systems, operations, and processes.
- 5) Interpret the output from continuous time simulations of system dynamics models as an aid for decision making.
- 6) Prepare and present the results of system dynamics analysis in oral and written form.
- 7) Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results.

d. Texts:

- 1) Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000.

e. Additional References.

- 1) Vensim User Guide: http://www.vensim.com/documentation/users_guide.html

f. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and practice problems.

g. Performance Assessment.

- 1) The course consists of 1000 points.
 - a) Written Partial Review. (Two at **150 points**, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
 - b) Graded Homework. (Four at **75 points**, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
 - c) Quizzes. (Two at **25 points**, Individual Work). The purpose of these Quizzes is to provide additional assessment opportunity of cadets' understanding and ability to apply the concepts taught, but under more time constraints than a full homework assignment.
 - d) Project. (Separate submissions for a total of **300 Points**, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.
 - e) Instructor Points. (**50 points**, Individual Work). These instructor points will be used at the discretion of the individual instructor and can factor in class participation and checks-on-learning. Instructors will inform cadets NLT lesson 6 on how instructor points will be assigned.

2) Student performance assessment summary:

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	2	8	Homework 1 - Vensim Lab	Individual	75	7.5%
2	10	12	Quiz 1	Individual	25	2.5%
3	8	14	Homework 2 - CLD	Individual	75	7.5%
4	14	22	Homework 3 - Stock and Flow	Individual	75	7.5%
5	16	16	WPR 1	Individual	150	15.0%
6	23	25	Quiz 2	Individual	25	2.5%
7	29	29	WPR 2	Individual	150	15.0%
8	28	34	Homework 4 - Fishbanks Policy	Individual	75	7.5%
9	7		Course Project			0.0%
9.1		23	<i>Project Part I - CLD</i>	Group	50	5.0%
9.2		31	<i>Project Part II - Stock and Flow</i>	Group	50	5.0%
9.3		36	<i>Project Presentation</i>	Group	100	10.0%
9.4		40	<i>Project Report</i>	Group	100	10.0%
10	6	40	Instructor Points	Individual	50	5.0%
TOTAL					1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Complete the course with at least 65% of the point total and 50% on the Course Project.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take the WPRs, complete all exercises and projects, and participate in all briefs. Cadets who will miss an event must contact the instructor as soon they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed. Graded Events will be administered IAW USMA and DSE policy—Cadets will report to the designated room for testing.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period IAW reference 2.a Department of Systems Engineering Admin Memo 1-1) and may result in disciplinary action. If you incorrectly/fail to include a Cover Page with an Acknowledgement Statement [with each submitted graded event], your assignment will be considered late until proper documentation is submitted.
- d. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).
- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references MUST BE PROPERLY DOCUMENTED. Documentation procedures are outlined in references 2a through 2d.

Generative AI is welcome for use, but assistance must be documented according to the DAAW, and remember you are responsible for checking the correctness of generated material. Generally acceptable use of AI is brainstorming, using it to clarify a specific unclear point, help with generating ideas for writing. Generally unacceptable use of AI is using it to either write from scratch, or completely edit large portions of your paper. This use must still be documented, but will receive the same deduction as if you had some other person outside your group do that same work.

f. Collaboration.

1) ALL collaboration must be documented. Any discussion of assignments with anyone other than an SM484 instructor requires documentation. Documentation must be specific and detail the topics discussed and actions taken.

2) Sharing of electronic files via email or any other electronic means is strongly discouraged.

3) Collaboration will result in a minor deduction of points in accordance with the following constructed scale:

- a) I did this problem myself with help only from web-based resources or resources provided in class. **(100% credit)**
- b) I received verbal help from another person that did not include access to or copying their work. I did all typing myself. **(95% credit)**
- c) I received help from another person that included looking at and copying their work. I did all the typing myself. **(85% credit)**
- d) I received help from another person that included electronic access to their work. I did all the typing myself. **(75% credit)**
- e) I received help from another person that included copying portions of their solution into my work. **(50% credit)**

4) You must be very specific (which problem, etc.) when explaining any assistance used in your documentation or you will be deducted at a higher penalty. This specific verbiage should be included within your e-Acknowledgement Statement depending on the type of assistance you received on each problem.

g. Additional Instruction. AI is available at mutually agreed upon times between a cadet and the instructor. Cadets are expected to have specific questions for any AI session. General review sessions for groups of cadets prior to the WPRs will not be provided outside of regularly scheduled class periods. AI closes 24 hours prior to homework submission deadlines, so be proactive with your studies and don't wait until the last minute to ask for help. Be prepared to show your Challenge Problems at any AI session.

h. Challenge Problems. Challenge Problems are assigned along with select reading assignments. These problems are intended to enhance understanding of text material. Although these problems will not always be collected and graded, you

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are expected to work them, particularly if you are struggling with the material.

- i. Class. You are expected to attend all synchronous lessons. If events arise and teaching must move to a virtual environment, you will attend class via your Instructor's online chosen online platform (Canvas, Teams, Zoom, etc.). When attending class in a remote status, you must turn on your camera for the duration of the lesson. If you are quarantined due to COVID-19, you will (1) inform your Instructor and (2) attend class via the online platform. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.
- j. Canvas: All course material, including the syllabus, graded events, and lesson material, will be posted in your course specific Canvas site. Please note that all sections use the same Canvas site, however, certain assignments and materials may only be viewable to a specific section if your instructor chooses to share specific additional material. Your instructor may choose to post supplemental materials under their specific folder. It is your responsibility to routinely check Canvas for announcements and course material.

FOR THE HEAD OF THE DEPARTMENT:

JACOB T LUEDERS
MAJ, EN
Course Director

STEPHEN GILLESPIE
LTC, IN (FA49)
SDS Program Director

DISTRIBUTION:

1 each instructor, SM484

1 each cadet, SM484

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AY 24-1 Course Assessment SM484 (14 December 2023)

Enclosure 3: Course Syllabus

Lsn	Date	Adjacent Events	Title	Assignment (Complete Prior to Class)	Graded Event	Lesson Objectives
Intro to SD	1	14-Aug		Introduction to System Dynamics	Read: Business Dynamics (BD) Sections 1.1 - 1.3 (p 3 - 20) Discussion Board: Challenge Problem (p 14)	Define and describe the field of System Dynamic Describe the applications of System Dynamics Articulate a System Dynamics Problem
	2	16-Aug		Dynamic Complexity	Read: BD Sections 1.3.1 - 1.5 (p 21 - 39)	Articulate a System Dynamics Problem Understand the primary elements of Dynamic Complexity and how it arises Recognize sources of Policy Resistance Design and evaluate policy based on System Dynamics Mod
	3	21-Aug		No Scheduled Class		
	4	23-Aug		Dynamics Lab: Beer Game	Read: Skim BD Chapter 2 - Focus on one example	Articulate a System Dynamics Problem Model the behavior of the system using a management flight simulator Evaluate the effectiveness of your policy
	5	25-Aug	Ring Wknd	The SD Modeling Process	Read: BD Section 2.5 (p 79-81) and Ch 3 (p 83 - 105)	List and describe the 5 steps of the System Dynamics Modeling Process Articulate a System Dynamics Problem Formulate a Dynamic Hypothesis Understand the role of testing in the System Dynamics Modeling Process Understand how to utilize System Dynamics within the SDI
	6	28-Aug		Introduction to Vensim	Read: Vensim Guide (VG) Chapter 2 (p 11-22) Read: Project Guidance	Create a stock and flow diagram in Vensim Define and Develop Reference Modes Design and evaluate policy based on System Dynamics Mod
	7	30-Aug		Causal Loop Diagrams (CLD)	Read: BD Chapter 5 (p 137-158) Discussion Board: Challenge Problem (p 147)	Create a causal loop diagram Describe the cause and effect relationship in complex systems Describe the difference between causation and correlation Identify link and loop polarity
System Dynamics Tools	8	5-Sep		Structure and Behavior of Dynamic Systems	Read: BD Chapter 4 (p 107-133)	List and describe the fundamental modes of behavior and describe the feedback structure Define and Develop Reference Modes Formulate a Dynamic Hypothesis
	9	7-Sep	Modified (Branch Week)	No Scheduled Class		
	10	11-Sep		Causal Loop Diagram Workshop	Read: BD Chapter 5 (p 159-176) Discussion Board: Challenge Problem 2 (p 168) Skim: VG Chapter 4 (p 41-62)	Create a causal loop diagram in Vensim Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
	11	13-Sep		Stocks and Flows	Read: BD Chapter 6 (p 191-210) Read: VG Chapter 5 (p 63 - 74) Discussion Board: Challenge Problem (p 211 - pick one 1, 2, or 3)	Define a Stock Define a Flow Classify an object as a stock or flow Apply the principles of stock and flow diagramming to represent a dynamic system Convert a causal loop diagram to a stock and flow diagram
	12	15-Sep		Stocks and Flows Workshop	Read: https://climate.nasa.gov/news/2365/seven-case-studies-in-carbon-and-climate/ Read: BD Chapter 6 (p 210-230) Review: VG Chapter 5 (p 63-74)	Convert a causal loop diagram to a stock and flow diagram Create a stock and flow diagram in Vensim
	13	18-Sep		Dynamics of Stocks and Flows	Read: BD Chapter 7 (p 231-241) Discussion Board: Challenge Problem (p 239)	Understand the primary elements of Dynamic Complexity and how it arise Apply graphical integration and differentiation Determine Net Flow
	14	20-Sep		Dynamics of Stocks and Flows Part II	Review: VG Chapters 2 (p 11-22), 4 (p 41-62) , & 5 (p 63 - 74) Read: HW 3 Guidance	Describe the cause and effect relationship in complex system Apply graphical integration and differentiation Determine Net Flow Create a causal loop diagram in Vensim Articulate a System Dynamics Problem
	15	25-Sep		No Scheduled Class		
	16	27-Sep		WPR1	WPR 1	
	17	29-Sep		Dynamics of Simple Structures: Intro	Read: BD Chapter 8 (p 263-280) Discussion Board: Challenge Problem (p 281- part 1)	Understand the characteristics of a first-order, linear feedback system Understand the relationship between a first-order, positive feedback loop and exponential growth Understand the relationship between a first-order, negative feedback loop and goal-seeking behavior
Basic Modeling	18	2-Oct		Dynamics of Simple Structures: S-shaped Growth	Read: BD Chapter 9 (p 295 - 309, 323 - 334)	Understand the dynamics and behavior of systems that experience S-Shaped growth Understand how the Logistic Growth model is used to model S-shaped growth Understand the SIR Model Understand the relationship between a nonlinear, first-order system and S-shaped grow
	19	4-Oct		Delays - Material	Read: BD Chapter 11 (p 409-425) Discussion Board: Challenge (p 425 - part 1)	Define a Delay Describe a material delay Describe a pipeline delay Determine the order of a delay Understand the structure and behavior of material delays Model a Material Delay
	20	10-Oct		Delays - Information	Read: BD Chapter 11 (p 426-448); Skim Section 11.6; p. 449-467 optional	Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delay
	21	12-Oct		Dynamics of Growth: Path Dependence	Read: BD Chapter 10 (p 349-406) Discussion Board: Challenge Problem (p 353)	Understand the characteristics and behaviors of multiple-loop system Understand the concept of loop dominance and how it impacts behavior in multiple-loop systems Understand the dynamics and behaviors that experience path dependence Understand how positive feedback drives path dependence
	22	16-Oct		Project Working Session		HW3 Due by 2359 Create a System Dynamics Model
	23	18-Oct		Dynamics of Simple Structures: Aging Chains	Read: BD Chapter 12 (p 469 - 497) Discussion Board: Challenge (p 495 - part 3)	Describe an aging chain Understand and structure and behavior of an aging chain, as well as when they are appropriate for use Model an Aging Chair
	24	20-Oct	Modified	No Scheduled Class		
	25	23-Oct		Modeling Decision Making	Read: BD Chapter 13 (p 513-550)	Understand the principles for modeling decision making Be familiar with common formulations for rate equations Understand the common pitfalls of modeling decision rules that should be avoided
	26	25-Oct		Formulating Nonlinear Relationships	Read: BD Chapter 14 (p 551-595) Read: VG Chapter 8 (p 107-120) Discussion Board: Challenge (p 566)	Understand the guidelines for formulating table functions to capture nonlinear relationships in System Dynamics models Understand the common pitfalls in the specification and use of nonlinear functions in System Dynamics models
	27	30-Oct		Project Modeling Working Session	Read: Project Guidance	Quiz 2 due NLT Class Create a System Dynamics Model
	28	1-Nov		Modeling Human Behavior	Read: BD Chapter 15 (pp 597-630)	Understand the cognitive limitations of decision makers Incorporate human behavior into a model
	29	3-Nov		Project Working Session	Review: Project Guidance	Create a Stock and Flow Model
	30	6-Nov		WPR 2	WPR 2	
	31	8-Nov		Model Testing	Read: BD Chapter 21 (p 845 - 891) Read: VG Chapters 14 and 15 (p 193 - 226) Discussion Board: Challenge (p 871)	Project Part II Due by 2359 List the types of assessment tests for dynamic model: Produce a calibrated model to given data Conduct goodness of fit test using Vensim
	32	13-Nov		Policy Design and Evaluation Workshop	Review: BD Section 3.5.5 (p 103 - 104) Read: HW 4 Guidance	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	33	15-Nov		Promotion Lab (Part 1)	Read: BD Chapter 12 (p 469-496) Focus: BD p. 485-490	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	34	20-Nov		Promotion Lab (Part 2)	Review: BD Chapter 12 (p 469-496) Review: BD p 485-490	HW4 Due by 2359 Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	35	28-Nov		Challenges in System Dynamics	Read: System Dynamics and Decision Making in the Context of Dynamically Complex Multi-Dimensional Societal Issues	Project Presentations Due by 2359 Understand the challenges faced by systems dynamics modelers Understand the complexity of communicating a systems dynamics model
	36	30-Nov		Project Working Session	Review: Project Guidance, Part II Feedback	
	37	30-Nov		Project Presentations	Review: Project Guidance	Project Presentations Effectively communicate analysis and findings to a decision maker through oral presentations
	38	6-Dec		Project Presentations	Review: Project Guidance	Project Presentations Effectively communicate analysis and findings to a decision maker through oral presentations
	39	8-Dec	Modified (Armc-Navy)	No Scheduled Class		
	40	11-Dec		Course Wrap-up / Feedback	Complete end of course survey online	Project Report Due by 2359 Create a System Dynamics Mode Describe the applications of System Dynamics Articulate a System Dynamics Problem Effectively communicate analysis and findings to a decision maker through written repor

Enclosure 4: SM484 Course Objective Assessment Table

	Apply system dynamics methodology to the solution of large scale, complex problems	Create models of dynamic systems, operations, and processes	Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes	Develop policies and controls to improve systems, operations, and processes	Interpret the output from continuous time simulations of system dynamics models as an aid for decision making	Prepare and present the results of system dynamics analysis in oral and written form	Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results	
Homework 1 - Vensim Lab	1	1	1					81.7%
Homework 2 - CLD	1	1						88.8%
Homework 3 - Stock and Flow	1	1						85.9%
Homework 4 - Fishbanks Policy			1	1			1	87.3%
Quiz 1	1	1	1		1			81.7%
Quiz 2	1	1	1		1			66.8%
WPR 1	1	1	1		1			79.4%
WPR 2	1	1	1		1			82.7%
Course Project								
Project Part I - CLD	1	1	1					84.2%
Project Part II - Stock and Flow	1	1	1					80.9%
Project Presentation			1	1	1	1		88.1%
Project Report			1	1	1	1	1	86.8%
Instructor Points -	1	1	1				1	91.0%
	82.3%	82.3%	82.8%	87.4%	80.9%	87.5%	88.4%	
Direct Assessment	3.3	3.3	3.3	3.5	3.2	3.5	3.5	3.4
Course Director Assessment	3.5	3.5	4	3.5	3	4	3.5	3.6

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AY 24-1 Course Assessment SM484 (14 December 2023)

Enclosure 5: Course End Feedback

TBP



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB D

AY2023 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996



MADN-SE

14 December 2022

MEMORANDUM FOR COL Rich Morales, Professor and Department Head, Department of Systems Engineering

SUBJECT: AY23-1 Course Assessment and Final Grade Approval – SM484: System Dynamics Simulation

1. The attached report summarizes the SM484 assessment and curriculum change recommendations from AY23-1.
2. POC for this action is the undersigned.

JACOB T. LUEDERS
CPT, EN
Course Director, System Dynamics Simulation

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

Enclosure 1: Course Assessment**Section I – Course Information and History****1. Historic Course Offerings and Enrollments.****a. Offered.**

AY18		AY19		AY20		AY21			AY22		AY23	
18-1	18-2	19-1	19-2	20-1	20-1	20-1	20-2	21-3	22-1	21-2	23-1	23-2
X		X		X	X	X		X	X		X	

Table 1: Historical Course Offerings

b. Enrollments.

Academic Term	Enrollment
18-1	42
19-1	51
20-1	67
20-3	20
21-1	46
22-1	52
23-1	62

Table 2: Historical Enrollment

c. Historical Course Final Grade Averages.

Academic Term	Course Average
18-1	86.1%
19-1	84.7%
20-1	86.3%
20-3	91.5%
21-1	87.1%
22-1	84.0%
23-1	86.0%

Table 3: Historical Course Average

d. **Standard Grade Distribution Table**

Grade	Standard Cut	Recommended Cut	Approved Cut	# CDTs Std (Rec.)[App.]	% of Course (App)
A+	97.50%	94.50%	94.50%	0 (5) [5]	8.1%
A	92.50%	91.90%	91.90%	10 (9) [9]	14.5%
A-	90.00%	88.90%	88.90%	7 (7) [7]	11.3%
B+	87.50%	86.50%	86.00%	12 (10) [12]	19.4%
B	82.50%	82.50%	81.90%	14 (12) [12]	19.4%
B-	80.00%	80.00%	80.00%	8 (8) [6]	9.7%
C+	77.50%	77.5%	76.50%	5 (5) [7]	11.3%
C	72.50%	72.50%	72.50%	5 (5) [3]	4.8%
C-	70.00%	70.00%	70.00%	0 (0) [0]	0.0%
D	65.00%	65.00%	65.00%	1 (1) [1]	1.6%
F	< 65.00%	< 65.00%	<65.00%	0 (0) [0]	0%

Table 4: Standard Grade Distribution Table

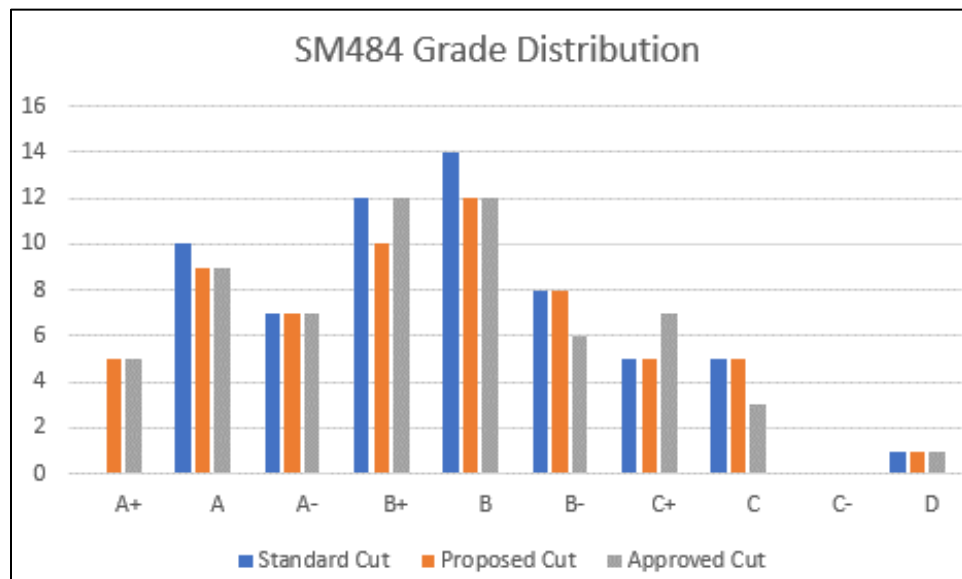


Figure 1: Grade Distribution

2. **Course Strategy.**

a. This course is a simulation elective for the Engineering Management, Systems Engineering, Operations Research, and Systems and Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These

systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive.

b. Cadets will use the properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, implement these models, and perform an analysis of the results. Topics include applications of systems dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting design results in written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers. Cadets will spend several lessons in a computer lab environment.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback.** Enclosure 5

2. Course Content.

a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge through the identification and application of knowledge, skills and behaviors. This course teaches the tools necessary to understand System Dynamics concepts, tools, and models. Block I focused on an introduction to System Dynamics. Block II focused on System Dynamics Tools. Block III focused on System Dynamics modeling, Block IV focused on policy analysis. These blocks adequately covered the topics and course objectives. The cadets enjoyed the applications of System Dynamics block as it provided cadets with a means to gain a deeper understanding of System Dynamics as they see potential applications.

b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provides an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. Course Processes.

a. **Textbook.** Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000. *Business Dynamics: Thinking and Modeling for a Complex World* by John Sterman is the foremost text book on System Dynamics. Recommend continued use of this text book for future classes in conjunction with appropriate case studies.

b. **Case Studies.** For this semester, we continued the use of one of the case studies from last year (Fishbanks), but adjusted the course project, forcing the cadets to explore the application of System Dynamics to one of 22 global challenges facing the United Nations. Additionally, we

used the Fishbanks case study as a means to teach cadets how to develop policies and conduct policy analysis. The two separate exercises reinforced each other and fostered growth that would allow students to develop models for a case of their choosing for the final project. For WPR questions, graded homework, and in class labs, it is important to seek current case studies through formal and informal channels (e.g. System Dynamics Review and periodicals/news media, respectively) to ensure the cadets understand the relevance of System Dynamics and how it can be applied to analyze and develop solutions to real world problems.

c. **Graded Requirements.** The graded requirements in SM484 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Based on practices in AY22-1, we maintained four graded homework assignments to assess students learning in a distributed environment. Term 23-1 graded events in the course consisted of the following:

- i. Written Partial Review. (Two at 150 points, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
- ii. Graded Homework. (Four at 75 points, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
- iii. Project. (Separate submissions for a total of 300 Points, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.

The 11 graded events in 23-1 are summarized in the following table:

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	3	9	Homework 1 - Vensim Lab	Individual	75	7.5%
2	9	14	Homework 2 - CLD	Individual	75	7.5%
3	15	22	Homework 3 - Stock and Flow	Individual	75	7.5%
4		16	WPR 1	Individual	150	15.0%
5		29	WPR 2	Individual	150	15.0%
6	28	34	Homework 4 - Fishbanks Policy	Individual	75	7.5%
	7		Course Project	Group	300	30.0%
7		23	<i>Project Part I - CLD</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
8		31	<i>Project Part II - Stock and Flow</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
9		36	<i>Project Presentation</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
10		39	<i>Project Report</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
11		40	<i>Instructor Points</i>	<i>Individual</i>	<i>100</i>	<i>10.0%</i>
Total Points					1000	

4. Course Pedagogy.

- a. **Feedback Emphasis:** The course is centered on the completion of deliverables in each block with nearly all deliverables building upon previous submissions. This is meant to reinforce learning by allowing students to adjust previous submissions. It also takes some focus off of grades and puts emphasis on growth. As Sterman writes in *Business Dynamics*, "Just as dynamics arise from feedback, so too all learning depends on feedback."
- b. **Group Work:** At the beginning of the semester, cadets were allowed to select project teams. In order to demonstrate the applicability of System Dynamics to a broad range of problems, the course project allowed for cadets to choose the topic. However, the topic had to come from one of 22 case studies found on the United Nations website or an instructor approved topic. In keeping with previous practices, the assignment built on individual work that was completed for Homework 2. I strongly recommend publishing the course project guidance early in the semester to force cadets to start work early.
- c. **Learning Modality.** In AY23-1, SM484 was taught completely in-person over the entire semester.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** None
- b. **Textbook.** None
- c. **Curriculum.** Leveraged multiple Kahoot! Quizzes to reinforce key learning, and continually highlight areas that cadets were struggling.
- d. **Graded Requirements.** None
- e. **Communications.** All communication occurred over Teams.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None
- b. **Textbook.** Continue to look for interesting new course project topics and case studies. The Vensim Guide that is currently used in class is out of date, and the Vensim Guide is now available online. Restructure the Vensim Tutorial with a basic tutorial for in class, and a more in-depth tutorial for out of class. Review assigned readings and focus in on the important pieces versus reading chapters in their entirety.
- c. **Curriculum.** Reorganize the course to build the basics earlier, there was some confusion among cadets when we jumped into some more complex material early. Add a dedicated review session prior to both WPR 1 and 2. Adjust the project working sessions to be half working session and half instruction, this will allow more sessions to occur earlier in the course. Cadets requested more practice in Vensim early, so look at incorporating more Vensim desk and challenge problems to build more than just the basics. Look at MIT System Dynamics curriculum and try to incorporate both some of that lesson structure and more current examples. Consider incorporating a singular example that builds throughout the semester so that cadets can see how many of the different learning objectives could combine into their project.
- d. **Graded Requirements.**
 - i. **Project:** Consider changing the point allocation for the project so that the final presentation is reduced from 100 points to 50 points while project parts 1 and 2

are increased to 75 points each. Many teams did not devote enough time to the project in the initial stages which caused issues later. Look at restructuring the focus from the report to the model building and the process itself. Consider structuring the due-outs as an iterative process so that the groups go through the SDMP twice, once with broad strokes and then once to add additional detail. This would also enforce simulating early and often to ensure their model functions as desired.

- ii. Homework: Consider bringing homework 4 earlier in the semester as it highlights policy well and they can apply what they learn to the projects. Also consider updating it to a more current policy example.

MADN-SE

AY 23-1 Course Assessment SM484 (14 December 2022)

Enclosure 2: Instructional Memorandum



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

08 September 2022

INSTRUCTIONAL MEMORANDUM
NO. 484-23-1

SYSTEM DYNAMICS SIMULATION

1. **PURPOSE.** The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete SM484: System Dynamics Simulation.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 29 July 2022.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 29 July 2022.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 29 July 2022.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2017.
3. **GENERAL.**
 - a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, practice problem guidance, and criteria for passing the course.
 - b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. This course is a simulation elective for the Systems Engineering, Engineering Management, Operations Research, and System Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Design Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, Complex

Adaptive Systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive. Cadets use properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, produce general estimates of model parameters, and perform an analysis of the results to inform decision-makers and policy outcomes. Topics include applications of System Dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting their design results in both written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers.

- b. Course Strategy. This course consists of four major blocks.
- 1) Block 1, ***Introduction to System Dynamics***, consists of 6 lessons introducing the system dynamics methodology and explains the basic concepts of modeling dynamic systems. One graded modeling homework will introduce students to system dynamics concepts and then link those concepts to the underlying equations in the models. In this block, two labs will be used to reinforce lesson objectives. The first is a dynamics lab that will demonstrate the concepts of dynamic complexity. The second will be a simulation lab lesson with a tutorial and will introduce the simulation software used in this course, Vensim PLE.
 - 2) Block 2, ***System Dynamics Tools***, consists of 7 lessons that extend cadets' abilities to understand and model dynamic systems. This block focuses on the system dynamics basics of causal loop diagrams, stock and flow models, and an understanding of the basic behaviors of dynamic systems. Cadets will begin to use Vensim PLE to model and analyze complex systems. Two graded homework assignments supplement the in-class learning of model structures and simulation tools. This block concludes with a WPR.
 - 3) Block 3, ***Basic Modeling***, consists of 12 lessons focusing on the development and analysis of continuous time models. The block starts the course project part 1, where each cadet communicates a system dynamics problem and a dynamic hypothesis for a topic of his/her choice. This writing will form the initial pool of problems that the cadets will fully model by the end of the course. Cadets will practice their modeling skills in both the classroom and through technical writing and initial model development for their final

projects. Cadets will have multiple lab sessions, culminating in the development of a full stock and flow model in support of their semester project. This block concludes with a WPR.

- 4) Block 4, ***Model Validation & Policy Analysis***, consists of 10 lessons focusing on model testing (verification and validation) and policy analysis. The Promotion simulation exercise will be used as a lab to help cadets better understand how to formally test, describe, and explain policy alternatives with a system dynamics model. Cadets will complete and present their semester project during this block.

c. Course Objectives:

- 1) Apply system dynamics methodology to the solution of large scale, complex problems
- 2) Create models of dynamic systems, operations, and processes
- 3) Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes
- 4) Develop policies and controls to improve systems, operations, and processes
- 5) Interpret the output from continuous time simulations of system dynamics models as an aid for decision making
- 6) Prepare and present the results of system dynamics analysis in oral and written form
- 7) Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results

d. Texts:

- 1) Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000.
- 2) Ventana Systems, Inc, *Vensim User's Guide Version 5*, Ventana Systems, 2007.

e. Additional References. None

f. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and practice problems.

g. Performance Assessment.

- 1) The course consists of 1000 points.
- a) Written Partial Review. (Two at **150 points**, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
 - b) Graded Homework. (Four at **75 points**, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
 - c) Project. (Separate submissions for a total of **300 Points**, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.
 - d) Instructor Points. (**100 points**, Individual Work). These instructor points will be used at the discretion of the individual instructor and can factor in class participation and checks-on-learning.
- 2) Student performance assessment summary:

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	3	9	Homework 1 - Vensim Lab	Individual	75	7.5%
2	9	14	Homework 2 - CLD	Individual	75	7.5%
3	15	22	Homework 3 - Stock and Flow	Individual	75	7.5%
4		16	WPR 1	Individual	150	15.0%
5		29	WPR 2	Individual	150	15.0%
6	28	34	Homework 4 - Fishbanks Policy	Individual	75	7.5%
	7		Course Project	Group	300	30.0%
7		23	<i>Project Part I - CLD</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
8		31	<i>Project Part II - Stock and Flow</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
9		36	<i>Project Presentation</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
10		39	<i>Project Report</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
11		40	<i>Instructor Points</i>	<i>Individual</i>	<i>100</i>	<i>10.0%</i>
Total Points					1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Complete the course with at least 65% of the point total and 50% on the Course Project.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take the WPRs, complete all exercises and projects, and participate in all briefs. Cadets who will miss an event must contact the instructor as soon they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed. Graded Events will be administered IAW USMA and DSE policy—Cadets will report to the designated room for testing. The chosen room will allow Cadets to maintain social distance.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period IAW reference 2.a Department of Systems Engineering Admin Memo 1-1) and may result in disciplinary action. If you incorrectly/fail to include a Cover Page with an Acknowledgement Statement [with each submitted graded event], your assignment will be considered late until proper documentation is submitted.
- d. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).
- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references MUST BE PROPERLY DOCUMENTED. Documentation procedures are outlined in references 2a through 2d.
- f. Additional Instruction. AI is available at mutually agreed upon times between a cadet and the instructor. Cadets are expected to have specific questions for any AI session. General review sessions for groups of cadets prior to the WPRs will not be provided outside of regularly scheduled class periods. AI closes 24 hours prior to homework submission deadlines, so be proactive with your studies and don't wait until the last minute to ask for help. Be prepared to show your Challenge Problems at any AI session.
- g. Challenge Problems. Challenge Problems are assigned along with select reading assignments. These problems are intended to enhance understanding of text material. Although these problems will not always be collected and graded, you are expected to work them, particularly if you are struggling with the material.
- h. Class. You are expected to attend all synchronous lessons. If events arise and teaching must move to a virtual environment, you will attend class via your Instructor's online chosen online platform (Blackboard, Teams, Zoom, etc.). When attending class in a remote status, you must turn on your camera for the duration of the lesson. If you are quarantined due to COVID-19, you will (1) inform your Instructor and (2) attend class via the online platform. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.

- i. Microsoft Teams. All course material, including the syllabus, graded events, and lesson material, will be posted in your course specific MS Team under an appropriately labeled Teams link. Your instructor may choose to post supplemental materials under their specific folder. It is your responsibility to routinely check Teams for announcements and course material.

FOR THE HEAD OF THE DEPARTMENT:

JACOB T LUEDERS
CPT, EN
Course Director

DR. VIKRAM MITTAL
Systems Decision Sciences
Program Director

DISTRIBUTION:

1 each instructor, SM484

1 each cadet, SM484

MADN-SE

AY 23-1 Course Assessment SM484 (14 December 2022)

Enclosure 3: Course Syllabus

	Lsn	Date	Adjacent Events	Title	Assignment (Complete Prior to Class)	Graded Event	Lesson Objectives
Intro to SD	1	15-Aug		Introduction to System Dynamics	Read: Business Dynamics (BD) Sections 1.1 - 1.3 (p 3 - 20) Discussion Board: Challenge Problem (p 14)		Define and describe the field of System Dynamics Describe the applications of System Dynamics Articulate a System Dynamics Problem
	2	17-Aug		Dynamic Complexity	Read: BD Sections 1.3.1 - 1.5 (p 21 - 39)		Articulate a System Dynamics Problem Understand the primary elements of Dynamic Complexity and how it arises Recognize sources of Policy Resistance Design and evaluate policy based on System Dynamics Model
	3	19-Aug		Dynamics Lab: Beer Game	Read: Skim BD Chapter 2 - Focus on one example	HW1 Issued	Articulate a System Dynamics Problem Model the behavior of the system using a management flight simulator Evaluate the effectiveness of your policies
	4	22-Aug		The SD Modeling Process	Read: BD Section 2.5 (p 79-81) and Ch 3 (p 83 - 105)		List and describe the 5 steps of the System Dynamics Modeling Process Articulate a System Dynamics Problem Formulate a Dynamic Hypothesis Understand the role of testing in the System Dynamics Modeling Process Understand how to utilize System Dynamics within the SDP
	5	24-Aug		Introduction to Vensim	Read: Vensim Guide (VG) Chapter 2 (p 11-22) Read: Project Guidance		Create a stock and flow diagram in Vensim Define and Develop Reference Modes Design and evaluate policy based on System Dynamics Model
	6	26-Aug	Ring Wknd	DROP			
	7	29-Aug		Causal Loop Diagrams (CLD)	Read: BD Chapter 5 (p 137-158) Discussion Board: Challenge Problem (p 147)	Project Issued	Create a causal loop diagram Describe the cause and effect relationship in complex systems Describe the difference between causation and correlation Identify link and loop polarity
System Dynamics Tools	8	1-Sep		Structure and Behavior of Dynamic Systems	Read: BD Chapter 4 (p 107-133)		List and describe the fundamental modes of behavior and describe the feedback structure Define and Develop Reference Modes Formulate a Dynamic Hypothesis
	9	7-Sep	Modified (Branch Week)	DROP		HW1 Due by 2359 HW2 Issued	
	10	9-Sep		Causal Loop Diagram Workshop	Read: BD Chapter 5 (p 159-176) Discussion Board: Challenge Problem 2 (p 169) Skim: VG Chapter 4 (p 41-62)		Create a causal loop diagram in Vensim Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
	11	12-Sep		Stocks and Flows	Read: BD Chapter 6 (p 191-210) Read: VG Chapter 5 (p 63 - 74) Discussion Board: Challenge Problem (p 211 - pick one 1, 2, or 3)		Define a Stock Define a Flow Classify an object as a stock or flow Apply the principles of stock and flow diagramming to represent a dynamic system Convert a causal loop diagram to a stock and flow diagram
	12	15-Sep		Stocks and Flows Workshop	Read: https://climate.nasa.gov/news/2365/seven-case-studies-in-carbon-and-climate/ Read: BD Chapter 6 (p 210-230) Review: VG Chapter 5 (p 63-74)		Convert a causal loop diagram to a stock and flow diagram Create a stock and flow diagram in Vensim
	13	19-Sep		Dynamics of Stocks and Flows	Read: BD Chapter 7 (p 231-241) Discussion Board: Challenge Problem (p 239)		Understand the primary elements of Dynamic Complexity and how it arises Apply graphical integration and differentiation Determine Net Flow
	14	21-Sep		DROP		HW2 Due by 2359	
	15	23-Sep		Dynamics of Stocks and Flows Part II	Review: VG Chapters 2 (p 11-22), 4 (p 41-62) , & 5 (p 63 - 74) Read: HW 3 Guidance	HW3 Issued	Describe the cause and effect relationship in complex systems Apply graphical integration and differentiation Determine Net Flow Create a causal loop diagram in Vensim Articulate a System Dynamics Problem
	16	26-Sep		WPR1	WPR 1	WPR 1	
Basic Modeling	17	29-Sep		Dynamics of Simple Structures: Intro	Read: BD Chapter 8 (p 263-280) Discussion Board: Challenge Problem (p 281- part 1)		Understand the characteristics of a first-order, linear feedback system Understand the relationship between a first-order, positive feedback loop and exponential growth Understand the relationship between a first-order, negative feedback loop and goal-seeking behavior
	18	3-Oct		Dynamics of Simple Structures: S-shaped Growth	Read: BD Chapter 9 (p 295 - 309, 323 - 334)		Understand the dynamics and behavior of systems that experience S-Shaped growth Understand how the Logistic Growth model is used to model S-shaped growth Understand the SIR Model Understand the relationship between a nonlinear, first-order system and S-shaped growth
	19	5-Oct		Delays - Material	Read: BD Chapter 11 (p 409-425) Discussion Board: Challenge (p 425 - part 1)		Define a Delay Describe a material delay Describe a pipeline delay Determine the order of a delay Understand the structure and behavior of material delays Model a Material Delay
	20	7-Oct		Delays - Information	Read: BD Chapter 11 (p 426-448); Skim Section 11.6; p. 449-467 optional		Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delays
	21	12-Oct		Dynamics of Growth: Path Dependence	Read: BD Chapter 10 (p 349-406) Discussion Board: Challenge Problem (p 353)		Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delays
	22	14-Oct		Project Working Session		HW3 Due by 2359	Create a System Dynamics Model
	23	17-Oct		Dynamics of Simple Structures: Aging Chains	Read: BD Chapter 12 (p 469 - 497) Discussion Board: Challenge (p 495 - part 3)	Project Part I Due by 2359	Describe an aging chain Understand and structure and behavior of an aging chain, as well as when they are appropriate for use Model an Aging Chain
	24	20-Oct		Modeling Decision Making	Read: BD Chapter 13 (p 513-550)		Understand the principles for modeling decision making Be familiar with common formulations for rate equations Understand the common pitfalls of modeling decision rules that should be avoided
	25	24-Oct		Formulating Nonlinear Relationships	Read: BD Chapter 14 (p 551-595) Read: VG Chapter 8 (p 107-120) Discussion Board: Challenge (p 566)		Understand the guidelines for formulating table functions to capture nonlinear relationships in System Dynamics models Understand the common pitfalls in the specification and use of nonlinear functions in System Dynamics models
	26	26-Oct		Project Modeling Working Session	Read: Project Guidance		Create a System Dynamics Model
	27	28-Oct	Modified	DROP			
	28	31-Oct		Modeling Human Behavior	Read: BD Chapter 15 (pp 597-630)	HW4 Issued	Understand the cognitive limitations of decision makers Incorporate human behavior into a model
	29	3-Nov		WPR 2	WPR 2	WPR 2	
Model Validation & Policy Analysis	30	7-Nov		Model Testing	Read: BD Chapter 21 (p 845 - 891) Read: VG Chapters 14 and 15 (p 193 - 226) Discussion Board: Challenge (p 871)		List the types of assessment tests for dynamic models Produce a calibrated model to given data Conduct goodness of fit test using Vensim
	31	9-Nov		Policy Design and Evaluation Workshop	Review: BD Section 3.5.5 (p 103 - 104) Read: HW 4 Guidance	Project Part II Due by 2359	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	32	14-Nov		Promotion Lab (Part 1)	Read: BD Chapter 12 (p 469-496) Focus: BD p. 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	33	17-Nov		Promotion Lab (Part 2)	Review: BD Chapter 12 (p 469-496) Review: BD p 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	34	21-Nov		Challenges in System Dynamics	Read: System Dynamics and Decision Making in the Context of Dynamically Complex Multi-Dimensional Societal Issues	HW4 Due by 2359	Understand the challenges faced by systems dynmics modelers Understand the complexity of communicating a systems dynamics model
	35	28-Nov		Project Working Session	Review: Project Guidance		
	36	30-Nov		Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	37	2-Dec		Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	38	5-Dec		Project Working Session	Review: Project Guidance		
	39	7-Dec		Course Wrap-up / Feedback	Complete end of course survey online	Project Report Due by 2359	Create a System Dynamics Model Describe the applications of System Dynamics Articulate a System Dynamics Problem Effectively communicate analysis and findings to a decision maker through written reports
	40	9-Dec	Modified (Army-Navy)	DROP			

Enclosure 4: SM484 Course Objective Assessment Table

	Apply system dynamics methodology to the solution of large scale, complex problems	Create models of dynamic systems, operations, and processes	Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes	Develop policies and controls to improve systems, operations, and processes	Interpret the output from continuous time simulations of system dynamics models as an aid for decision making	Prepare and present the results of system dynamics analysis oral and written form	Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results	
Homework 1 - Vensim Lab	1	1	1					80.8%
Homework 2 - CLD	1	1						83.1%
Homework 3 - Stock and Flow	1	1						84.8%
WPR 1	1	1	1		1			78.9%
WPR 2	1	1	1		1			80.2%
Homework 4 - Fishbanks Policy			1	1			1	87.6%
Course Project								
Project Part I - CLD	1	1	1					78.5%
Project Part II - Stock and Flow	1	1	1					88.8%
Project Presentation			1	1	1	1		88.3%
Project Report			1	1	1	1	1	89.3%
Instructor Points - Discussion Board							1	87.2%
	82.2%	82.2%	84.1%	88.4%	84.2%	88.8%	88.0%	
Direct Assessment	3.3	3.3	3.4	3.5	3.4	3.6	3.5	3.4
Course Director Assessment	3.5	3.5	4	4	3	3.5	4	3.6

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AY 23-1 Course Assessment SM484 (14 December 2022)

Enclosure 5: Course End Feedback

TBP



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB E AY2022 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996



MADN-SE

17 December 2021

MEMORANDUM FOR COL Rich Morales, Professor and Department Head, Department of Systems Engineering

SUBJECT: AY22-1 Course Assessment and Final Grade Approval – SM484: System Dynamics Simulation

1. The attached report summarizes the SM484 assessment and curriculum change recommendations from AY22-1.
2. POC for this action is the undersigned.

JAE H KIM
MAJ, FC/49
Course Director, System Dynamics Simulation

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY17		AY18		AY19		AY20		AY21			AY22	
17-1	17-2	18-1	18-2	19-1	19-2	20-1	20-2	21-1	20-2	21-3	22-1	21-2
X		X		X		X		X		X	X	

b. Enrollments.

Academic Term	Enrollment
17-1	49
18-1	42
19-1	51
20-1	67
20-3	20
21-1	46
22-1	52

Table 1: Historical Enrollment

c. Historical Course Final Grade Averages.

Academic Term	Course Average
17-1	85.9%
18-1	86.1%
19-1	84.7%
20-1	86.3%
20-3	91.5%
21-1	87.1%
22-1	84.0%

Table 2: Historical Course Average

d. **Standard Grade Distribution Table**

Grade	Standard Cut	Recommended Cut	# CDTs Std (Rec.)	% of Course
A+	97.50%	93.43%	0 (2)	3.8%
A	92.50%	90.54%	3 (6)	11.5%
A-	90.00%	88.04%	6 (6)	11.5%
B+	87.50%	86.66%	5 (7)	13.5%
B	82.50%	82.24%	18 (12)	23.1%
B-	80.00%	80.00%	10 (9)	17.3%
C+	77.50%	77.5%	5 (5)	9.6%
C	72.50%	72.50%	4 (1)	1.9%
C-	70.00%	70.00%	1 (4)	7.7%
D	65.00%	65.00%	0 (0)	0%
F	< 65.0%	< 65.0%	0 (0)	0%

Table 3: Standard Grade Distribution Table

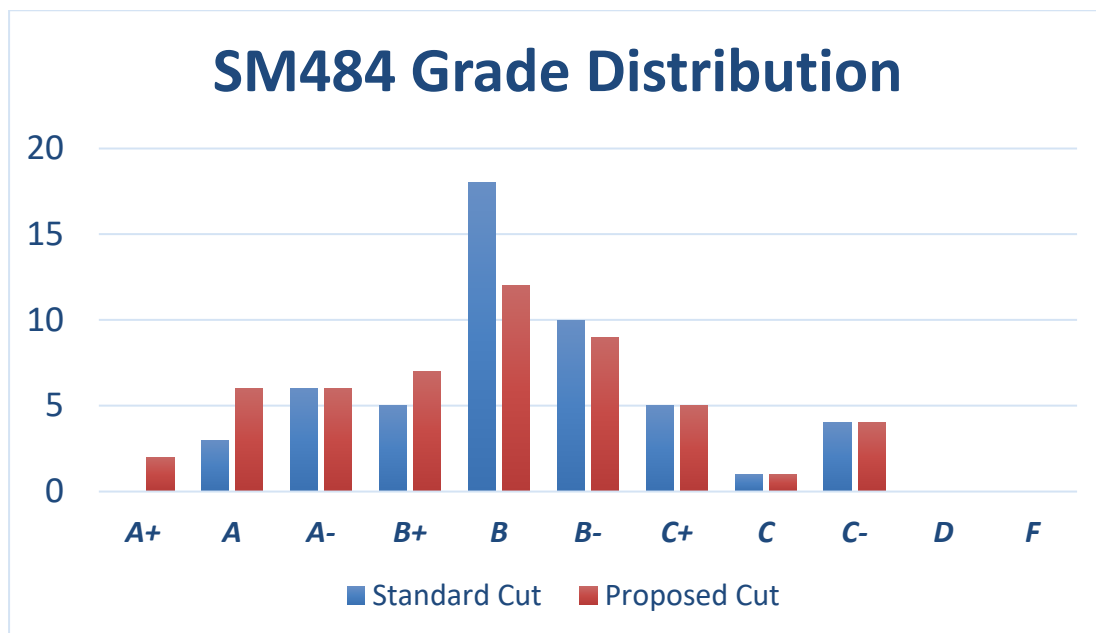


Figure 1: Standard Grade Distribution

2. **Course Strategy.**

a. This course is a simulation elective for the Engineering Management, Systems Engineering, Operations Research, and Systems and Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These

systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive.

b. Cadets will use the properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, implement these models, and perform an analysis of the results. Topics include applications of systems dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting design results in written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers. Cadets will spend several lessons in a computer lab environment.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback.** Enclosure 5

2. Course Content.

a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge through the identification and application of knowledge, skills and behaviors. This course teaches the tools necessary to understand System Dynamics concepts, tools, and models. Block I focused on an introduction to System Dynamics. Block II focused on System Dynamics Tools. Block III focused on System Dynamics modeling, Block IV focused on policy analysis. These blocks adequately covered the topics and course objectives. The cadets enjoyed the applications of System Dynamics block as it provided cadets with a means to gain a deeper understanding of System Dynamics as they see potential applications.

b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provides an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. Course Processes.

a. **Textbook.** Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000. *Business Dynamics: Thinking and Modeling for a Complex World* by John Sterman is the foremost text book on System Dynamics. Recommend continued use of this text book for future classes in conjunction with appropriate case studies.

b. **Case Studies.** For this semester, we continued the use of one of the case studies from last year (Fishbanks), but adjusted the course project, forcing the cadets to explore the application of System Dynamics to one of 22 global challenges facing the United Nations. Additionally, we

used the Fishbanks case study as a means to teach cadets how to develop policies and conduct policy analysis. The two separate exercises reinforced each other and fostered growth that would allow students to develop models for a case of their choosing for the final project. For WPR questions, graded homework, and in class labs, it is important to seek current case studies through formal and informal channels (e.g. System Dynamics Review and periodicals/news media, respectively) to ensure the cadets understand the relevance of System Dynamics and how it can be applied to analyze and develop solutions to real world problems.

c. **Graded Requirements.** The graded requirements in SM484 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Based on practices in AY21-1, we maintained four graded homework assignments to assess students learning in a distributed environment. Term 22-1 graded events in the course consisted of the following:

- i. Written Partial Review. (Two at 150 points, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
- ii. Graded Homework. (Four at 75 points, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
- iii. Project. (Separate submissions for a total of 300 Points, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.

The 11 graded events in 22-1 are summarized in the following table:

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	3	8	Homework 1 - Vensim Lab	Individual	75	7.5%
2	8	14	Homework 2 - CLD	Individual	75	7.5%
3	13	21	Homework 3 - Stock and Flow	Individual	75	7.5%
4		15	WPR 1	Individual	150	15.0%
5		28	WPR 2	Individual	150	15.0%
6	30	33	Homework 4 - Fishbanks Policy	Individual	75	7.5%
	6		Course Project	Group	300	30.0%
7		23	Project Part I - CLD	Group	50	5.0%
8		30	Project Part II - Stock and Flow	Group	50	5.0%
9		35	Project Presentation	Group	100	10.0%
10		39	Project Report	Group	100	10.0%
11	1	38	Instructor Points	Individual	100	10.0%

Total Points

1000

4. Course Pedagogy.

- a. **Feedback Emphasis:** The course is centered on the completion of deliverables in each block with nearly all deliverables building upon previous submissions. This is meant to reinforce learning by allowing students to adjust previous submissions. It also takes some focus off of grades and puts emphasis on growth. As Sterman writes in *Business Dynamics*, "Just as dynamics arise from feedback, so too all learning depends on feedback."
- b. **Group Work:** At the beginning of the semester, cadets were allowed to select project teams. In order to demonstrate the applicability of System Dynamics to a broad range of problems, the course project allowed for cadets to choose the topic. However, the topic had to come from one of 22 case studies found on the United Nations website. In keeping with previous practices, the assignment built on individual work that was completed for Homework 2. I strongly recommend publishing the course project guidance early in the semester to force cadets to start work early.
- c. **Learning Modality.** In AY22-1, SM484 was taught completely in-person over the entire semester.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Assessment of Changes to Current Academic Term (based on previous Course Assessments).**
 - a. **Course Objectives.** None
 - b. **Textbook.** Used new case studies for course project, such as a COVID related problem set for HW 1. This was very helpful in demonstrating the applicability of System Dynamics to a broad range of problems.
 - c. **Curriculum.** Minor changes were made to the syllabus to incorporate previous feedback. The Beer Game lesson was reintroduced to explore the emergent behavior of systems. Allocate one lesson to an in-class Vensim tutorial (Lesson 2). We also allocated one lesson to an in-class Vensim tutorial (Lesson 5) and devoted one lesson to the System Dynamics Modeling Process (Lesson 4) as recommended by the previous course director.
 - d. **Graded Requirements.** The Aging Chain lab replaced the Fishbanks problem.
 - e. **Communications.** All communication occurred over Teams. This worked out well, and cadets
2. **Proposed Change for Next Academic Term.**
 - a. **Course Objectives.** None
 - b. **Textbook.** Continue to look for interesting new course project topics and case studies. The Vensim Guide that is currently used in class is out of date, so I recommend that a new Vensim tutorial be provided when available for the new Vensim version that was published this year.
 - c. **Curriculum.** Consider using a Lesson Drop to spend an entire class reviewing and working problems prior to WPR 1. WPR 1 results for the past few years have been very low, so finding the time to work actual problems similar to the ones on the WPR will better prepare the cadets.
 - d. **Graded Requirements.** Consider changing the point allocation for the project so that the final presentation is reduced from 100 points to 50 points while project parts 1 and 2 are increased to 75 points each. Many teams did not devote enough time to the project in the initial stages which caused issues later.

Enclosure 2: Instructional Memorandum

INSTRUCTIONAL MEMORANDUM
NO. 484-22-1

SYSTEM DYNAMICS SIMULATION

1. PURPOSE. The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete SM484: System Dynamics Simulation.

2. REFERENCES.

- a. Department of Systems Engineering Administrative Memo I-I, "General Administrative Instructions," 10 July 2021.
- b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 10 July 2021.
- c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 10 July 2021.
- d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2017.

3. GENERAL.

- a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, practice problem guidance, and criteria for passing the course.
- b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

- a. Course Scope. This course is a simulation elective for the Systems Engineering, Engineering Management, Operations Research, and System Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Design Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive. Cadets use properties of dynamic systems and analytical techniques to design continuous models of

complex systems or processes, produce general estimates of model parameters, and perform an analysis of the results to inform decision-makers and policy outcomes. Topics include applications of System Dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting their design results in both written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers.

b. Course Strategy. This course consists of four major blocks.

- 1) Block 1, **Introduction to System Dynamics**, consists of 4 lessons introducing the system dynamics methodology and explains the basic concepts of modeling dynamic systems. One graded modeling homework will introduce students to system dynamics concepts and then link those concepts to the underlying equations in the models. One simulation lab lesson and a tutorial will introduce the simulation software used in this course, Vensim PLE.
- 2) Block 2, **System Dynamics Tools**, consists of 9 lessons that extend cadets' abilities to understand and model dynamic systems. This block focuses on the system dynamics basics of causal loop diagrams, stock and flow models, and an understanding of the basic behaviors of dynamic systems. Cadets will begin to use Vensim PLE to model and analyze complex systems. Two graded homework assignments supplement the in-class learning of model structures and simulation tools. This block concludes with a WPR.
- 3) Block 3, **Basic Modeling**, consists of 12 lessons focusing on the development and analysis of continuous time models. The block starts the course project part 1, where each cadet communicates a system dynamics problem and a dynamic hypothesis for a topic of his/her choice. This writing will form the initial pool of problems that the cadets will fully model by the end of the course. Cadets will practice their modeling skills in both the classroom and through technical writing and initial model development for their final projects. Cadets will have multiple lab sessions, culminating in the development of a full stock and flow model in support of their semester project. This block concludes with a WPR.
- 4) Block 4, **Model Validation & Policy Analysis**, consists of 10 lessons focusing on model testing (verification and validation) and policy analysis. The Fishbanks simulation exercise will be used as a lab to help cadets' better

understand how to formally test, describe, and explain policy alternatives with a system dynamics model. Cadets will complete and present their semester project during this block.

c. Course Objectives:

- 1) Apply system dynamics methodology to the solution of large scale, complex problems
- 2) Create models of dynamic systems, operations, and processes
- 3) Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes
- 4) Develop policies and controls to improve systems, operations, and processes
- 5) Interpret the output from continuous time simulations of system dynamics models as an aid for decision making
- 6) Prepare and present the results of system dynamics analysis in oral and written form
- 7) Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results

d. Texts:

- 1) Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000.
- 2) Ventana Systems, Inc, *Vensim User's Guide Version 5*, Ventana Systems, 2007.

e. Additional References. None

f. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and practice problems.

g. Performance Assessment.

- 1) The course consists of 1000 points.
 - a) Written Partial Review. (Two at **150 points**, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.

- b) Graded Homework. (Four at **75 points**, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
- c) Project. (Separate submissions for a total of **300 Points**, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.
- d) Instructor Points. (**100 points**, Individual Work). These instructor points will be used to assess cadet's participation in blackboard discussions and attendance and participation in synchronous lessons.

2) Student performance assessment summary:

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	3	8	Homework 1 - Vensim Lab	Individual	75	7.5%
2	8	14	Homework 2 - CLD	Individual	75	7.5%
3	13	21	Homework 3 - Stock and Flow	Individual	75	7.5%
4		15	WPR 1	Individual	150	15.0%
5		28	WPR 2	Individual	150	15.0%
6			Homework 4 - Fishbanks Policy	Individual	75	7.5%
	26		Course Project	Group	300	30.0%
7	32		<i>Project Part I - CLD</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
8	6		<i>Project Part II - Stock and Flow</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
9		18	<i>Project Presentation</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
10			<i>Project Report</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
11	30		Instructor Points	Individual	100	10.0%
Total Points					1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Complete the course with at least 65% of the point total and 50% on the Course Project.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take the WPRs, complete all exercises and projects, and participate in all briefs. Cadets who will miss an event must contact the instructor as soon they know

about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed.

- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (10% per 24 hour period) and may result in disciplinary action. If you incorrectly/fail to include a Cover Page with an Acknowledgement Statement [with each submitted graded event], your assignment will be considered late until proper documentation is submitted.
- d. Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).
- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references MUST BE PROPERLY DOCUMENTED. Documentation procedures are outlined in references 2a through 2f.
- f. Additional Instruction. AI is available at mutually agreed upon times between a cadet and the instructor. Cadets are expected to have specific questions for any AI session. General review sessions for groups of cadets prior to the WPRs will not be provided outside of regularly scheduled class periods. Be prepared to show your Challenge Problems at any AI session.
- g. Challenge Problems. Challenge Problems are assigned along with select reading assignments. These problems are intended to enhance understanding of text material. Although these problems will not always be collected and graded, you are expected to work them, particularly if you are struggling with the material. These challenge problems will provide a basis for the discussion forms on blackboard.
- h. Class. You are expected to attend all synchronous lessons. If events arise and teaching must move to a virtual environment, you will attend class via your Instructor's online chosen online platform (Blackboard, Teams, Zoom, etc.). When attending class in a remote status, you must turn on your camera for the duration of the lesson. If you are quarantined due to COVID-19, you will (1) inform your Instructor and (2) attend class via the online platform. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.
- i. Microsoft Teams. All course material, including the syllabus, graded events, and lesson material, will be posted in your course specific MS Team under an appropriately labeled Teams link. Your instructor may choose to post supplemental materials under their specific folder. It is your responsibility to routinely check Teams for announcements and course material.

Enclosure 3: Course Syllabus

	Ln	Date	Adjacent Events	Title	Assignment (Complete Prior to Class)	Graded Event	Lesson Objectives
Intro to SD	1	17-Aug		Introduction to System Dynamics	Read: Business Dynamics (BD) Sections 1.1 - 1.5 (p 3 - 39) Discussion Board: Challenge Problem (p 14)		Define and describe the field of System Dynamics Describe the applications of System Dynamics Articulate a System Dynamics Problem
	2	19-Aug		Dynamic Complexity	Read: BD Sections 1.3.1 - 1.5 (p 21 - 39) Read: Project Guidance	Project Issued	Articulate a System Dynamics Problem Understand the primary elements of Dynamic Complexity and how it arises Recognize sources of Policy Resistance Design and evaluate policy based on System Dynamics Model
	3	21-Aug		The SD Modeling Process -OR- The System Dynamics Beer Game	Read: BD Section 2.5 (p 79-81) and Ch 3 (p 83 - 105)	HW1 Issued	List and describe the 5 steps of the System Dynamics Modeling Process Articulate a System Dynamics Problem Formulate a Dynamic Hypothesis Understand the role of testing in the System Dynamics Modeling Process Understand how to utilize System Dynamics within the SDP
	4	25-Aug		Introduction to Vensim	Read: Vensim Guide (VG) Chapter 2 (p 11-22)		Create a stock and flow diagram in Vensim Define and Develop Reference Modes Design and evaluate policy based on System Dynamics Model
	5	27-Aug		Causal Loop Diagrams (CLD)	Read: BD Chapter 5 (p 137-158) Discussion Board: Challenge Problem (p 147)		Create a causal loop diagram Describe the cause and effect relationship in complex systems Describe the difference between causation and correlation Identify link and loop polarity
System Dynamics Tools	6	31-Aug	Ring Wknd	Structure and Behavior of Dynamic Systems	Read: BD Chapter 4 (p 107-133)		List and describe the fundamental modes of behavior and describe the feedback structure Define and Develop Reference Modes Formulate a Dynamic Hypothesis
	7	2-Sep	Modified	Causal Loop Diagram Workshop	Read: BD Chapter 5 (p 159-176) Discussion Board: Challenge Problem 2 (p 169) Skin: VG Chapter 4 (p 41-62)	HW2 Issued	Create a causal loop diagram in Vensim Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
	8	4-Sep	Modified			HW1 Due by 2359	
	9	8-Sep		Homework 2 Working Session	Read: HW 2 Guidance		Create a causal loop diagram in Vensim Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
	10	10-Sep		Stocks and Flows	Read: BD Chapter 6 (p 191-210) Read: VG Chapter 5 (p 63 - 74) Discussion Board: Challenge Problem (p 211 - pick one 1, 2, or 3)		Define a Stock Define a Flow Classify an object as a stock or flow Apply the principles of stock and flow diagramming to represent a dynamic system Convert a causal loop diagram to a stock and flow diagram
	11	14-Sep		Stocks and Flows Workshop	https://climate.nasa.gov/news/2305/seven-case-studies-in-carbon-and-climate/ Read: BD Chapter 6 (p 210-230) Review: VG Chapter 5 (p 63-74)		Convert a causal loop diagram to a stock and flow diagram Create a stock and flow diagram in Vensim
	12	17-Sep		Dynamics of Stocks and Flows	Read: BD Chapter 7 (p 231-241) Discussion Board: Challenge Problem (p 239)		Understand the primary elements of Dynamic Complexity and how it arises Apply graphical integration and differentiation Determine Net Flow
	13	21-Sep		Dynamics of Stocks and Flows Part II	Review: VG Chapters 2 (p 11-22), 4 (p 41-62), & 5 (p 63 - 74) Read: HW 3 Guidance	HW2 Due by 2359 HW3 Issued	Describe the cause and effect relationship in complex systems Apply graphical integration and differentiation Determine Net Flow Create a causal loop diagram in Vensim Articulate a System Dynamics Problem
	14	25-Sep					
	15	29-Sep		WPR1	WPR 1	WPR 1	
	16	1-Oct		Dynamics of Simple Structures: Intro	Read: BD Chapter 8 (p 263-280) Discussion Board: Challenge Problem (p 281- part 1)		Understand the characteristics of a first-order, linear feedback system Understand the relationship between a first-order, positive feedback loop and exponential growth Understand the relationship between a first-order, negative feedback loop and goal-seeking behavior
	17	5-Oct		Dynamics of Simple Structures: S-shaped Growth	Read: BD Chapter 9 (p 295 - 309, 323 - 334)		Understand the dynamics and behavior of systems that experience S-Shaped growth Understand how the Logistic Growth model is used to model S-shaped growth Understand the SIR Model Understand the relationship between a nonlinear, first-order system and S-shaped growth
	18	7-Oct		Delays - Material	Read: BD Chapter 11 (p 409-425) Discussion Board: Challenge (p 425 - part 1)		Define a Delay Describe a material delay Describe a pipeline delay Determine the order of a delay Understand the structure and behavior of material delays Model a Material Delay
	19	9-Oct	Columbus Day Weekend				
Basic Modeling	20	14-Oct		Delays - Information	Read: BD Chapter 11 (p 426-448); Skim Section 11.6; p. 449-467 optional		Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delays
	21	16-Oct		Dynamics of Growth: Path Dependence	Read: BD Chapter 10 (p 349-406) Discussion Board: Challenge Problem (p 353)	HW3 Due by 2359	Understand the characteristics and behaviors of multiple-loop systems Understand the concept of loop dominance and how it impacts behavior in multiple-loop systems Understand the dynamics and behaviors that experience path dependence Understand how positive feedback drives path dependence
	22	20-Oct		Project Working Session	Read: Project Guidance		
	23	22-Oct		Dynamics of Simple Structures: Aging Chains	Read: BD Chapter 12 (p 469 - 497) Discussion Board: Challenge (p 495 - part 3)	Project Part I Due by 2359	Describe an aging chain Understand and structure and behavior of an aging chain, as well as when they are appropriate for use Model an Aging Chain
	24	26-Oct		Modeling Decision Making	Read: BD Chapter 13 (p 513-550)		Understand the principles for modeling decision making Be familiar with common formulations for rate equations Understand the common pitfalls of modeling decision rules that should be avoided
	25	28-Oct		Formulating Nonlinear Relationships	Read: BD Chapter 14 (p 551-595) Read: VG Chapter 8 (p 107-120) Discussion Board: Challenge (p 566)		Understand the guidelines for formulating table functions to capture nonlinear relationships in System Dynamics models Understand the common pitfalls in the specification and use of nonlinear functions in System Dynamics models
	26	30-Oct		Project Modeling Working Session	Read: Project Guidance		Create a System Dynamics Model
	27	2-Nov		Modeling Human Behavior	Read: BD Chapter 15 (pp 597-630)		Understand the cognitive limitations of decision makers Incorporate human behavior into a model
	28	6-Nov		WPR 2	WPR 2	WPR 2	
	29	9-Nov		Model Testing	Read: BD Chapter 21 (p 845 - 891) Read: VG Chapters 14 and 15 (p 193 - 226) Discussion Board: Challenge (p 871)		List the types of assessment tests for dynamic models Produce a calibrated model to given data Conduct goodness of fit test using Vensim
Model Validation & Policy Analysis	30	12-Nov		Policy Design and Evaluation Workshop	Review: BD Section 3.5.5 (p 103 - 104) Read: HW 4 Guidance	Project Part II Due by 2359 HW4 Issued	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	31	16-Nov		Promotion Lab (Part 1)	Read: BD Chapter 12 (p 469-496) Focus: BD p. 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	32	18-Nov		Promotion Lab (Part 2)	Review: BD Chapter 12 (p 469-496) Review: BD p 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	33	20-Nov		Challenges in System Dynamics	Read: System Dynamics and Decision Making in the Context of Dynamically Complex Multi-Dimensional Societal Issues	HW4 Due by 2359	Understand the challenges faced by systems dynamics modelers Understand the complexity of communicating a systems dynamics model
	34	23-Nov		Project Working Session	Review: Project Guidance		
	35	25-Nov					
	36	30-Nov		Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	37	1-Dec		Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	38	4-Dec		Course Closeout	Course Closeout	Project Report Due 7 DEC by 2359	Create a System Dynamics Model Describe the applications of System Dynamics Articulate a System Dynamics Problem Effectively communicate analysis and findings to a decision maker through written reports

Enclosure 4: SM484 Course Objective Assessment Table

	Apply system dynamics methodology to the solution of large scale, complex problems	Create models of dynamic systems, operations, and processes	Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes	Develop policies and controls to improve systems, operations, and processes	Interpret the output from continuous time simulations of system dynamics models as an aid for decision making	Prepare and present the results of system dynamics analysis oral and written form	Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results	
Homework 1 - Vensim Lab	1	1	1					80.8%
Homework 2 - CLD	1	1						83.1%
Homework 3 - Stock and Flow	1	1						84.8%
WPR 1	1	1	1		1			78.9%
WPR 2	1	1	1		1			80.2%
Homework 4 - Fishbanks Policy			1	1			1	87.6%
Course Project								
Project Part I - CLD	1	1	1					78.5%
Project Part II - Stock and Flow	1	1	1					88.8%
Project Presentation			1	1	1	1		88.3%
Project Report			1	1	1	1	1	89.3%
Instructor Points - Discussion Board							1	87.2%
	82.2%	82.2%	84.1%	88.4%	84.2%	88.8%	88.0%	
Direct Assessment	3.3	3.3	3.4	3.5	3.4	3.6	3.5	3.4
Course Director Assessment	3.5	3.5	4	4	3	3.5	4	3.6

MADN-SE

AY 22-1 Course Assessment SM484 (17 December 2021)

Enclosure 5: Course End Feedback

TBP



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB F

AY2021 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

9 December 2020

MEMORANDUM FOR LTC Jim Schreiner, Program Director, Department of Systems Engineering,
United States Military Academy.

SUBJECT: AY21-1 Course Assessment – SM484: System Dynamics Simulation

1. The attached report summarizes the SM484 assessment and curriculum change recommendations from AY21-1.
2. POC for this action is the undersigned.

ARTHUR J. MIDDLEBROOKS
CPT, AR/49
Course Director, System Dynamics Simulation

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

AY16		AY17		AY18		AY19		AY20			AY21	
16-1	16-2	17-1	17-2	18-1	18-2	19-1	19-2	20-1	20-2	20-3	21-1	21-2
X		X		X		X		X		X	X	

b. Enrollments.

Academic Term	Enrollment
16-1	80
17-1	49
18-1	42
19-1	51
20-1	67
20-3	20
21-1	46

c. Historical Course Final Grade Averages.

Academic Term	Course Average
16-1	83.2%
17-1	85.9%
18-1	86.1%
19-1	84.7%
20-1	86.3%
20-3	91.5%
21-1	87.1%

2. Course Strategy.

a. This course is a simulation elective for the Engineering Management, Systems Engineering, Operations Research, and Systems and Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive.

b. Cadets will use the properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, implement these models, and perform an analysis of the results. Topics include applications of systems dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock

and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting design results in written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers. Cadets will spend several lessons in a computer lab environment.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback.** Enclosure 5

2. Course Content.

- a. **Body of Knowledge.** The course objectives adequately cover the body of knowledge through the identification and application of knowledge, skills and behaviors. This course teaches the tools necessary to understand System Dynamics concepts, tools, and models. Block I focused on an introduction to System Dynamics. Block II focused on System Dynamics Tools. Block III focused on System Dynamics modeling, Block IV focused on policy analysis. These blocks adequately covered the topics and course objectives. The cadets enjoyed the applications of System Dynamics block as it provided cadets with a means to gain a deeper understanding of System Dynamics as they see potential applications.
- b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provides an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. Course Processes.

- a. **Textbook.** Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000. *Business Dynamics: Thinking and Modeling for a Complex World* by John Sterman is the foremost text book on System Dynamics. Recommend continued use of this text book for future classes in conjunction with appropriate case studies.
- b. **Case Studies.** For this semester, we continued the use of one of the case studies from last year (Fishbanks), but adjusted the course project, forcing the cadets to explore the application of System Dynamics to one of 22 global challenges facing the United Nations. Additionally, we used the Fishbanks case study as a means to teach cadets how to develop policies and conduct policy analysis. The two separate exercises reinforced each other and fostered growth that would allow students to develop models for a case of their choosing for the final project. For WPR questions, graded homework, and in class labs, it is important to seek current a case studies through formal and informal channels (e.g. System Dynamics Review and periodicals/news media, respectively) to ensure the cadets understand the relevance of System Dynamics and how it can be applied to analyze and develop solutions to real world problems.

AY 21-1 Course Assessment SM484 (9 December 2020)

c. **Graded Requirements.** The graded requirements in SM484 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Based on practices in AY20-3, we maintained four graded homework assignments to assess students learning in a distributed environment. Term 21-1 graded events in the course consisted of the following:

- i. Written Partial Review. (Two at 150 points, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
- ii. Graded Homework. (Four at 75 points, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
- iii. Project. (Separate submissions for a total of 300 Points, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.

The 11 graded events in 21-1 are summarized in the following table:

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	3	8	Homework 1 - Vensim Lab	Individual	75	7.5%
2	7	13	Homework 2 - CLD	Individual	75	7.5%
3	13	21	Homework 3 - Stock and Flow	Individual	75	7.5%
4		15	WPR 1	Individual	150	15.0%
5		28	WPR 2	Individual	150	15.0%
6	30	33	Homework 4 - Fishbanks Policy	Individual	75	7.5%
	2		Course Project	Group	300	30.0%
7		23	<i>Project Part I - CLD</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
8		30	<i>Project Part II - Stock and Flow</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
9		35	<i>Project Presentation</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
10		38	<i>Project Report</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
11	1	38	Instructor Points	Individual	100	10.0%
Total Points					1000	

4. Course Pedagogy.

- a. **Feedback Emphasis:** The course is centered on the completion of deliverables in each block with nearly all deliverables building upon previous submissions. This is meant to reinforce learning by allowing students to adjust previous submissions. It also takes some focus off of grades and puts emphasis on growth. As Sterman writes in *Business Dynamics*, "Just as dynamics arise from feedback, so too all learning depends on feedback."
- b. **Group Work:** At the beginning of the semester, cadets were assigned group project partners, based on Major CQPA entering the semester. In order to demonstrate the applicability of System Dynamics to a broad range of problems, the course project allowed for cadets to choose the topic. However, the topic had to come from one of 22 case studies

found on the United Nations website. In keeping with previous practices, the assignment built on individual work that completed for Homework 2. I strongly recommend publishing the course project guidance early in the semester in order to force cadets to start work early.

- c. **Learning Modality.** In AY21-1, SM484 was taught in a hybrid fashion—alternating between in-person and online modes, based on classroom restrictions. Anecdotally, cadets very much prefer the in-person lessons, as it allows them more access to instructor assistance with modeling in Vensim. One other notable challenge with teaching this course remotely is the reduced capacity for board work. To mitigate this issue, some instructors created Microsoft Teams assignments using a blank PowerPoint. The cadets would then complete the exercise, submit, and then discuss during class. I recommend sustaining this practice, especially early in the course when building Causal Loop Diagrams.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** None
- b. **Textbook.** Used new case studies for course project. This was very helpful in demonstrating the applicability of System Dynamics to a broad range of problems.
- c. **Curriculum.** We made minor changes to the syllabus to support recommendations from previous terms as we included several lab lessons to develop Vensim skills and address project development. Due to COVID-19, lessons 39 and 40 were cancelled. This resulted in minimal issues for the course, but it did move up the submission date of the project presentations.
- d. **Graded Requirements.** While we utilized the Fishbanks lab for Homework #4, we changed the graded lab to an Aging Chain exercise. Given the broad application of the aging chain model, especially for many of the course projects, we wanted to ensure this topic received additional focus.
- e. **Teams/Blackboard.** Based on instructor preference, two sections were taught using Teams, with the other using Blackboard. Every lesson involved an online component on either medium. The course maintained a course-wide Blackboard page for announcements and assignments. Cadets appreciated assignments being due at midnight versus 1600 as it afforded them a little more time to work on the project.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None
- b. **Textbook.** Continue to find new course project topics and case studies. I recommend also reviewing the *System Dynamics* textbook on SharePoint, which was published in 2017. This could act as a supplement or replacement for Sterman's text.
- c. **Curriculum.** While cancelled this year due to COVID, I recommend playing the Beer Game to explore the emergent behavior of systems. Allocate one lesson to an in-class Vensim tutorial (Lesson 2). Ensure one lesson is devoted to the System Dynamics Modeling Process.
- d. **Graded Requirements.** Consider updating the Fishbanks lab to a new, more relevant case. Stress the applicability of Aging Chains to a variety of problems.
- e. **Other.** Use one digital medium—Teams or Blackboard—for course-wide announcements and assignments. The system has a lot of capability that can help in a modeling class like SD. I recommend recording all lessons so cadets can watch outside of class.

Enclosure 2: Instructional Memorandum

INSTRUCTIONAL MEMORANDUM

NO. 484-20-3

SYSTEM DYNAMICS SIMULATION

1. PURPOSE. The purpose of this memorandum is to provide each cadet with the administrative details necessary to successfully complete SM484: System Dynamics Simulation.

2. REFERENCES.

- a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 8 July 2020.
- b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 8 July 2020.
- c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 8 July 2020.
- d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2017.
- e. Course Website can be found at: <https://usma.blackboard.com/>

3. GENERAL.

- a. This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, practice problem guidance, and criteria for passing the course.
- b. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.

4. SPECIFIC.

- a. Course Scope. This course is a simulation elective for the Systems Engineering, Engineering Management, Operations Research, and System Decision Sciences (SDS) majors. Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Design Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive. Cadets use properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, produce general estimates of model parameters, and

perform an analysis of the results to inform decision-makers and policy outcomes. Topics include applications of System Dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting their design results in both written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers.

b. Course Strategy. This course consists of four major blocks.

- 1) Block 1, ***Introduction to System Dynamics***, consists of 4 lessons introducing the system dynamics methodology and explains the basic concepts of modeling dynamic systems. One graded modeling homework will introduce students to system dynamics concepts and then link those concepts to the underlying equations in the models. One simulation lab lesson and a tutorial will introduce the simulation software used in this course, Vensim PLE.
- 2) Block 2, ***System Dynamics Tools***, consists of 9 lessons that extend cadets' abilities to understand and model dynamic systems. This block focuses on the system dynamics basics of causal loop diagrams, stock and flow models, and an understanding of the basic behaviors of dynamic systems. Cadets will begin to use Vensim PLE to model and analyze complex systems. Two graded homework assignments supplement the in-class learning of model structures and simulation tools. This block concludes with a WPR.
- 3) Block 3, ***Basic Modeling***, consists of 12 lessons focusing on the development and analysis of continuous time models. The block starts the course project part 1, where each cadet communicates a system dynamics problem and a dynamic hypothesis for a topic of his/her choice. This writing will form the initial pool of problems that the cadets will fully model by the end of the course. Cadets will practice their modeling skills in both the classroom and through technical writing and initial model development for their final projects. Cadets will have multiple lab sessions, culminating in the development of a full stock and flow model in support of their semester project. This block concludes with a WPR.
- 4) Block 4, ***Model Validation & Policy Analysis***, consists of 10 lessons focusing on model testing (verification and validation) and policy analysis. The Fishbanks simulation exercise will be used as a lab to help cadets' better understand how to formally test, describe, and explain policy alternatives with a system dynamics model. Cadets will complete and present their semester project during this block.

c. Course Objectives:

- 1) Apply system dynamics methodology to the solution of large scale, complex problems
 - 2) Create models of dynamic systems, operations, and processes
 - 3) Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes
 - 4) Develop policies and controls to improve systems, operations, and processes
 - 5) Interpret the output from continuous time simulations of system dynamics models as an aid for decision making
 - 6) Prepare and present the results of system dynamics analysis in oral and written form
 - 7) Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results
- d. Texts:
- 1) Sterman, John D. *Business Dynamics: Thinking and Modeling for a Complex World*, Boston: Irwin McGraw-Hill, 2000.
 - 2) Ventana Systems, Inc, *Vensim User's Guide Version 5*, Ventana Systems, 2007.
- e. Additional References. None
- f. Lesson Assignments. See Enclosure 1 (Syllabus) for specific readings and practice problems.
- g. Performance Assessment.
- 1) The course consists of 1000 points.
 - a) Written Partial Review. (Two at **150 points**, Individual Work). The purpose of this exercise is to reinforce those concepts taught during the course.
 - b) Graded Homework. (Four at **75 points**, Individual Work). The purpose of these assignments is to assess cadets' understanding and ability to apply the concepts taught in their respective blocks.
 - c) Project. (Separate submissions for a total of **300 Points**, Individual and Group Work). The purpose of the group project is for cadets to model a complex system using System Dynamics tools to address a specific problem and to assess the cadets' understanding and ability to perform the basic tasks associated with system dynamics modeling. Additional information and guidance will be provided during the course of the semester.

- d) Instructor Points. (**100 points**, Individual Work). These instructor points will be used to assess cadet's participation in blackboard discussions and attendance and participation in synchronous lessons.

2) Student performance assessment summary:

Event	Lesson Issued	Lesson Due	Event Title	Eval Type	Course Points	% of Total
1	3	8	Homework 1 - Vensim Lab	Individual	75	7.5%
2	7	13	Homework 2 - CLD	Individual	75	7.5%
3	13	21	Homework 3 - Stock and Flow	Individual	75	7.5%
4		15	WPR 1	Individual	150	15.0%
5		28	WPR 2	Individual	150	15.0%
6	30	33	Homework 4 - Fishbanks Policy	Individual	75	7.5%
	2		Course Project	Group	300	30.0%
7		23	<i>Project Part I - CLD</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
8		30	<i>Project Part II - Stock and Flow</i>	<i>Group</i>	<i>50</i>	<i>5.0%</i>
9		35	<i>Project Presentation</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
10		38	<i>Project Report</i>	<i>Group</i>	<i>100</i>	<i>10.0%</i>
11	1	38	Instructor Points	Individual	100	10.0%
Total Points					1000	

5. ADMINISTRATIVE GUIDANCE.

- Minimum Requirements. Complete the course with at least 65% of the point total and 50% on the Course Project.
- Graded Events. Unless there are exceptional circumstances, cadets will take the WPRs, complete all exercises and projects, and participate in all briefs. Cadets who will miss an event must contact the instructor as soon they know about the conflict. It is the cadet's responsibility to coordinate with the instructor to make up any material/requirements missed.
- Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24 hour period) and may result in disciplinary action. If you incorrectly/fail to include a Cover Page with an Acknowledgement Statement [with each submitted graded event], your assignment will be considered late until proper documentation is submitted.
- Final Grades. Final grades will be awarded IAW with reference 2.a (Department of Systems Engineering Admin Memo 1-1).
- Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED.** Documentation procedures are outlined in references 2a through 2f.

- f. Additional Instruction. AI is available at mutually agreed upon times between a cadet and the instructor. Cadets are expected to have specific questions for any AI session. General review sessions for groups of cadets prior to the WPRs will not be provided outside of regularly scheduled class periods. Be prepared to show your Challenge Problems at any AI session.
- g. Challenge Problems. Challenge Problems are assigned along with select reading assignments. These problems are intended to enhance understanding of text material. Although these problems will not always be collected and graded, you are expected to work them, particularly if you are struggling with the material. These challenge problems will provide a basis for the discussion forms on blackboard.
- h. Class. You are expected to attend all synchronous lessons. In order to comply with USMA and Department policy, no more than 12 Cadets will be allowed to attend class in person. Your Instructor will assign you specific lessons in which you will attend class in person. If you are not scheduled to attend class in person, you will attend class via you Instructor's online chosen online platform (Blackboard, Teams, Zoom, etc.). When attending class in a remote status, you must turn on your camera for the duration of the lesson. If you are quarantined due to COVID-19, you will (1) inform your Instructor and (2) attend class via the online platform. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.
- i. Blackboard. All course material, including the syllabus, graded events, and lesson material, will be posted on the course Blackboard page. Your Instructor may choose to post supplemental materials under their specific folder. It is your responsibility to routinely check Blackboard for announcements and course material.

Enclosure 3: Course Syllabus

	Lsn	Date	Adjacent Events	Title	Assignment (Complete Prior to Class)	Graded Event	Lesson Objectives
Intro to SD	1	17-Aug		Introduction to System Dynamics	Read: Business Dynamics (BD) Sections 1.1 - 1.5 (p 3 - 39) Discussion Board: Challenge Problem (p 14)		Define and describe the field of System Dynamics Describe the applications of System Dynamics Articulate a System Dynamics Problem
	2	19-Aug		Dynamic Complexity	Read: BD Sections 1.3.1 - 1.5 (p 21 - 39) Read: Project Guidance	Project Issued	Articulate a System Dynamics Problem Understand the primary elements of Dynamic Complexity and how it arises Recognize sources of Policy Resistance Design and evaluate policy based on System Dynamics Model
	3	21-Aug		The SD Modeling Process -OR- The System Dynamics Beer Game	Read: BD Section 2.5 (p 79-81) and Ch 3 (p 83 - 105)	HW1 Issued	List and describe the 5 steps of the System Dynamics Modeling Process Articulate a System Dynamics Problem Formulate a Dynamic Hypothesis Understand the role of testing in the System Dynamics Modeling Process Understand how to utilize System Dynamics within the SDP
	4	25-Aug		Introduction to Vensim	Read: Vensim Guide (VG) Chapter 2 (p 11-22)		Create a stock and flow diagram in Vensim Define and Develop Reference Modes Design and evaluate policy based on System Dynamics Model
	5	27-Aug		Causal Loop Diagrams (CLD)	Read: BD Chapter 5 (p 137-158) Discussion Board: Challenge Problem (p 147)		Create a causal loop diagram Describe the cause and effect relationship in complex systems Describe the difference between causation and correlation Identify link and loop polarity
System Dynamics Tools	6	31-Aug	Ring Wind	Structure and Behavior of Dynamic Systems	Read: BD Chapter 4 (p 107-133)		List and describe the fundamental modes of behavior and describe the feedback structure Define and Develop Reference Modes Formulate a Dynamic Hypothesis
	7	2-Sep	Modified	Causal Loop Diagram Workshop	Read: BD Chapter 5 (p 159-176) Discussion Board: Challenge Problem 2 (p 169) Skim: VG Chapter 4 (p 41-62)	HW2 Issued	Create a causal loop diagram in Vensim Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
	8	4-Sep	Modified			HW1 Due by 2359	
	9	8-Sep		Homework 2 Working Session	Read: HW 2 Guidance		Create a causal loop diagram in Vensim Apply the principles of causal loop diagramming to represent a process Formulate a Dynamic Hypothesis Identify link and loop polarity
	10	10-Sep		Stocks and Flows	Read: BD Chapter 6 (p 191-210) Read: VG Chapter 5 (p 63 - 74) Discussion Board: Challenge Problem (p 211 - pick one 1, 2, or 3)		Define a Stock Define a Flow Classify an object as a stock or flow Apply the principles of stock and flow diagramming to represent a dynamic system Convert a causal loop diagram to a stock and flow diagram
	11	14-Sep		Stocks and Flows Workshop	Read: https://climate.nasa.gov/news/2365/seven-case-studies-in-carbon-and-climate/ Read: BD Chapter 6 (p 210-230) Review: VG Chapter 5 (p 63-74)		Convert a causal loop diagram to a stock and flow diagram Create a stock and flow diagram in Vensim
	12	17-Sep		Dynamics of Stocks and Flows	Read: BD Chapter 7 (p 231-241) Discussion Board: Challenge Problem (p 239)		Understand the primary elements of Dynamic Complexity and how it arises Apply graphical integration and differentiation Determine Net Flow
	13	21-Sep		Dynamics of Stocks and Flows Part II	Review: VG Chapters 2 (p 11-22), 4 (p 41-62), & 5 (p 63 - 74) Read: HW 3 Guidance	HW2 Due by 2359 HW3 Issued	Describe the cause and effect relationship in complex systems Apply graphical integration and differentiation Determine Net Flow Create a causal loop diagram in Vensim Articulate a System Dynamics Problem
	14	25-Sep					
	15	29-Sep		WPR1	WPR 1	WPR 1	
Basic Modeling	16	1-Oct		Dynamics of Simple Structures: Intro	Read: BD Chapter 8 (p 263-280) Discussion Board: Challenge Problem (p 281- part 1)		Understand the characteristics of a first-order, linear feedback system Understand the relationship between a first-order, positive feedback loop and exponential growth Understand the relationship between a first-order, negative feedback loop and goal-seeking behavior
	17	5-Oct		Dynamics of Simple Structures: S-shaped Growth	Read: BD Chapter 9 (p 295 - 309, 323 - 334)		Understand the dynamics and behavior of systems that experience S-Shaped growth Understand how the Logistic Growth model is used to model S-shaped growth Understand the SIR Model Understand the relationship between a nonlinear, first-order system and S-shaped growth
	18	7-Oct		Delays - Material	Read: BD Chapter 11 (p 409-425) Discussion Board: Challenge (p 425 - part 1)		Define a Delay Describe a material delay Determine the order of a delay Understand the structure and behavior of material delays Model a Material Delay
	19	9-Oct	Columbus Day Weekend				
	20	14-Oct		Delays - Information	Read: BD Chapter 11 (p 426-448); Skim Section 11.6; p. 449-467 optional		Describe an information delay Understand the structure and behavior of an information delay Model an Information Delay Describe the methods for estimating the duration and distribution of delays
	21	16-Oct		Dynamics of Growth: Path Dependence	Read: BD Chapter 10 (p 349-406) Discussion Board: Challenge Problem (p 353)	HW3 Due by 2359	Understand the characteristics and behaviors of multiple-loop systems Understand the concept of loop dominance and how it impacts behavior in multiple-loop systems Understand the dynamics and behaviors that experience path dependence Understand how positive feedback drives path dependence
	22	20-Oct		Project Working Session	Read: Project Guidance		
	23	22-Oct		Dynamics of Simple Structures: Aging Chains	Read: BD Chapter 12 (p 469 - 497) Discussion Board: Challenge (p 495 - part 3)	Project Part I Due by 2359	Describe an aging chain Understand the structure and behavior of an aging chain, as well as when they are appropriate for use Model an Aging Chain
	24	26-Oct		Modeling Decision Making	Read: BD Chapter 13 (p 513-550)		Understand the principles for modeling decision making Be familiar with common formulations for rate equations Understand the common pitfalls of modeling decision rules that should be avoided Understand the guidelines for formulating table functions to capture nonlinear relationships in System Dynamics models Understand the common pitfalls in the specification and use of nonlinear functions in System Dynamics models
	25	28-Oct		Formulating Nonlinear Relationships	Read: BD Chapter 14 (p 551-595) Read: VG Chapter 8 (p 107-120) Discussion Board: Challenge (p 566)		
Model Validation & Policy Analysis	26	30-Oct		Project Modeling Working Session	Read: Project Guidance		Create a System Dynamics Model
	27	2-Nov		Modeling Human Behavior	Read: BD Chapter 15 (pp 597-630)		Understand the cognitive limitations of decision makers Incorporate human behavior into a model
	28	6-Nov		WPR 2	WPR 2	WPR 2	
	29	9-Nov		Model Testing	Read: BD Chapter 21 (p 845 - 891) Read: VG Chapters 14 and 15 (p 193 - 226) Discussion Board: Challenge (p 871)		List the types of assessment tests for dynamic models Produce a calibrated model to given data Conduct goodness of fit test using Vensim
	30	12-Nov		Policy Design and Evaluation Workshop	Review: BD Section 3.5.5 (p 103 - 104) Read: HW 4 Guidance	Project Part II Due by 2359 HW4 Issued	Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	31	16-Nov		Promotion Lab (Part 1)	Read: BD Chapter 12 (p 469-496) Focus: BD p. 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	32	18-Nov		Promotion Lab (Part 2)	Review: BD Chapter 12 (p 469-496) Review: BD p 485-490		Design and evaluate policy based on System Dynamics Model Identify critical leverage points in a SD Model
	33	20-Nov		Challenges in System Dynamics	Read: System Dynamics and Decision Making in the Context of Dynamically Complex Multi-Dimensional Societal Issues	HW4 Due by 2359	Understand the challenges faced by systems dynamics modelers Understand the complexity of communicating a systems dynamics model
	34	23-Nov		Project Working Session	Review: Project Guidance		
	35	25-Nov					
	36	30-Nov		Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	37	1-Dec		Project Presentations	Review: Project Guidance	Project Presentations	Effectively communicate analysis and findings to a decision maker through oral presentations
	38	4-Dec		Course Closeout	Course Closeout	Project Report Due 7 DEC by 2359	Create a System Dynamics Model Describe the applications of System Dynamics Articulate a System Dynamics Problem Effectively communicate analysis and findings to a decision maker through written reports

Enclosure 4: SM484 Course Objective Assessment Table

	Apply system dynamics methodology to the solution of large scale, complex problems	Create models of dynamic systems, operations, and processes	Apply system dynamics modeling techniques to simulate behavior over time of systems, operations, and processes	Develop policies and controls to improve systems, operations, and processes	Interpret the output from continuous time simulations of system dynamics models as an aid for decision making	Prepare and present the results of system dynamics analysis in oral and written form	Recognize the ethical considerations involved with gathering and analyzing data, using quantitative models, validating assumptions, and reporting results	
Homework 1 - Vensim Lab	1	1	1					86.2%
Homework 2 - CLD	1	1						84.8%
Homework 3 - Stock and Flow	1	1						90.4%
WPR 1	1	1	1		1			85.7%
WPR 2	1	1	1		1			81.5%
Homework 4 - Fishbanks Policy			1	1			1	89.6%
Course Project								
Project Part I - CLD	1	1	1					87.1%
Project Part II - Stock and Flow	1	1	1					87.0%
Project Presentation			1	1	1	1		89.8%
Project Report			1	1	1	1	1	89.9%
Instructor Points - Discussion Board							1	90.0%
	86.1%	86.1%	87.1%	89.8%	86.7%	89.9%	89.8%	
Direct Assessment	3.4	3.4	3.5	3.6	3.5	3.6	3.6	3.5
Course Director Assessment	3.5	3.5	4	4	3	4	4	3.7

Enclosure 5: Course End Feedback

TBP



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION II

EXAMINATIONS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB G

Written Partial Review 1

Instructor Solution



System Dynamics Simulation
SM 484 – AT 26-1
MAJ Jake Lueders
MAJ Pat Lupfer

SOLUTION

WPR 1 (Primary)

Name: _____

Section: _____

Finish Time: _____

1. This is a **70-minute** exam worth **150 points**.
2. This is a closed book/notes exam, **ONE side of an 8-1/2x11 sheet of paper note sheet is allowed**. NO COLLABORATION allowed.
3. The only items that you may bring into the classroom are: one sided note sheet, pencil, straight edge, and calculator. **DO NOT BRING INK PENS!!!**
4. **Place your name NEATLY on each page in the space provided**. Feel free to disassemble the exam; a stapler is available to reattach the pages when you are finished. Make sure you have all 14 pages (including this cover page).
5. **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
6. After you finish the exam, fill in the finish time at the top of this page and sign the honor acknowledgement statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security at a time announced by the Course Director, no earlier than 06OCT25. Prior to that time, I may not discuss any aspects of this exam with anyone except an SM484 instructor." **PRINT IN BLOCK LETTERS NEATLY**

Printed Name (**FIRST LAST**): _____ Signature: _____

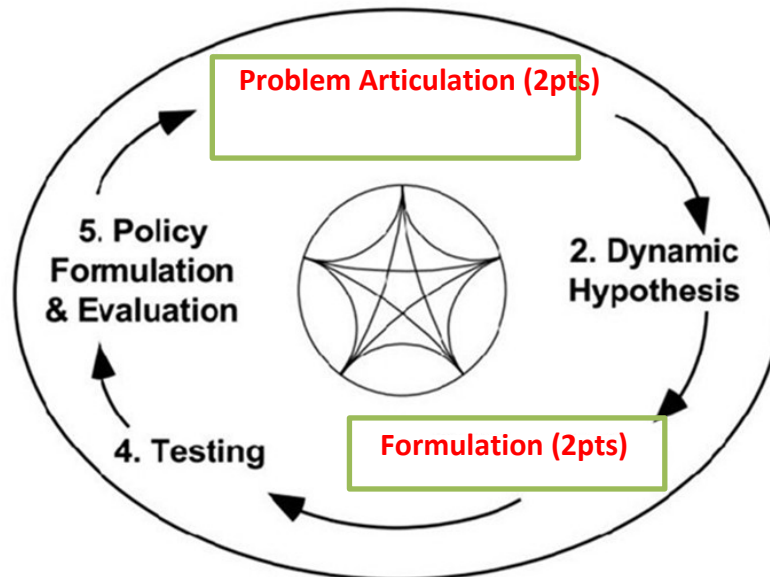
Date Signed: _____

Grading summary

Question	Possible Points	Actual Points
1. Short Answer/Multiple Choice	35	
2. Fundamental Modes of Behavior	25	
3. Diagram Notation	20	
4. Stock and Flow Diagram	30	
5. Dynamics of Stocks and Flows	40	
Total	150	

A. Short Answer/Multiple Choice Questions (30 Points)

1. **(12 Points)** The System Dynamics Modeling Process is shown below. Identify **BOTH** missing steps (in the green boxes) and briefly describe ONE of the missing steps of the SD Modeling Process below.



8pts: PA: Boundary Selection, why it's a problem, what is the problem, who is client, Key variables time horizon, reference mode, pattern of behavior

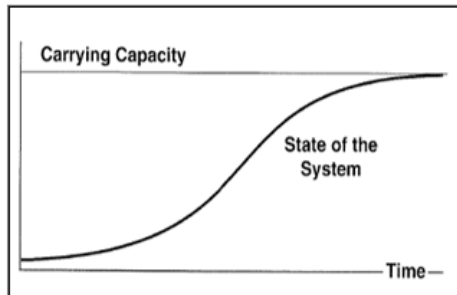
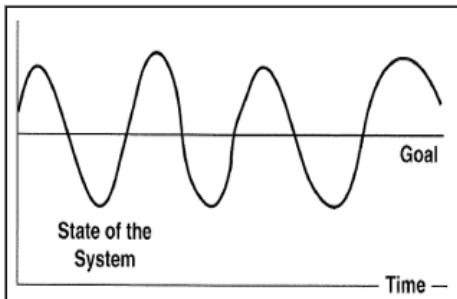
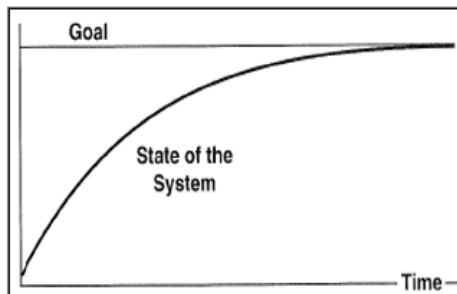
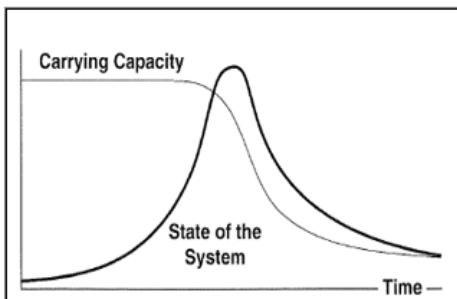
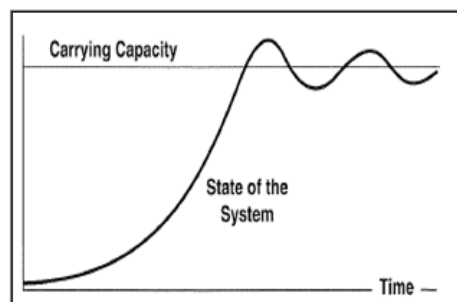
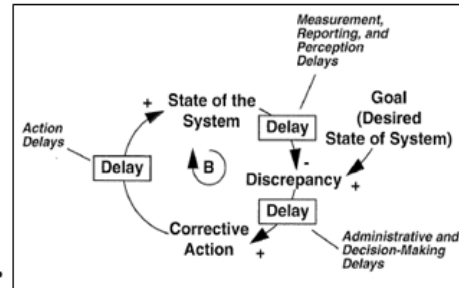
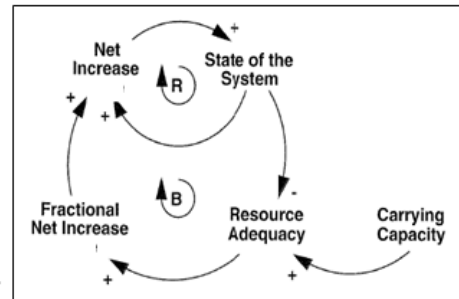
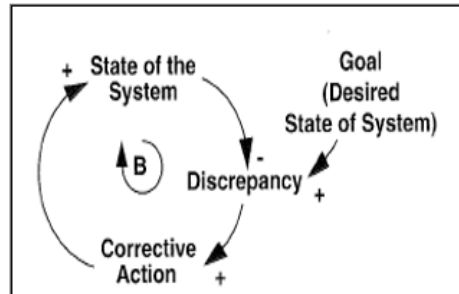
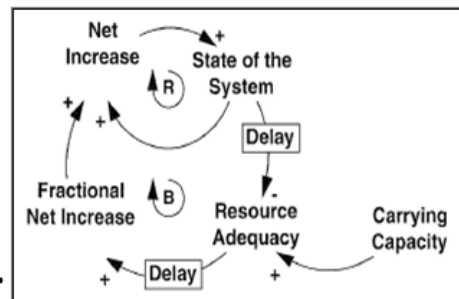
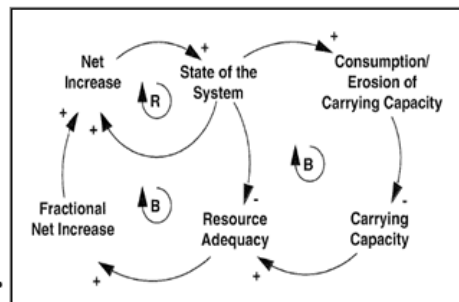
OR Form: developing a simulation model using stock and flow, estimating parameters relationships and initial conditions, add units as you build, simulate often, translate causal theory into a model

2. **(3 Points)** Which of the following are elements that contribute to **dynamic complexity**? (Select all that apply) **(1pt not selecting reference mode, .5pts each correctly selected)**

- ☒ a. Policy-Resistant
- ☐ b. Reference Mode
- ☒ c. History-Dependent
- ☒ d. Tightly Coupled
- ☒ e. Self-Organizing

3. **(5 Points)** Which of the following are characteristics of **exogenous variables**? **(Select all that apply)** **(1pt each correctly selected or not selected)**
- a. They can be changed by the system
 - ☒ b. They are constants
 - c. They originate inside the boundary of the system
 - ☒ d. They originate outside the boundary of the system
 - e. They are critical drivers of system feedback
4. **(5 Points)** Which of the following statements about **stocks** are true? **(Select all that apply)** **(1pt each correctly selected or not selected)**
- ☒ a. Stocks decouple rates of flow and create disequilibrium dynamics
 - ☒ b. Stocks are the sources of delay
 - c. Stocks can be directly changed by auxiliary variables
 - ☒ d. Stocks characterize the state of the system and provide the basis for actions
 - ☒ e. Stocks provide systems with inertia and memory

5. **(10 Points)** Match the system behavior on the left, to the structure on the right. **(Fill in the blank on the behavior, with the letter from the structure) 2PT EACH CORRECT**

B**A****C****E****D****A.****B.****C.****D.****E.**

B. Fundamental Modes of Behavior (25 Points)

Read the following description about the Federal Reserve and answer the subsequent questions.

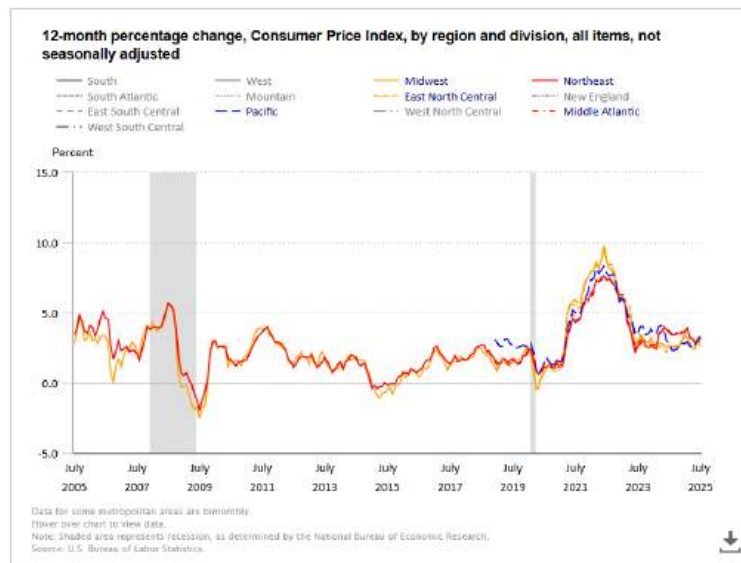


Figure 1: Image Credit <https://www.bls.gov/charts/consumer-price-index/consumer-price-index-by-region.htm>

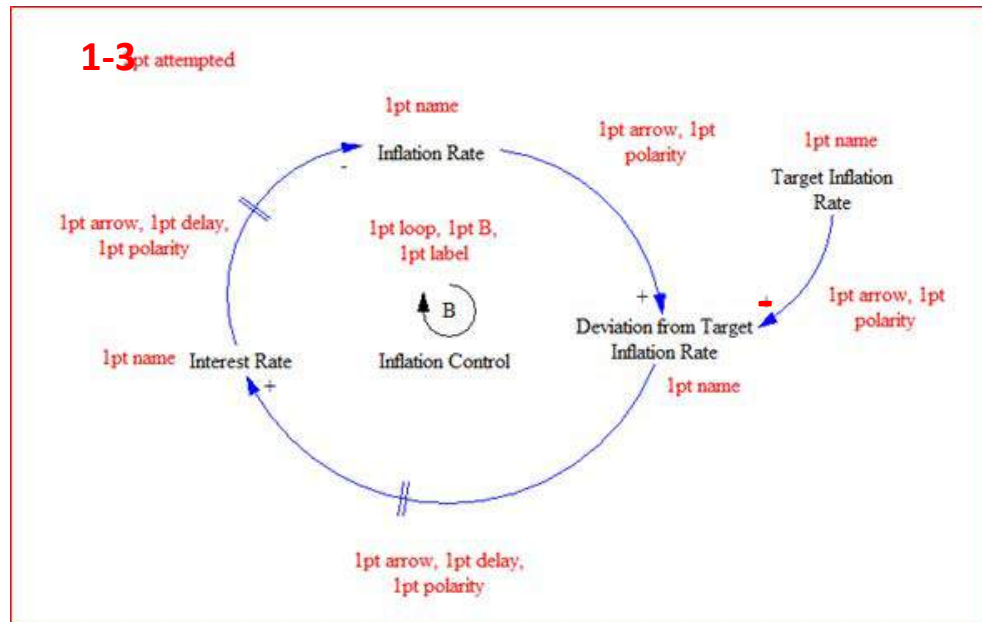
The Federal Reserve enacts policy with **the goal of keeping inflation at 2%**, measured by the Consumer Price Index shown in Figure 1 above. **If inflation is significantly below the inflation target, the Federal Reserve may cut interest rates** to spur economic activity and bring inflation back to the 2% target. **When inflation exceeds target inflation, the Federal Reserve will react and adjust the interest rate higher** in an attempt to check inflation.

To this end, the Federal Reserve increased interest rates several times in 2022 through 2023 to combat the rise in inflation, but the effect is not instantaneous. The figure above shows how the inflation rate continued to rise through 2022, even with the Federal Reserve already adjusting the interest rate, and it didn't start to drop until 2023. Though it does not show when interest rate hikes have occurred, those increases can be inferred from the inflation history. Answer the following questions based on the above image and description of the Federal Reserve's role in managing the inflation rate.

6. **(5 Points)** What fundamental mode of behavior is being described in the previous description and shown in Figure 1 above? **(5pts oscillation, 3pts S shaped with overshoot, 2pts goal seeking)**
- OSCILLATION**

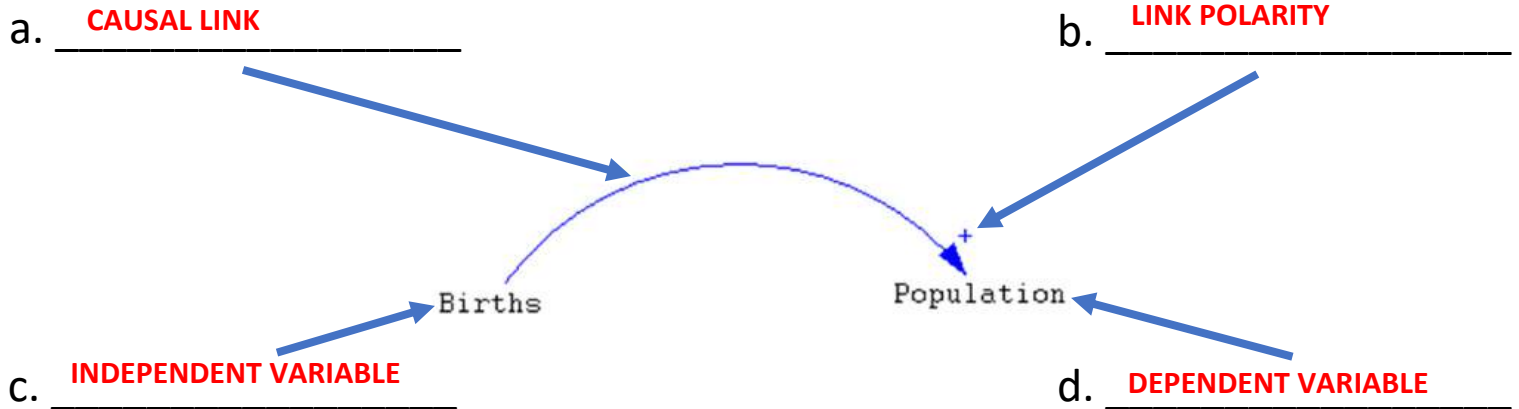
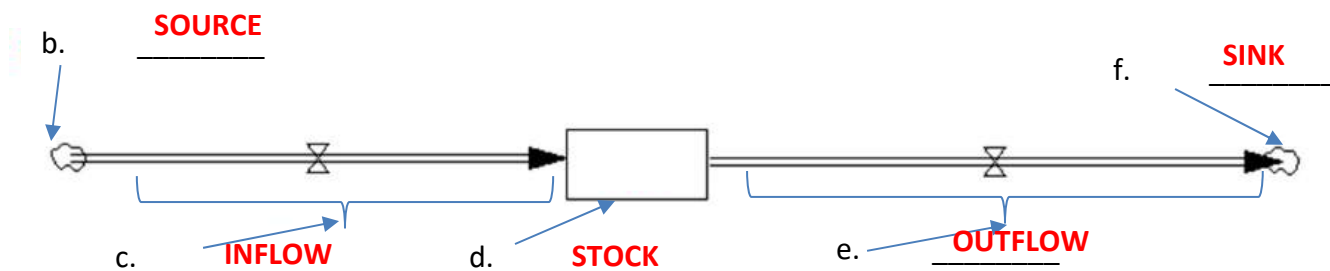
(20 Points) The Federal Reserve implements monetary policy tools, such as the interest rate, to control inflation. In this scenario, you are facing a situation where the inflation rate is *above the target rate* and corrective actions are being taken by the Federal Reserve to bring the inflation back down. In the space below, draw a CLD that depicts the system structure corresponding to the fundamental mode of behavior that you identified previously. Use the following variables to create an abstracted CLD: **Deviation from Target Inflation Rate, Interest Rate, Inflation Rate, and Target Inflation Rate**. Remember that the system structure dictates the system behavior. Be sure to specify (1) Link Polarities, (2) Loop Directions, and (3) Loop Types with Loop name

use judgment for partial credit



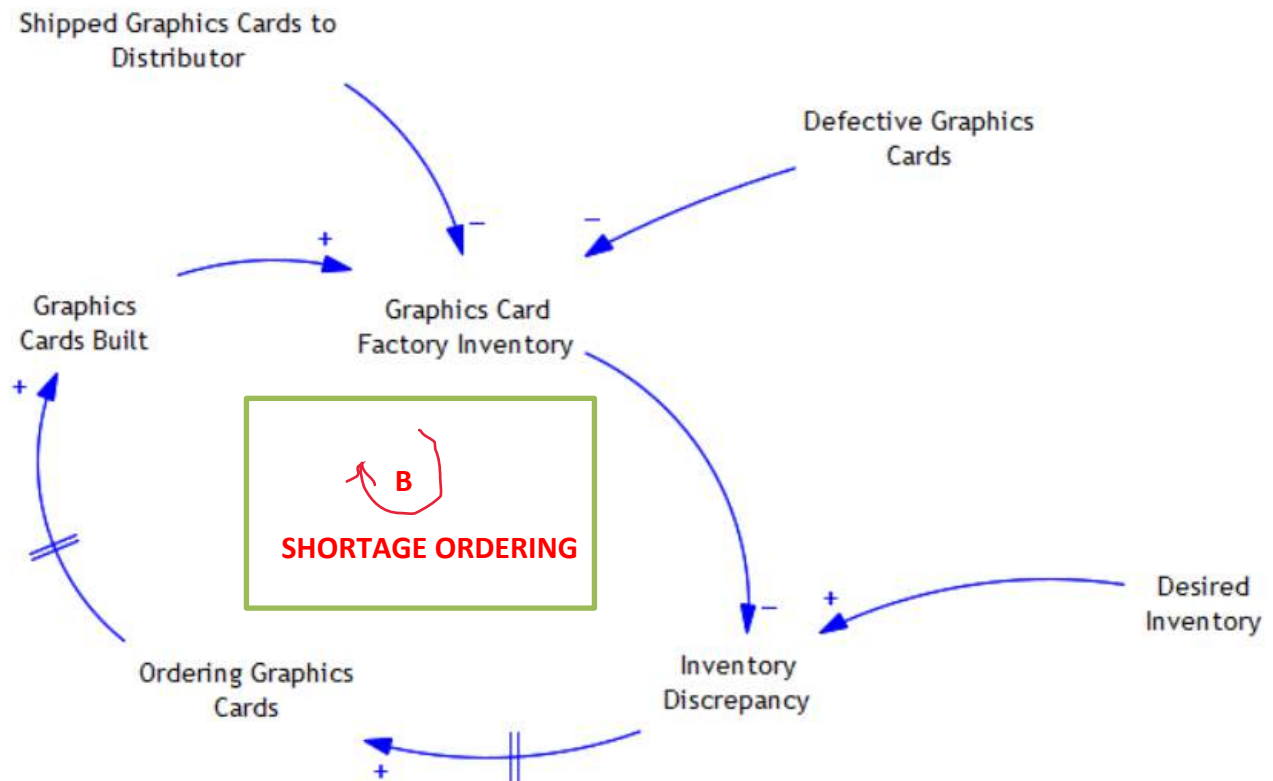
C. Diagram Notation (20 Points)**2PTS EACH CORRECT**

7. (8 Points) Correctly label each entity on the diagram below:

8. (12 Points) Correctly label each entity on the diagram below: **2PT EACH CORRECT**a. What is this diagram called? **STOCK AND FLOW DIAGRAM**

D. CLD → S&F (30 Points)

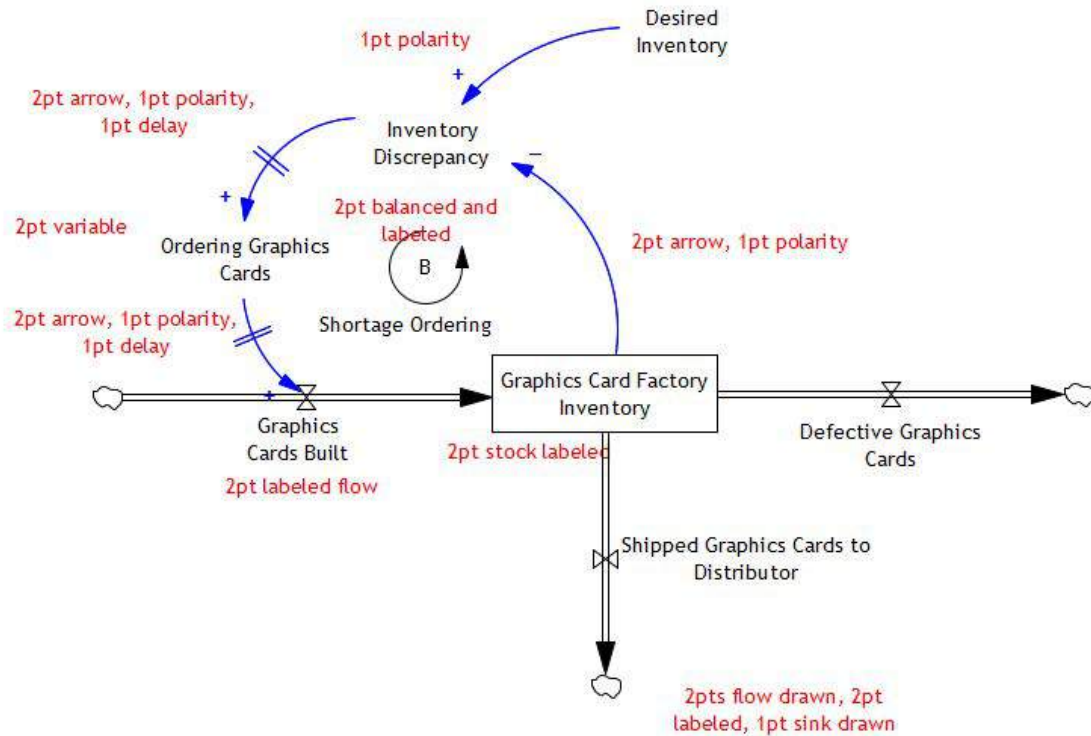
The diagram below depicts a simple structure of a graphics card factory. The factory needs to maintain a certain inventory so that they can meet shipping requirements to their distributors. Sometimes during testing prior to shipping, they also identify some defective graphics cards that can't be shipped. When the factory inventory falls too low, the warehouse manager will order additional graphics cards to be built to bring them back up to their desired inventory.



9. (5 Points) Identify the type of feedback loop in the diagram above and clearly **depict** and **label** and **name** this feedback on the diagram. **Write your answer in the green box above.**

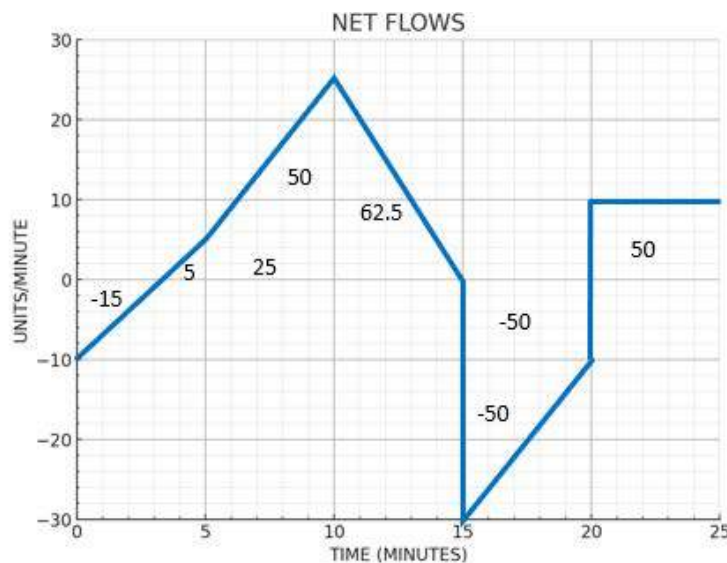
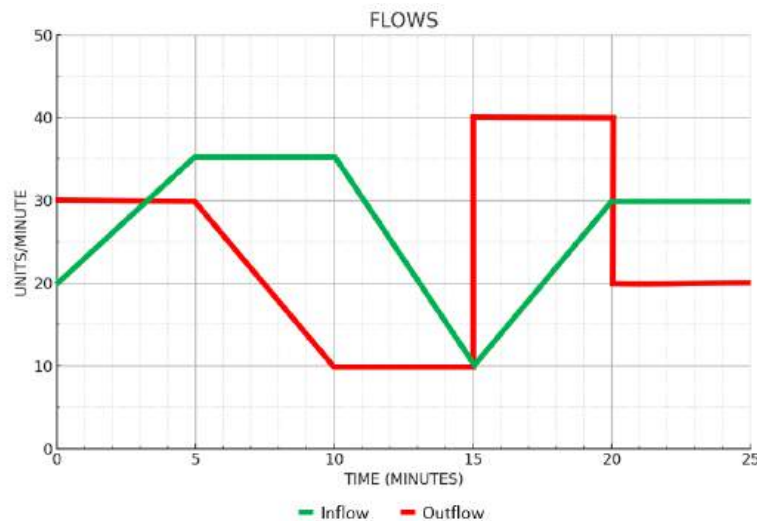
2PT CLOCKWISE ARROW, 2PT BALANCING, 1PT NAME

10. (25 Points) Using **only the variables** listed in the diagram on the previous page, draw a complete Stock and Flow Diagram to represent this system in the space below. Build on the basic structure that has been given to you below. Use only the CLD to construct your Stock and Flow diagram.



E. Dynamics of Stocks and Flows (40 Points)

11. (17 points) Consider a stock with the single inflow rate and the single outflow rate shown below. Complete the net flow of the stock and complete the behavior of the stock on the graphs provided. *The initial value of the stock is 20 units.*



3PT ATTEMPT NET FLOW
2PT ATTEMPT STOCK
2PT PER SEGMENT (SLOPE/SHAPE & ENDPOINTS)

Show your work here!

0-3:

$$.5 \cdot -10 \cdot 3 = -15$$

3-5:

$$.5 \cdot 2 \cdot 5 = 5$$

5-10:

$$5 \cdot 5 + .5 \cdot 5 \cdot 20 = 25 + 50 = 75$$

10-15:

$$.5 \cdot 25 \cdot 5 = 62.5$$

15-20:

$$-10 \cdot 5 + .5 \cdot -20 \cdot 5 = -50$$

$$+50 = -100$$

20-25:

$$10 \cdot 5 = 50$$

0:20

3:5

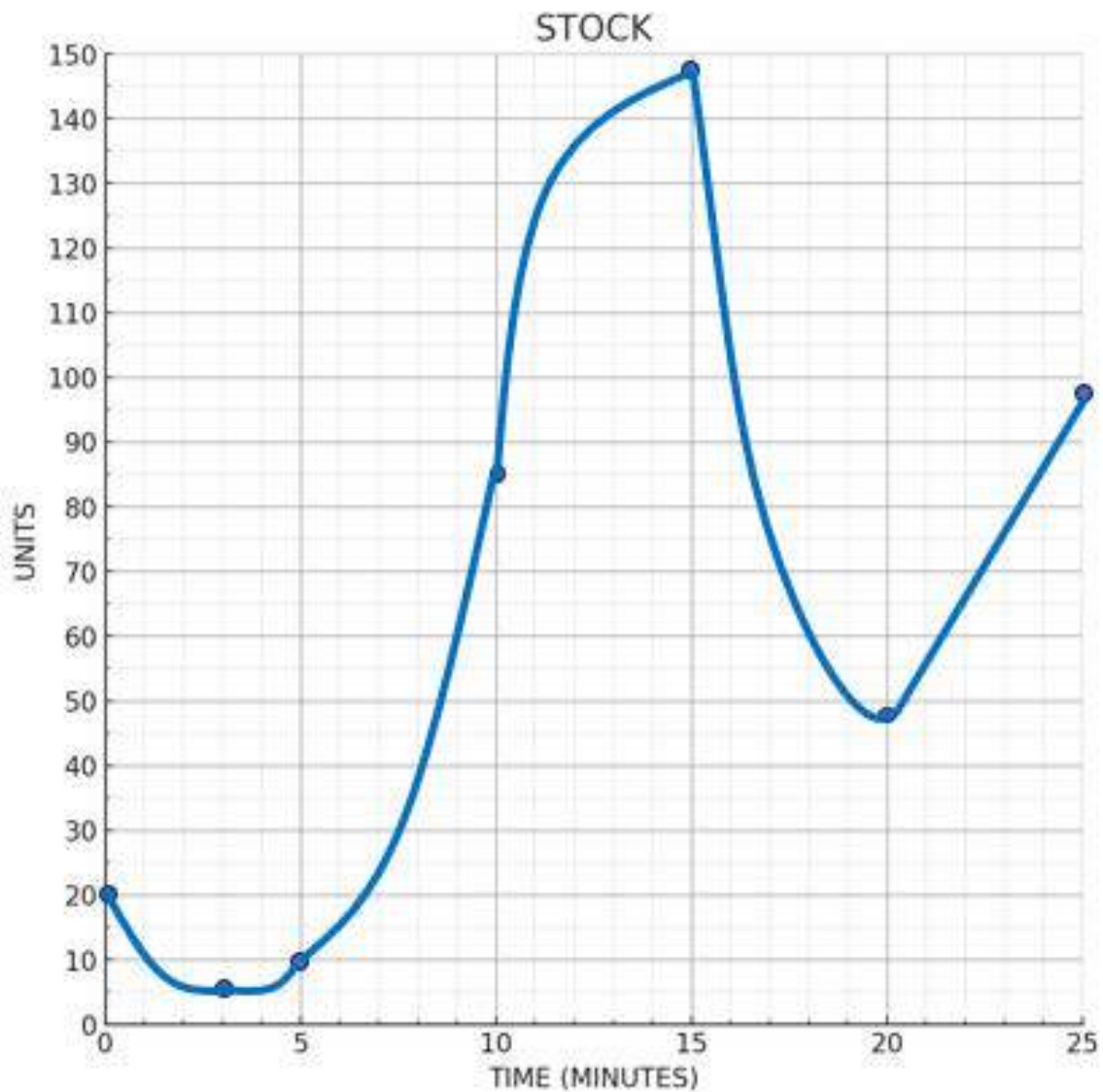
5:10

10:85

15:147.5

20:47.5

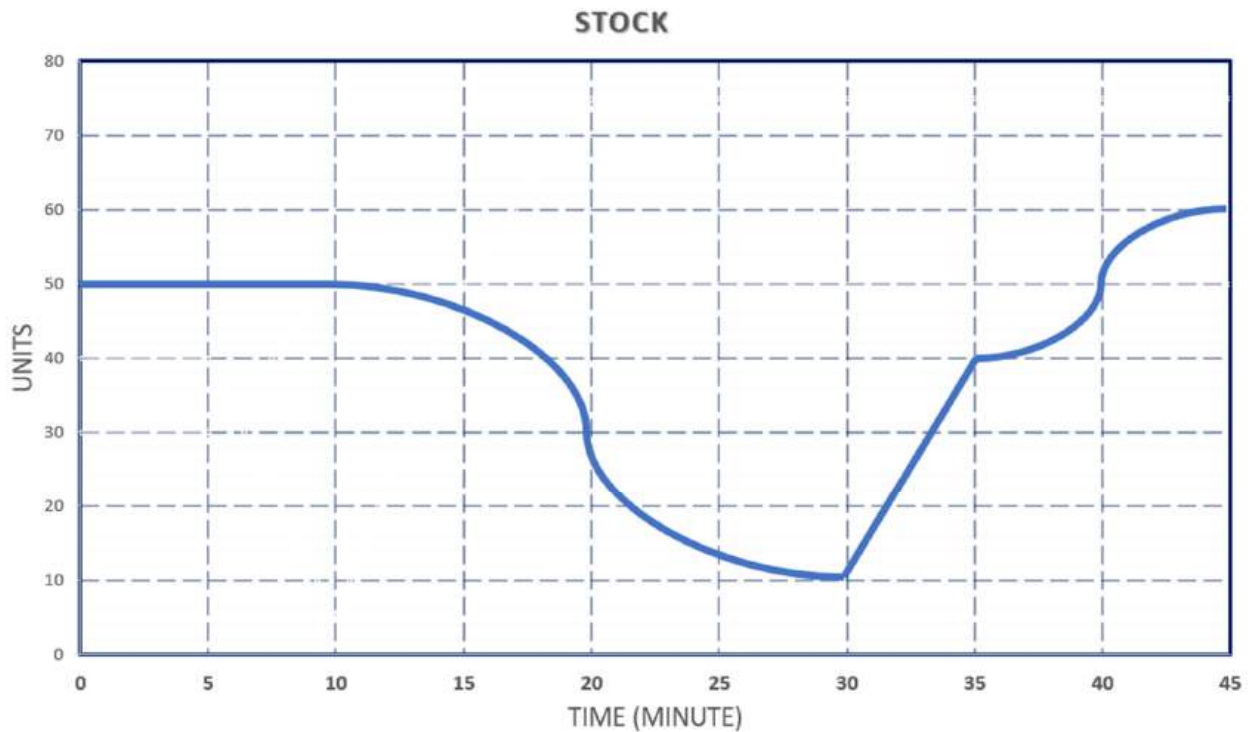
25:97.5



12. (3 points) What is the value of the stock at a time of 20 minutes?

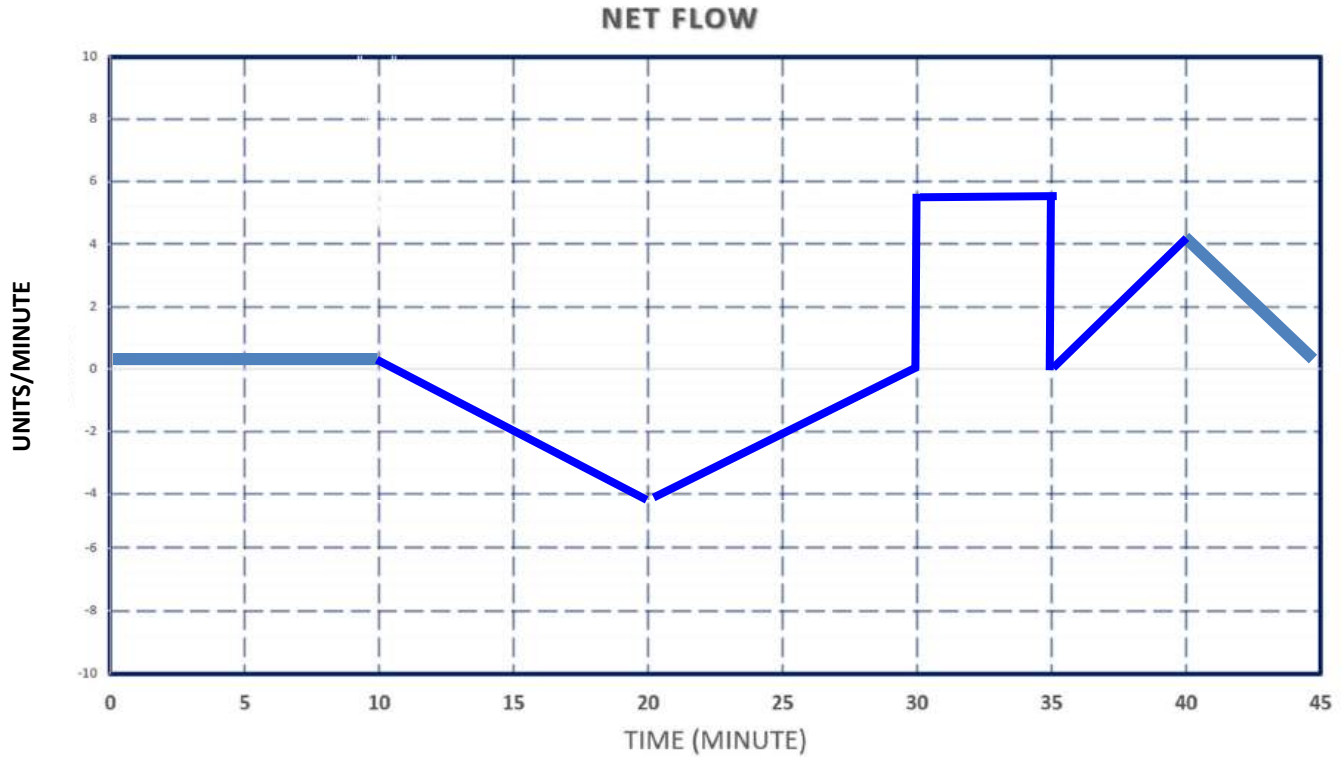
47.5 (WITHIN REASON OK DUE TO ROUNDING FROM READING GRAPH)

13. **(17 points)** Consider the Stock illustrated below. Fill in the missing time segments to graph the Net Flow (on page 13) that would result in the following stock behavior.



Show your work here!

(3pts for attempting, 2pts neatness, 3pts per each segment (10-20, 20-30, 30-35, 35-40: Partial credit if math error but shape is correct, or if endpoints are correct but shape draw incorrectly)



14. (3 points) What is the Net Flow at 20 Minutes?

-4

Extra Work Page

Outstanding Student Work

WPR1

● Graded

Student

Jackson Meyer

Total Points

148 / 150 pts

Question 1

Q1

12 / 12 pts

✓ - 0 pts Correct

- 1 pt PA missed key variables
- 1 pt PA missed time horizon
- 1 pt PA missed reference mode/pattern of behavior
- 1 pt PA missed "why problem, what is the problem, who is the client, or boundary selection"
- 1 pt Answer lacking correct detailed description.
- 1 pt Form missed simulation model/stock and flow
- 1 pt Form missed estimating parameters, relationships, and initial conditions
- 1 pt Form missed adding units/simulate often
- 1 pt Form missed translating causal theory into a model
- 2 pts Problem Articulation Incorrect
- 2 pts Formulation Incorrect

Question 2

Q2

3 / 3 pts

✓ - 0 pts Correct

- 0.5 pts missed a
- 1 pt selected b
- 0.5 pts missed c
- 0.5 pts missed d
- 0.5 pts missed e

Question 3

Q3

5 / 5 pts

✓ - 0 pts Correct

- 1 pt selected a
- 1 pt missed b
- 1 pt selected c
- 1 pt missed d
- 1 pt selected e

Question 4

Q4

5 / 5 pts

✓ - 0 pts Correct

- 1 pt missed a
- 1 pt missed b
- 1 pt selected c
- 1 pt missed d
- 1 pt missed e

Question 5

Q5

10 / 10 pts

✓ - 0 pts Correct

- 2 pts missed B
- 2 pts Missed A
- 2 pts missed C
- 2 pts missed E
- 2 pts Missed D

Question 6

Q6

5 / 5 pts

✓ - 0 pts Correct-Oscillation

- 2 pts S shaped with overshoot
- 3 pts goal seeking
- 2 pts mentions something else primarily, but secondary mention of oscillation
- 5 pts completely wrong answer
- 4 pts mentions/describes only balancing (not goal seek or balancing with delay), or overshoot and collapse

Question 7

Q CLD

19 / 20 pts

- 0 pts Correct
- 1 pt no/incorrect loop name
- 1 pt Missing arrows on links
- 1 pt missing delay between interest rate and inflation rate

✓ - 1 pt missing delay from deviation...to interest rate

- 1 pt missing/incorrect one polarity
- 2 pts missing/incorrect two polarities
- 3 pts missing/incorrect three polarities
- 4 pts missing/incorrect four polarities
- 1 pt target inflation rate not exogenous
- 1 pt Interest Rate not endogenous
- 1 pt missing balancing loop 'B'
- 1 pt missing or incorrect loop/direction
- 2 pts two variables switched
- 1 pt one incorrect causal link
- 2 pts interest rate not linked to inflation rate
- 2 pts neatness/hard to read
- 2 pts multiple incorrect causal links
- 2 pts missing one of the variables

Question 8

Q7

8 / 8 pts

✓ - 0 pts Correct

- 2 pts missed causal link
- 2 pts missed link polarity
- 2 pts missed independent variable
- 2 pts missed dependent variable
- 1 pt missed "Link Polarity" but put "polarity"
- 1 pt missed "variable"
- 1 pt missed "causal link" but included "link"
- 1 pt positive polarity instead of link polarity

Question 9

Q8

12 / 12 pts

✓ - 0 pts Correct

- 2 pts missed source
- 2 pts missed inflow
- 2 pts missed stock
- 2 pts missed outflow
- 2 pts missed sink
- 2 pts missed stock and flow diagram

Question 10

Q9

5 / 5 pts

✓ - 0 pts Correct

- 2 pts no clockwise arrow
- 2 pts no 'B'
- 1 pt no name/incorrect naming convention (i.e. calling a loop a CLD)
- 4 pts partial credit-some other incorrect effort

Question 11

Q10

24 / 25 pts

– 0 pts Correct

– 5 pts no shipped graphics cards outflow

– 2 pts outflow represented as causal link

– 2 pts incorrect stock (not graphics card factory inventory)

– 2 pts 1 incorrect/missing causal link

– 4 pts 2 incorrect/missing causal links

– 6 pts 3 incorrect/missing causal links

– 1 pt 1 incorrect/missing polarity

– 2 pts 2 incorrect/missing polarity

– 3 pts 3 incorrect/missing polarity

– 4 pts 4 incorrect/missing polarities

– 1 pt minor loop errors

– 2 pts major loop errors

– 2 pts incorrect inflow

– 2 pts additional general formulation errors

– 1 pt missing one delay

– 2 pts missing two delays

✓ – 1 pt missing shipped graphics outflow sink (cloud)

– 1 pt missed variable

Question 12

Q11

17 / 17 pts

12.1 Q11 FLOW

9 / 9 pts

✓ - 0 pts Correct

- 1 pt 10-15 partially wrong
- 2 pts 10-15 completely wrong
- 1 pt 15-20 partially wrong
- 2 pts 15-20 completely wrong
- 1 pt 20-25 partially wrong
- 2 pts 20-25 completely wrong

12.2 Q11 STOCK

8 / 8 pts

✓ - 0 pts Correct

- 1 pt 10-15 partially wrong
- 2 pts 10-15 completely wrong
- 1 pt 15-20 partially wrong
- 2 pts 15-20 completely wrong
- 1 pt 20-25 partially wrong
- 2 pts 20-25 completely wrong
- 0 pts credit awarded, work follows incorrect work in Net Flow. Other deductions may still be present.
- 8 pts no attempt

Question 13

Q12

3 / 3 pts

✓ - 0 pts Correct

- 3 pts completely wrong
- 0 pts Wrong answer but follows incorrect graph

Question 14

Q13

17 / 17 pts

14.1 Q13 WORK

0 / 0 pts

✓ - 0 pts No points here, questions only for grading purposes to see work in gradescope

14.2 Q13 FLOW

17 / 17 pts

✓ - 0 pts Correct

- 1.5 pts 10-20 partially wrong

- 3 pts 10-20 completely wrong

- 1.5 pts 20-30 partially wrong

- 3 pts 20-30 completely wrong

- 1.5 pts 30-35 partially wrong

- 3 pts 30-35 completely wrong

- 1.5 pts 35-40 partially wrong

- 3 pts 35-40 completely wrong

- 2 pts neatness (if your lines look curved, or extremely smudged from poor erasing)

- 4 pts minimal attempt

Question 15

Q14

3 / 3 pts

✓ - 0 pts Correct

- 0 pts wrong answer, but correct based on incorrect graph

- 3 pts wrong or no answer

Question 16

EXTRA WORK

0 / 0 pts

✓ - 0 pts No points associated with this, just for gradescope to see if you needed extra space



Department of Systems Engineering

System Dynamics Simulation

SM 484 – AT 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

WPR 1 (Primary)

Name: Jackson Meyer

Section: I1

Finish Time: 1007

1. This is a **70-minute** exam worth **150 points**.
2. This is a closed book/notes exam, **ONE side of an 8-1/2x11 sheet of paper note sheet is allowed**. NO COLLABORATION allowed.
3. The only items that you may bring into the classroom are: one sided note sheet, pencil, straight edge, and calculator. **DO NOT BRING INK PENS!!!**
4. **Place your name NEATLY on each page in the space provided**. Feel free to disassemble the exam; a stapler is available to reattach the pages when you are finished. Make sure you have all 14 pages (including this cover page).
5. **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
6. After you finish the exam, fill in the finish time at the top of this page and sign the honor acknowledgement statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security at a time announced by the Course Director, no earlier than 06OCT25. Prior to that time, I may not discuss any aspects of this exam with anyone except an SM484 instructor." **PRINT IN BLOCK LETTERS NEATLY**

Printed Name (**FIRST LAST**): Jackson Meyer Signature: Jake

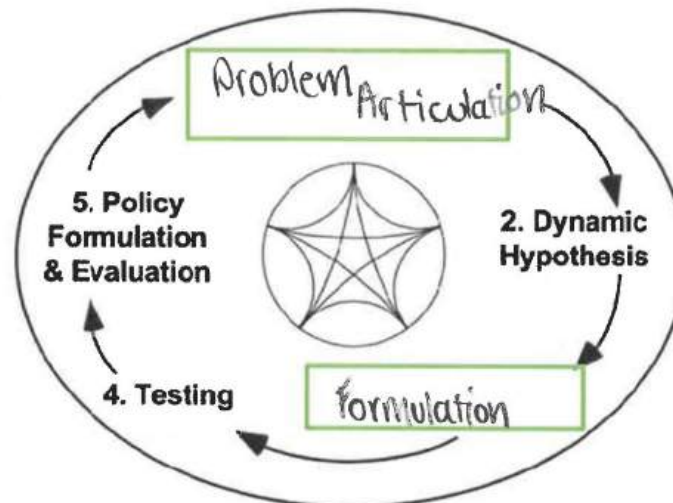
Date Signed: 10/06/2025

Grading summary

Question	Possible Points	Actual Points
1. Short Answer/Multiple Choice	35	
2. Fundamental Modes of Behavior	25	
3. Diagram Notation	20	
4. Stock and Flow Diagram	30	
5. Dynamics of Stocks and Flows	40	
Total	150	

A. Short Answer/Multiple Choice Questions (30 Points)

1. **(12 Points)** The System Dynamics Modeling Process is shown below. Identify BOTH missing steps (in the green boxes) and briefly describe ONE of the missing steps of the SD Modeling Process below.



Problem Articulation is defining the issue of the system. In this step you will define the issue behavior while in dynamic hypothesis you will form a hypothesis regarding what is causing the behavior or why it is happening. Formulation you create a C.D. with feedback loops, causal links, delays etc. this is a baseline where you establish your system boundaries and create your model.

2. **(3 Points)** Which of the following are elements that contribute to dynamic complexity? (Select all that apply)

- ☒ a. Policy-Resistant
- ☒ b. Reference Mode
- ☒ c. History-Dependent
- ☒ d. Tightly Coupled
- ☒ e. Self-Organizing

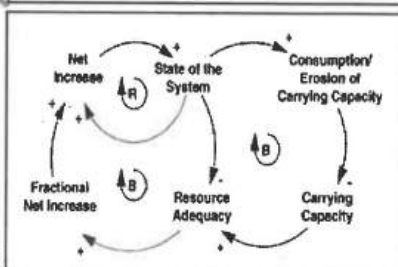
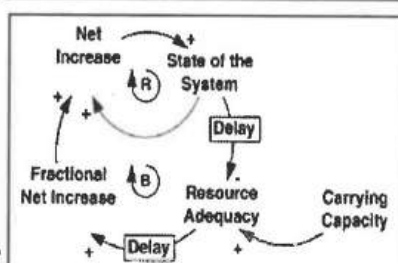
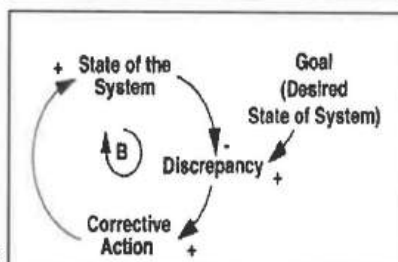
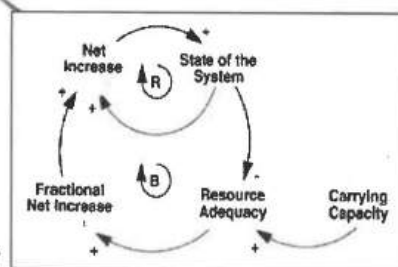
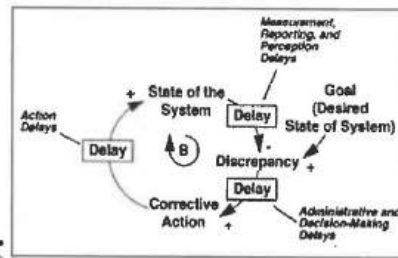
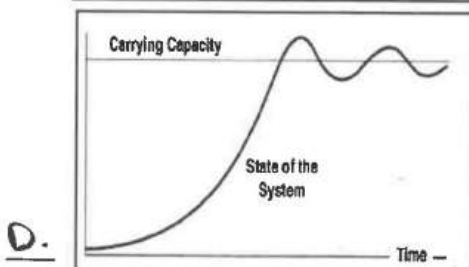
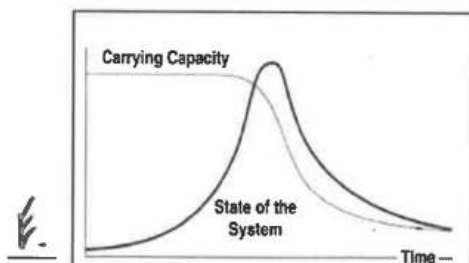
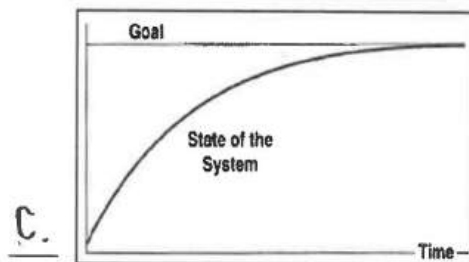
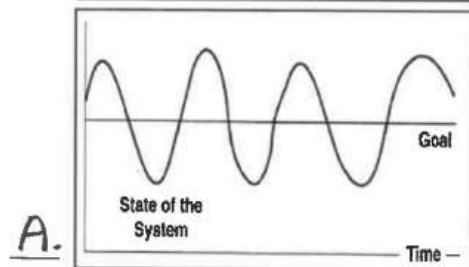
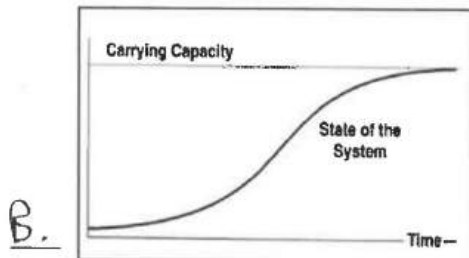
3. (5 Points) Which of the following are characteristics of **exogenous variables**? (Select all that apply)

- ☒ a. They can be changed by the system
- ☐ b. They are constants
- ☒ c. They originate inside the boundary of the system
- ☐ d. They originate outside the boundary of the system
- ☒ e. They are critical drivers of system feedback

4. (5 Points) Which of the following statements about **stocks** are true? (Select all that apply)

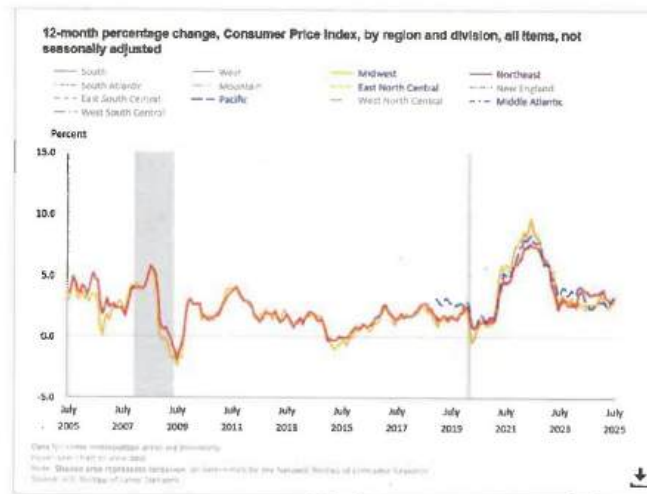
- ☐ a. Stocks decouple rates of flow and create disequilibrium dynamics
- ☐ b. Stocks are the sources of delay
- ☐ c. Stocks can be directly changed by auxiliary variables
- ☒ d. Stocks characterize the state of the system and provide the basis for actions
- ☒ e. Stocks provide systems with inertia and memory

5. (10 Points) Match the system behavior on the left, to the structure on the right. (Fill in the blank on the behavior, with the letter from the structure)



B. Fundamental Modes of Behavior (25 Points)

Read the following description about the Federal Reserve and answer the subsequent questions.



Cut interest rates
to raise inflation

The Federal Reserve enacts policy with the goal of keeping inflation at 2%, measured by the Consumer Price Index shown in Figure 1 above. If inflation is significantly below the inflation target, the Federal Reserve may cut interest rates to spur economic activity and bring inflation back to the 2% target. When inflation exceeds target inflation, the Federal Reserve will react and adjust the interest rate higher in an attempt to check inflation.

interest rates ↑ to lower inflation

delays

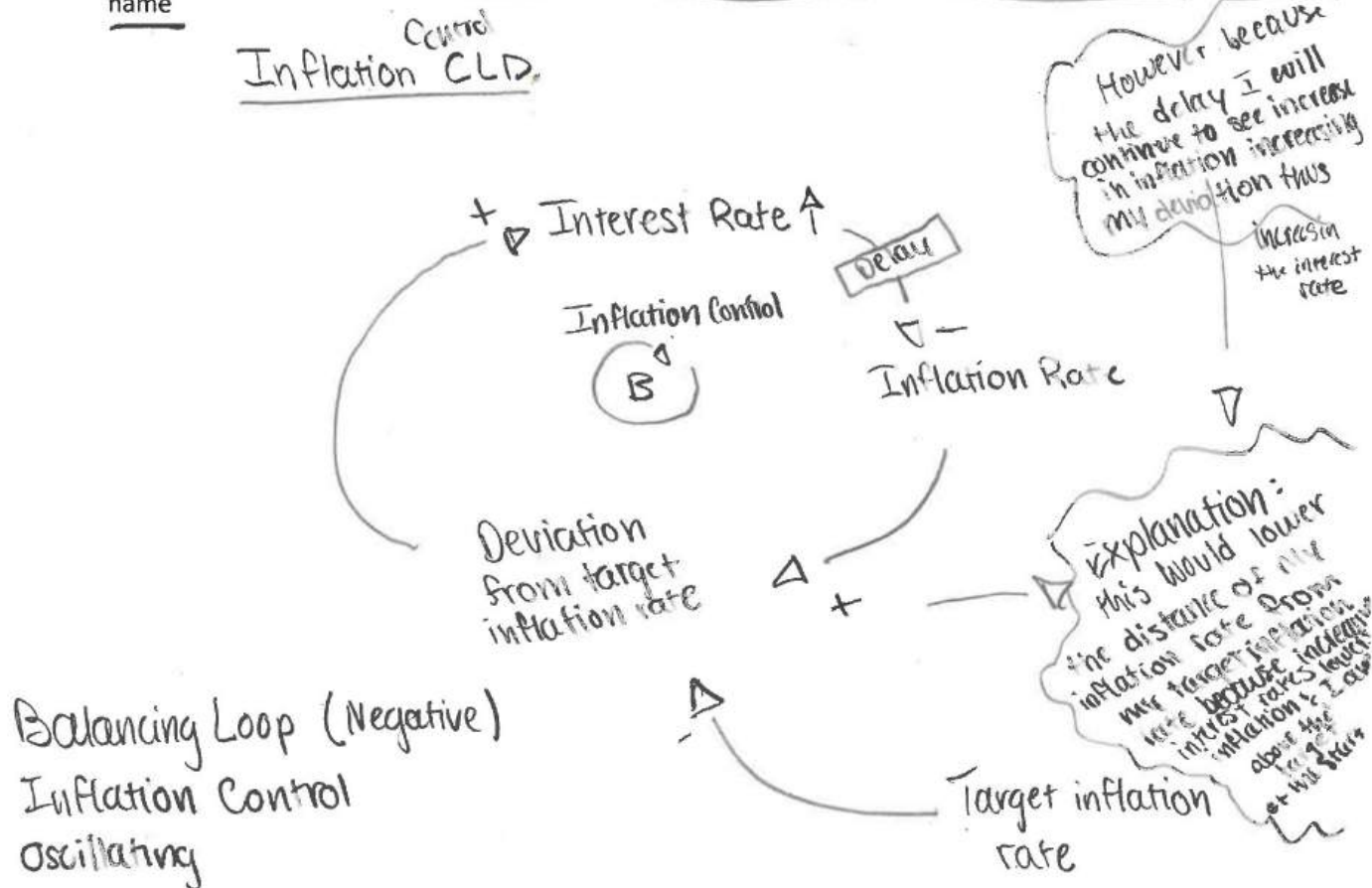
To this end, the Federal Reserve increased interest rates several times in 2022 through 2023 to combat the rise in inflation, but the effect is not instantaneous. The figure above shows how the inflation rate continued to rise through 2022, even with the Federal Reserve already adjusting the interest rate, and it didn't start to drop until 2023. Though it does not show when interest rate hikes have occurred, those increases can be inferred from the inflation history. Answer the following questions based on the above image and description of the Federal Reserve's role in managing the inflation rate.

6. (5 Points) What fundamental mode of behavior is being described in the previous description and shown in Figure 1 above?

oscillations

(20 Points) The Federal Reserve implements monetary policy tools, such as the interest rate, to control inflation. In this scenario, you are facing a situation where the inflation rate is above the target rate and corrective actions are being taken by the Federal Reserve to bring the inflation back down. In the space below, draw a CLD that depicts the system structure corresponding to the fundamental mode of behavior that you identified previously. Use the following variables to create an abstracted CLD: Deviation from Target Inflation Rate, Interest Rate, Inflation Rate, and Target Inflation Rate. Remember that the system structure dictates the system behavior. Be sure to specify (1) Link Polarities, (2) Loop Directions, and (3) Loop Types with Loop name

Inflation CLD

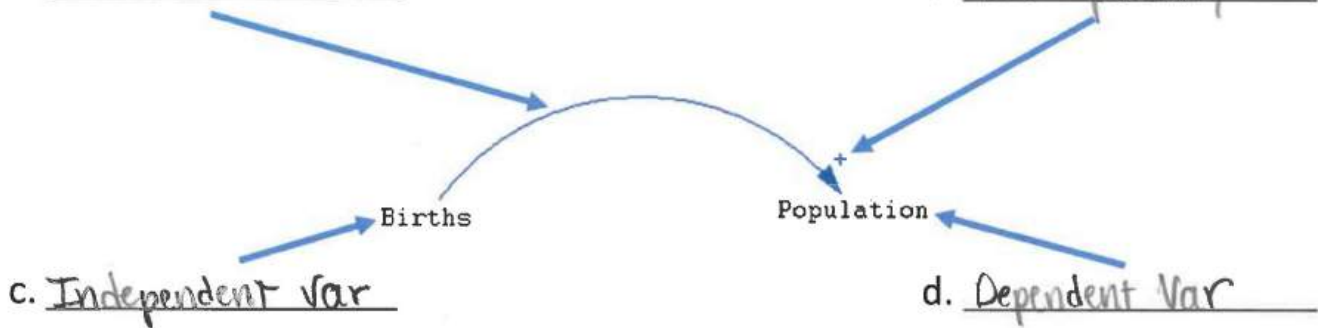


C. Diagram Notation (20 Points)

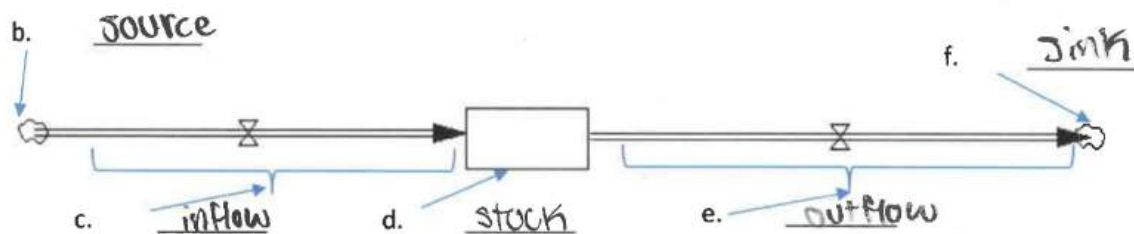
7. (8 Points) Correctly label each entity on the diagram below:

a. Causal Link

b. Link polarity



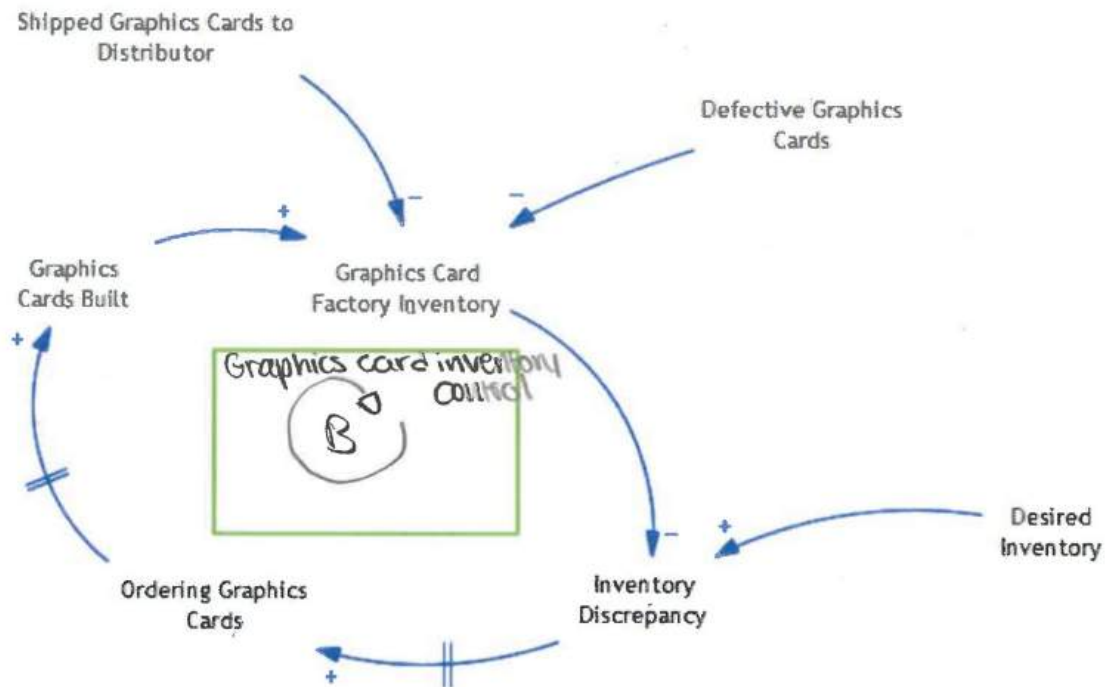
8. (12 Points) Correctly label each entity on the diagram below:



a. What is this diagram called? Stock & Flow Diagram

D. CLD → S&F (30 Points)

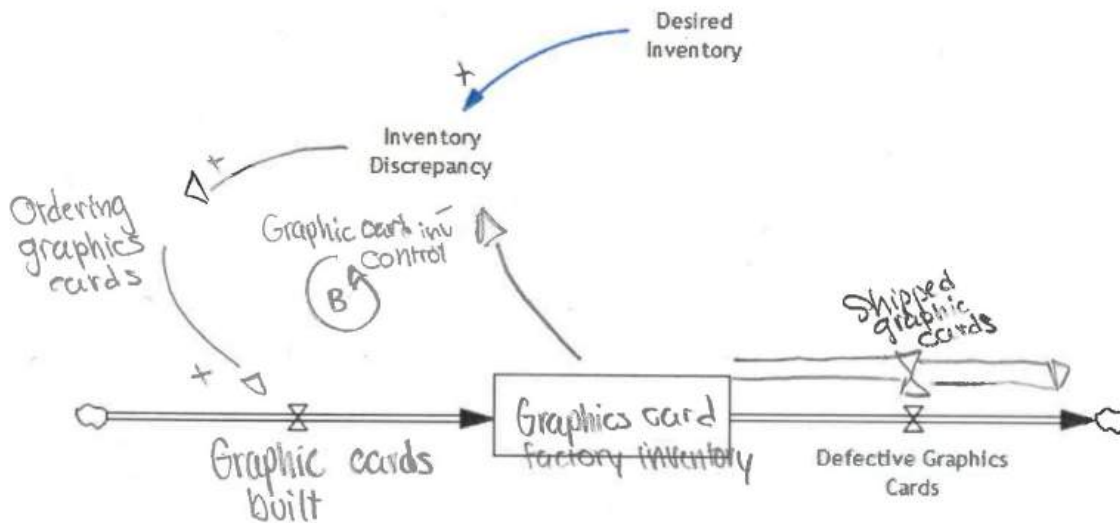
The diagram below depicts a simple structure of a graphics card factory. The factory needs to maintain a certain inventory so that they can meet shipping requirements to their distributors. Sometimes during testing prior to shipping, they also identify some defective graphics cards that can't be shipped. When the factory inventory falls too low, the warehouse manager will order additional graphics cards to be built to bring them back up to their desired inventory.



9. (5 Points) Identify the type of feedback loop in the diagram above and clearly **depict** and **label** and name this feedback on the diagram. Write your answer in the green box above.

Balancing (Negative) Feedback loop
Graphics card inventory control

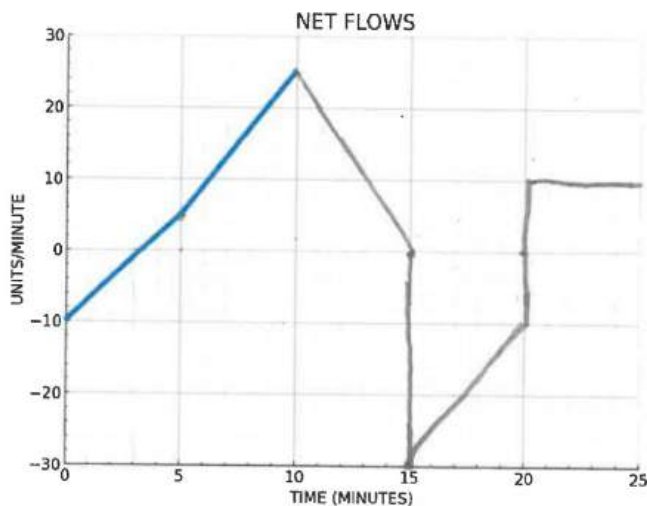
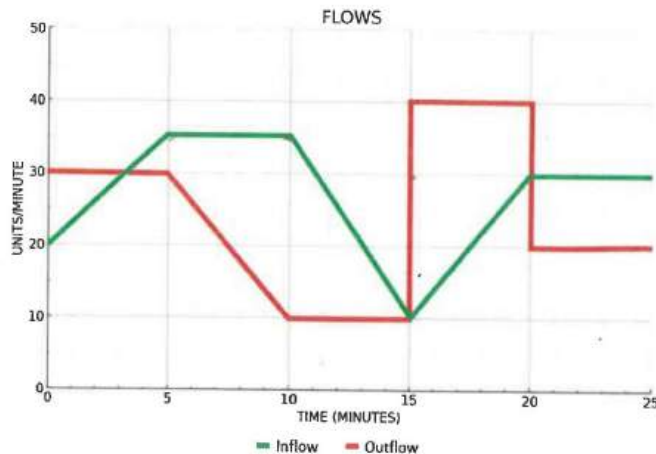
10. (25 Points) Using **only the variables** listed in the diagram on the previous page, draw a complete Stock and Flow Diagram to represent this system in the space below. Build on the basic structure that has been given to you below. Use only the CLD to construct your Stock and Flow diagram.



Name: Jackson Meyer

E. Dynamics of Stocks and Flows (40 Points)

11. (17 points) Consider a stock with the single inflow rate and the single outflow rate shown below. Complete the net flow of the stock and complete the behavior of the stock on the graphs provided. The initial value of the stock is 20 units.



Show your work here!

$$10 - 15 :$$

$$bh/2$$

$$5(25) = 125 / 2 = 62.5$$

$$25$$

$$50 = 75$$

$$85 + 62.5 = 147.5$$

$$20(5) = 100 \neq 50$$

$$10(5) = 50$$

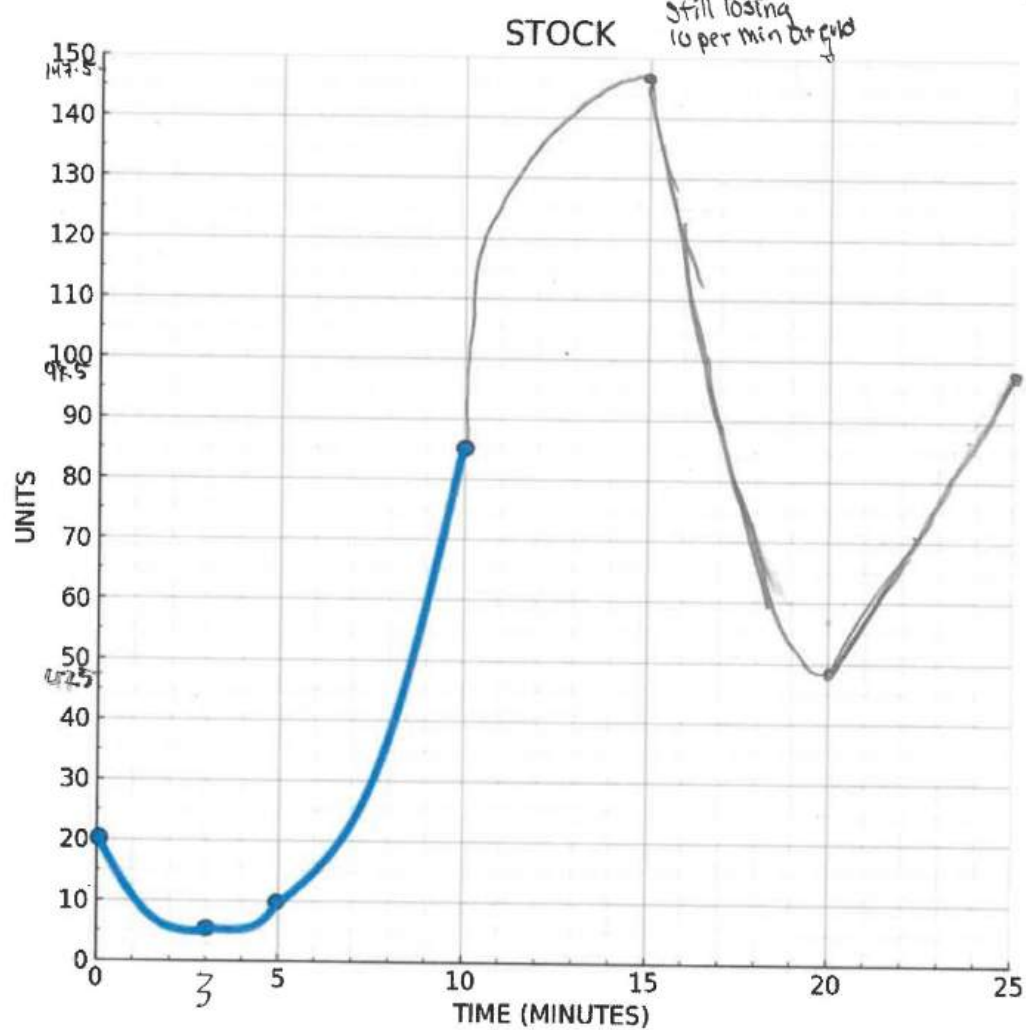
$$= 100$$

$$147.5 - 100 = 47.5$$

$$10(5) = 50 +$$

$$97.5$$

Name: Jarvison Meyer

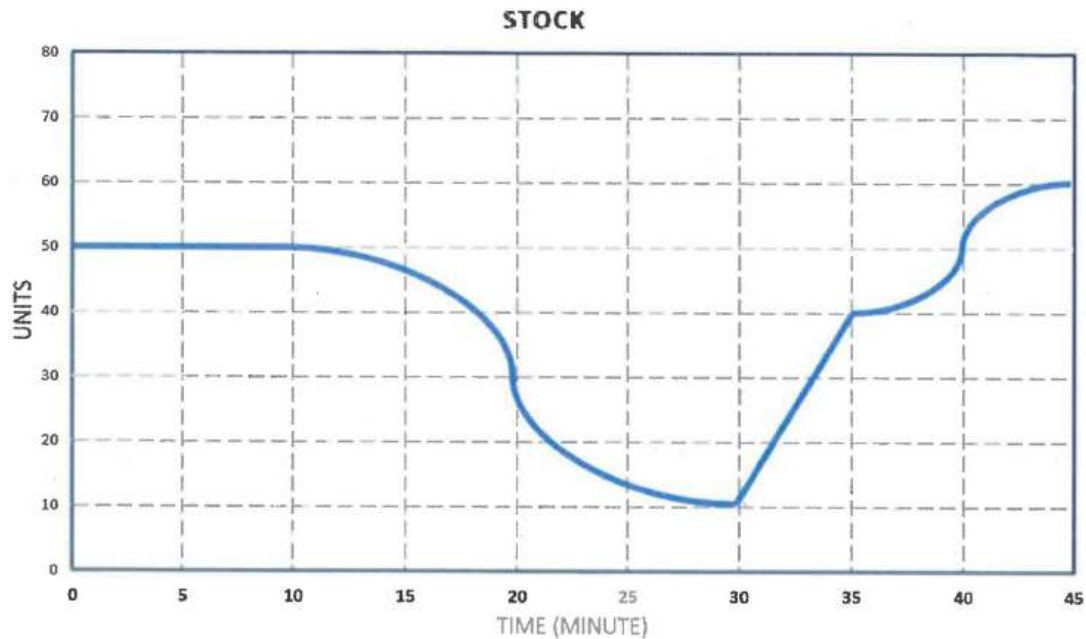


12. (3 points) What is the value of the stock at a time of 20 minutes?

47.5 units

Name: Jackson Meyer

13. (17 points) Consider the Stock illustrated below. Fill in the missing time segments to graph the Net Flow (on page 13) that would result in the following stock behavior.



Show your work here!

0-10:

—

10-20:

50 → 30
lose 20
increasing at end

↓
 $bh/2 = 20$
 $bh = 40$
 $n = 4$

20-30:

30 → 10
lose 20
increasing at beginning

↓
 $bh/2 = 20$
 $bh = 40$
 $n = 4$

30-35:

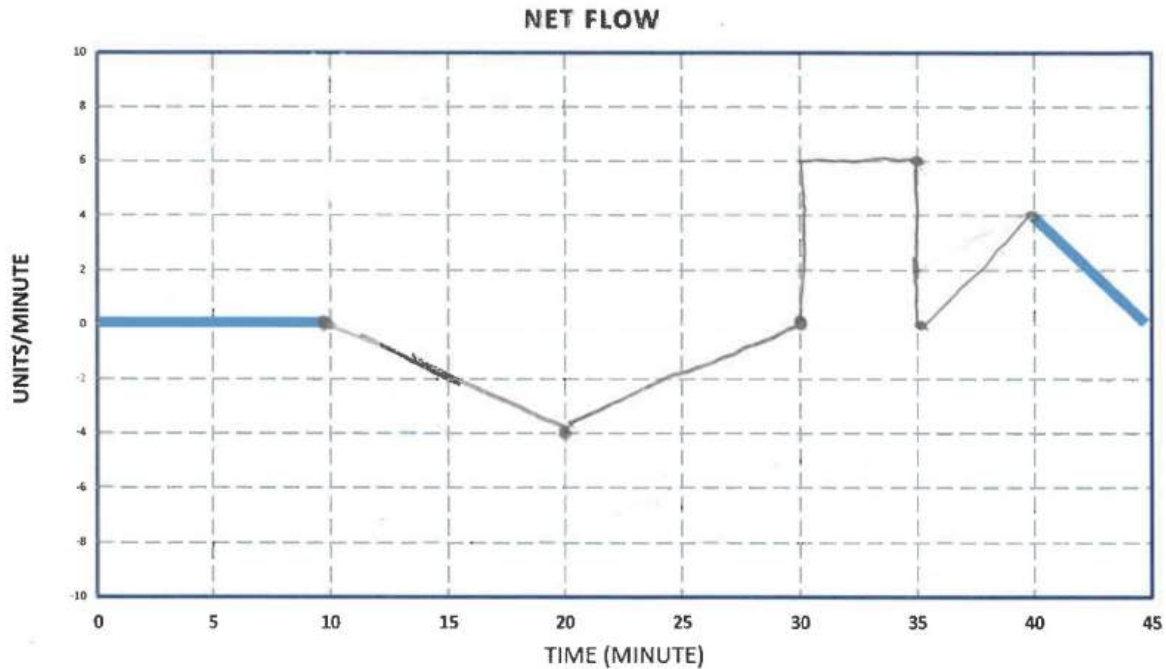
10 → 40
gain 30
constant

↓
 $bh = 30$
 $n = 6$

35-40:

40 → 50
gain 10
increasing at end

↓
 $bh/2 = 10$
 $bh = 20$
 $n = 4$

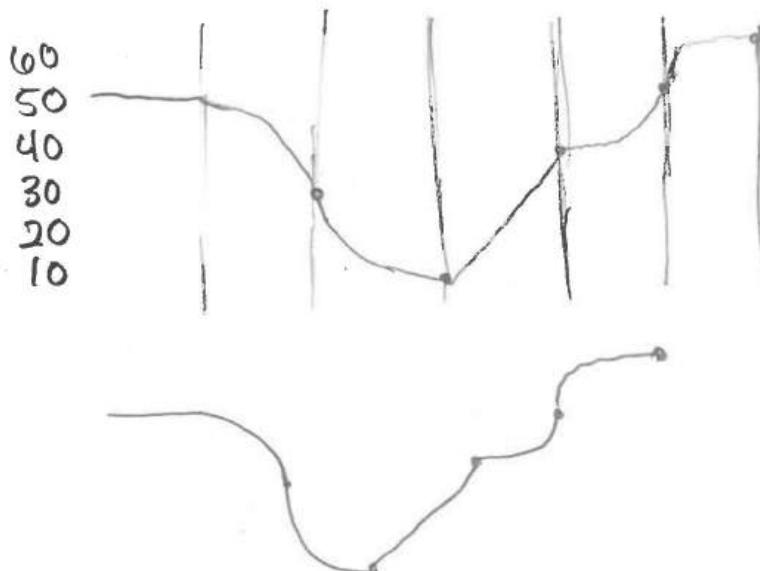


14. (3 points) What is the Net Flow at 20 Minutes?

-4 units per minute

$5(6) = 30$

$10(4)/2 = 20$
 $10(4)/2 = 20$
 $10(4)/2 = 20$



SM484: System Dynamics Simulation – AT 26-1
WPR 1 (Primary)

Name: _____

Extra Work Page

Fair-to-Good Student Work

WPR1

● Graded

Student

Lucas Sorensen

Total Points

132.5 / 150 pts

Question 1

Q1

12 / 12 pts

✓ - 0 pts Correct

- 1 pt PA missed key variables
- 1 pt PA missed time horizon
- 1 pt PA missed reference mode/pattern of behavior
- 1 pt PA missed "why problem, what is the problem, who is the client, or boundary selection"
- 1 pt Answer lacking correct detailed description.
- 1 pt Form missed simulation model/stock and flow
- 1 pt Form missed estimating parameters, relationships, and initial conditions
- 1 pt Form missed adding units/simulate often
- 1 pt Form missed translating causal theory into a model
- 2 pts Problem Articulation Incorrect
- 2 pts Formulation Incorrect

Question 2

Q2

2.5 / 3 pts

- 0 pts Correct
- 0.5 pts missed a
- 1 pt selected b
- 0.5 pts missed c
- 0.5 pts missed d

✓ - 0.5 pts missed e

Question 3

Q3

3 / 5 pts

– 0 pts Correct

– 1 pt selected a

✓ – 1 pt missed b

– 1 pt selected c

– 1 pt missed d

✓ – 1 pt selected e

Question 4

Q4

3 / 5 pts

– 0 pts Correct

✓ – 1 pt missed a

✓ – 1 pt missed b

– 1 pt selected c

– 1 pt missed d

– 1 pt missed e

Question 5

Q5

10 / 10 pts

✓ – 0 pts Correct

– 2 pts missed B

– 2 pts Missed A

– 2 pts missed C

– 2 pts missed E

– 2 pts Missed D

Question 6

Q6

5 / 5 pts

✓ - 0 pts Correct-Oscillation

- 2 pts S shaped with overshoot
- 3 pts goal seeking
- 2 pts mentions something else primarily, but secondary mention of oscillation
- 5 pts completely wrong answer
- 4 pts mentions/describes only balancing (not goal seek or balancing with delay), or overshoot and collapse

Question 7

Q CLD

19 / 20 pts

- 0 pts Correct
- 1 pt no/incorrect loop name
- 1 pt Missing arrows on links
- 1 pt missing delay between interest rate and inflation rate
- 1 pt missing delay from deviation...to interest rate

✓ - 1 pt missing/incorrect one polarity

- 2 pts missing/incorrect two polarities
- 3 pts missing/incorrect three polarities
- 4 pts missing/incorrect four polarities
- 1 pt target inflation rate not exogenous
- 1 pt Interest Rate not endogenous
- 1 pt missing balancing loop 'B'
- 1 pt missing or incorrect loop/direction
- 2 pts two variables switched
- 1 pt one incorrect causal link
- 2 pts interest rate not linked to inflation rate
- 2 pts neatness/hard to read
- 2 pts multiple incorrect causal links
- 2 pts missing one of the variables

Question 8

Q7

8 / 8 pts

✓ - 0 pts Correct

- 2 pts missed causal link
- 2 pts missed link polarity
- 2 pts missed independent variable
- 2 pts missed dependent variable
- 1 pt missed "Link Polarity" but put "polarity"
- 1 pt missed "variable"
- 1 pt missed "causal link" but included "link"
- 1 pt positive polarity instead of link polarity

Question 9

Q8

12 / 12 pts

✓ - 0 pts Correct

- 2 pts missed source
- 2 pts missed inflow
- 2 pts missed stock
- 2 pts missed outflow
- 2 pts missed sink
- 2 pts missed stock and flow diagram

Question 10

Q9

5 / 5 pts

✓ - 0 pts Correct

- 2 pts no clockwise arrow
- 2 pts no 'B'
- 1 pt no name/incorrect naming convention (i.e. calling a loop a CLD)
- 4 pts partial credit-some other incorrect effort

Question 11

Q10

13 / 25 pts

– 0 pts Correct

✓ – 5 pts no shipped graphics cards outflow

– 2 pts outflow represented as causal link

– 2 pts incorrect stock (not graphics card factory inventory)

✓ – 2 pts 1 incorrect/missing causal link

– 4 pts 2 incorrect/missing causal links

– 6 pts 3 incorrect/missing causal links

✓ – 1 pt 1 incorrect/missing polarity

– 2 pts 2 incorrect/missing polarity

– 3 pts 3 incorrect/missing polarity

– 4 pts 4 incorrect/missing polarities

– 1 pt minor loop errors

– 2 pts major loop errors

– 2 pts incorrect inflow

✓ – 2 pts additional general formulation errors

– 1 pt missing one delay

✓ – 2 pts missing two delays

– 1 pt missing shipped graphics outflow sink (cloud)

– 1 pt missed variable

Question 12

Q11

17 / 17 pts

12.1 Q11 FLOW

9 / 9 pts

✓ - 0 pts Correct

- 1 pt 10-15 partially wrong
- 2 pts 10-15 completely wrong
- 1 pt 15-20 partially wrong
- 2 pts 15-20 completely wrong
- 1 pt 20-25 partially wrong
- 2 pts 20-25 completely wrong

12.2 Q11 STOCK

8 / 8 pts

✓ - 0 pts Correct

- 1 pt 10-15 partially wrong
- 2 pts 10-15 completely wrong
- 1 pt 15-20 partially wrong
- 2 pts 15-20 completely wrong
- 1 pt 20-25 partially wrong
- 2 pts 20-25 completely wrong
- 0 pts credit awarded, work follows incorrect work in Net Flow. Other deductions may still be present.
- 8 pts no attempt

Question 13

Q12

3 / 3 pts

✓ - 0 pts Correct

- 3 pts completely wrong
- 0 pts Wrong answer but follows incorrect graph

Question 14

Q13

17 / 17 pts

14.1 Q13 WORK

0 / 0 pts

✓ - 0 pts No points here, questions only for grading purposes to see work in gradescope

14.2 Q13 FLOW

17 / 17 pts

✓ - 0 pts Correct

- 1.5 pts 10-20 partially wrong

- 3 pts 10-20 completely wrong

- 1.5 pts 20-30 partially wrong

- 3 pts 20-30 completely wrong

- 1.5 pts 30-35 partially wrong

- 3 pts 30-35 completely wrong

- 1.5 pts 35-40 partially wrong

- 3 pts 35-40 completely wrong

- 2 pts neatness (if your lines look curved, or extremely smudged from poor erasing)

- 4 pts minimal attempt

Question 15

Q14

3 / 3 pts

✓ - 0 pts Correct

- 0 pts wrong answer, but correct based on incorrect graph

- 3 pts wrong or no answer

Question 16

EXTRA WORK

0 / 0 pts

✓ - 0 pts No points associated with this, just for gradescope to see if you needed extra space



Department of Systems Engineering

System Dynamics Simulation

SM 484 - AT 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

WPR 1 (Primary)

Name: LUCAS SORENSEN

Section: J1

Finish Time: 1459

1. This is a **70-minute** exam worth **150 points**.
2. This is a closed book/notes exam, **ONE side of an 8-1/2x11 sheet of paper note sheet is allowed**. NO COLLABORATION allowed.
3. The only items that you may bring into the classroom are: one sided note sheet, pencil, straight edge, and calculator. **DO NOT BRING INK PENS!!!**
4. **Place your name NEATLY on each page in the space provided**. Feel free to disassemble the exam; a stapler is available to reattach the pages when you are finished. Make sure you have all 14 pages (including this cover page).
5. **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
6. After you finish the exam, fill in the finish time at the top of this page and sign the honor acknowledgement statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security at a time announced by the Course Director, no earlier than 06OCT25. Prior to that time, I may not discuss any aspects of this exam with anyone except an SM484 instructor." **PRINT IN BLOCK LETTERS NEATLY**

Printed Name (FIRST LAST): LUCAS SORENSEN Signature: [Signature]

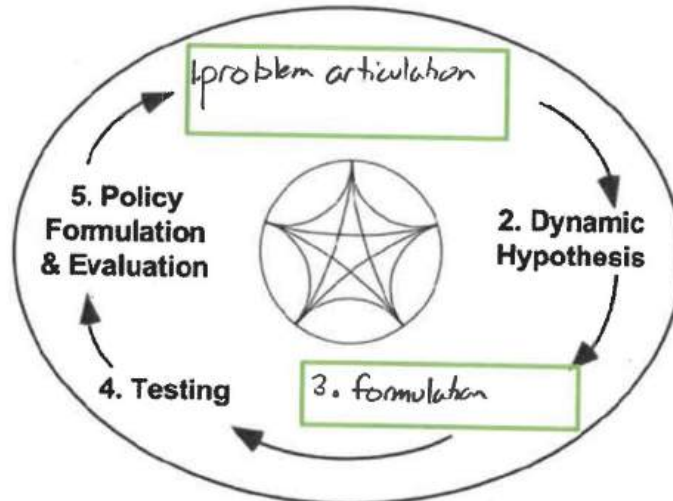
Date Signed: 22SEP2025

Grading summary

Question	Possible Points	Actual Points
1. Short Answer/Multiple Choice	35	
2. Fundamental Modes of Behavior	25	
3. Diagram Notation	20	
4. Stock and Flow Diagram	30	
5. Dynamics of Stocks and Flows	40	
Total	150	

A. Short Answer/Multiple Choice Questions (30 Points)

1. **(12 Points)** The System Dynamics Modeling Process is shown below. Identify BOTH missing steps (in the green boxes) and briefly describe ONE of the missing steps of the SD Modeling Process below.



Problem Articulation involves defining key variables, setting a time horizon & reference mode, as well as setting boundaries to what you are modeling.

2. **(3 Points)** Which of the following are elements that contribute to dynamic complexity? (Select all that apply)

- ☒ a. Policy-Resistant
- ☐ b. Reference Mode
- ☒ c. History-Dependent
- ☒ d. Tightly Coupled
- ☐ e. Self-Organizing

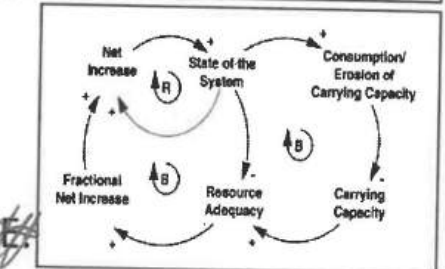
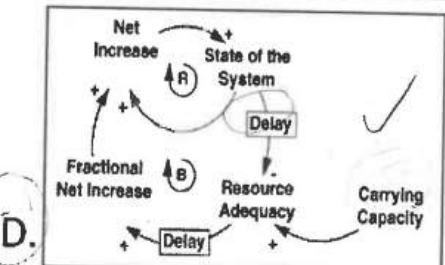
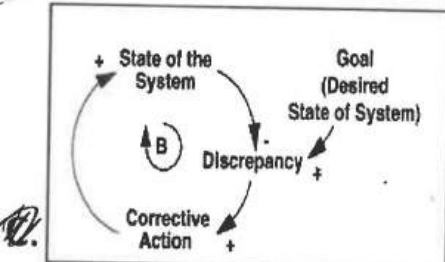
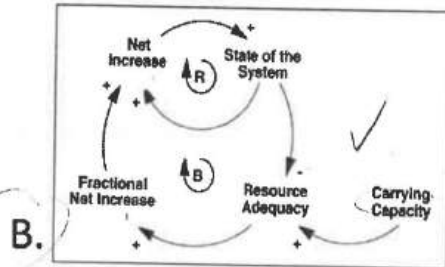
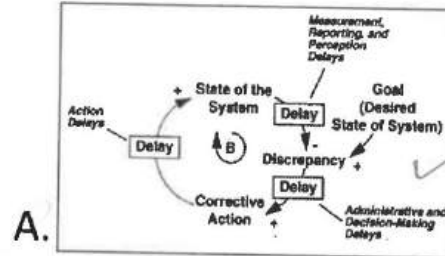
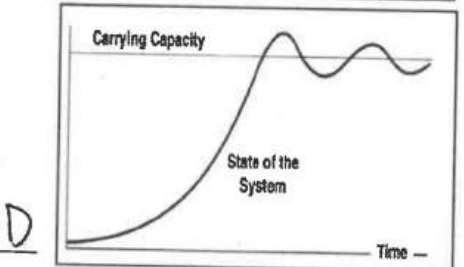
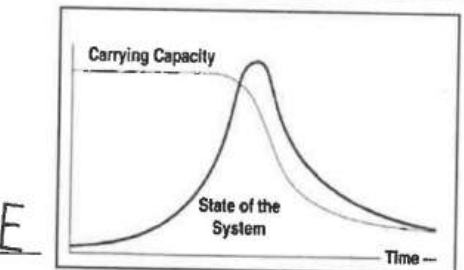
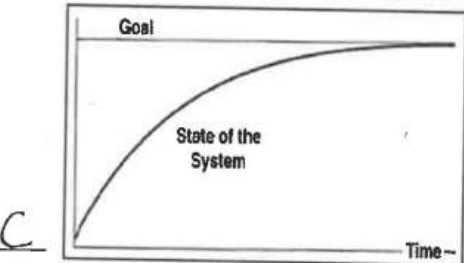
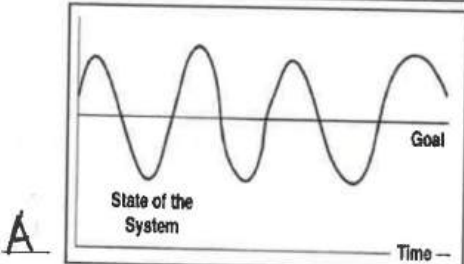
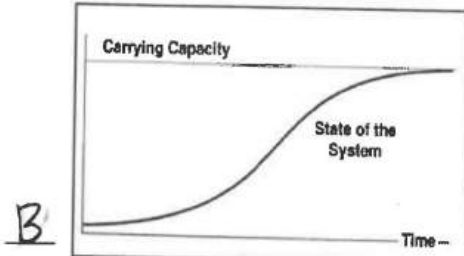
3. (5 Points) Which of the following are characteristics of exogenous variables? (Select all that apply)

- ☒ a. They can be changed by the system
- b. They are constants *can be modified by something outside of system*
- c. They originate inside the boundary of the system
- ☒ d. They originate outside the boundary of the system
- ☒ e. They are critical drivers of system feedback *goal setters*

4. (5 Points) Which of the following statements about stocks are true? (Select all that apply)

- a. Stocks decouple rates of flow and create disequilibrium dynamics
- b. Stocks are the sources of delay
- c. Stocks can be directly changed by auxiliary variables
- ☒ d. Stocks characterize the state of the system and provide the basis for actions
- ☒ e. Stocks provide systems with inertia and memory

5. (10 Points) Match the system behavior on the left, to the structure on the right. (Fill in the blank on the behavior, with the letter from the structure)



B. Fundamental Modes of Behavior (25 Points)

Read the following description about the Federal Reserve and answer the subsequent questions.

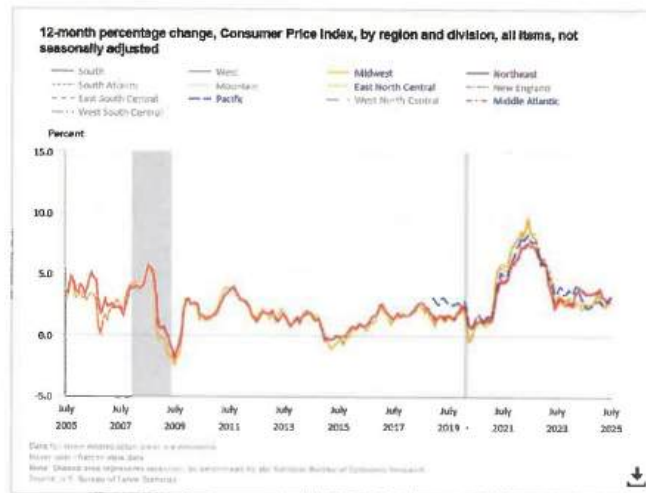


Figure 1: Image Credit <https://www.bls.gov/charts/consumer-price-index/consumer-price-index-by-region.htm>

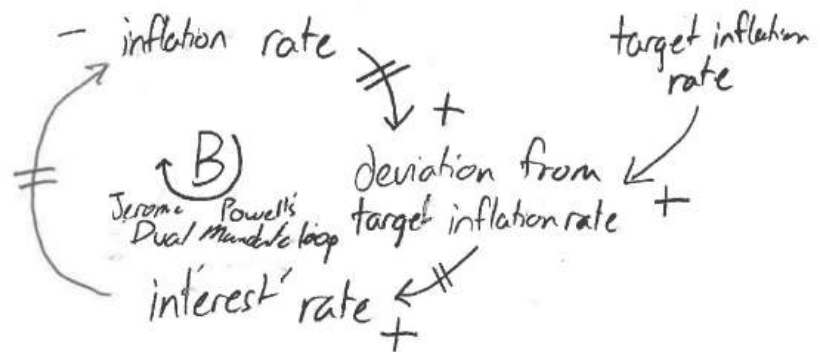
The Federal Reserve enacts policy with **the goal of keeping inflation at 2%**, measured by the Consumer Price Index shown in Figure 1 above. **If inflation is significantly below the inflation target, the Federal Reserve may cut interest rates** to spur economic activity and bring inflation back to the 2% target. **When inflation exceeds target inflation, the Federal Reserve will react and adjust the interest rate higher** in an attempt to check inflation.

To this end, the Federal Reserve increased interest rates several times in 2022 through 2023 to combat the rise in inflation, but the effect is not instantaneous. The figure above shows how the inflation rate continued to rise through 2022, even with the Federal Reserve already adjusting the interest rate, and it didn't start to drop until 2023. Though it does not show when interest rate hikes have occurred, those increases can be inferred from the inflation history. Answer the following questions based on the above image and description of the Federal Reserve's role in managing the inflation rate.

6. (5 Points) What fundamental mode of behavior is being described in the previous description and shown in Figure 1 above?

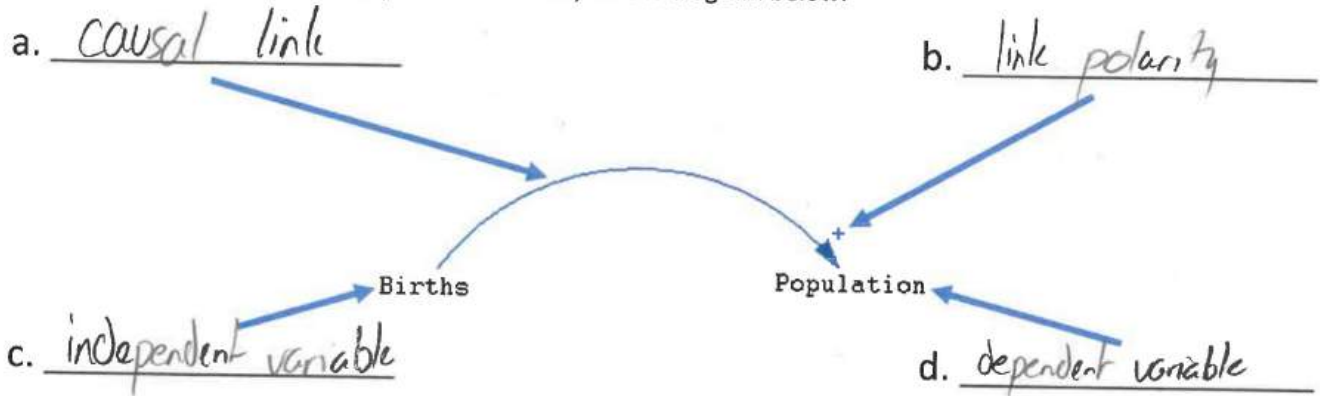
Oscillation is being shown. There is an exogenous goal, balancing loop, & delays.

(20 Points) The Federal Reserve implements monetary policy tools, such as the interest rate, to control inflation. In this scenario, you are facing a situation where the inflation rate is *above the target rate* and corrective actions are being taken by the Federal Reserve to bring the inflation back down. In the space below, draw a CLD that depicts the system structure corresponding to the fundamental mode of behavior that you identified previously. Use the following variables to create an abstracted CLD: **Deviation from Target Inflation Rate, Interest Rate, Inflation Rate, and Target Inflation Rate**. Remember that the system structure dictates the system behavior. Be sure to specify (1) Link Polarities, (2) Loop Directions, and (3) Loop Types with Loop name

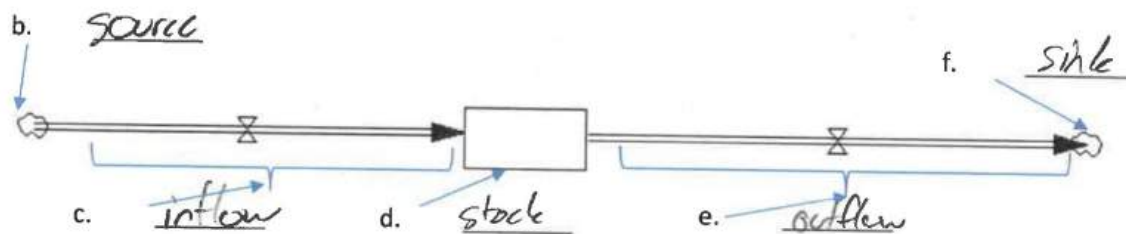


C. Diagram Notation (20 Points)

7. (8 Points) Correctly label each entity on the diagram below:



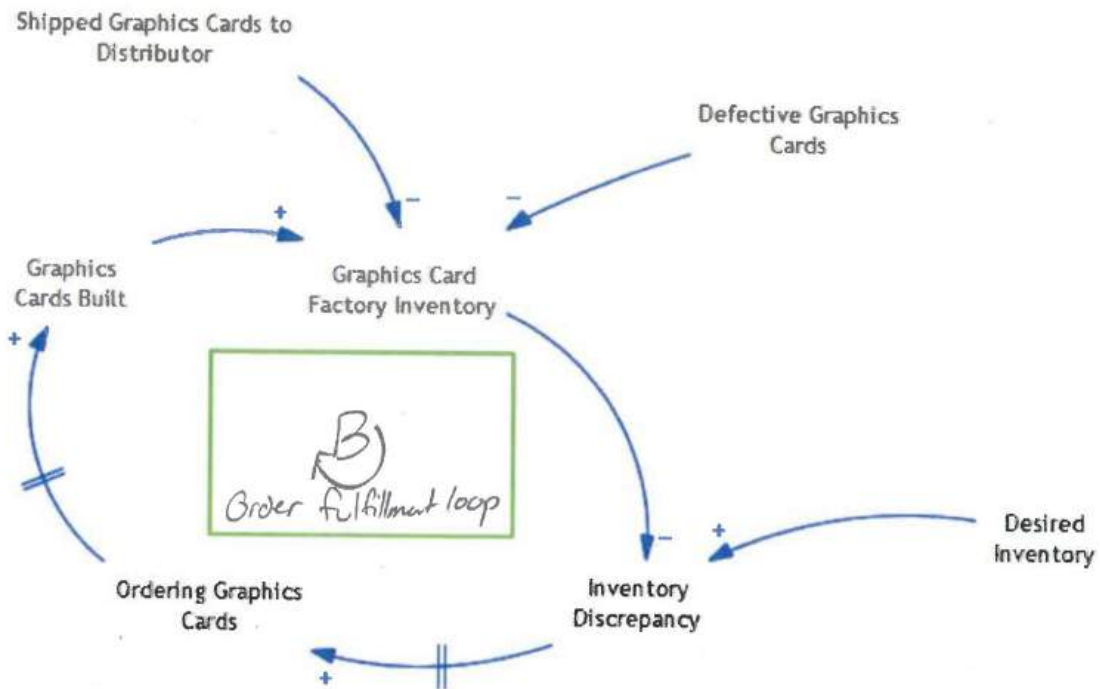
8. (12 Points) Correctly label each entity on the diagram below:



a. What is this diagram called? stocks & flows diagram

D. CLD → S&F (30 Points)

The diagram below depicts a simple structure of a graphics card factory. The factory needs to maintain a certain inventory so that they can meet shipping requirements to their distributors. Sometimes during testing prior to shipping, they also identify some defective graphics cards that can't be shipped. When the factory inventory falls too low, the warehouse manager will order additional graphics cards to be built to bring them back up to their desired inventory.



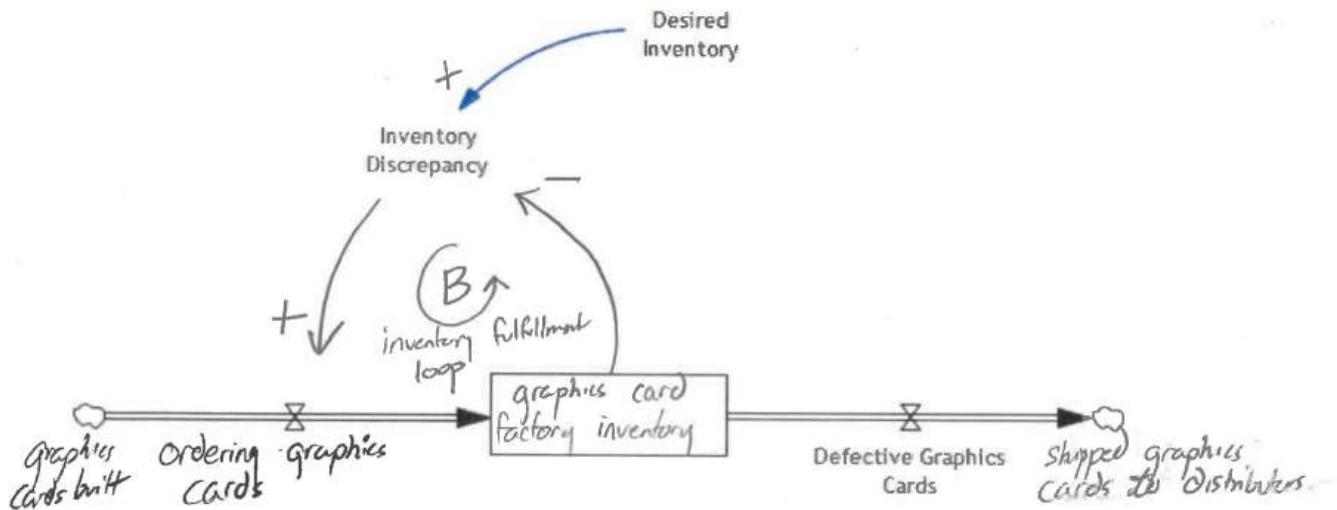
9. (5 Points) Identify the type of feedback loop in the diagram above and clearly *depict* and *label* and name this feedback on the diagram. Write your answer in the green box above. ✓

SM484: System Dynamics Simulation – AT 26-1

WPR 1 (Primary)

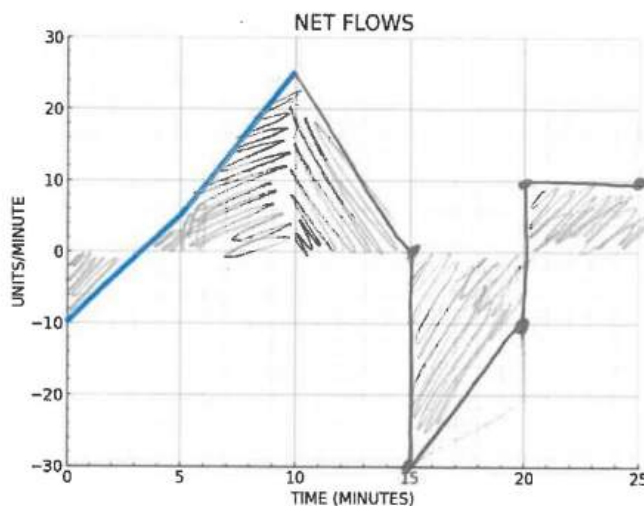
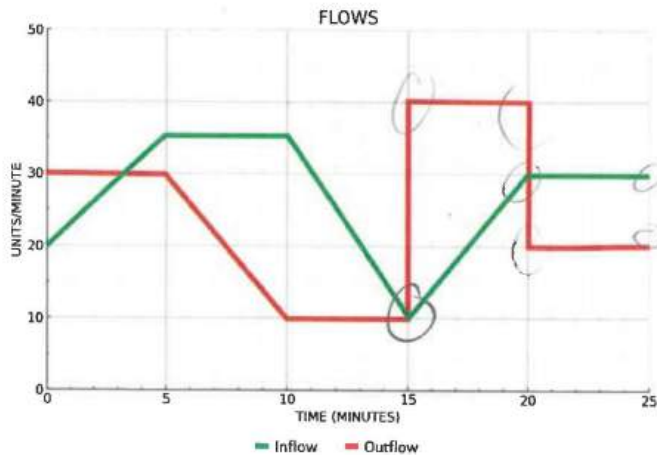
Name: LUCAS SORENSEN

10. (25 Points) Using **only the variables** listed in the diagram on the previous page, draw a complete Stock and Flow Diagram to represent this system in the space below. Build on the basic structure that has been given to you below. Use only the CLD to construct your Stock and Flow diagram.



E. Dynamics of Stocks and Flows (40 Points)

11. (17 points) Consider a stock with the single inflow rate and the single outflow rate shown below. Complete the net flow of the stock and complete the behavior of the stock on the graphs provided. *The initial value of the stock is 20 units.*



Show your work here!

$$15: 10 - 10 = 0$$

$$10 - 40 = -30$$

$$20: 30 - 40 = -10$$

$$30 - 20 = 10$$

$$25: 30 - 20 = 10$$

stock

$$10: 85$$

$$15: 5 \cdot 25 / 2 = 62.5$$

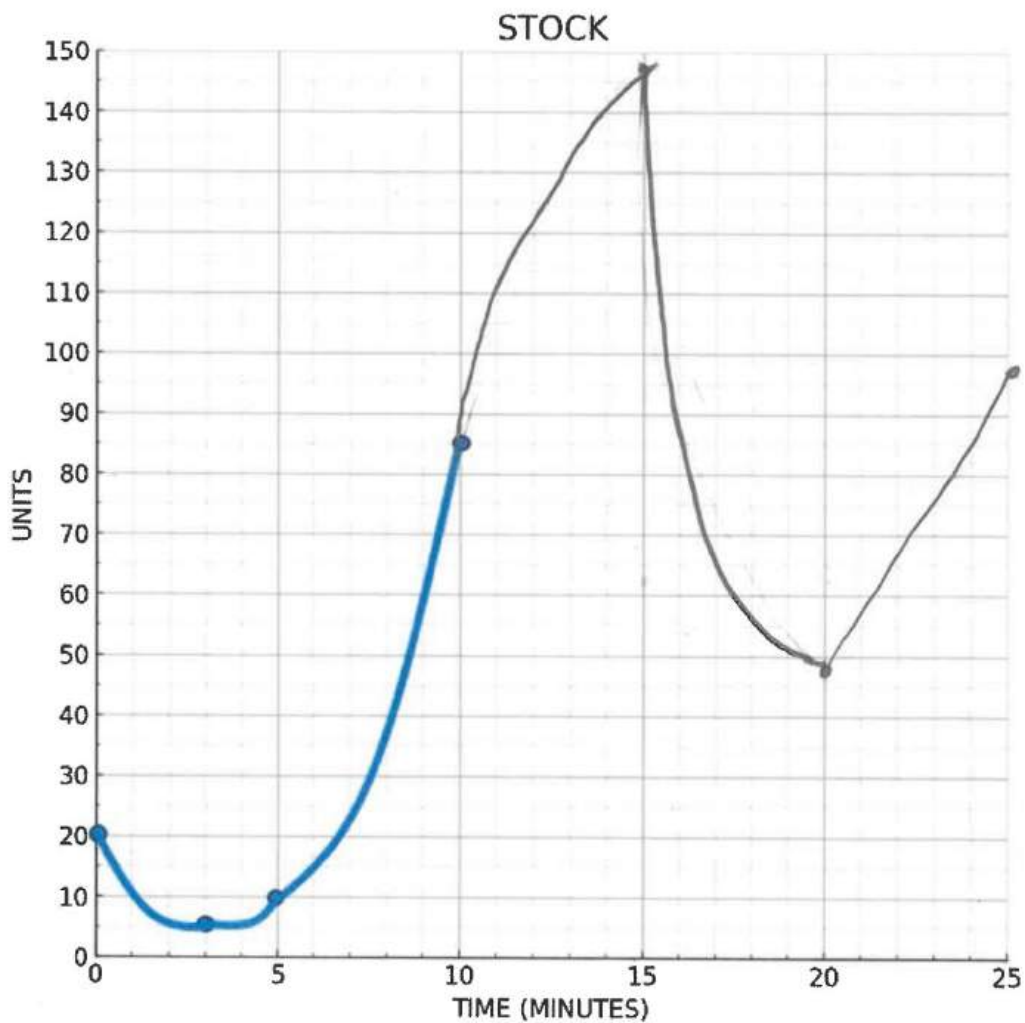
$$85 + 62.5 = 147.5$$

$$20: (5 \cdot 10) + (-5 \cdot 20 / 2) = -100$$

$$147.5 - 100 = 47.5$$

$$25: 5 \cdot 10 = 50$$

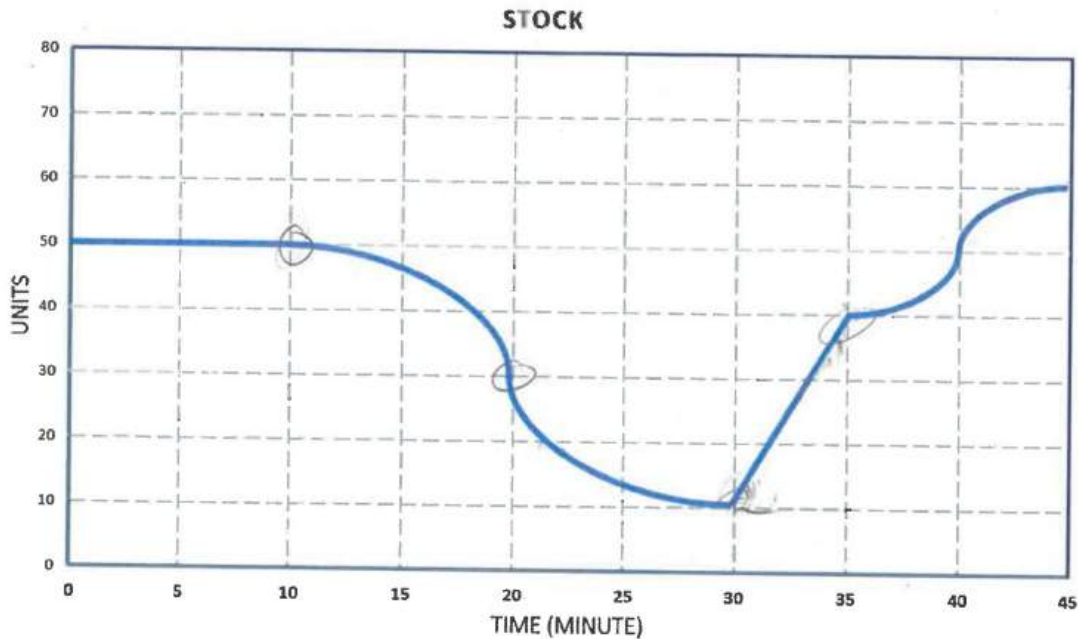
$$50 + 47.5 = 97.5$$



12. (3 points) What is the value of the stock at a time of 20 minutes?

47.5

13. (17 points) Consider the Stock illustrated below. Fill in the missing time segments to graph the Net Flow (on page 13) that would result in the following stock behavior.



Show your work here!

$$0-10 : NF = 0 \text{ —}$$

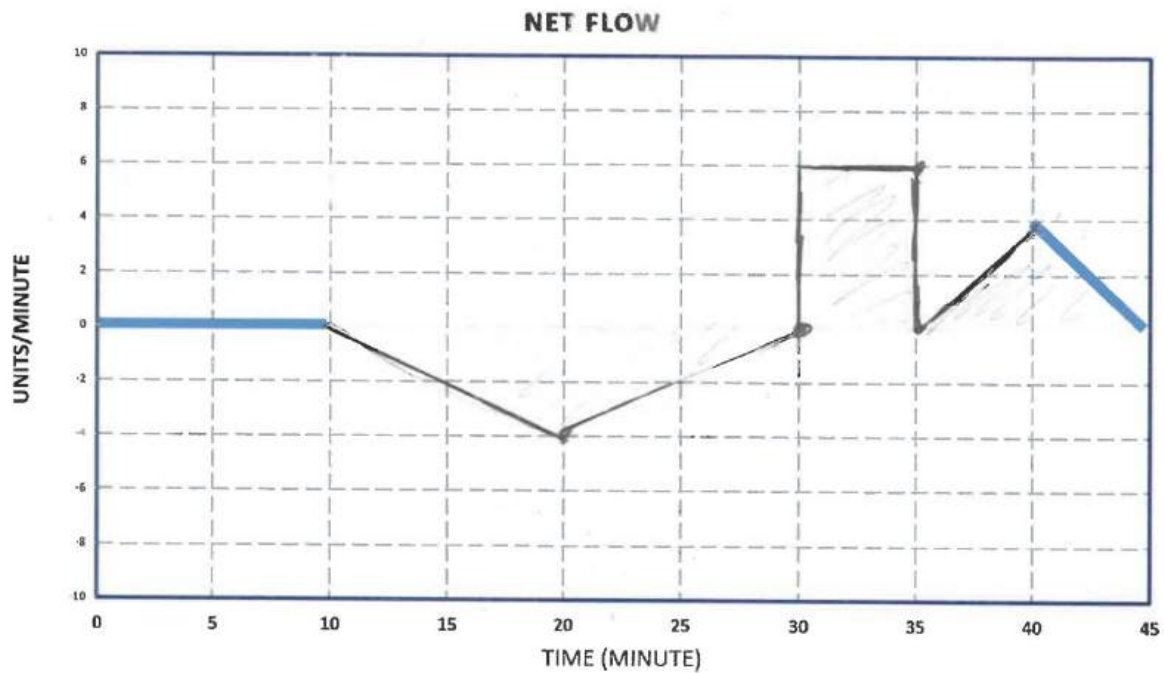
$$10-20 : NF = \frac{30-50}{20-10} = -2 \rightarrow \backslash \text{ dec}$$

$$20-30 : NF = \frac{10-30}{30-20} = -2 \rightarrow / \text{ inc}$$

$$30-35 : NF = \frac{40-10}{35-30} = 6 \rightarrow \text{— con}$$

$$35-40 : NF = \frac{50-40}{40-35} = 2 \rightarrow / \text{ inc}$$

$$40-45 : NF = \frac{60-50}{45-40} = 2 \rightarrow \backslash \text{ dec}$$



14. (3 points) What is the Net Flow at 20 Minutes?

-4 units per minute

Extra Work Page

BEAT NAVY
ALL SYSTEMS GO

Poor Student Work

Student
Henry Okeke

Total Points
86 / 150 pts

Question 1

Q1

10 / 12 pts

- 0 pts Correct
- 1 pt PA missed key variables
- 1 pt PA missed time horizon
- 1 pt PA missed reference mode/pattern of behavior
- 1 pt PA missed "why problem, what is the problem, who is the client, or boundary selection"
- 1 pt Answer lacking correct detailed description.

✓ - 1 pt Form missed simulation model/stock and flow

✓ - 1 pt Form missed estimating parameters, relationships, and initial conditions

- 1 pt Form missed adding units/simulate often
- 1 pt Form missed translating causal theory into a model
- 2 pts Problem Articulation Incorrect
- 2 pts Formulation Incorrect

Question 2

Q2

1.5 / 3 pts

- 0 pts Correct
- 0.5 pts missed a

✓ - 1 pt selected b

- 0.5 pts missed c
- 0.5 pts missed d

✓ - 0.5 pts missed e

Question 3

Q3

5 / 5 pts

✓ - 0 pts Correct

- 1 pt selected a
- 1 pt missed b
- 1 pt selected c
- 1 pt missed d
- 1 pt selected e

Question 4

Q4

1 / 5 pts

- 0 pts Correct

✓ - 1 pt missed a

✓ - 1 pt missed b

✓ - 1 pt selected c

✓ - 1 pt missed d

- 1 pt missed e

Question 5

Q5

10 / 10 pts

✓ - 0 pts Correct

- 2 pts missed B
- 2 pts Missed A
- 2 pts missed C
- 2 pts missed E
- 2 pts Missed D

Question 6

Q6

1 / 5 pts

- 0 pts Correct-Oscillation
- 2 pts S shaped with overshoot
- 3 pts goal seeking
- 2 pts mentions something else primarily, but secondary mention of oscillation
- 5 pts completely wrong answer

✓ - 4 pts mentions/describes only balancing (not goal seek or balancing with delay), or overshoot and collapse

Question 7

Q CLD

14 / 20 pts

- 0 pts Correct

✓ - 1 pt no/incorrect loop name

- 1 pt Missing arrows on links

- 1 pt missing delay between interest rate and inflation rate

- 1 pt missing delay from deviation...to interest rate

- 1 pt missing/incorrect one polarity

✓ - 2 pts missing/incorrect two polarities

- 3 pts missing/incorrect three polarities

- 4 pts missing/incorrect four polarities

✓ - 1 pt target inflation rate not exogenous

✓ - 1 pt Interest Rate not endogenous

- 1 pt missing balancing loop 'B'

- 1 pt missing or incorrect loop/direction

- 2 pts two variables switched

✓ - 1 pt one incorrect causal link

- 2 pts interest rate not linked to inflation rate

- 2 pts neatness/hard to read

- 2 pts multiple incorrect causal links

- 2 pts missing one of the variables

Question 8

Q7

8 / 8 pts

✓ - 0 pts Correct

- 2 pts missed causal link

- 2 pts missed link polarity

- 2 pts missed independent variable

- 2 pts missed dependent variable

- 1 pt missed "Link Polarity" but put "polarity"

- 1 pt missed "variable"

- 1 pt missed "causal link" but included "link"

- 1 pt positive polarity instead of link polarity

Question 9

Q8

12 / 12 pts

✓ - 0 pts Correct

- 2 pts missed source

- 2 pts missed inflow

- 2 pts missed stock

- 2 pts missed outflow

- 2 pts missed sink

- 2 pts missed stock and flow diagram

Question 10

Q9

1 / 5 pts

- 0 pts Correct

- 2 pts no clockwise arrow

- 2 pts no 'B'

- 1 pt no name/incorrect naming convention (i.e. calling a loop a CLD)

✓ - 4 pts partial credit-some other incorrect effort

Question 11

Q10

6 / 25 pts

- 0 pts Correct

✓ - 5 pts no shipped graphics cards outflow

- 2 pts outflow represented as causal link

- 2 pts incorrect stock (not graphics card factory inventory)

- 2 pts 1 incorrect/missing causal link

- 4 pts 2 incorrect/missing causal links

✓ - 6 pts 3 incorrect/missing causal links

- 1 pt 1 incorrect/missing polarity

- 2 pts 2 incorrect/missing polarity

- 3 pts 3 incorrect/missing polarity

✓ - 4 pts 4 incorrect/missing polarities

- 1 pt minor loop errors

✓ - 2 pts major loop errors

- 2 pts incorrect inflow

- 2 pts additional general formulation errors

- 1 pt missing one delay

✓ - 2 pts missing two delays

- 1 pt missing shipped graphics outflow sink (cloud)

- 1 pt missed variable

Question 12

Q11

4 / 17 pts

12.1

Q11 FLOW

4 / 9 pts

- 0 pts Correct

✓ - 1 pt 10-15 partially wrong

- 2 pts 10-15 completely wrong

- 1 pt 15-20 partially wrong

✓ - 2 pts 15-20 completely wrong

- 1 pt 20-25 partially wrong

✓ - 2 pts 20-25 completely wrong

12.2

Q11 STOCK

0 / 8 pts

- 0 pts Correct

- 1 pt 10-15 partially wrong

- 2 pts 10-15 completely wrong

- 1 pt 15-20 partially wrong

- 2 pts 15-20 completely wrong

- 1 pt 20-25 partially wrong

- 2 pts 20-25 completely wrong

- 0 pts credit awarded, work follows incorrect work in Net Flow. Other deductions may still be present.

✓ - 8 pts no attempt

Question 13

Q12

0 / 3 pts

- 0 pts Correct

✓ - 3 pts completely wrong

- 0 pts Wrong answer but follows incorrect graph

Question 14

Q13

9.5 / 17 pts

14.1

Q13 WORK

0 / 0 pts

✓ - 0 pts No points here, questions only for grading purposes to see work in gradescope

14.2

Q13 FLOW

9.5 / 17 pts

- 0 pts Correct

- 1.5 pts 10-20 partially wrong

- 3 pts 10-20 completely wrong

- 1.5 pts 20-30 partially wrong

✓ - 3 pts 20-30 completely wrong

- 1.5 pts 30-35 partially wrong

✓ - 3 pts 30-35 completely wrong

✓ - 1.5 pts 35-40 partially wrong

- 3 pts 35-40 completely wrong

- 2 pts neatness (if your lines look curved, or extremely smudged from poor erasing)

- 4 pts minimal attempt

Question 15

Q14

3 / 3 pts

✓ - 0 pts Correct

- 0 pts wrong answer, but correct based on incorrect graph

- 3 pts wrong or no answer

Question 16

EXTRA WORK

0 / 0 pts

✓ - 0 pts No points associated with this, just for gradescope to see if you needed extra space



Department of Systems Engineering

System Dynamics Simulation

SM 484 - AT 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

WPR 1 (Primary)

Name: OKELVE

Section: II

Finish Time: _____

1. This is a **70-minute** exam worth **150 points**.
2. This is a closed book/notes exam, **ONE side of an 8-1/2x11 sheet of paper note sheet is allowed**. NO COLLABORATION allowed.
3. The only items that you may bring into the classroom are: one sided note sheet, pencil, straight edge, and calculator. **DO NOT BRING INK PENS!!!**
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6. After you finish the exam, fill in the finish time at the top of this page and sign the honor acknowledgement statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security at a time announced by the Course Director, no earlier than 06OCT25. Prior to that time, I may not discuss any aspects of this exam with anyone except an SM484 instructor." **PRINT IN BLOCK LETTERS NEATLY**

Printed Name (**FIRST LAST**): HENRY OKELVE Signature: [Signature]

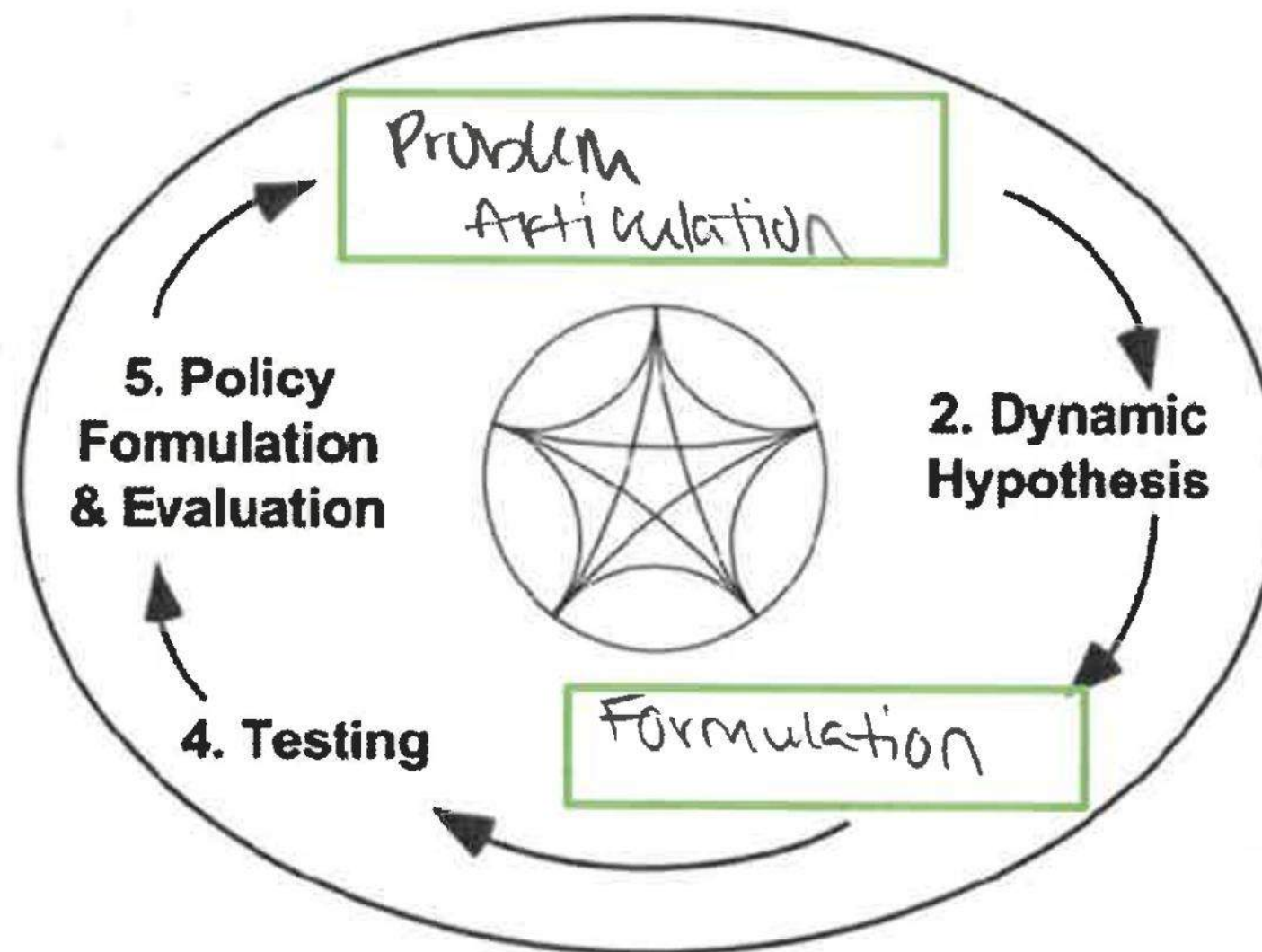
Date Signed: 0922 25

Grading summary

Question	Possible Points	Actual Points
1. Short Answer/Multiple Choice	35	
2. Fundamental Modes of Behavior	25	
3. Diagram Notation	20	
4. Stock and Flow Diagram	30	
5. Dynamics of Stocks and Flows	40	
Total	150	

A. Short Answer/Multiple Choice Questions (30 Points)

1. **(12 Points)** The System Dynamics Modeling Process is shown below. Identify **BOTH** missing steps (in the green boxes) and briefly describe **ONE** of the missing steps of the SD Modeling Process below.



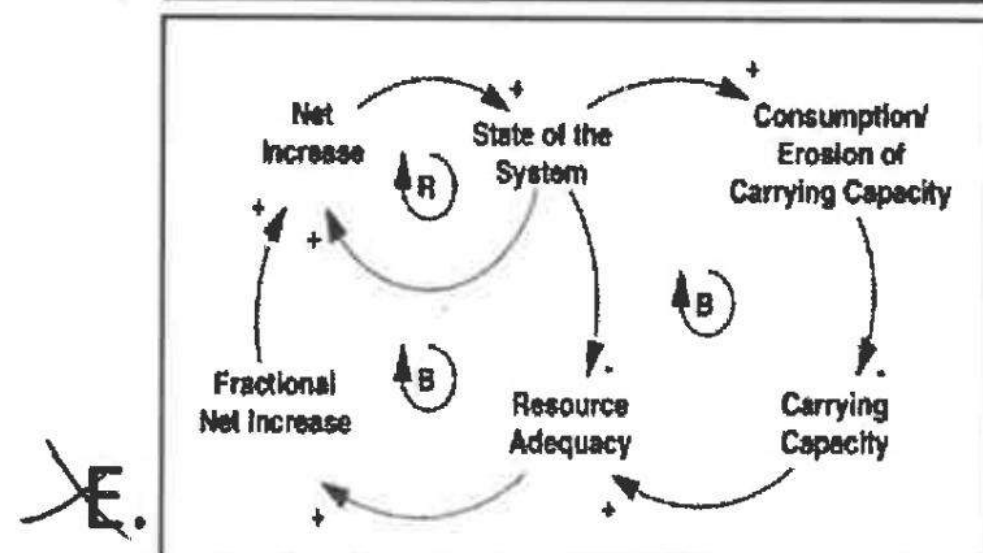
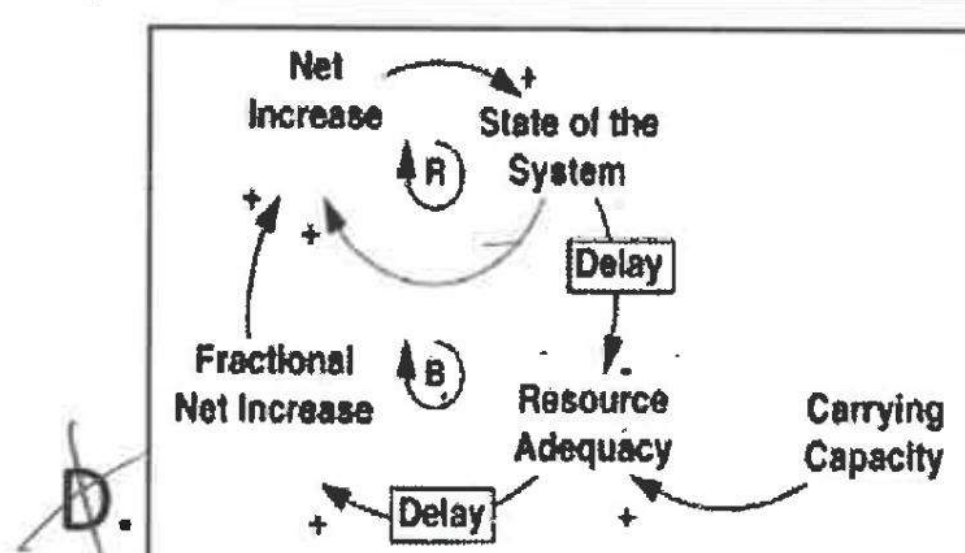
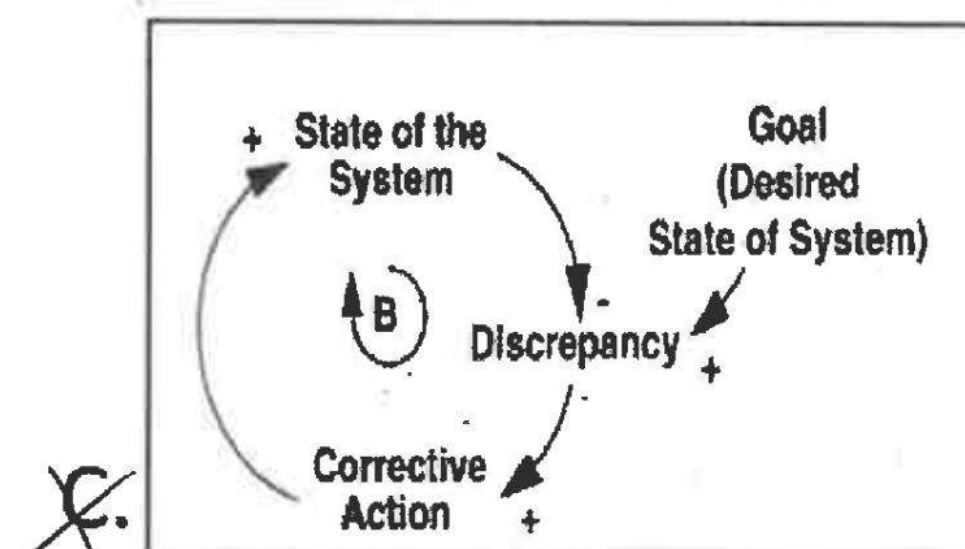
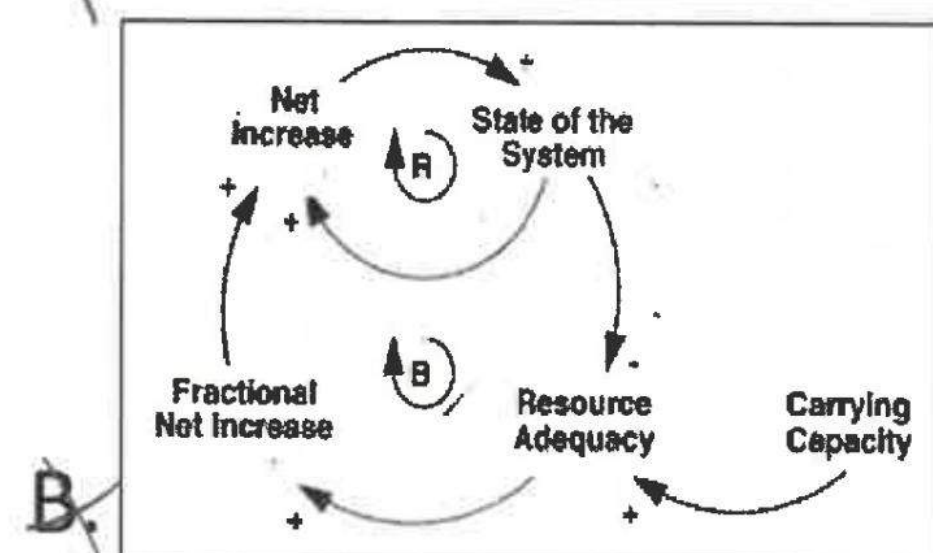
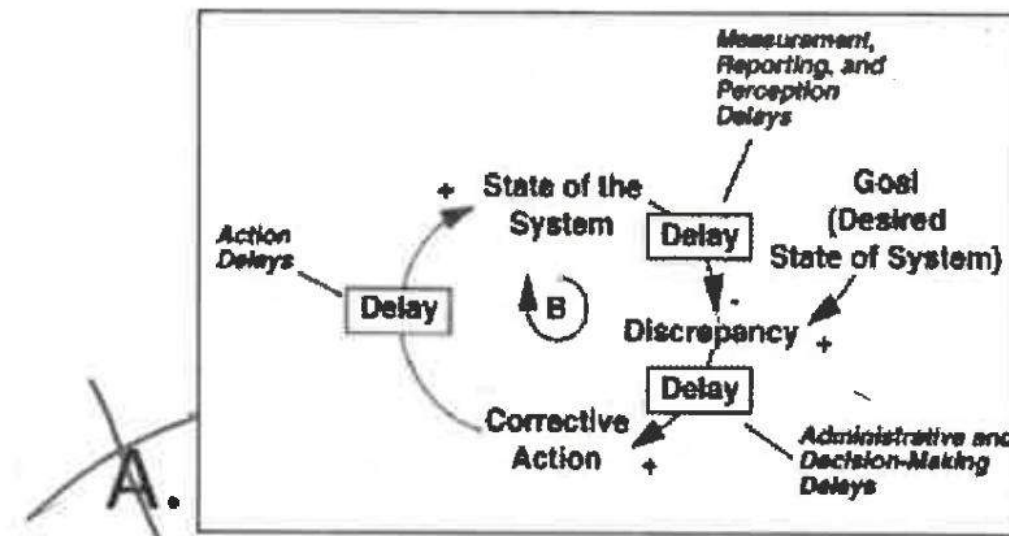
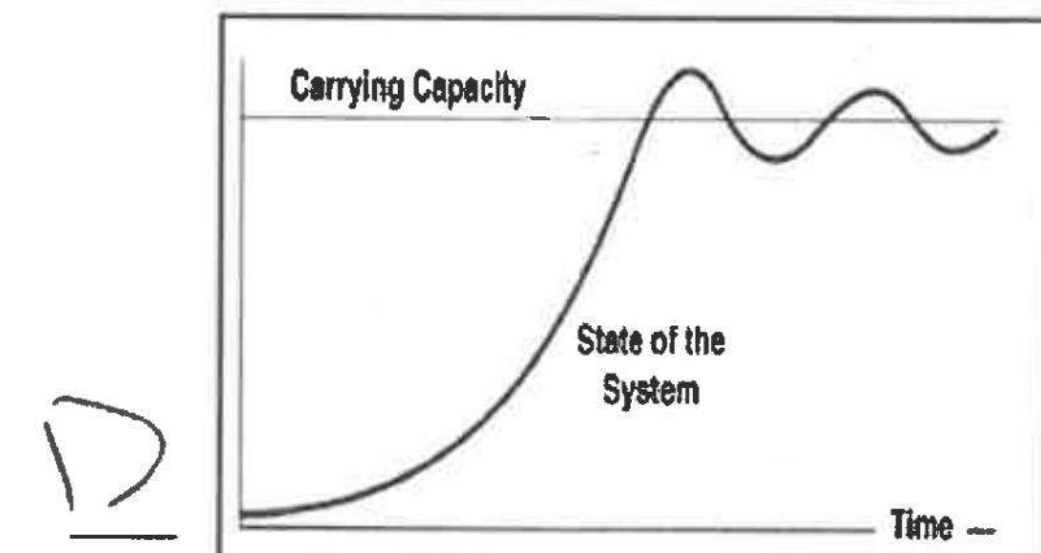
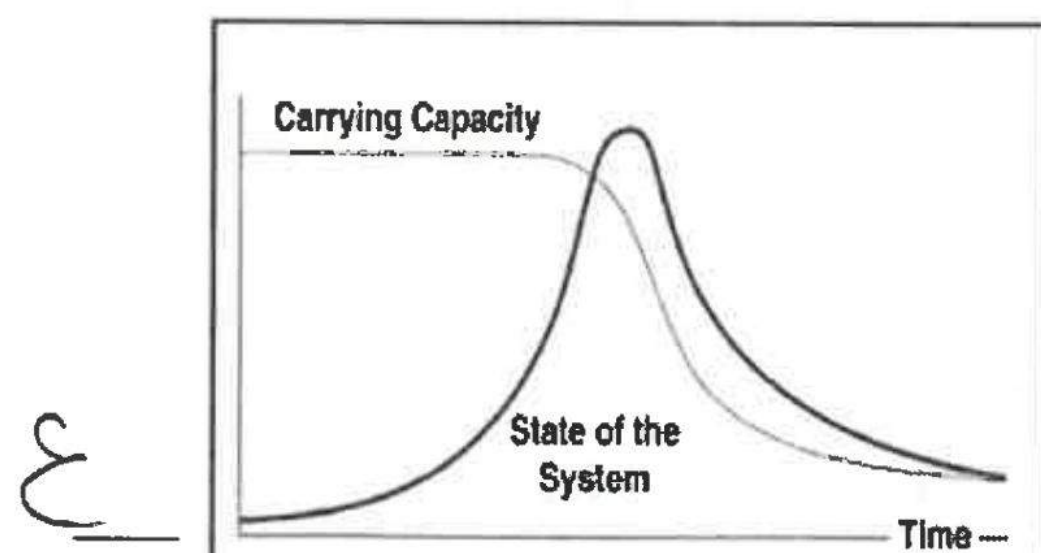
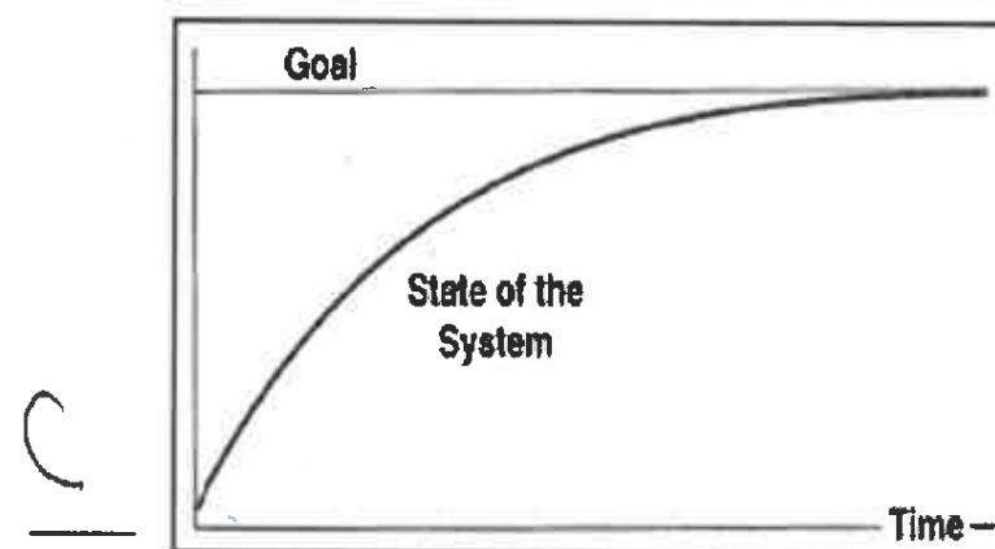
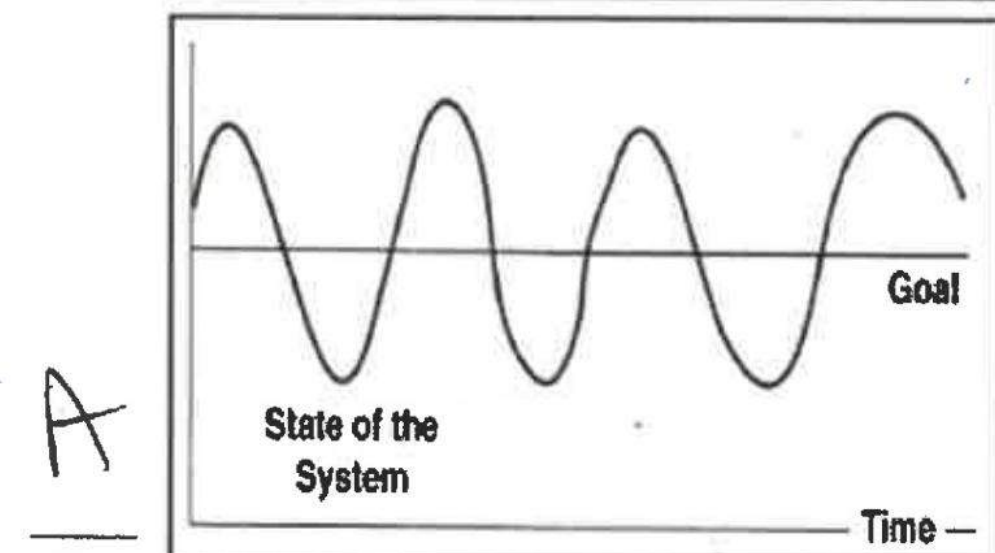
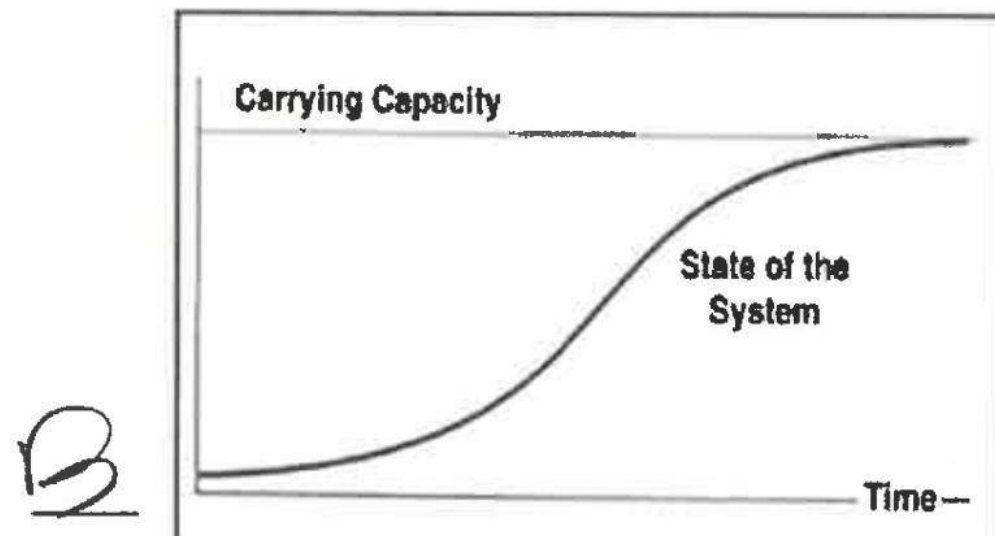
Formulation = Simulate early & often, & formulate a conceptual model that reflects the hypothesis that was created

2. **(3 Points)** Which of the following are elements that contribute to dynamic complexity? (Select all that apply)

- ☒ a. Policy-Resistant
- ☒ b. Reference Mode
- ☒ c. History-Dependent
- ☒ d. Tightly Coupled
- ☐ e. Self-Organizing

3. (5 Points) Which of the following are characteristics of **exogenous variables**? (Select all that apply)
- a. They can be changed by the system
 - ☒ b. They are constants
 - c. They originate inside the boundary of the system
 - ☒ d. They originate outside the boundary of the system
 - e. They are critical drivers of system feedback
4. (5 Points) Which of the following statements about **stocks** are true? (Select all that apply)
- a. Stocks decouple rates of flow and create disequilibrium dynamics
 - b. Stocks are the sources of delay
 - ☒ c. Stocks can be directly changed by auxiliary variables
 - d. Stocks characterize the state of the system and provide the basis for actions
 - ☒ e. Stocks provide systems with inertia and memory

5. (10 Points) Match the system behavior on the left, to the structure on the right. (Fill in the blank on the behavior, with the letter from the structure)



B. Fundamental Modes of Behavior (25 Points)

Read the following description about the Federal Reserve and answer the subsequent questions.

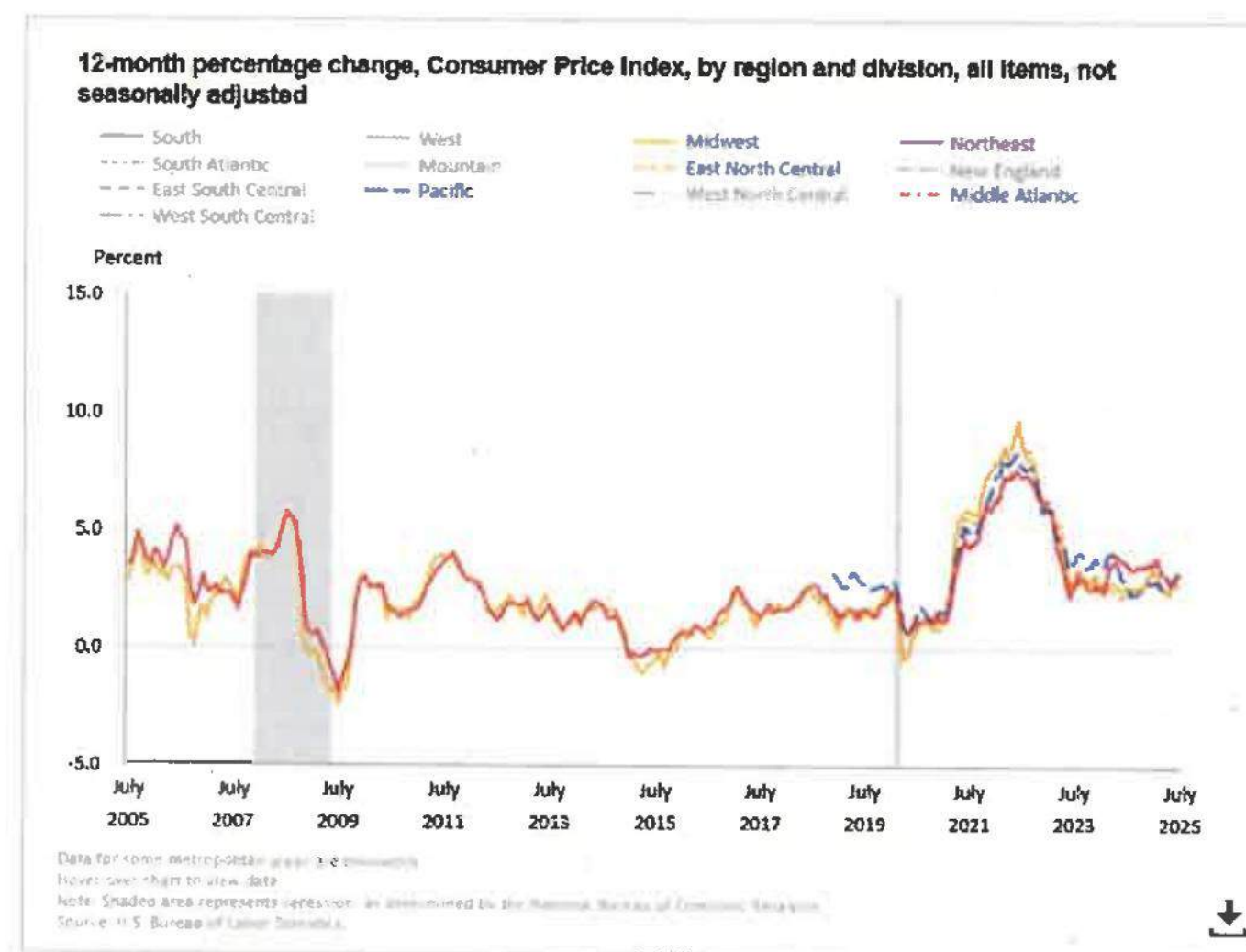


Figure 1: Image Credit <https://www.bls.gov/charts/consumer-price-index/consumer-price-index-by-region.htm>

The Federal Reserve enacts policy with the goal of keeping inflation at 2%, measured by the Consumer Price Index shown in Figure 1 above. **If inflation is significantly below the inflation target, the Federal Reserve may cut interest rates** to spur economic activity and bring inflation back to the 2% target. **When inflation exceeds target inflation, the Federal Reserve will react and adjust the interest rate higher** in an attempt to check inflation.

To this end, the Federal Reserve increased interest rates several times in 2022 through 2023 to combat the rise in inflation, but the effect is not instantaneous. The figure above shows how the inflation rate continued to rise through 2022, even with the Federal Reserve already adjusting the interest rate, and it didn't start to drop until 2023. Though it does not show when interest rate hikes have occurred, those increases can be inferred from the inflation history. Answer the following questions based on the above image and description of the Federal Reserve's role in managing the inflation rate.

6. (5 Points) What fundamental mode of behavior is being described in the previous description and shown in Figure 1 above?

Balancing (Negative) Feedback = self correcting / counteracts change

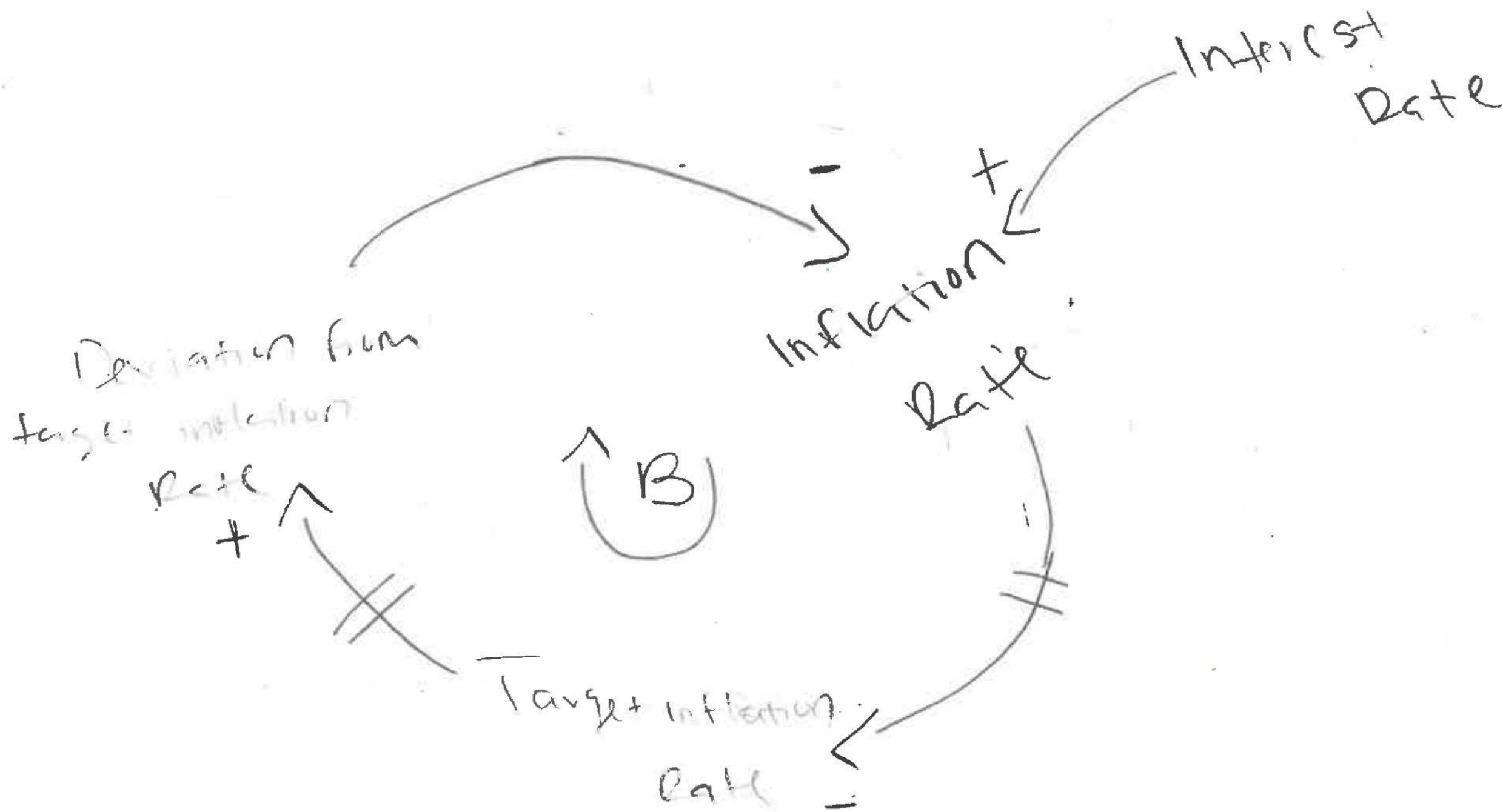
SM484: System Dynamics Simulation - AT 26-1

WPR 1 (Primary)

Name:

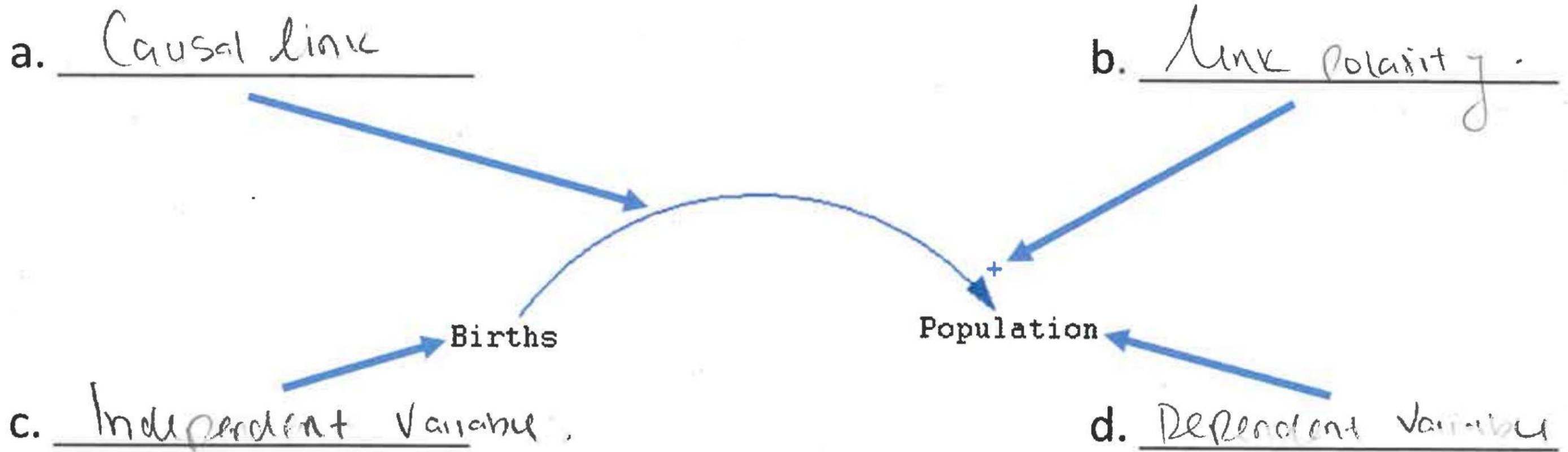
G/25/22

(20 Points) The Federal Reserve implements monetary policy tools, such as the interest rate, to control inflation. In this scenario, you are facing a situation where the inflation rate is *above the target rate* and corrective actions are being taken by the Federal Reserve to bring the inflation back down. In the space below, draw a CLD that depicts the system structure corresponding to the fundamental mode of behavior that you identified previously. Use the following variables to create an abstracted CLD: **Deviation from Target Inflation Rate, Interest Rate, Inflation Rate, and Target Inflation Rate**. Remember that the system structure dictates the system behavior. Be sure to specify (1) Link Polarities, (2) Loop Directions, and (3) Loop Types with Loop name

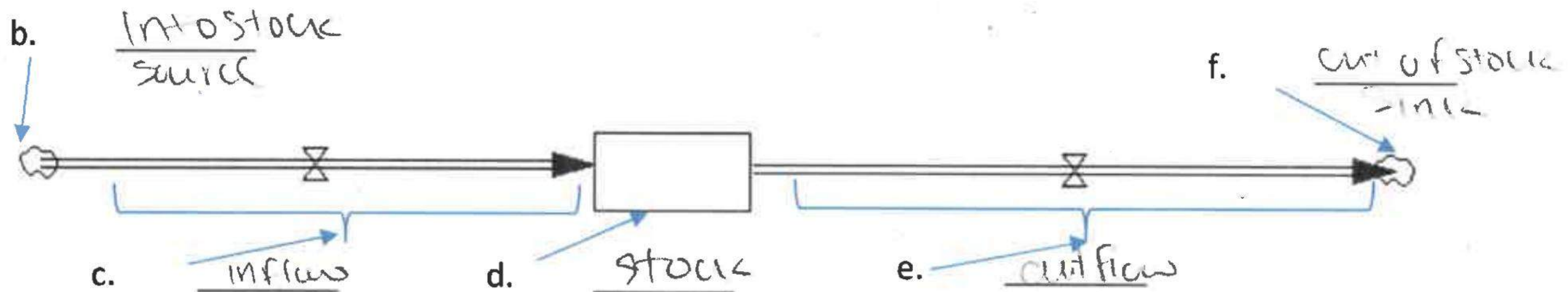


C. Diagram Notation (20 Points)

7. (8 Points) Correctly label each entity on the diagram below:



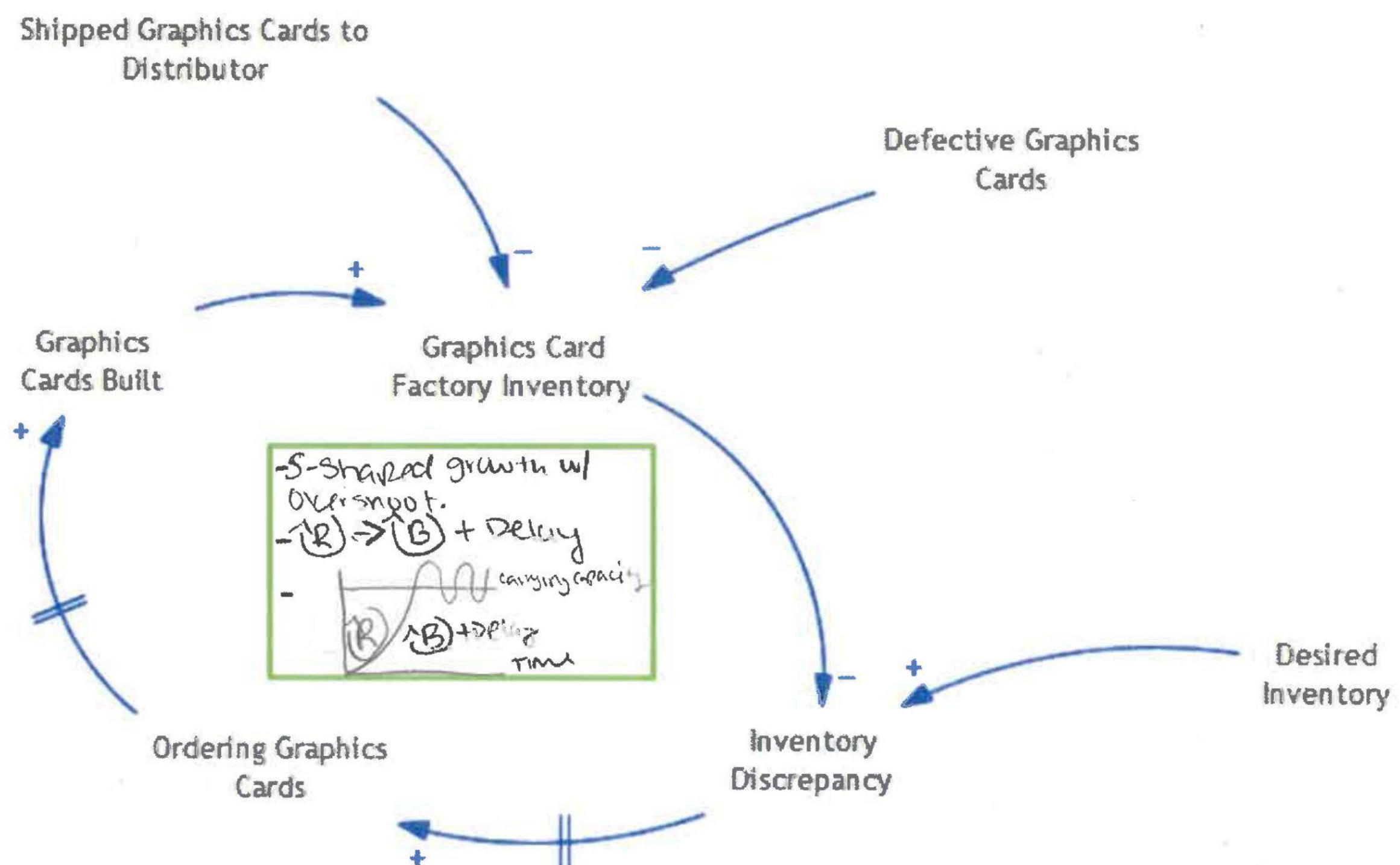
8. (12 Points) Correctly label each entity on the diagram below:



a. What is this diagram called? Stock & flow diagram

D. CLD → S&F (30 Points)

The diagram below depicts a simple structure of a graphics card factory. The factory needs to maintain a certain inventory so that they can meet shipping requirements to their distributors. Sometimes during testing prior to shipping, they also identify some defective graphics cards that can't be shipped. When the factory inventory falls too low, the warehouse manager will order additional graphics cards to be built to bring them back up to their desired inventory.

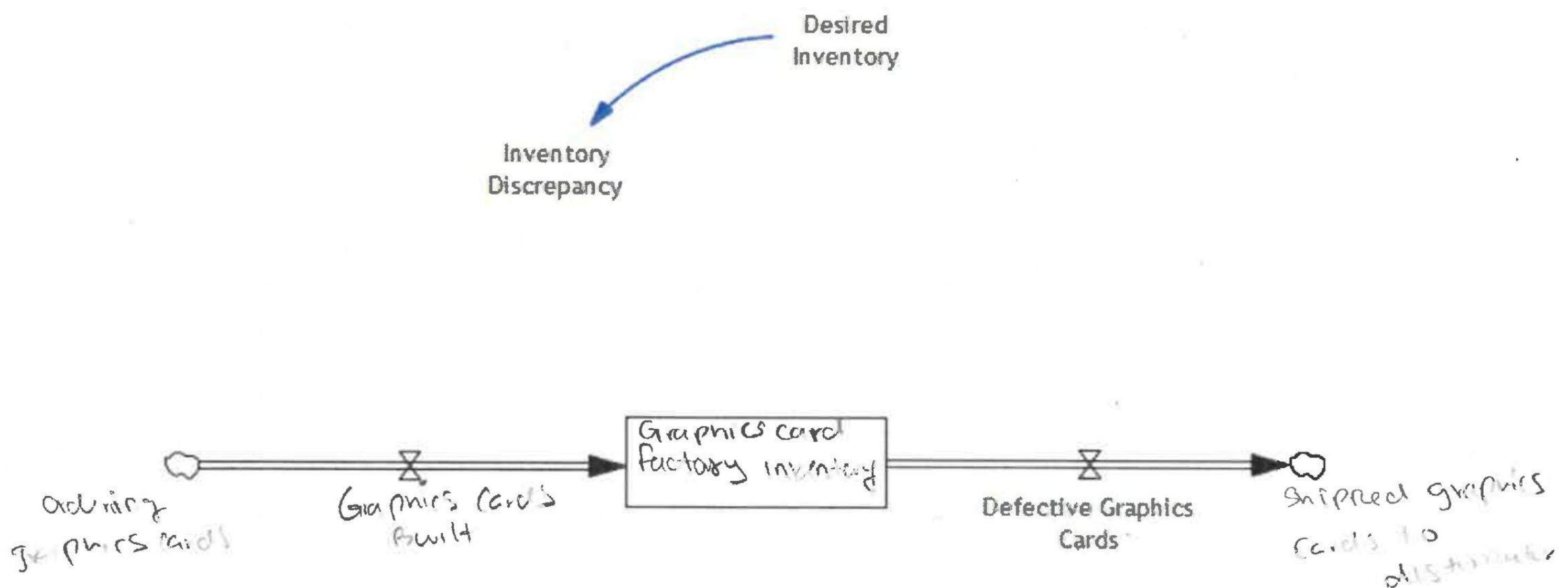


9. (5 Points) Identify the type of feedback loop in the diagram above and clearly **depict** and **label** and name this feedback on the diagram. Write your answer in the green box above.

SM484: System Dynamics Simulation - AT 26-1
WPR 1 (Primary)

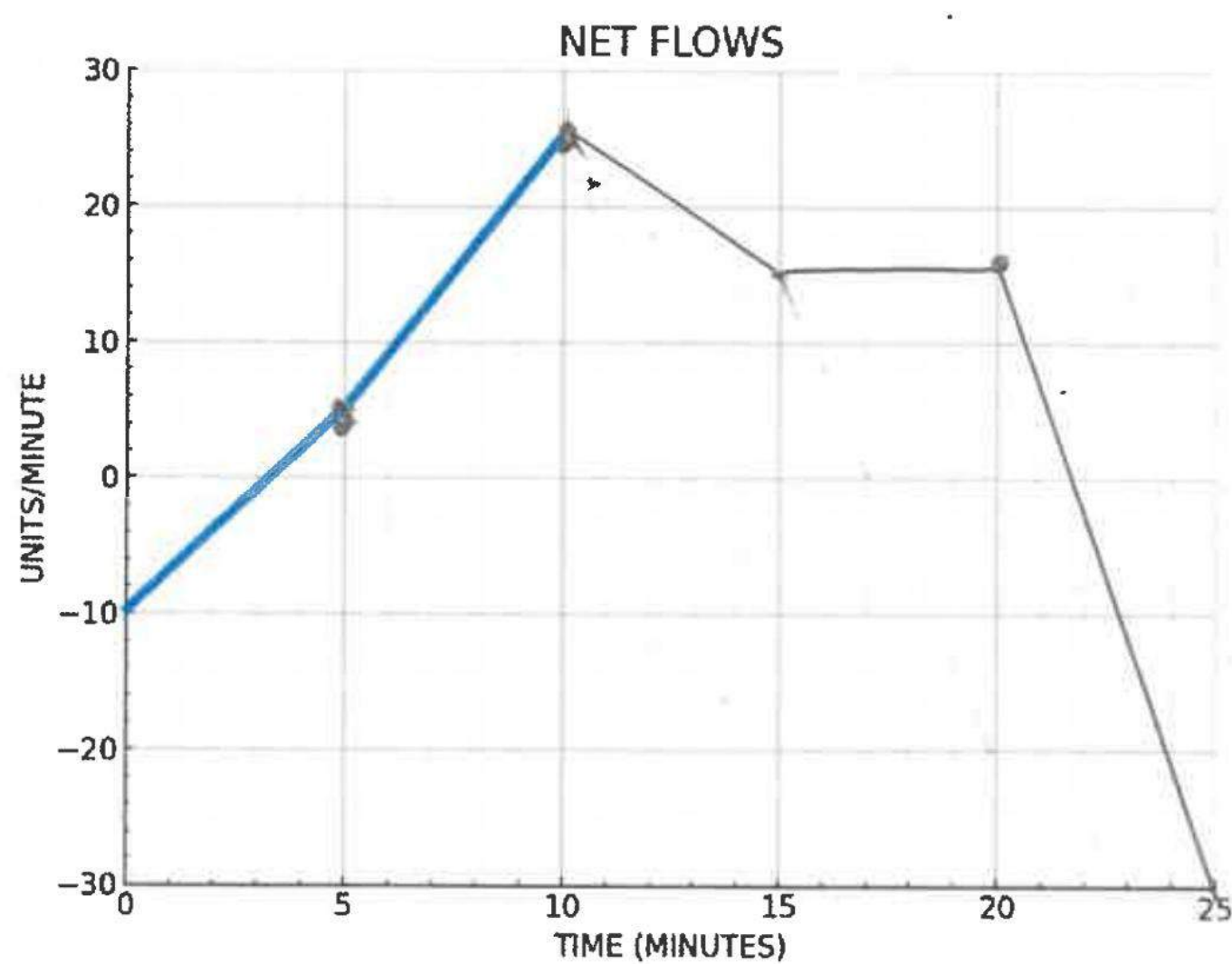
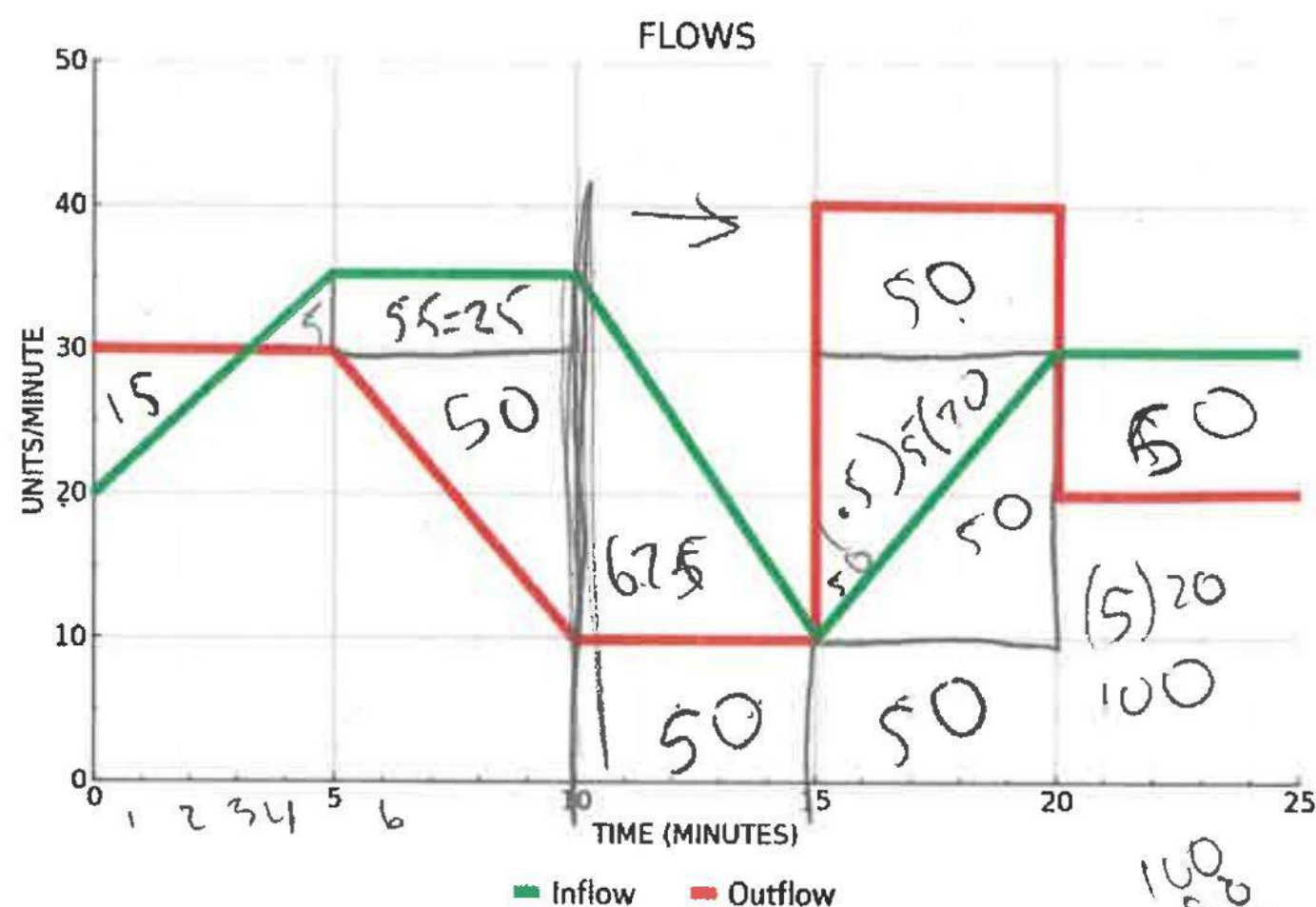
Name: OVERE

10. (25 Points) Using **only the variables** listed in the diagram on the previous page, draw a complete Stock and Flow Diagram to represent this system in the space below. Build on the basic structure that has been given to you below. Use only the CLD to construct your Stock and Flow diagram.



E. Dynamics of Stocks and Flows (40 Points)

11. (17 points) Consider a stock with the single inflow rate and the single outflow rate shown below. Complete the net flow of the stock and complete the behavior of the stock on the graphs provided. *The initial value of the stock is 20 units.*



Show your work here!

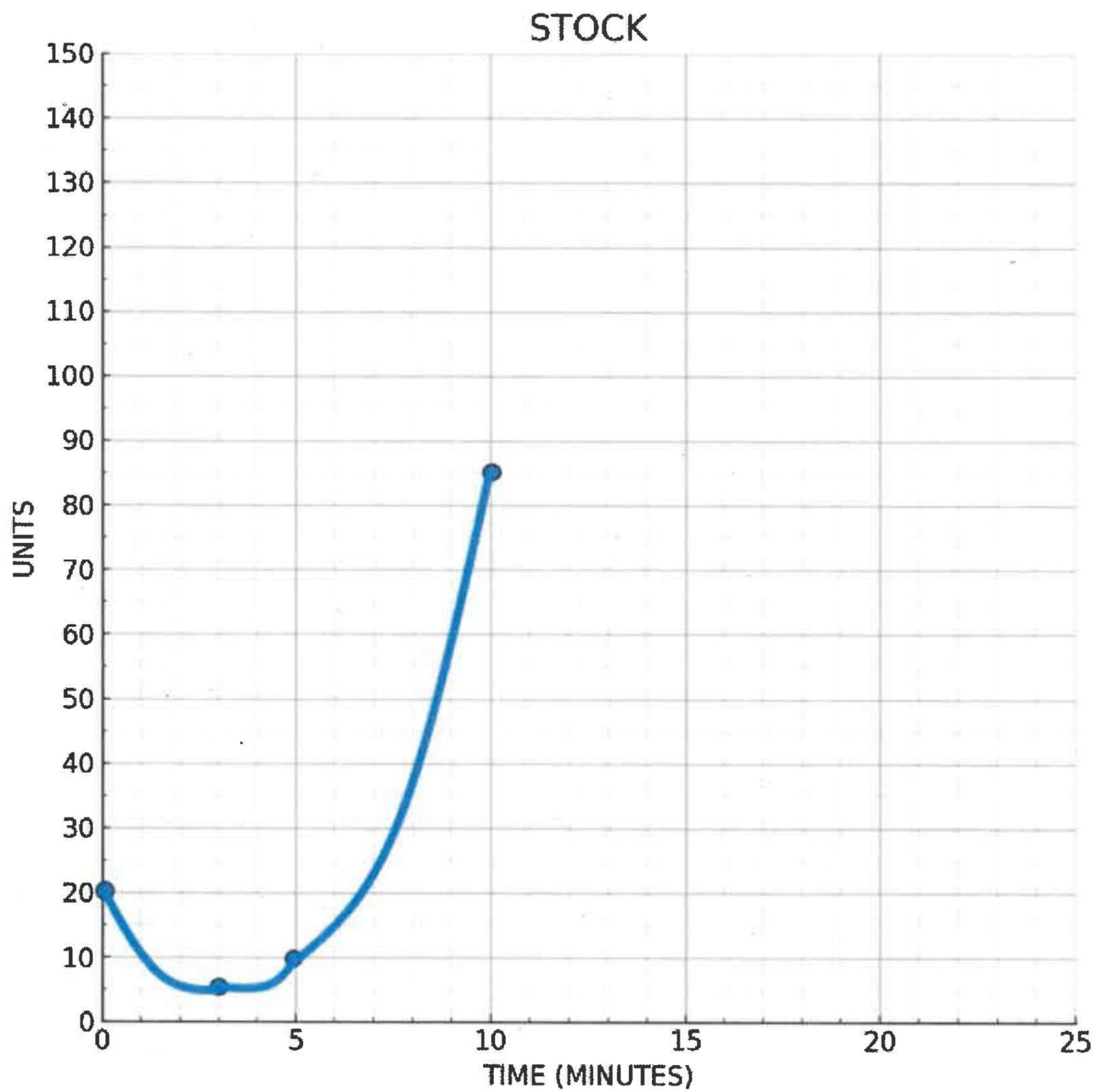
$$\begin{aligned} &1\frac{1}{2} \times 25.5 = 38.25 \\ &38.25 - 50 = -11.75 \end{aligned}$$

$$15 \rightarrow 20$$

$$5 \times 10 = 50$$

$$\frac{1}{2} \times 5 \times 20 = 50$$

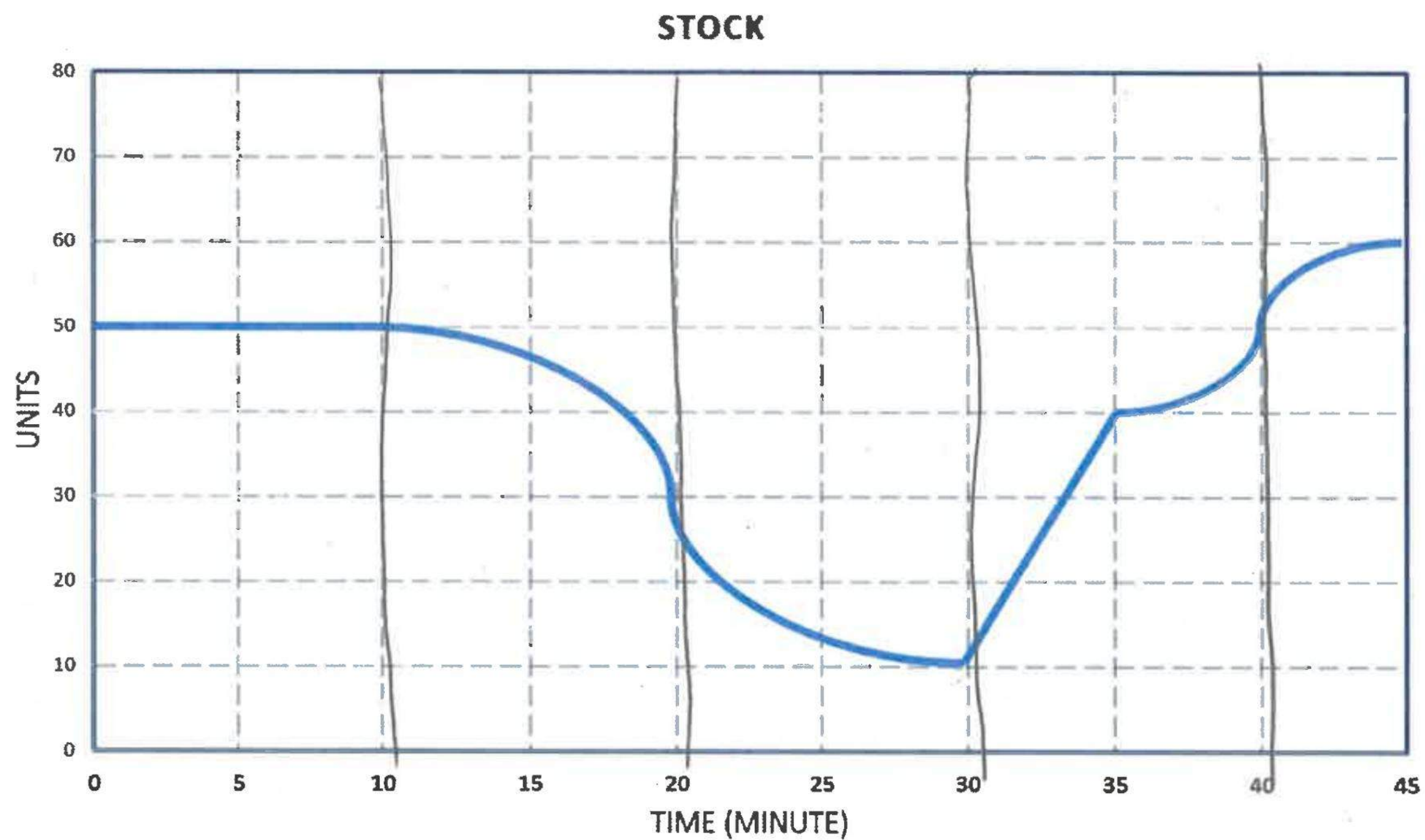
$$12.5 - 50 = -37.5$$



12. (3 points) What is the value of the stock at a time of 20 minutes?

120

13. (17 points) Consider the Stock illustrated below. Fill in the missing time segments to graph the Net Flow (on page 13) that would result in the following stock behavior.

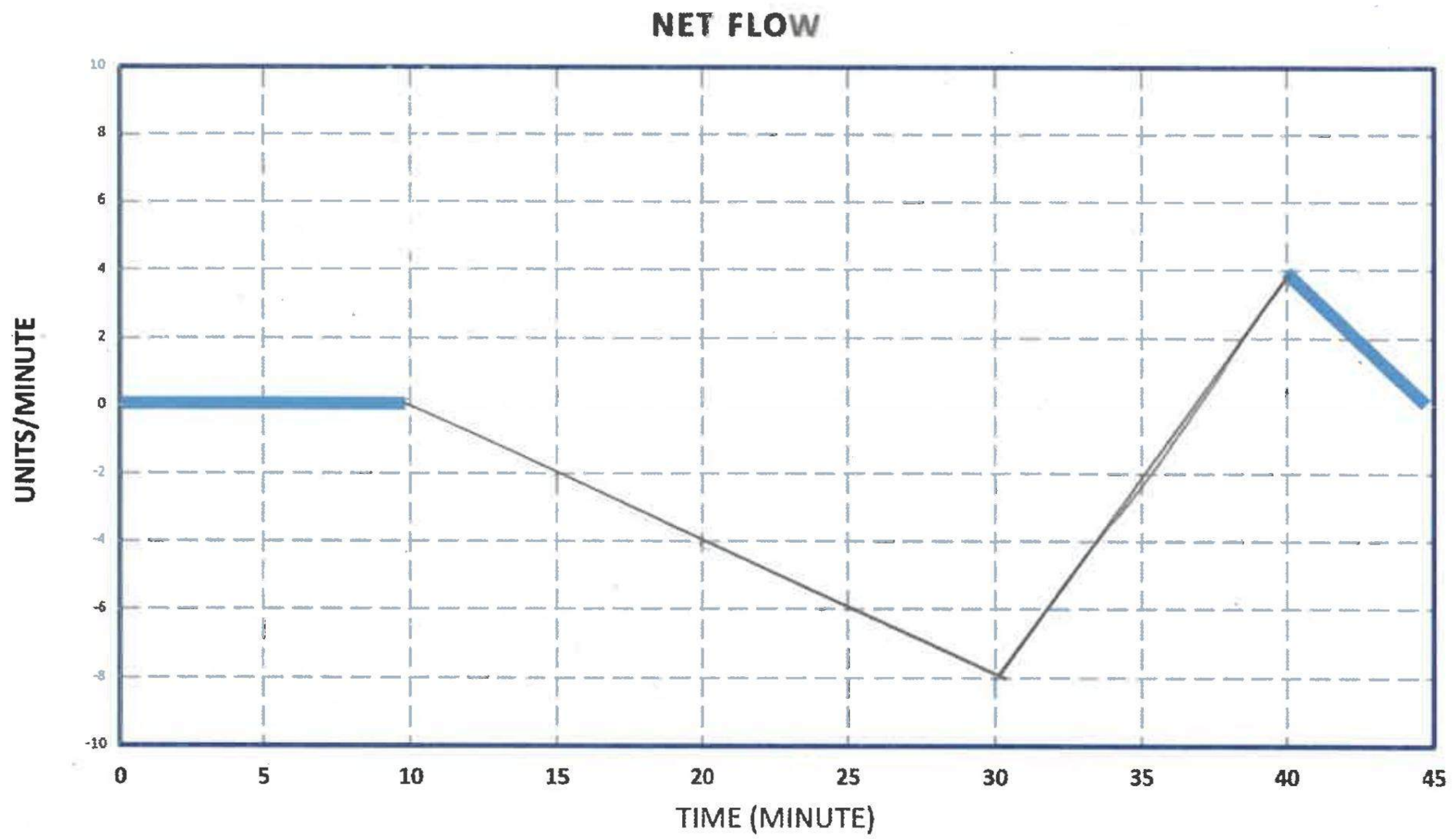


Show your work here!

$$\frac{2(30-50)}{10} = -4$$

$$\frac{2(10-30)}{10} = -4$$

$$\frac{2(50-10)}{10} = 8$$



14. (3 points) What is the Net Flow at 20 Minutes?

-4

SM484: System Dynamics Simulation - AT 26-1
WPR 1 (Primary)

Name: ONKAR

Extra Work Page



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H

Written Partial Review 2

Instructor Solution



System Dynamics Simulation

SM484 – AY 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

WPR2 (Primary)

Name: **SOLUTION** _____

Section: _____

Finish time: _____

1. This is a **70-minute** exam worth **150 points**.
2. This is a closed book/notes exam, **ONE side of an 8-1/2x11 sheet of paper note sheet is allowed**. NO COLLABORATION allowed.
3. The only items that you may bring into the classroom are: one sided note sheet, pencil, straight edge, and calculator. **DO NOT BRING INK PENS!!!**
4. **Place your name NEATLY on each page in the space provided**. Feel free to disassemble the exam; a stapler is available to reattach the pages when you are finished. Make sure you have all 12 pages (including this cover page).
5. **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
6. After you finish the exam, fill in the finish time at the top of this page and sign the honor acknowledgement statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security at a time announced by the Course Director, no earlier than 17NOV25. Prior to that time, I may not discuss any aspects of this exam with anyone except an SM484 instructor." **PRINT IN BLOCK LETTERS NEATLY**

Printed Name (**FIRST LAST**): **SOLUTION** _____ Signature: _____

Date Signed: _____

Grading summary

Question	Possible Points	Actual Points
I. Multiple Choice/Short Answer	30	
II. Dynamics of Growth	22	
III. Dynamics of Delays	27	
IV. Structure and Behavior of Delays	36	
V. Aging Chains	35	
Total	150	

I. Multiple Choice/Short Answer Questions (30 Points)

1. **(6 Points)** There are several different types of delays. List and briefly describe **TWO (2)** different types of delays.

1:

Material Delays (Pipeline, 1st Order, Higher Order), Information Delays (1st Order, Higher Order), Aging Chains are all acceptable answers with a short description of the type.

2:

2. **(5 Points)** Which of the following is true about path dependence? **(Fill in the bubble of all that apply)**

- ☒ The eventual end state in path dependent systems depends on the starting point and small, unpredictable perturbations early in its history.
- ☐ The system starts with locally stable equilibria.
- ☐ In determining design standards, such as the QWERTY keyboard, it ultimately allows the best design to dominate.
- ☐ Path-dependence only occurs when one equilibria is initially more likely than all others.
- ☒ It arises in systems dominated by positive feedback.

3. **(5 Points)** One of the Five Formulation Fundamentals described in your course text is “The Baker Criterion: The inputs to all decision rules in models must be restricted to information actually available to the real decision makers.” Which of the following statements are true with respect to this statement? **(Fill in the bubble of all that apply)**

- ☒ The future is not known to anyone. All expectations and beliefs about the future are based on historical information. Expectations and beliefs may therefore be incorrect.
- ☐ The use of arbitrary scaling factors (fudge factors) is sometimes necessary to ensure that the units of measure in all equations balance.
- ☐ Decision making should not be assumed to conform to any prior theory but should be investigated firsthand.
- ☒ Actual conditions and perceived conditions differ due to measurement and reporting delays, and beliefs are not updated immediately on receipt of new information. Perceptions often differ from the actual situation.
- ☐ Equilibrium and model stability should be assumed if your model equations are correct and conform to managerial practice.

4. **(8 Points)** What are **three** key structural components in a **stock and flow diagram** that must be present for a system to show S-Shaped behavior, and how does each structural component generate part of the behavior?

1: _____
2pts attempt
2pts for each component (1pt listing, 1pt explanation), reinforcing loop, a balancing loop, an exogenous goal. Sensible explanation of why involving those the structure. Could also include structure relating to a non-linear rate equation

2: _____

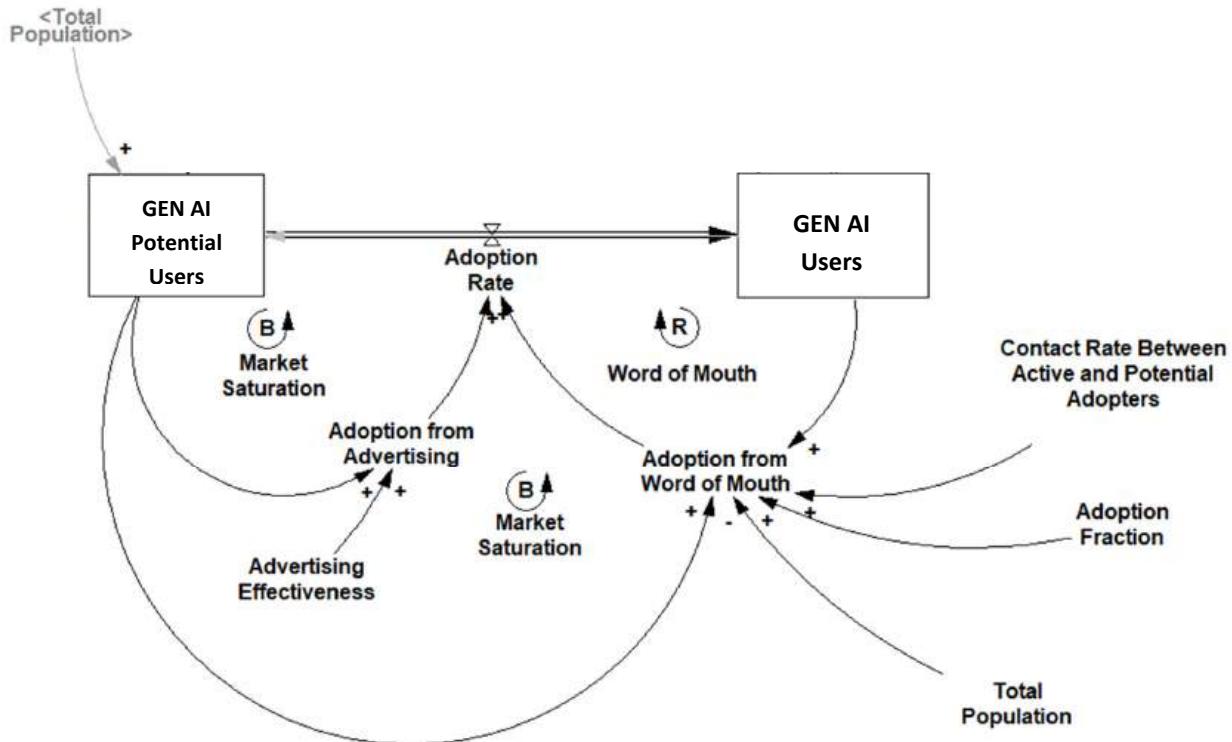
3: _____

5. **(6 Points)** Provide justification for whether you agree or disagree with the following statement:
Path dependency involves feedback mechanisms that will lead to an optimal solution, that is where the best product/design/system dominates in the end.

Agree/Disagree^x (circle one)

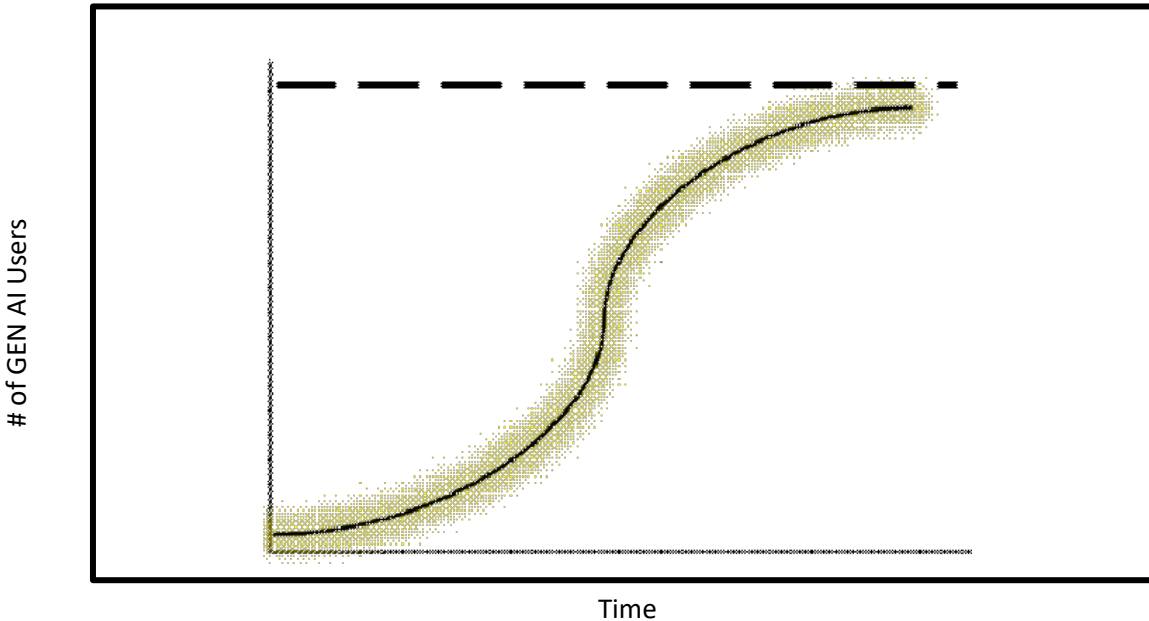
Justification:

Disagree. Path dependence is a result of positive feedback loops and arises from the initial conditions plus small perturbations early in the history. Once a path starts it is difficult to change, even if it is not the most optimal solution.

II. Dynamics of Growth (22 Points)

Generative AI (GEN AI) use has increased significantly since ChatGPT released its first model in 2022, with GEN AI being incorporated directly into many browsers, computers, phones, and the like. To predict the future number of users for GEN AI, we have constructed a simplified model that examines the various market factors driving its adoption.

6. (6 Points) On the plot below, draw the behavior of the model by representing the behavior (growth) of *GEN AI* over time based on the structure shown in the stock and flow diagram. Assume the time horizon starts in 2022 with the release of ChatGPT as a commercially available GEN AI.



Points Awarded
 6=S-Shaped
 5=S w/overshoot
 4= goal seeking
 3=exponential
 2 something else
 showing some effort

*cadets may make assumptions that create a different reference mode (i.e. preorders, which would cause a sudden increase at time 0), if reference mode matches their assumptions can get full credit

7. (6 Points) What assumption(s) did we make in the model, that are key to generating the behavior you drew?

At least one assumption listed that makes sense, some options for s-shaped: All users will eventually adopt, users don't change away from iPhone 15 once adopted, there are 0 initial adopters, the advertising effectiveness is >0 , assume no one will refuse to change, assume no pre-orders (or yes pre-orders), etc

8. (6 Points) If the initial value of *GEN AI Users* = 0 when it is first introduced, what variable in the model would provide the only source of adoption and why?

Adoption from Advertising: if there are no users, there's no way for the word of mouth loop to create adopters

9. (4 Points) This model is an example of a *first-order linear feedback* system. (Fill in the bubble)

☐ True

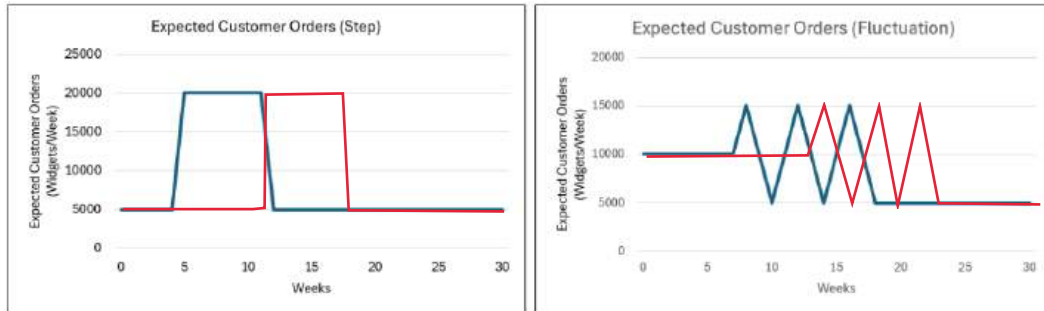
☒ False

False: This is a first order, non-linear feedback system because the total stock of the system is exactly $N = P + A$, and the rates are not just dependent on stocks and exogenous variables

III. Dynamics of Delays (27 Points)

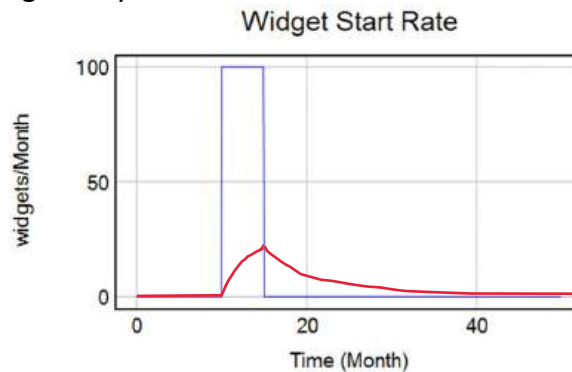
10. (6 Points) On each of the plots below, sketch the response of a **Pipeline Delay** to each of these inputs, assuming a 6-week average delay time. (Assume at $t=0$ the inflow and outflow are equal)

+1pt attempt
+2pt correct components but not perfect
+3pts shape, size, location all correct



acceptable range if it looks like the pipeline is ~5-8 weeks delayed, due to no minor axis marks

11. (3 Points) On the plot below, sketch the response (outflow) of a **1st order material delay** to the input, assuming a 10-week average delay time and zero stock at time zero.



3= correct shape/size/location
2= correct size/shape
1=correct shape

12. (6 Points) Given a **1st Order Material Delay**, the data in the table below, and an average delay time of 7 weeks, what is the Outflow at time 7? Write down your **answer in the GREEN box** below. Write down what **base equation you used in the BLUE box** below. Use additional work blank space as required.

Round to the Nearest .1

avg delay	7	which week?	7
-----------	---	-------------	---

Time (weeks)	0	1	2	3	4	5	6	7	8	9	10
Stock (rounded)	100	91	89	80	80	99	88	110	108	106	133
Inflow	5	11	4	11	30	3	35	14	13	42	3
Outflow								15.7			

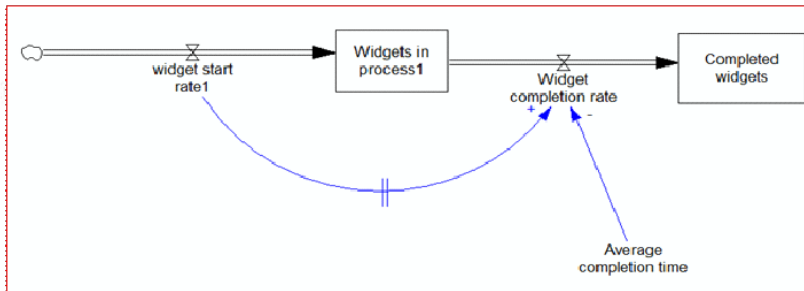
Equation Used:

out = MIT/D

Show additional work here if necessary:

6pts for correct equation and answer
5pts for correct equation with math error answer, or correct answer only without equation
3pts correct equation only

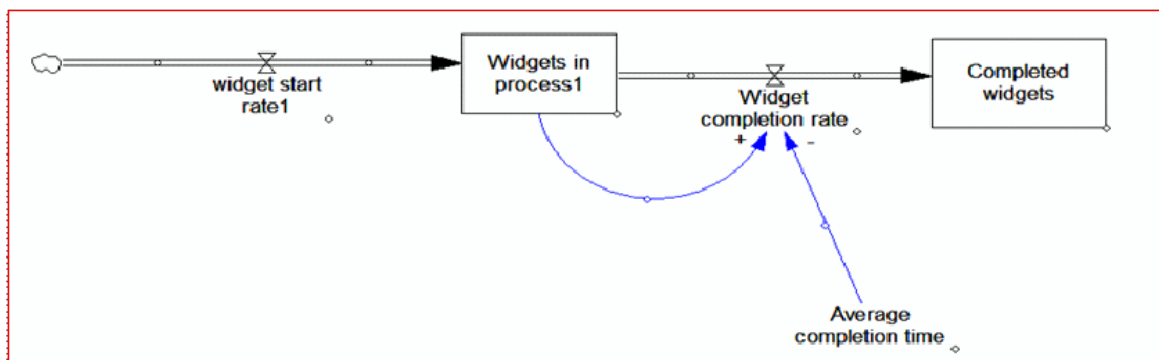
13. **(6 Points)** In the space below, draw and label the complete stock and flow structure for a **pipeline** delay (the basic starting point is provided for a delay, you will have to add structure):



Points awarded for either delay time, or a delay mark on the causal link from inflow to outflow

Stocks (1), Inflow (1), Outflow (1), Causal Link Inflow to Outflow (1), Delay time/tik marks (1), Some reasonable effort (1). Can go to a sink instead of completed widgets stock.

14. **(6 Points)** In the space below, draw and label the complete stock and flow structure for a **1st Order Material** delay (the basic starting point is provided for a delay, you will have to add structure):



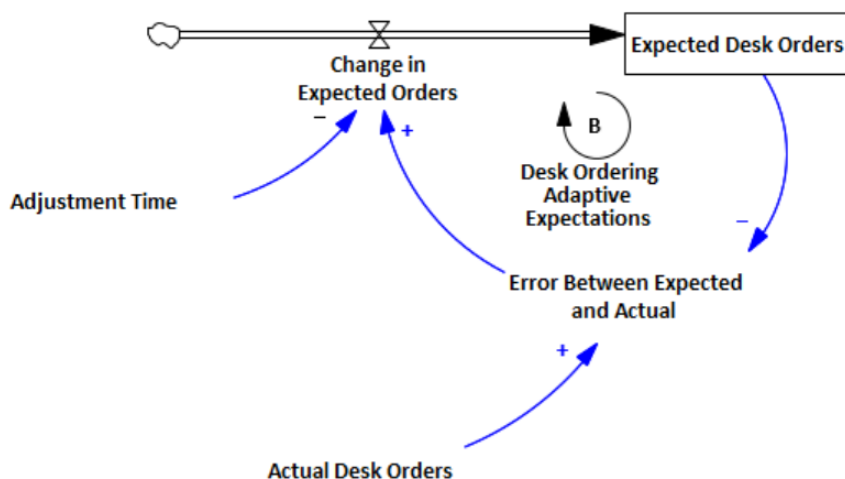
Stocks (1), Inflow (1), Outflow (1), Causal Link Stock to Outflow (1), Delay time (1), Some reasonable effort (1)

IV. Structure and Behavior of Delays (36 Points)

As back pain increases, standing desks offer relief for some people, with many companies capitalizing on the growing standing desk market. A fictitious standing desk company, desks-r-us, decides to invest and join the market. Prior to starting the company, desks-r-us conducts a break-even analysis to determine if the product would be profitable and worth undertaking. As part of this analysis, the team developed projections for their expected orders of their desk model. The model below illustrates what happens to the expected orders as actual desk order information begins to arrive.



15. **(15 Points)** Convert the Causal Loop Diagram (CLD) above to a stock and flow model (using **ONLY** the variables depicted).



Cut Sheet: 4pts some reasonable effort, 2pt expected orders stock, 2pts change in expected orders flow, 1pt each additional variable (3pt total), 3pts awarded for the correct causal links, polarities, and balancing loop (can take off in .5 increments)

16. **(4 Points)** Identify the order and type of delay represented.

Order: First Order Type: Information Delay

17. (9 Points) Define the equation and units for each variable depicted in the CLD. If the variable is exogenous, assign it a value that you believe is reasonable for the model:

Variable Name	Units	Equation or Constant Value
Change in Expected Orders	desks/day	error/AT
Error Between Expected and Actual	desks	Actual - Expected
Adjustment Time	days	any reasonable (5-20 days expected)

18. (8 Points) Assume the adjustment time is **24 days**. If the *Actual Desk Orders* increases by a one-time step function and stays at that higher value, how many days would it take for the *Expected Orders of Desks* to account for **50%** and **75%** of the change to the *Actual Desk Orders*?

Time to adjust to 50% of actual Desk Orders: 16.8 days

Time to adjust to 75% of actual Desk Orders: 33.6 days

What is the name of the rule you used to calculate these? rule of 70 or 72

Show additional work here:

$$\text{Half Life} = .7 * \text{AT}$$

$$50\% = .7 * 24 = 16.8 \text{ days}$$

Double that again for 75%, or

$$75\% = .7 * 24 + 16.8 = 33.6 \text{ days}$$

20. (10 Points) Provide the units and equations for the following (this is not an all-inclusive list for your model):

Variable Name	Units	Equation or Value if Constant
Promotion Rate to LTC	persons/year or month	LTC Review Board Rate * LTC Promotion Fraction
LTC Review Board Rate	Persons/Year	MAJ/MAJ Average Time in Grade
Average MAJ Time in Grade	years	5
LTC Promotion Fraction	dmnl	reasonable (.4 to .7)

21. (5 Points) What order system did you develop? Explain your response.

2nd Order System-two independent stocks (or could be first order if they don't have any inflows added).
2pts attempted answer, 3pts correct with explanation

Include Extra Work below for your test if needed, and list the corresponding question number if applicable.

Include Extra Work below for your test if needed, and list the corresponding question number if applicable.

Outstanding Student Work

Student

William Parker

Total Points

146 / 150 pts

Question 1

1

6 / 6 pts

✓ - 0 pts Correct

- 1 pt one delay lacking some detail
- 2 pts One delay type correct but explanation incorrect
- 2 pts both delays have some errors/lacking detail in explanations
- 1 pt messy and hard to read

Question 2

2

5 / 5 pts

✓ - 0 pts Correct

- 1 pt 1 & 5 Correctly selected, but one additional selected
- 2 pts missed one correct answer (1 & 5) and had one additional selection
- 1 pt missed one correct answer (1 & 5)
- 3 pts missed one correct answer (1 & 5) and had two additional selected

Question 3

3

5 / 5 pts

✓ - 0 pts Correct

- 1 pt selected 1 & 4 correctly, one additional incorrectly selected
- 2 pts one not selected (1 & 4 correct), one additional incorrectly selected
- 3 pts one not selected (1 & 4 correct), two additional incorrectly selected
- 1 pt one not selected (1 & 4 correct)
- 0 pts Click here to replace this description.

Question 4

4

8 / 8 pts

✓ - 0 pts Correct

- 2 pts missed balancing loop
- 2 pts missed reinforcing loop
- 2 pts incorrect/missing explanation on two components
- 2 pts missed exogenous goal/carrying capacity
- 1 pt incorrect explanation on correct structure (balancing loop, reinforcing loop, exogenous goal)
- 6 pts gave answers for oscillation behavior, not s-shaped behavior

Question 5

5

6 / 6 pts

✓ - 0 pts Correct

- 1 pt Explanation has missing/incorrect claims
- 2 pts Explanation lacking significant justification
- 4 pts selected agree-path dependency does not lead to the optimal solution
- 1 pt didn't circle agree/disagree

Question 6

6

6 / 6 pts

✓ - 0 pts Correct

- 0 pts correct, but should also show the carrying capacity
- 1 pt s-shaped drawn incorrectly (exponential at end, goal seek at beginning, exceeds carrying capacity, etc)
- 3 pts should be s-shaped, not exponential
- 4 pts should be s-shaped

Question 7

7

6 / 6 pts

✓ - 0 pts Correct, AT LEAST ONE valid assumption listed (all users will adopt, won't change once adopted, 0 initial adopters, population won't change, advertising effectiveness >0, no delays, no market competitors, these are just some of the assumptions that may be present)

- 1.5 pts Assumptions stated not key to generating the behavior, or the assumptions simply just restate the structure required to generate S shaped behavior
- 4 pts Answered question regarding phones, not regarding GEN AI
- 4 pts partial credit for attempt, but incorrect/incomplete discussion

Question 8

8

6 / 6 pts

✓ - 0 pts Correct, adoption from advertising

- 4 pts answer other than adoption from advertising

Question 9

9

4 / 4 pts

✓ - 0 pts Correct

- 4 pts false, this is a first-order, non-linear feedback system, because the rates are not just dependent on stocks and exogenous variables.

Question 10

10

6 / 6 pts

✓ - 0 pts Correct, same shape as the input, delayed by 6 weeks (we gave some leeway on where 6 weeks was drawn due to the difficulty in determining on these graphs)

- 1 pt first partially wrong

- 2 pts first incorrect

- 1 pt second partially wrong

- 2 pts second incorrect

- 6 pts no answer

- 3 pts one graph no answer

Question 11

11

2 / 3 pts

- 0 pts Correct, the shape and size of the output behavior are roughly correct. We gave some leeway on the scale as we didn't expect a mathematical calculation of this output, more generalized behavior.

- 2.5 pts incorrectly drew pipeline delay behavior

✓ - 1 pt partial errors (shape, size, location, etc)

- 3 pts incorrect

Question 12

12

5 / 6 pts

– 0 pts Correct, both equation $\text{Outflow} = \text{MIT}/\text{Delay}$ and the value 15.7 correct

✓ – 1 pt incorrect rounding (not to 0.1 accuracy)

– 6 pts No answer, or incorrect answer and didn't list equation $\text{Outflow} = \text{MIT}/\text{D}$

– 3 pts didn't list equation $\text{Outflow} = \text{MIT}/\text{D}$

– 3 pts incorrect answer

– 1 pt math error only

Question 13

13

6 / 6 pts

✓ – 0 pts Correct

– 0.5 pts one minor error (no sink or secondary stock, one incorrect polarity, one missing label, only delay hash mark not delay variable)

– 1 pt no causal link from inflow (widget start rate) to outflow (widget completion rate)

– 1 pt no average delay time into outflow

– 1 pt didn't label appropriately

– 1 pt No Outflow

– 1 pt two errors (no sink or secondary stock, incorrect polarity, missing label, only delay hash mark not delay variable)

Question 14

14

6 / 6 pts

✓ – 0 pts Correct

– 1 pt Minor Error

– 2 pts Two Minor Errors

– 4 pts Major Errors

Question 15

15

15 / 15 pts

✓ - 0 pts Correct

- 2 pts Minor Error
- 4 pts Two Minor Errors
- 6 pts Major Errors
- 8 pts Completely Wrong
- 15 pts Blank

Question 16

16

4 / 4 pts

✓ - 0 pts Correct

- 2 pts One Incorrect

Question 17

17

9 / 9 pts

✓ - 0 pts Correct

- 1 pt One Incorrect
- 2 pts Two Incorrect
- 3 pts Three Incorrect
- 4 pts Four Incorrect
- 5 pts Five Incorrect
- 9 pts Blank

Question 18

18

8 / 8 pts

✓ - 0 pts Correct

- 2 pts Correct Rule, One Incorrect Answer
- 4 pts Correct Rule, Incorrect Application
- 5 pts Correct Rule, No Attempt
- 5 pts Incorrect Rule with Attempt

Question 19

19

20 / 20 pts

✓ - 0 pts Correct

- 2 pts Minor Error

- 4 pts Multiple Minor Errors

- 6 pts Major Errors

- 8 pts Completely incorrect

Question 20

20

8 / 10 pts

- 0 pts Correct

- 1 pt One Incorrect

✓ - 2 pts Two Incorrect

- 3 pts Three Incorrect

- 4 pts Four Incorrect

- 5 pts Five Incorrect

- 6 pts Six Incorrect

Question 21

21

5 / 5 pts

✓ - 0 pts Correct

- 1.5 pts Correct with No Explanation

- 1.5 pts Incorrect with Good Detail

- 3 pts Incorrect

Question 22

Extra Work 1

0 / 0 pts

✓ - 0 pts Correct, no points associated with this, just space to review if extra work was completed for any previous questions

- 0 pts correct, extra work found and applied to previous questions

Question 23

Extra Work 2

0 / 0 pts

✓ - 0 pts Correct



Department of Systems Engineering

System Dynamics Simulation

SM484 – AY 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

WPR2 (Primary)

Name: William Parker

Section: H1

Finish time: _____

1. This is a **70-minute** exam worth **150 points**.
2. This is a closed book/notes exam, **ONE side of an 8-1/2x11 sheet of paper note sheet is allowed**. NO COLLABORATION allowed.
3. The only items that you may bring into the classroom are: one sided note sheet, pencil, straight edge, and calculator. **DO NOT BRING INK PENS!!!**
4. **Place your name NEATLY on each page in the space provided**. Feel free to disassemble the exam; a stapler is available to reattach the pages when you are finished. Make sure you have all 12 pages (including this cover page).
5. **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
6. After you finish the exam, fill in the finish time at the top of this page and sign the honor acknowledgement statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security at a time announced by the Course Director, no earlier than 17NOV25. Prior to that time, I may not discuss any aspects of this exam with anyone except an SM484 instructor." **PRINT IN BLOCK LETTERS NEATLY**

Printed Name (FIRST LAST): WILLIAM PARKER Signature: [Signature]

Date Signed: 2025-11-03

Grading summary

Question	Possible Points	Actual Points
I. Multiple Choice/Short Answer	30	
II. Dynamics of Growth	22	
III. Dynamics of Delays	27	
IV. Structure and Behavior of Delays	36	
V. Aging Chains	35	
Total	150	

I. Multiple Choice/Short Answer Questions (30 Points)

1. (6 Points) There are several different types of delays. List and briefly describe **TWO (2)** different types of delays.

1: Material: The delay of material or products experience while in transit, products go in, stay for sometime, possible extra's of making occur then outflows

2: Information: The delay of adjustments of perceptions or beliefs to the actual variable, how no outflow, perceptions don't leave just gradually adjust

2. (5 Points) Which of the following is true about path dependence? (Fill in the bubble of all that apply)

- ☒ The eventual end state in path dependent systems depends on the starting point and small, unpredictable perturbations early in its history.
- ☐ The system starts with locally stable equilibria.
- ☐ In determining design standards, such as the QWERTY keyboard, it ultimately allows the best design to dominate.
- ☐ Path-dependence only occurs when one equilibria is initially more likely than all others.
- ☒ It arises in systems dominated by positive feedback.

3. (5 Points) One of the Five Formulation Fundamentals described in your course text is "The Baker Criterion: The inputs to all decision rules in models must be restricted to information actually available to the real decision makers." Which of the following statements are true with respect to this statement? (Fill in the bubble of all that apply)

- ☒ The future is not known to anyone. All expectations and beliefs about the future are based on historical information. Expectations and beliefs may therefore be incorrect.
- ☐ The use of arbitrary scaling factors (fudge factors) is sometimes necessary to ensure that the units of measure in all equations balance.
- ☐ Decision making should not be assumed to conform to any prior theory but should be investigated firsthand.
- ☒ Actual conditions and perceived conditions differ due to measurement and reporting delays, and beliefs are not updated immediately on receipt of new information. Perceptions often differ from the actual situation.
- ☐ Equilibrium and model stability should be assumed if your model equations are correct and conform to managerial practice.

4. (8 Points) What are **three** key structural components in a **stock and flow diagram** that must be present for a system to **show S-Shaped behavior**, and how does each structural component generate part of the behavior?

- 1: Reinforcing loops: For an S shape you need a reinforcing loop in order to have the rapid increase for the S shape.
- 2: Balancing loops: In order to make the S shape you need to stop that rapid increase with a balancing loop to flatten out the curve which creates the S.
- 3: Exogenous Variables: You need these variables so the Stock and Flow has something to build a loop around to control the stock in order to create reinforcing and balancing behaviors.

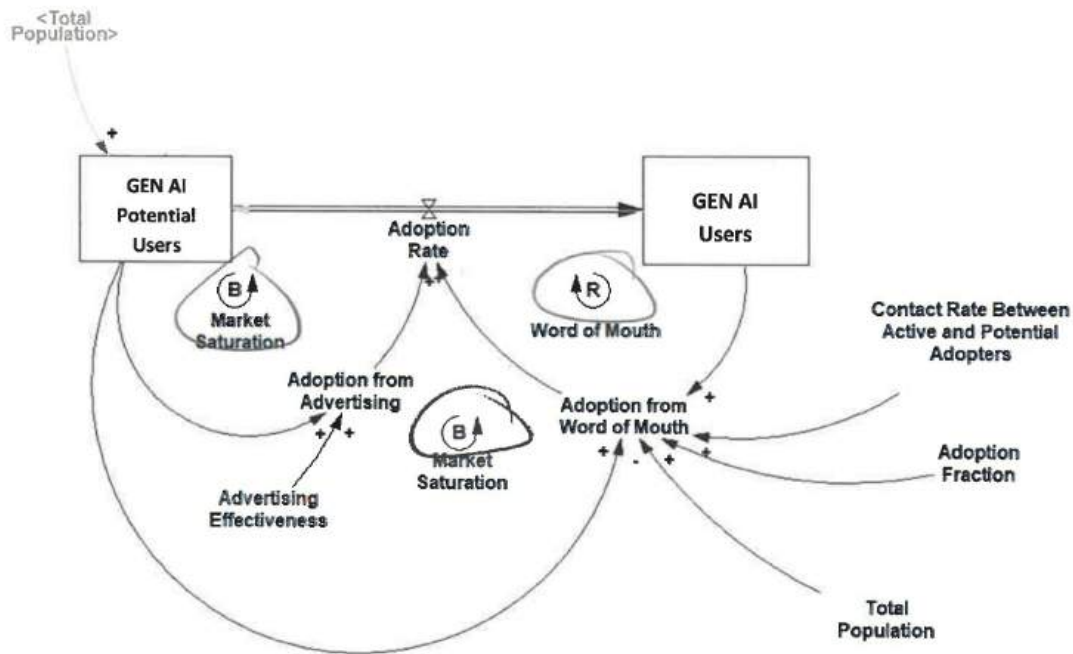
5. (6 Points) Provide justification for whether you agree or disagree with the following statement:
Path dependency involves feedback mechanisms that will lead to an optimal solution, that is where the best product/design/system dominates in the end.

Agree/Disagree (circle one)

Justification:

Path dependency doesn't always lead to the best solution. Equilibrium depends on initial conditions and certain stocks. We see this with Quercy. Once something has been used for a while (history) the optimal solution may be impractical.

II. Dynamics of Growth (22 Points)

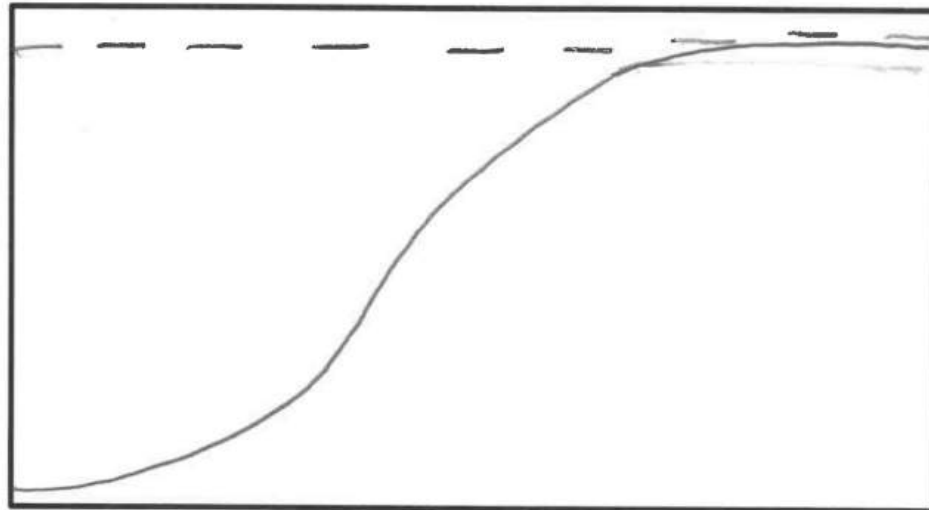


Generative AI (GEN AI) use has increased significantly since ChatGPT released its first model in 2022, with GEN AI being incorporated directly into many browsers, computers, phones, and the like. To predict the future number of users for GEN AI, we have constructed a simplified model that examines the various market factors driving its adoption.

6. (6 Points) On the plot below, draw the behavior of the model by representing the behavior (growth) of GEN AI over time based on the structure shown in the stock and flow diagram. Assume the time horizon starts in 2022 with the release of ChatGPT as a commercially available GEN AI.

Total Pop

of GEN AI Users



Time

7. (6 Points) What assumption(s) did we make in the model, that are key to generating the behavior you drew?

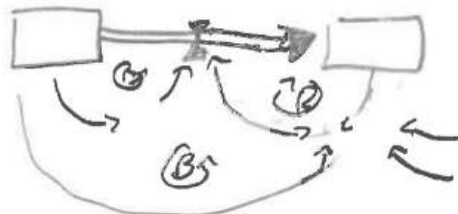
My Assumptions: When ChatGPT was first introduced not too many people used it, but as soon as it was added to computers, laptops, phones and etc. the total # of people using it skyrocketed until it reached its capacity of total population. This generates an S shape curve with a capacity.

8. (6 Points) If the initial value of GEN AI Users = 0 when it is first introduced, what variable in the model would provide the only source of adoption and why?

Adoption from advertising, since users = 0, there is no word of mouth, so potential users will only hear of it through advertising.

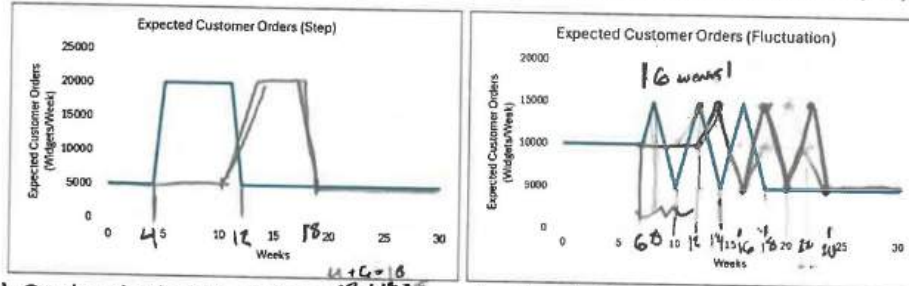
- ? 9. (4 Points) This model is an example of a **first-order linear feedback** system. (Fill in the bubble)

☐ True

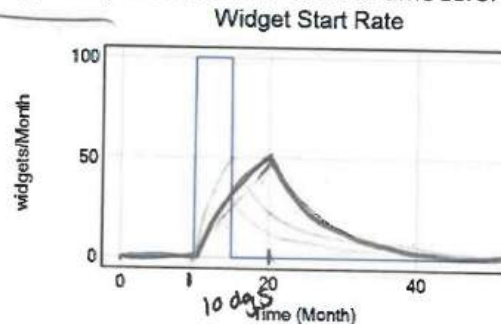
☒ False


III. Dynamics of Delays (27 Points)

10. (6 Points) On each of the plots below, sketch the response of a **Pipeline Delay** to each of these inputs, assuming a 6-week average delay time. (Assume at $t=0$ the inflow and outflow are equal)



11. (3 Points) On the plot below, sketch the response (outflow) of a **1st order material delay** to the input, assuming a 10-week average delay time and zero stock at time zero.



12. (6 Points) Given a **1st Order Material Delay**, the data in the table below, and an average delay time of 7 weeks, what is the Outflow at time 7? Write down your answer in the **GREEN** box below. Write down what **base equation** you used in the **BLUE** box below. Use additional work blank space as required.
Round to the Nearest .1

avg delay	7	which week?	7
-----------	---	-------------	---

Time (weeks)	0	1	2	3	4	5	6	7	8	9	10
Stock (rounded)	100	91	89	80	80	99	88	110	108	106	133
Inflow	5	11	4	11	30	3	35	14	13	42	3
Outflow								15.714			

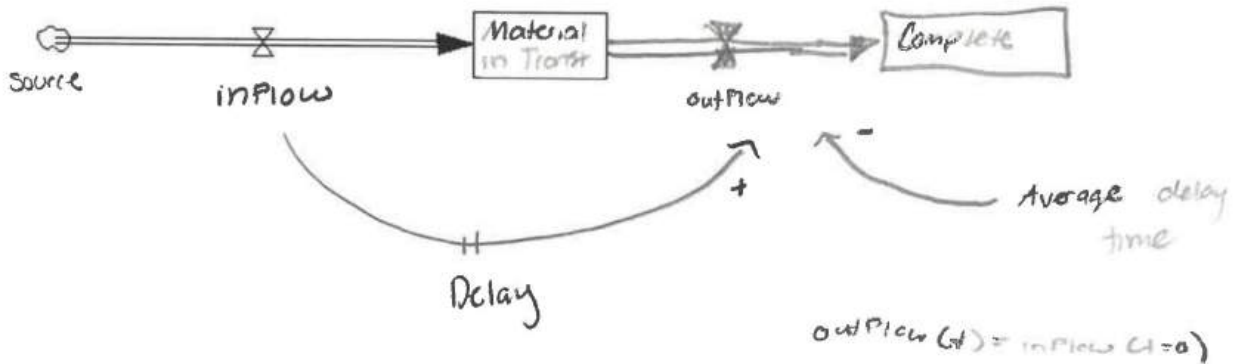
does it say rounded
Equation Used:

Outflow = $\frac{MI(t+\Delta t)}{\Delta t}$
Outflow = $\frac{\text{Stock}}{\text{delay}}$

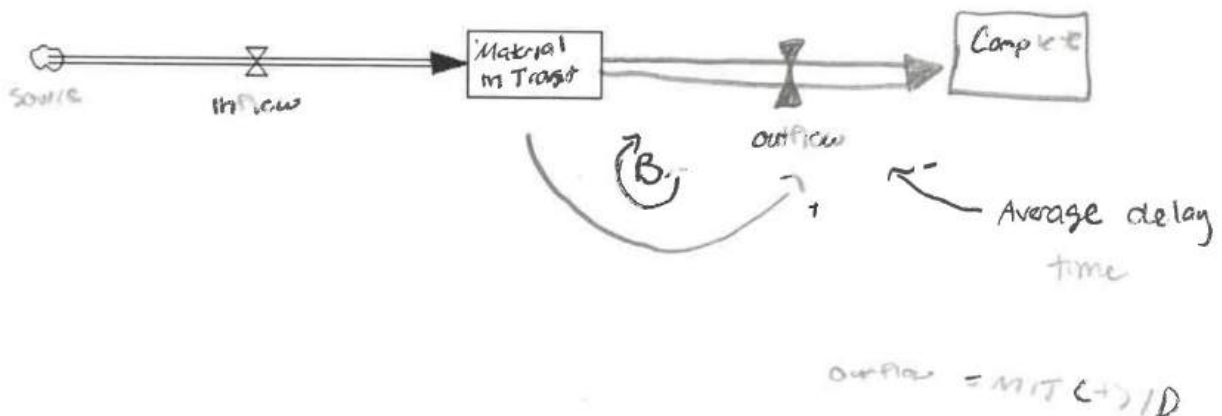
Show additional work here if necessary:

$\frac{110 \text{ (stock)}}{7 \text{ (delay)}} = 15.714$

13. (6 Points) In the space below, draw and label the complete stock and flow structure for a pipeline delay (the basic starting point is provided for a delay, you will have to add structure):



14. (6 Points) In the space below, draw and label the complete stock and flow structure for a 1st Order Material delay (the basic starting point is provided for a delay, you will have to add structure):

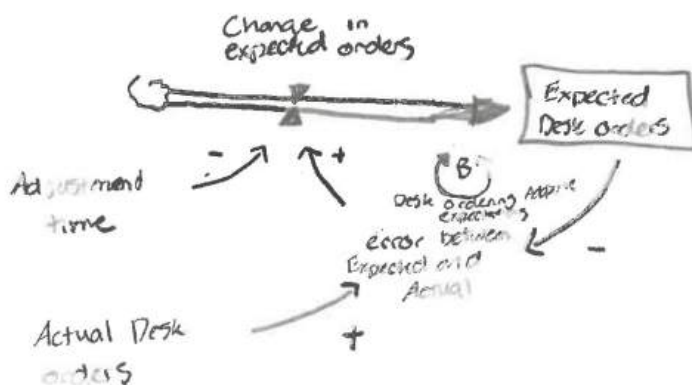


IV. Structure and Behavior of Delays (36 Points)

As back pain increases, standing desks offer relief for some people, with many companies capitalizing on the growing standing desk market. A fictitious standing desk company, desks-r-us, decides to invest and join the market. Prior to starting the company, desks-r-us conducts a break-even analysis to determine if the product would be profitable and worth undertaking. As part of this analysis, the team developed projections for their expected orders of their desk model. The model below illustrates what happens to the expected orders as actual desk order information begins to arrive.



15. (15 Points) Convert the Causal Loop Diagram (CLD) above to a stock and flow model (using **ONLY** the variables depicted).



16. (4 Points) Identify the order and type of delay represented.

Order: 1st order Type: Information delay

17. (9 Points) Define the equation and units for each variable depicted in the CLD. If the variable is exogenous, assign it a value that you believe is reasonable for the model:

Variable Name	Units	Equation or Constant Value
Change in Expected Orders	$\frac{\text{Desks}}{\text{days}}$	$\frac{\text{error between expected and actual}}{\text{adjustment time}}$
Error Between Expected and Actual	desks	Actual Value - Expected value
Adjustment Time	days	24 days

18. (8 Points) Assume the adjustment time is **24 days**. If the *Actual Desk Orders* increases by a one-time step function and stays at that higher value, how many days would it take for the *Expected Orders of Desks* to account for **50%** and **75%** of the change to the *Actual Desk Orders*?

Time to adjust to 50% of actual Desk Orders: 16.2

Time to adjust to 75% of actual Desk Orders: 33.6

What is the name of the rule you used to calculate these? Rule of 70

Show additional work here:

- Ads = 24 days
- one time

- Delay time = .7

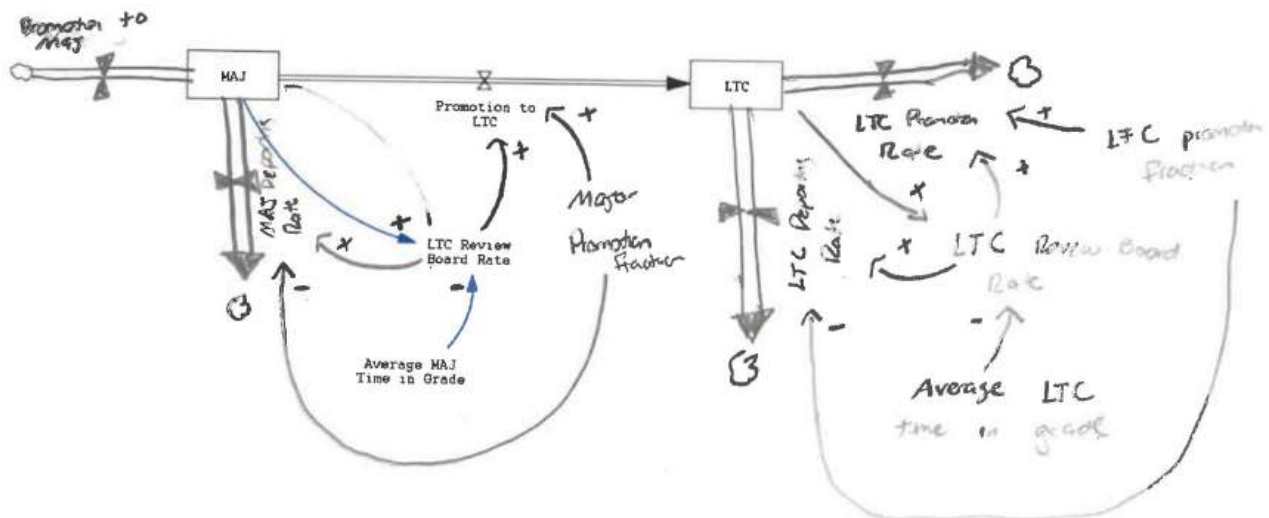
$$.7 \cdot 24 = 16.8$$

$$50\% = 16.8$$

$$16.8 \times 2$$

$$75\% = 33.6$$

- V. **Aging Chains (35 Points)** HRC conducts a board review yearly to determine those selected for promotion. The terminology for review boards and promotions follows that a current MAJ undergoes a LTC review board, and if selected, they undergo a LTC Promotion (i.e., the promotion name follows the rank of the stock they are promoted into, not from). Assume that if either a Major (MAJ) or Lieutenant Colonel (LTC) is not selected for promotion, they are chaptered out of the Army. For this problem, map out the officer promotion system for Field Grade Officers, promoting to a Major through promotion to Colonel and out of the Field Grade cohort (just use a sink, do not need to include a stock for Colonel and higher ranks). There is a delay for time in grade requirements that occur after each promotion before you can promote to the next level.
19. (20 Points) Use the base diagram below to develop an Aging Chain Model showing the progression of a Field Grade Officer. Show promoting to MAJ through being promoted from LTC (you do not need to include a COL stock, just an outflow from LTC to the COL cohort). *Hint: To model the flow of personnel from MAJ to COL, you need to include things like MAJ promotion rate, Average MAJ Time in Grade, LTC Promotion Fraction, and LTC Departing Army Rate. THIS LIST IS NOT ALL INCLUSIVE. YOU WILL NEED TO ADD OTHER STRUCTURE.* Remember to label all polarities of causal links. You may have to add stocks, flows, variables, causal links, sources, and sinks.



SM484: System Dynamics Simulation - AY 26-1

WPR 2

Name: WILLIAM PAIZ KEK

20. (10 Points) Provide the units and equations for the following (this is not an all-inclusive list for your model):

Promoting Majors - Majors Departing

Variable Name	Units	Equation or Value if Constant
Promotion to LTC	Persons	Major Promotion - Maj Departing
LTC Review Board Rate	Persons/Year	LTC / Avg LTC time in grade
Average MAJ Time in Grade	Year	5
LTC Promotion Fraction	Dim 1	.9 dim 1

21. (5 Points) What order system did you develop? Explain your response.

We created a second order system because we created a stock and flow with two stocks, exogenous variables affecting the stocks. Also it has to flow through the system, for example you need to be promoted to Major then promoted to LTC so the flows flow through and affect each other in that order there will never be LTC less than Majors. So it's in ORDER through 2 stocks

Include Extra Work below for your test if needed, and list the corresponding question number if applicable.

SM484: System Dynamics Simulation – AY 26-1
WPR 2

Name: _____

Include Extra Work below for your test if needed, and list the corresponding question number if applicable.

Fair-to-Good Student Work

Student

Ryan Remaly

Total Points

129.5 / 150 pts

Question 1

1

6 / 6 pts

✓ - 0 pts Correct

- 1 pt one delay lacking some detail
- 2 pts One delay type correct but explanation incorrect
- 2 pts both delays have some errors/lacking detail in explanations
- 1 pt messy and hard to read

Question 2

2

5 / 5 pts

✓ - 0 pts Correct

- 1 pt 1 & 5 Correctly selected, but one additional selected
- 2 pts missed one correct answer (1 & 5) and had one additional selection
- 1 pt missed one correct answer (1 & 5)
- 3 pts missed one correct answer (1 & 5) and had two additional selected

Question 3

3

5 / 5 pts

✓ - 0 pts Correct

- 1 pt selected 1 & 4 correctly, one additional incorrectly selected
- 2 pts one not selected (1 & 4 correct), one additional incorrectly selected
- 3 pts one not selected (1 & 4 correct), two additional incorrectly selected
- 1 pt one not selected (1 & 4 correct)
- 0 pts Click here to replace this description.

Question 4

4

2 / 8 pts

– 0 pts Correct

– 2 pts missed balancing loop

– 2 pts missed reinforcing loop

– 2 pts incorrect/missing explanation on two components

– 2 pts missed exogenous goal/carrying capacity

– 1 pt incorrect explanation on correct structure (balancing loop, reinforcing loop, exogenous goal)

✓ – 6 pts gave answers for oscillation behavior, not s-shaped behavior

💬 Looks like you used last year's test to study. Note this question is asking about S-shaped behavior, not oscillation.

Question 5

5

5 / 6 pts

– 0 pts Correct

✓ – 1 pt Explanation has missing/incorrect claims

– 2 pts Explanation lacking significant justification

– 4 pts selected agree-path dependency does not lead to the optimal solution

– 1 pt didn't circle agree/disagree

Question 6

6

6 / 6 pts

✓ – 0 pts Correct

– 0 pts correct, but should also show the carrying capacity

– 1 pt s-shaped drawn incorrectly (exponential at end, goal seek at beginning, exceeds carrying capacity, etc)

– 3 pts should be s-shaped, not exponential

– 4 pts should be s-shaped

Question 7

7

4.5 / 6 pts

– 0 pts Correct, AT LEAST ONE valid assumption listed (all users will adopt, won't change once adopted, 0 initial adopters, population won't change, advertising effectiveness >0 , no delays, no market competitors, these are just some of the assumptions that may be present)

✓ – 1.5 pts Assumptions stated not key to generating the behavior, or the assumptions simply just restate the structure required to generate S shaped behavior

– 4 pts Answered question regarding phones, not regarding GEN AI

– 4 pts partial credit for attempt, but incorrect/incomplete discussion

Question 8

8

6 / 6 pts

✓ – 0 pts Correct, adoption from advertising

– 4 pts answer other than adoption from advertising

Question 9

9

4 / 4 pts

✓ – 0 pts Correct

– 4 pts false, this is a first-order, non-linear feedback system, because the rates are not just dependent on stocks and exogenous variables.

Question 10

10

5 / 6 pts

– 0 pts Correct, same shape as the input, delayed by 6 weeks (we gave some leeway on where 6 weeks was drawn due to the difficulty in determining on these graphs)

– 1 pt first partially wrong

– 2 pts first incorrect

✓ – 1 pt second partially wrong

– 2 pts second incorrect

– 6 pts no answer

– 3 pts one graph no answer

Question 11

11

3 / 3 pts

✓ - 0 pts Correct, the shape and size of the output behavior are roughly correct. We gave some leeway on the scale as we didn't expect a mathematical calculation of this output, more generalized behavior.

- 2.5 pts incorrectly drew pipeline delay behavior
- 1 pt partial errors (shape, size, location, etc)
- 3 pts incorrect

Question 12

12

6 / 6 pts

✓ - 0 pts Correct, both equation $\text{Outflow} = \text{MIT}/\text{Delay}$ and the value 15.7 correct

- 1 pt incorrect rounding (not to 0.1 accuracy)
- 6 pts No answer, or incorrect answer and didn't list equation $\text{Outflow} = \text{MIT}/\text{D}$
- 3 pts didn't list equation $\text{Outflow} = \text{MIT}/\text{D}$
- 3 pts incorrect answer
- 1 pt math error only

Question 13

13

6 / 6 pts

✓ - 0 pts Correct

- 0.5 pts one minor error (no sink or secondary stock, one incorrect polarity, one missing label, only delay hash mark not delay variable)
- 1 pt no causal link from inflow (widget start rate) to outflow (widget completion rate)
- 1 pt no average delay time into outflow
- 1 pt didn't label appropriately
- 1 pt No Outflow
- 1 pt two errors (no sink or secondary stock, incorrect polarity, missing label, only delay hash mark not delay variable)

Question 14

14

6 / 6 pts

✓ - 0 pts Correct

- 1 pt Minor Error
- 2 pts Two Minor Errors
- 4 pts Major Errors

Question 15

15

15 / 15 pts

✓ - 0 pts Correct

- 2 pts Minor Error
- 4 pts Two Minor Errors
- 6 pts Major Errors
- 8 pts Completely Wrong
- 15 pts Blank

Question 16

16

4 / 4 pts

✓ - 0 pts Correct

- 2 pts One Incorrect

Question 17

17

9 / 9 pts

✓ - 0 pts Correct

- 1 pt One Incorrect
- 2 pts Two Incorrect
- 3 pts Three Incorrect
- 4 pts Four Incorrect
- 5 pts Five Incorrect
- 9 pts Blank

Question 18

18

4 / 8 pts

- 0 pts Correct
- 2 pts Correct Rule, One Incorrect Answer

✓ - 4 pts Correct Rule, Incorrect Application

- 5 pts Correct Rule, No Attempt
- 5 pts Incorrect Rule with Attempt

Question 19

19

16 / 20 pts

– 0 pts Correct

– 2 pts Minor Error

✓ – 4 pts Multiple Minor Errors

– 6 pts Major Errors

– 8 pts Completely incorrect

Question 20

20

7 / 10 pts

– 0 pts Correct

– 1 pt One Incorrect

– 2 pts Two Incorrect

✓ – 3 pts Three Incorrect

– 4 pts Four Incorrect

– 5 pts Five Incorrect

– 6 pts Six Incorrect

Question 21

21

5 / 5 pts

✓ – 0 pts Correct

– 1.5 pts Correct with No Explanation

– 1.5 pts Incorrect with Good Detail

– 3 pts Incorrect

Question 22

Extra Work 1

0 / 0 pts

✓ – 0 pts Correct, no points associated with this, just space to review if extra work was completed for any previous questions

– 0 pts correct, extra work found and applied to previous questions

Question 23

Extra Work 2

0 / 0 pts

✓ – 0 pts Correct



Department of Systems Engineering

System Dynamics Simulation

SM484 - AY 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

WPR2 (Primary)

Name: Ryan Remaly

Section: I1

Finish time: 1140

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Printed Name (FIRST LAST): Ryan Remaly Signature: Ryan Remaly

Date Signed: 03 NOV 2025

Grading summary

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I. Multiple Choice/Short Answer	30	
II. Dynamics of Growth	22	
III. Dynamics of Delays	27	
IV. Structure and Behavior of Delays	36	
V. Aging Chains	35	
Total	150	

I. Multiple Choice/Short Answer Questions (30 Points)

1. (6 Points) There are several different types of delays. List and briefly describe TWO (2) different types of delays.

1: Material Delay → captures the physical flow of material. They are countable/physical units that move through the process. The delay implies a stock of material in process.

2: Information delay → The delay between the reception of new information. There is a gradual adjustment of perceptions and beliefs over time with information delays.

2. (5 Points) Which of the following is true about path dependence? (Fill in the bubble of all that apply)

- ☒ The eventual end state in path dependent systems depends on the starting point and small, unpredictable perturbations early in its history.
- ☐ The system starts with locally stable equilibria. X
- ☐ In determining design standards, such as the QWERTY keyboard, it ultimately allows the best design to dominate. X
- ☐ Path-dependence only occurs when one equilibria is initially more likely than all others. X
- ☒ It arises in systems dominated by positive feedback.

3. (5 Points) One of the Five Formulation Fundamentals described in your course text is "The Baker Criterion: The inputs to all decision rules in models must be restricted to information actually available to the real decision makers." Which of the following statements are true with respect to this statement? (Fill in the bubble of all that apply)

- ☒ The future is not known to anyone. All expectations and beliefs about the future are based on historical information. Expectations and beliefs may therefore be incorrect.
- ☐ The use of arbitrary scaling factors (fudge factors) is sometimes necessary to ensure that the units of measure in all equations balance. X 2
- ☐ Decision making should not be assumed to conform to any prior theory but should be investigated firsthand. X 2
- ☒ Actual conditions and perceived conditions differ due to measurement and reporting delays, and beliefs are not updated immediately on receipt of new information. Perceptions often differ from the actual situation.
- ☐ Equilibrium and model stability should be assumed if your model equations are correct and conform to managerial practice. S

Name: Ryan Remaly

4. (8 Points) What are three key structural components in a stock and flow diagram that must be present for a system to show S-Shaped behavior, and how does each structural component generate part of the behavior?

1: Delays, they cause overshoots and undershoots in relation to the target goal before adjustments take place.

2: A goal; the stock's inflow and outflow rates are controlled by how far the current value deviates from the goal.

3: Negative feedback loops; they attempt to reduce the gap between the current stock and goal. However, delays cause the correction to overshoot.

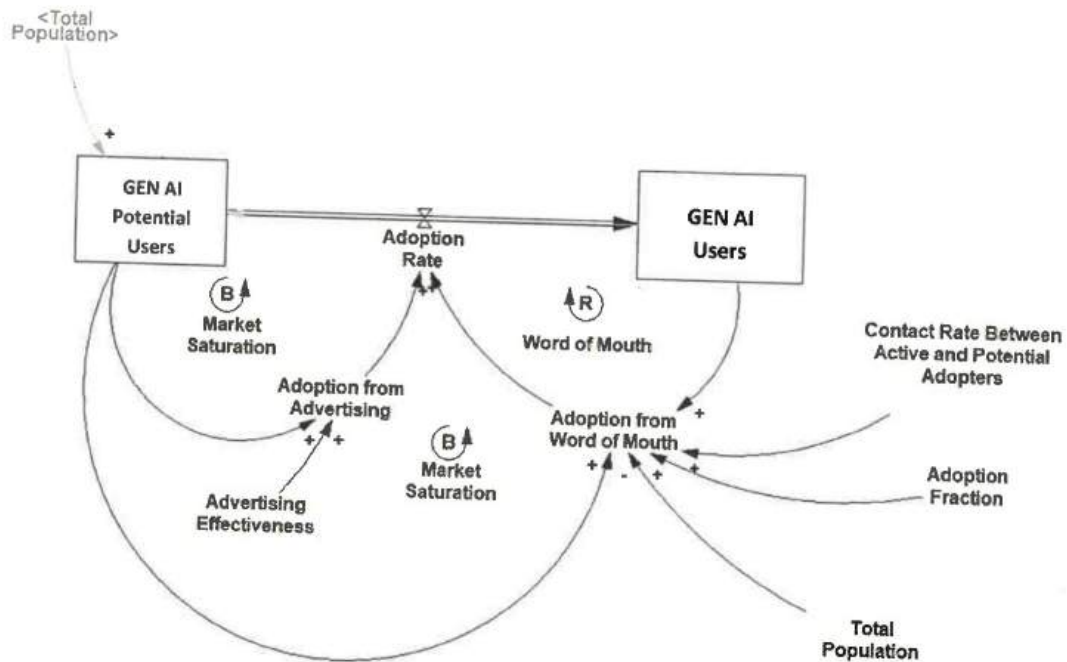
5. (6 Points) Provide justification for whether you agree or disagree with the following statement:
Path dependency involves feedback mechanisms that will lead to an optimal solution, that is where the best product/design/system dominates in the end.

Agree/Disagree (circle one)

Justification:

Sometimes, the solution is not the best design but only perceived that way due to people's feelings or comfort associated with that design. Examples of this are the iPhone and the QWERTY keyboard

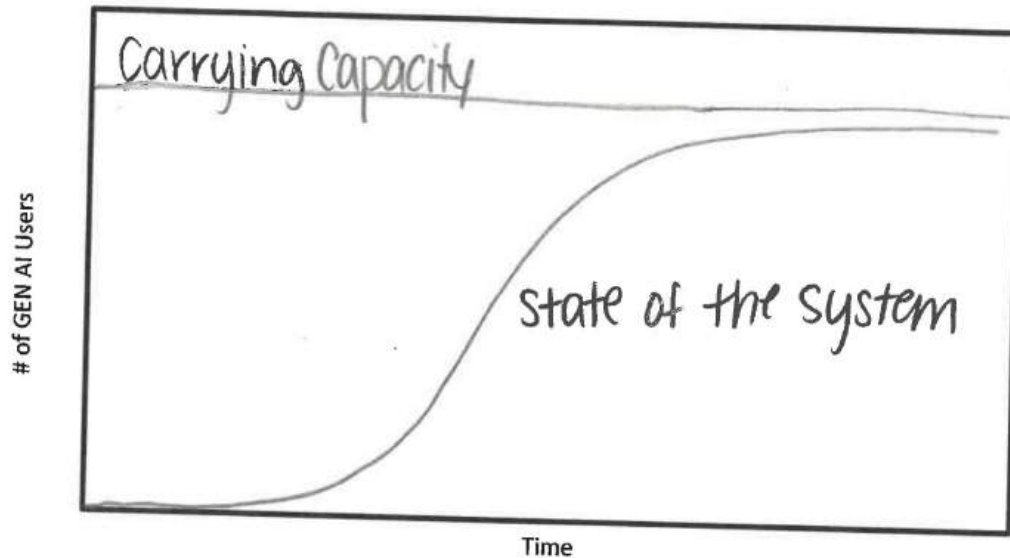
II. Dynamics of Growth (22 Points)



Generative AI (GEN AI) use has increased significantly since ChatGPT released its first model in 2022, with GEN AI being incorporated directly into many browsers, computers, phones, and the like. To predict the future number of users for GEN AI, we have constructed a simplified model that examines the various market factors driving its adoption.

Name: Ryan Remaly

6. (6 Points) On the plot below, draw the behavior of the model by representing the behavior (growth) of GEN AI over time based on the structure shown in the stock and flow diagram. Assume the time horizon starts in 2022 with the release of ChatGPT as a commercially available GEN AI.



7. (6 Points) What assumption(s) did we make in the model, that are key to generating the behavior you drew?

An assumption that I made was as more people and companies adopt GEN AI, the need for adopting AI will die down. The system will level off by the carrying capacity because you cannot have an infinite number of AI users.

8. (6 Points) If the initial value of GEN AI Users = 0 when it is first introduced, what variable in the model would provide the only source of adoption and why?

The only source of adoption would be advertising. This is because advertisement is not dependent on the number of users, whereas word of mouth is. Without users, it is impossible for word of mouth to be a contributing factor to the number of users.

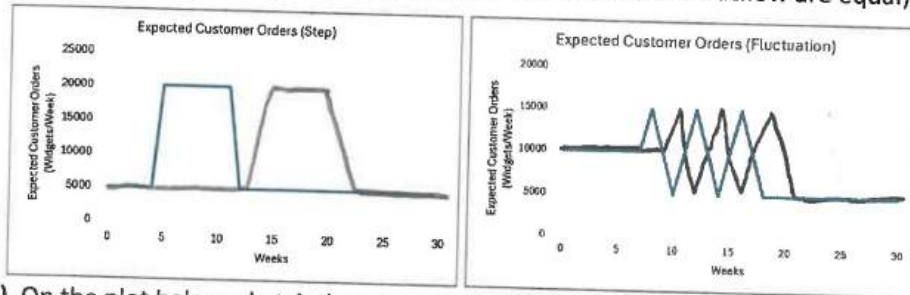
9. (4 Points) This model is an example of a **first-order linear feedback** system. (Fill in the bubble)

- ☐ True
☒ False

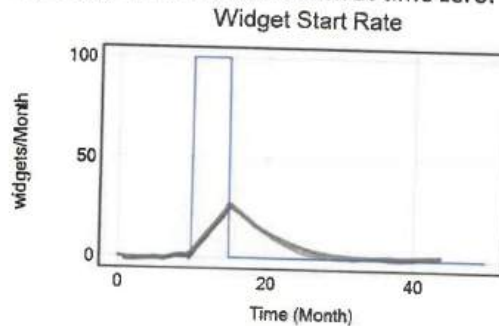
Name: Ryan Remaly

III. Dynamics of Delays (27 Points)

10. (6 Points) On each of the plots below, sketch the response of a **Pipeline Delay** to each of these inputs, assuming a 6-week average delay time. (Assume at $t=0$ the inflow and outflow are equal)



11. (3 Points) On the plot below, sketch the response (outflow) of a **1st order material delay** to the input, assuming a 10-week average delay time and zero stock at time zero.



12. (6 Points) Given a **1st Order Material Delay**, the data in the table below, and an average delay time of **7 weeks**, what is the Outflow at time 7? Write down your **answer in the GREEN box** below. Write down what **base equation you used in the BLUE box** below. Use additional work blank space as required.

Round to the Nearest .1

avg delay 7 which week? 7

Time (weeks)	0	1	2	3	4	5	6	7	8	9	10
Stock (rounded)	100	91	89	80	80	99	88	110	108	106	133
Inflow	5	11	4	11	30	3	35	14	13	42	3
Outflow											

15.7

Equation Used:

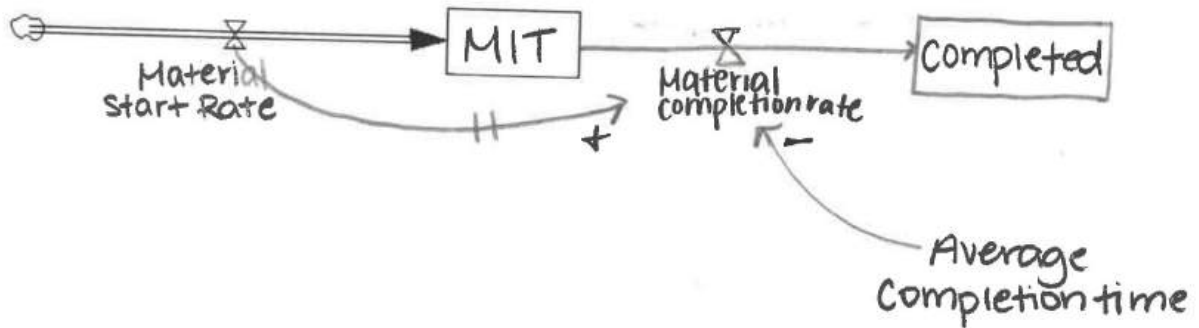
$$\text{Outflow} = \text{MIT}(t)/D$$

Show additional work here if necessary:

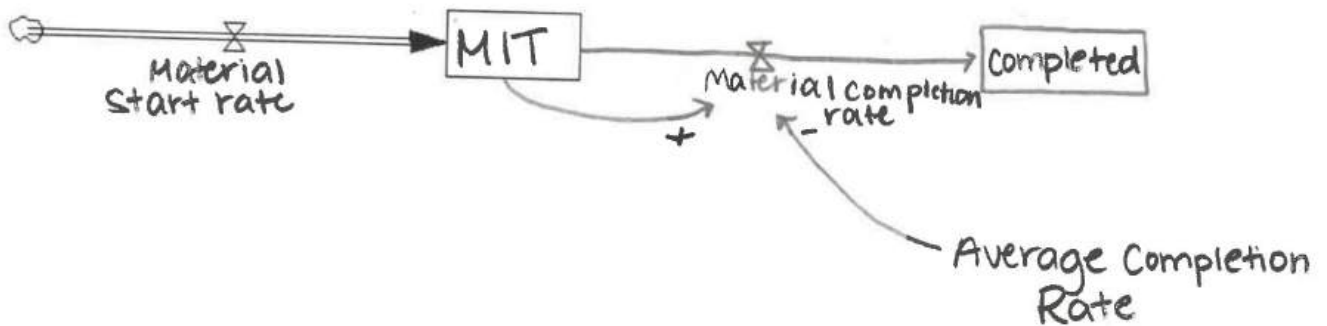
$$\text{Outflow} = 110/7 = 15.7$$

Name: Ryan Remaly

13. (6 Points) In the space below, draw and label the complete stock and flow structure for a pipeline delay (the basic starting point is provided for a delay, you will have to add structure):



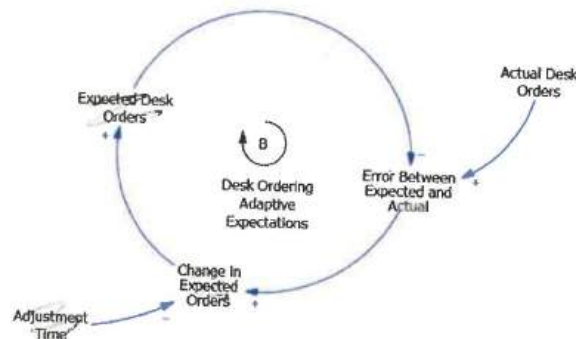
14. (6 Points) In the space below, draw and label the complete stock and flow structure for a 1st Order Material delay (the basic starting point is provided for a delay, you will have to add structure):



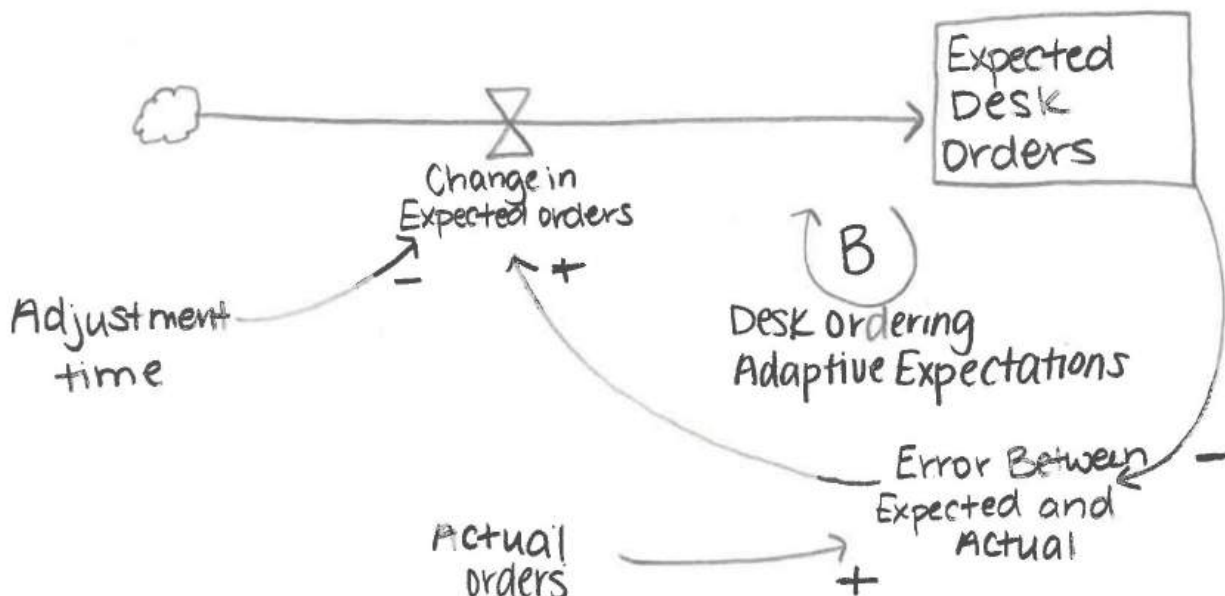
Name: Ryan Remaly

IV. **Structure and Behavior of Delays (36 Points)**

As back pain increases, standing desks offer relief for some people, with many companies capitalizing on the growing standing desk market. A fictitious standing desk company, desks-r-us, decides to invest and join the market. Prior to starting the company, desks-r-us conducts a break-even analysis to determine if the product would be profitable and worth undertaking. As part of this analysis, the team developed projections for their expected orders of their desk model. The model below illustrates what happens to the expected orders as actual desk order information begins to arrive.



15. (15 Points) Convert the Causal Loop Diagram (CLD) above to a stock and flow model (using **ONLY** the variables depicted).



16. (4 Points) Identify the order and type of delay represented.

Order: 1st Order Type: Information

Name: Ryan Remaly

17. (9 Points) Define the equation and units for each variable depicted in the CLD. If the variable is exogenous, assign it a value that you believe is reasonable for the model:

Variable Name	Units	Equation or Constant Value
Change in Expected Orders	desks / day	Error between Expected and actual Adjustment time
Error Between Expected and Actual	desks	Expected - Actual
Adjustment Time	days	24

18. (8 Points) Assume the adjustment time is **24 days**. If the *Actual Desk Orders* increases by a one-time step function and stays at that higher value, how many days would it take for the *Expected Orders of Desks* to account for **50%** and **75%** of the change to the *Actual Desk Orders*?

Time to adjust to 50% of actual Desk Orders: ≈ 6 days

Time to adjust to 75% of actual Desk Orders: ≈ 4 days

What is the name of the rule you used to calculate these? Rule of 70

Show additional work here:

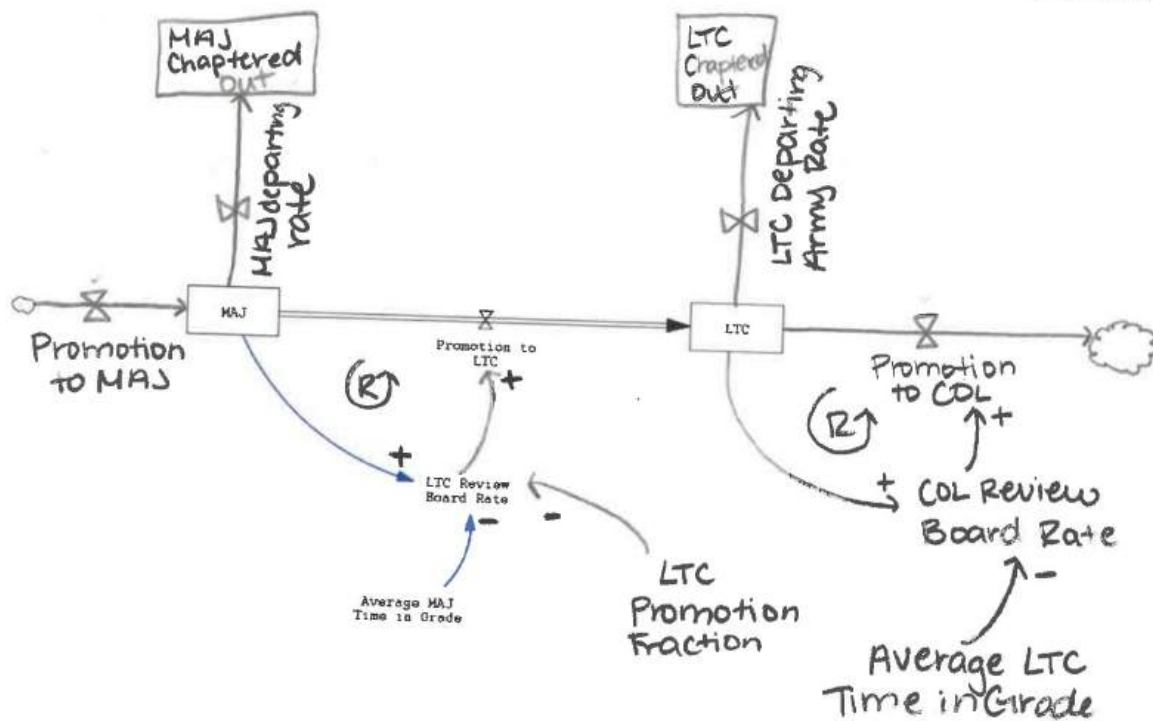
$$50\% \rightarrow 24(.5) = 12 \rightarrow 70/12 \rightarrow 5.8$$

$$75\% \rightarrow 24(.75) = 18 \rightarrow 70/18 \rightarrow 3.88$$

Name: Ryan Remaly

- V. **Aging Chains (35 Points)** HRC conducts a board review yearly to determine those selected for promotion. The terminology for review boards and promotions follows that a current MAJ undergoes a LTC review board, and if selected, they undergo a LTC Promotion (i.e., the promotion name follows the rank of the stock they are promoted into, not from). Assume that if either a Major (MAJ) or Lieutenant Colonel (LTC) is not selected for promotion, they are chaptered out of the Army. For this problem, map out the officer promotion system for Field Grade Officers, promoting to a Major through promotion to Colonel and out of the Field Grade cohort (just use a sink, do not need to include a stock for Colonel and higher ranks). There is a delay for time in grade requirements that occur after each promotion before you can promote to the next level.

19. (20 Points) Use the base diagram below to develop an Aging Chain Model showing the progression of a Field Grade Officer. Show promoting to MAJ through being promoted from LTC (you do not need to include a COL stock, just an outflow from LTC to the COL cohort. Hint: To model the flow of personnel from MAJ to COL, you need to include things like MAJ promotion rate, Average MAJ Time in Grade, LTC Promotion Fraction, and LTC Departing Army Rate. **THIS LIST IS NOT ALL INCLUSIVE. YOU WILL NEED TO ADD OTHER STRUCTURE.** Remember to label all polarities of causal links. You may have to add stocks, flows, variables, causal links, sources, and sinks.



Name: Ryan Remaly

20. (10 Points) Provide the units and equations for the following (this is not an all-inclusive list for your model):

Variable Name	Units	Equation or Value if Constant
Promotion to LTC	Persons / Year	MAJ / LTC Review Board Rate
LTC Review Board Rate	Persons / Year	MAJ / Average MAJ Time in Grade
Average MAJ Time in Grade	Years	5
LTC Promotion Fraction	Persons / Year	MAJ / LTC Board Review rate

21. (5 Points) What order system did you develop? Explain your response.

A 2nd order system. This stock and flow diagram contains two independent stocks.

Include Extra Work below for your test if needed, and list the corresponding question number if applicable.

SM484: System Dynamics Simulation - AY 26-1
WPR 2

Name: Ryan Remaly

Include Extra Work below for your test if needed, and list the corresponding question number if applicable.

Poor Student Work

Student

Aatavion Andrews

Total Points

92 / 150 pts

Question 1

1

5 / 6 pts

- 0 pts Correct

✓ - 1 pt one delay lacking some detail

- 2 pts One delay type correct but explanation incorrect

- 2 pts both delays have some errors/lacking detail in explanations

- 1 pt messy and hard to read

Question 2

2

4 / 5 pts

- 0 pts Correct

✓ - 1 pt 1 & 5 Correctly selected, but one additional selected

- 2 pts missed one correct answer (1 & 5) and had one additional selection

- 1 pt missed one correct answer (1 & 5)

- 3 pts missed one correct answer (1 & 5) and had two additional selected

Question 3

3

2 / 5 pts

- 0 pts Correct

- 1 pt selected 1 & 4 correctly, one additional incorrectly selected

- 2 pts one not selected (1 & 4 correct), one additional incorrectly selected

✓ - 3 pts one not selected (1 & 4 correct), two additional incorrectly selected

- 1 pt one not selected (1 & 4 correct)

- 0 pts [Click here to replace this description.](#)

Question 4

4

4 / 8 pts

– 0 pts Correct

✓ – 2 pts missed balancing loop

✓ – 2 pts missed reinforcing loop

– 2 pts incorrect/missing explanation on two components

– 2 pts missed exogenous goal/carrying capacity

– 1 pt incorrect explanation on correct structure (balancing loop, reinforcing loop, exogenous goal)

– 6 pts gave answers for oscillation behavior, not s-shaped behavior

Question 5

5

5 / 6 pts

– 0 pts Correct

✓ – 1 pt Explanation has missing/incorrect claims

– 2 pts Explanation lacking significant justification

– 4 pts selected agree-path dependency does not lead to the optimal solution

– 1 pt didn't circle agree/disagree

Question 6

6

6 / 6 pts

– 0 pts Correct

✓ – 0 pts correct, but should also show the carrying capacity

– 1 pt s-shaped drawn incorrectly (exponential at end, goal seek at beginning, exceeds carrying capacity, etc)

– 3 pts should be s-shaped, not exponential

– 4 pts should be s-shaped

Question 7

7

6 / 6 pts

✓ – 0 pts Correct, AT LEAST ONE valid assumption listed (all users will adopt, won't change once adopted, 0 initial adopters, population won't change, advertising effectiveness >0, no delays, no market competitors, these are just some of the assumptions that may be present)

– 1.5 pts Assumptions stated not key to generating the behavior, or the assumptions simply just restate the structure required to generate S shaped behavior

– 4 pts Answered question regarding phones, not regarding GEN AI

– 4 pts partial credit for attempt, but incorrect/incomplete discussion

Question 8

8

6 / 6 pts

✓ - 0 pts Correct, adoption from advertising

- 4 pts answer other than adoption from advertising

Question 9

9

0 / 4 pts

- 0 pts Correct

✓ - 4 pts false, this is a first-order, non-linear feedback system, because the rates are not just dependent on stocks and exogenous variables.

Question 10

10

2 / 6 pts

- 0 pts Correct, same shape as the input, delayed by 6 weeks (we gave some leeway on where 6 weeks was drawn due to the difficulty in determining on these graphs)

✓ - 1 pt first partially wrong

- 2 pts first incorrect

- 1 pt second partially wrong

- 2 pts second incorrect

- 6 pts no answer

✓ - 3 pts one graph no answer

Question 11

11

0.5 / 3 pts

- 0 pts Correct, the shape and size of the output behavior are roughly correct. We gave some leeway on the scale as we didn't expect a mathematical calculation of this output, more generalized behavior.

✓ - 2.5 pts incorrectly drew pipeline delay behavior

- 1 pt partial errors (shape, size, location, etc)

- 3 pts incorrect

Question 12

12

0 / 6 pts

– 0 pts Correct, both equation $\text{Outflow} = \text{MIT}/\text{Delay}$ and the value 15.7 correct

– 1 pt incorrect rounding (not to 0.1 accuracy)

✓ – 6 pts No answer, or incorrect answer and didn't list equation $\text{Outflow} = \text{MIT}/D$

– 3 pts didn't list equation $\text{Outflow} = \text{MIT}/D$

– 3 pts incorrect answer

– 1 pt math error only

Question 13

13

1.5 / 6 pts

– 0 pts Correct

✓ – 0.5 pts one minor error (no sink or secondary stock, one incorrect polarity, one missing label, only delay hash mark not delay variable)

✓ – 1 pt no causal link from inflow (widget start rate) to outflow (widget completion rate)

✓ – 1 pt no average delay time into outflow

✓ – 1 pt didn't label appropriately

✓ – 1 pt No Outflow

– 1 pt two errors (no sink or secondary stock, incorrect polarity, missing label, only delay hash mark not delay variable)

Question 14

14

2 / 6 pts

– 0 pts Correct

– 1 pt Minor Error

– 2 pts Two Minor Errors

✓ – 4 pts Major Errors

Question 15

15

13 / 15 pts

– 0 pts Correct

✓ – 2 pts Minor Error

– 4 pts Two Minor Errors

– 6 pts Major Errors

– 8 pts Completely Wrong

– 15 pts Blank

Question 16

16

2 / 4 pts

– 0 pts Correct

✓ – 2 pts One Incorrect

Question 17

17

6 / 9 pts

– 0 pts Correct

– 1 pt One Incorrect

– 2 pts Two Incorrect

✓ – 3 pts Three Incorrect

– 4 pts Four Incorrect

– 5 pts Five Incorrect

– 9 pts Blank

Question 18

18

4 / 8 pts

– 0 pts Correct

– 2 pts Correct Rule, One Incorrect Answer

✓ – 4 pts Correct Rule, Incorrect Application

– 5 pts Correct Rule, No Attempt

– 5 pts Incorrect Rule with Attempt

Question 19

19

14 / 20 pts

- 0 pts Correct
- 2 pts Minor Error
- 4 pts Multiple Minor Errors

✓ - 6 pts Major Errors

- 8 pts Completely incorrect

Question 20

20

7 / 10 pts

- 0 pts Correct
- 1 pt One Incorrect
- 2 pts Two Incorrect

✓ - 3 pts Three Incorrect

- 4 pts Four Incorrect

- 5 pts Five Incorrect

- 6 pts Six Incorrect

Question 21

21

2 / 5 pts

- 0 pts Correct
- 1.5 pts Correct with No Explanation
- 1.5 pts Incorrect with Good Detail

✓ - 3 pts Incorrect

Question 22

Extra Work 1

0 / 0 pts

✓ - 0 pts Correct, no points associated with this, just space to review if extra work was completed for any previous questions

- 0 pts correct, extra work found and applied to previous questions

Question 23

Extra Work 2

0 / 0 pts

✓ - 0 pts Correct



Department of Systems Engineering

System Dynamics Simulation

SM484 - AY 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

WPR2 (Primary)

Name: Antawion Andrew

Section: I 2

Finish time: _____

1. This is a **70-minute** exam worth **150 points**.
2. This is a closed book/notes exam, **ONE side of an 8-1/2x11 sheet of paper note sheet is allowed**. NO COLLABORATION allowed.
3. The only items that you may bring into the classroom are: one sided note sheet, pencil, straight edge, and calculator. **DO NOT BRING INK PENS!!!**
4. **Place your name NEATLY on each page in the space provided**. Feel free to disassemble the exam; a stapler is available to reattach the pages when you are finished. Make sure you have all 12 pages (including this cover page).
5. **SHOW ALL WORK!** Unsupported or missing work earns little or no partial credit.
6. After you finish the exam, fill in the finish time at the top of this page and sign the honor acknowledgement statement below:

"I did not use any sources, nor did I receive any assistance while completing this WPR. This WPR will be released from academic security at a time announced by the Course Director, no earlier than 17NOV25. Prior to that time, I may not discuss any aspects of this exam with anyone except an SM484 instructor." **PRINT IN BLOCK LETTERS NEATLY**

Printed Name (**FIRST LAST**): Antawion Andrew Signature: at

Date Signed: 11/03/25

Grading summary

Question	Possible Points	Actual Points
I. Multiple Choice/Short Answer	30	
II. Dynamics of Growth	22	
III. Dynamics of Delays	27	
IV. Structure and Behavior of Delays	36	
V. Aging Chains	35	
Total	150	

I. **Multiple Choice/Short Answer Questions (30 Points)**

1. (6 Points) There are several different types of delays. List and briefly describe **TWO (2)** different types of delays.

1: Information Delay: The gradual adjustment of perceptions or belief.
is the state of the system, delay between receipt of new information

2: Material Delay: material or product in transit from one place to another
an example is shipping.

2. (5 Points) Which of the following is true about path dependence? (Fill in the bubble of all that apply)

- ☒ The eventual end state in path dependent systems depends on the starting point and small, unpredictable perturbations early in its history.
- ☐ The system starts with locally stable equilibria.
- ☐ In determining design standards, such as the QWERTY keyboard, it ultimately allows the best design to dominate.
- ☒ Path-dependence only occurs when one equilibria is initially more likely than all others.
- ☒ It arises in systems dominated by positive feedback.

3. (5 Points) One of the Five Formulation Fundamentals described in your course text is "The Baker Criterion: The inputs to all decision rules in models must be restricted to information actually available to the real decision makers." Which of the following statements are true with respect to this statement? (Fill in the bubble of all that apply)

- ☒ The future is not known to anyone. All expectations and beliefs about the future are based on historical information. Expectations and beliefs may therefore be incorrect.
- ☐ The use of arbitrary scaling factors (fudge factors) is sometimes necessary to ensure that the units of measure in all equations balance.
- ☒ Decision making should not be assumed to conform to any prior theory but should be investigated firsthand.
- ☐ Actual conditions and perceived conditions differ due to measurement and reporting delays, and beliefs are not updated immediately on receipt of new information. Perceptions often differ from the actual situation.
- ☒ Equilibrium and model stability should be assumed if your model equations are correct and conform to managerial practice.

✓ (B5)

4. (8 Points) What are **three** key structural components in a **stock and flow diagram** that must be present for a system to show S-Shaped behavior, and how does each structural component generate part of the behavior?

- 1: Stock: this is the total amount of what is in the system we are counting
- 2: Flow: This is the rate at which the stock increases or decreases
- 3: Goal: this achieves the balancing effect in order for a system to plateau. There must be a goal or maximum the stock can reach

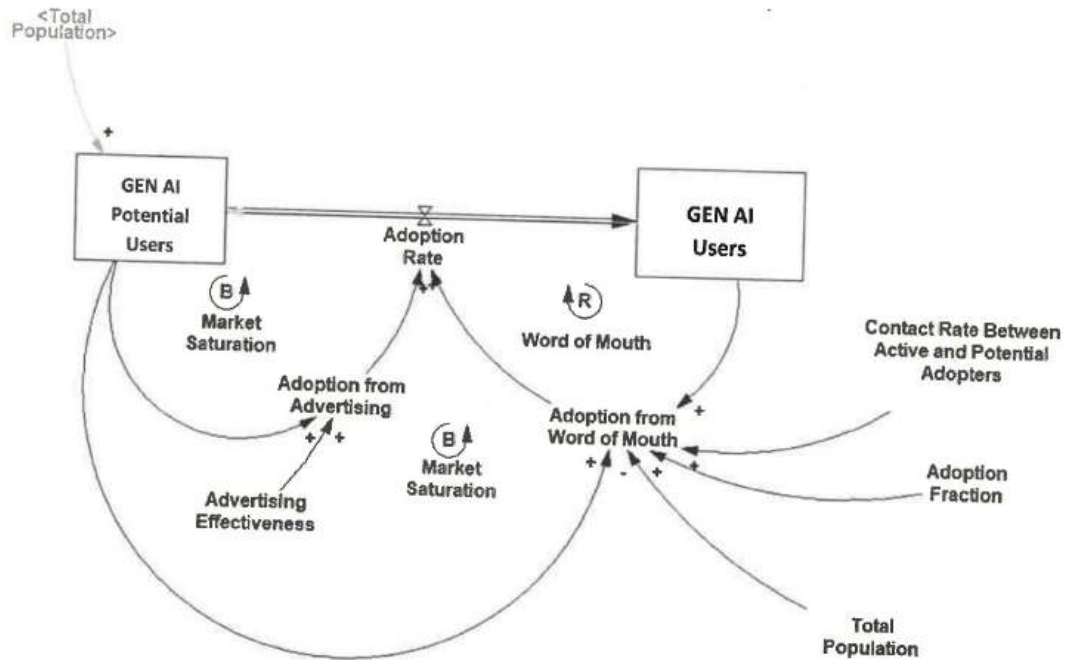
5. (6 Points) Provide justification for whether you agree or disagree with the following statement:
Path dependency involves feedback mechanisms that will lead to an optimal solution, that is where the best product/design/system dominates in the end.

Agree/Disagree (circle one)

Justification:

Path Dependence does not necessarily rely on feedback, it depends on initial conditions and random shocks. This is seen with QWERTY keyboards, it is not the most optimal, but yet is the most popular.

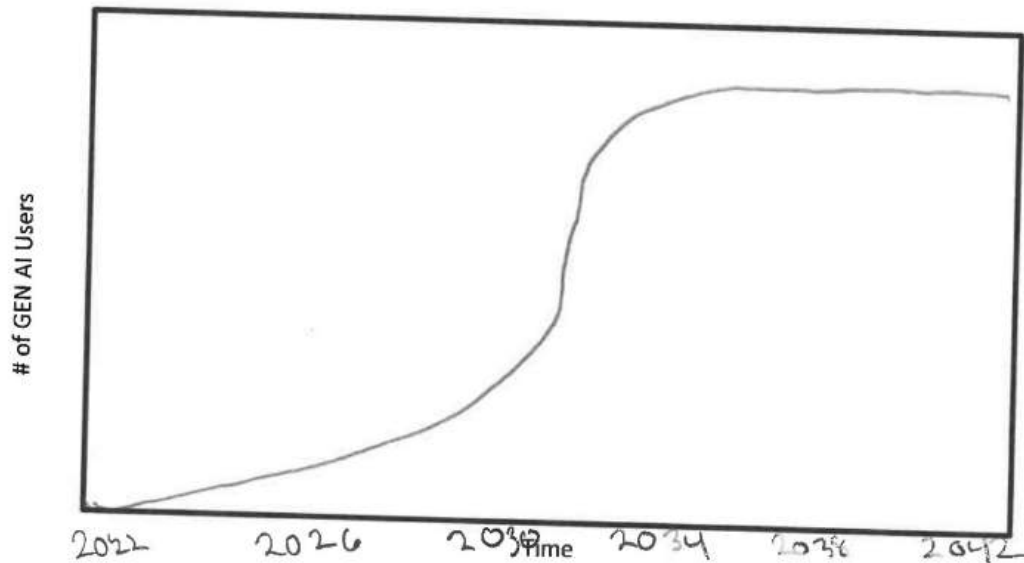
II. Dynamics of Growth (22 Points)



Generative AI (GEN AI) use has increased significantly since ChatGPT released its first model in 2022, with GEN AI being incorporated directly into many browsers, computers, phones, and the like. To predict the future number of users for GEN AI, we have constructed a simplified model that examines the various market factors driving its adoption.

Name: Antawion A

6. (6 Points) On the plot below, draw the behavior of the model by representing the behavior (growth) of GEN AI over time based on the structure shown in the stock and flow diagram. Assume the time horizon starts in 2022 with the release of ChatGPT as a commercially available GEN AI.



7. (6 Points) What assumption(s) did we make in the model, that are key to generating the behavior you drew?

We assume that Chat GPT users will increase at an exponential rate before eventually plateauing as everyone has adopted it. Potential users is exogenously linked to <Total Population> there is a finite # of population (this essentially a goal)

8. (6 Points) If the initial value of GEN AI Users = 0 when it is first introduced, what variable in the model would provide the only source of adoption and why?

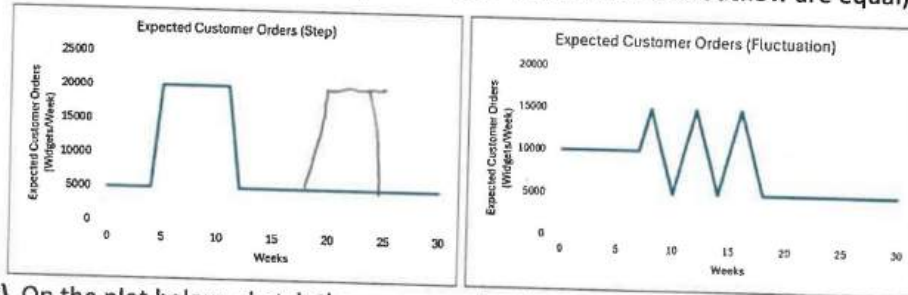
Adoption from advertising. If the GEN AI users = 0 Adoption from word of mouth which has a positive causal relationship with GEN AI users would be 0 as well, whereas adoption from advertising is not directly linked to GEN AI users, only the adoption rate.

9. (4 Points) This model is an example of a *first-order linear feedback system*. (Fill in the bubble)

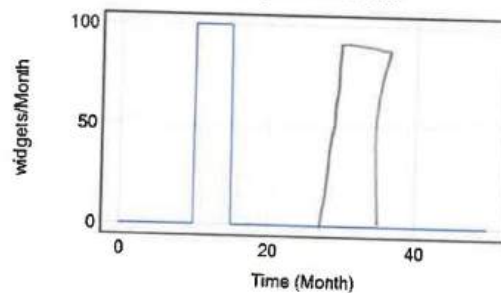
☒ True
☐ False

III. Dynamics of Delays (27 Points)

10. (6 Points) On each of the plots below, sketch the response of a **Pipeline Delay** to each of these inputs, assuming a 6-week average delay time. (Assume at $t=0$ the inflow and outflow are equal)



11. (3 Points) On the plot below, sketch the response (outflow) of a **1st order material delay** to the input, assuming a 10-week average delay time and zero stock at time zero.



12. (6 Points) Given a **1st Order Material Delay**, the data in the table below, and an average delay time of 7 weeks, what is the Outflow at time 7? Write down your **answer in the GREEN box** below. Write down what **base equation** you used in the **BLUE box** below. Use additional work blank space as required.

Round to the Nearest .1

avg delay

7

which week?

7

Time (weeks)	0	1	2	3	4	5	6	7	8	9	10
Stock (rounded)	100	91	89	80	80	99	88	110	108	106	133
Inflow	5	11	4	11	30	3	35	14	13	42	3
Outflow								1540			

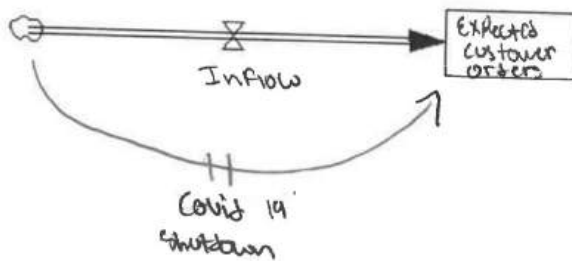
Equation Used:

Resource Production

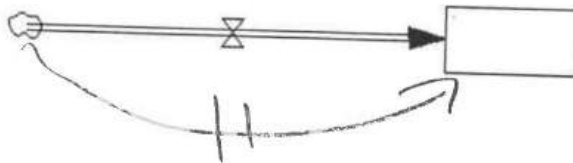
Show additional work here if necessary:

Resource Production

13. (6 Points) In the space below, draw and label the complete stock and flow structure for a **pipeline** delay (the basic starting point is provided for a delay, you will have to add structure):

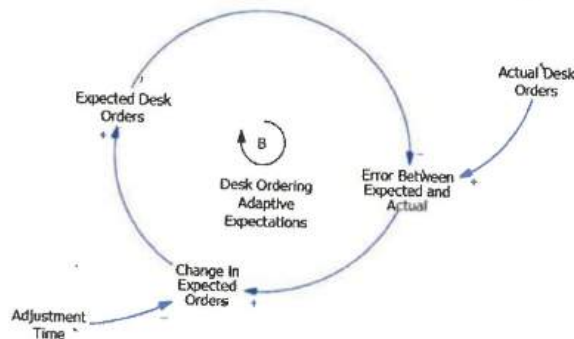


14. (6 Points) In the space below, draw and label the complete stock and flow structure for a **1st Order Material** delay (the basic starting point is provided for a delay, you will have to add structure):



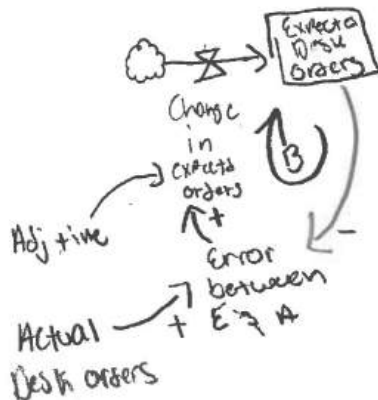
IV. **Structure and Behavior of Delays (36 Points)**

As back pain increases, standing desks offer relief for some people, with many companies capitalizing on the growing standing desk market. A fictitious standing desk company, desks-r-us, decides to invest and join the market. Prior to starting the company, desks-r-us conducts a break-even analysis to determine if the product would be profitable and worth undertaking. As part of this analysis, the team developed projections for their expected orders of their desk model. The model below illustrates what happens to the expected orders as actual desk order information begins to arrive.



15. (15 Points) Convert the Causal Loop Diagram (CLD) above to a stock and flow model (using **ONLY** the variables depicted).

Change in perceived value, $\frac{1}{\tau}$ (0)



16. (4 Points) Identify the order and type of delay represented.

Order: first Type: material

94: System Dynamics Simulation - AY 26-1

Name: Autumn A

... (3 Points) Define the equation and units for each variable depicted in the CLD. If the variable is exogenous, assign it a value that you believe is reasonable for the model:

Variable Name	Units	Equation or Constant Value
Change in Expected Orders	Desk / Days	# of Expected - Actual / Days
Error Between Expected and Actual	desks	Expected Desk orders / Change in expected orders
Adjustment Time	Days	DMNL

18. (8 Points) Assume the adjustment time is **24 days**. If the *Actual Desk Orders* increases by a one-time step function and stays at that higher value, how many days would it take for the *Expected Orders of Desks* to account for **50%** and **75%** of the change to the *Actual Desk Orders*?

Time to adjust to 50% of actual Desk Orders: 1.41 days

Time to adjust to 75% of actual Desk Orders: 0.93 days

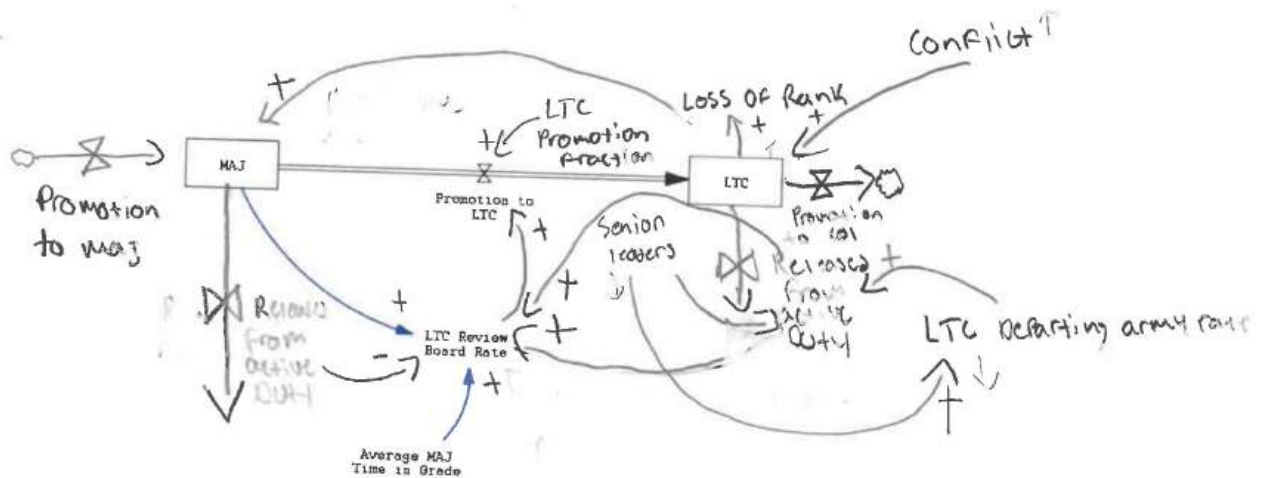
What is the name of the rule you used to calculate these? Rule of 70

Show additional work here:

$$\frac{70}{50} = 1.4 \quad \frac{70}{75} = .93$$

- V. **Aging Chains (35 Points)** HRC conducts a board review yearly to determine those selected for promotion. The terminology for review boards and promotions follows that a current MAJ undergoes a LTC review board, and if selected, they undergo a LTC Promotion (i.e., the promotion name follows the rank of the stock they are promoted into, not from). Assume that if either a Major (MAJ) or Lieutenant Colonel (LTC) is not selected for promotion, they are chaptered out of the Army. For this problem, map out the officer promotion system for Field Grade Officers, promoting to a Major through promotion to Colonel and out of the Field Grade cohort (just use a sink, do not need to include a stock for Colonel and higher ranks). There is a delay for time in grade requirements that occur after each promotion before you can promote to the next level.

19. (20 Points) Use the base diagram below to develop an Aging Chain Model showing the progression of a Field Grade Officer. Show promoting to MAJ through being promoted from LTC (you do not need to include a COL stock, just an outflow from LTC to the COL cohort. *Hint: To model the flow of personnel from MAJ to COL, you need to include things like MAJ promotion rate, Average MAJ Time in Grade, LTC Promotion Fraction, and LTC Departing Army Rate. THIS LIST IS NOT ALL INCLUSIVE. YOU WILL NEED TO ADD OTHER STRUCTURE. Remember to label all polarities of causal links. You may have to add stocks, flows, variables, causal links, sources, and sinks.*



SM484: System Dynamics Simulation – AY 26-1
WPR 2

Name: Antavion A

20. (10 Points) Provide the units and equations for the following (this is not an all-inclusive list for your model):

Variable Name	Units	Equation or Value if Constant
Promotion to LTC	Persons/Year	LTC Review Board rate * MAJ
LTC Review Board Rate	Persons/Year	Average MAJ in time grade / MAJ
Average MAJ Time in Grade	Years	5
LTC Promotion Fraction	Promotion/Year	LTC * Review Board rate

21. (5 Points) What order system did you develop? Explain your response.

First order system there is only one stock with independent in flow & outflow (MAJ) depletion, LTC depends on the MAJ stock to flow in to it

Include Extra Work below for your test if needed, and list the corresponding question number if applicable.

SM484: System Dynamics Simulation – AY 26-1
WPR 2

Name: Adrian Antares

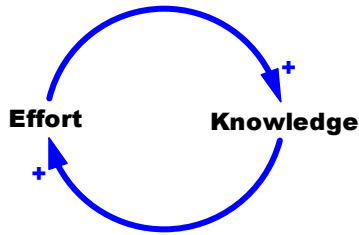
Include Extra Work below for your test if needed, and list the corresponding question number if applicable.



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I Course Project

PROJECT PROMPT

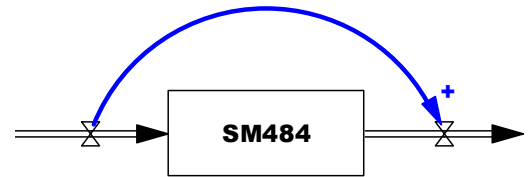


System Dynamics Simulation

SM 484 – AT 26-1

MAJ Jake Lueders

MAJ Pat Lupfer



Course Project (300 Points)

Assigned: Lesson 07

Due:

Project Proposal Lesson 09 (15SEP25 at 2359)

Part I Lesson 14 (01OCT25 at 2359) - 50pts

Part II Lesson 20 (20OCT25 at 2359) – 75pts

Part III Lesson 27 (07NOV25 at 2359) - 50pts

Presentation Lessons 33/34 (Due Lesson 32: 26NOV25 at 2359) - 50pts

Project Report Lesson 35 (10DEC25 at 1600) - 75pts

Submission guidance: All submissions will be via Canvas, with additional details in the assignment for what is required. All written submissions will be in .pdf format, and all vensim submissions will be the .mdl file and any simulation run .vdfx that you deem necessary.

Peer Review: Each major submission will contain a peer review, peer reviews will alter individual grades from the group grade if individuals do not put forth a team effort. Peer reviews should highlight effort discrepancy, not the capability discrepancy between members.

Project Proposal: The proposal is designed to verify that you have a plan for the course project. Submission is one page maximum (no DAAW requirement). Include team name, team members, project topic (see part I subsection b below), initial problem statement, and a short descriptive timeline/plan for project research through final submission.

Part I: Initial Causal Loop Diagram Brainstorm and Project Ideation (Group – 50 Points)

a. Project Ideation:

This initial submission ensures that your project is on track and provides an opportunity for instructor feedback early in your process. You will select a topic in alignment with section b (Topic Guidance) below. The initial submission shall include a cover page in accordance with the Dean's Documentation and Acknowledgment of Academic Work (DAAW) (June 2025), and all topics from the Part I Submission Checklist below. It is ok if your final project submission changes significantly from this Part I submission; modeling is iterative and should change as you continue working. Note that this project structure has changed from previous terms.

b. Topic Guidance

This assignment serves as the initial building block to support the term project. In this project, you will select a problem to model within **one** of the following focus areas: **Army Recruitment and Retention Focus, Army Maintenance Focus, or Enemy Troop Focus**. You have flexibility to determine your problem within these areas, check with your instructor early if you have questions. The overall output of your project will be a validated Vensim model with policy recommendations to help address your identified problem. In your research, you may find existing system dynamics models for the selected focus area that you can use as a base. Understanding an existing model and piece of work is not sufficient for a term project, so only investigate an issue for which you can add data, information, knowledge, analysis, and insight to the system dynamics model and investigation. Use the System Dynamics Modeling Process and the System's Decision Process to guide your work. Use the system descriptions to extract and document specific causal relationships, tell the dynamic story, build a dynamic hypothesis (conceptual model) based on those descriptions, build a formal model based on the conceptual model and other information, partially validate the model, and test part of that hypothesis. Only choose a topic for which you have good, detailed descriptions of the causal structure of the system that you will model and a description of system behavior. Finding the system descriptions is part of your proposal development. In other words, you need to demonstrate to the instructor that you have the descriptions needed in hand to build the model as part of your proposal.

Part I Submission Checklist:

- € Cover page IAW Dean's DAAW (June 2025)
- € Content includes the following (no required format, organize logically):
 - € Research Problem Statement **(5 points)**
 - € Initial CLD (can be screenshot from Vensim or hand drawn neatly) **(35 points)**
 - € CLD Image
 - € Short Description of why you included each variable (1-3 sentences per variable)
 - € Description of the System Boundary (2-4 sentences)
 - € Additional Questions **(10 points)**
 - € How your group intends to work together and handle dispute resolution (1-2 paragraphs)
 - € What you see as your biggest challenge(s) (1-2 challenges)
 - € How you plan to overcome each challenge (2-4 sentences per challenge)

Part II: Problem Articulation and Dynamic Hypothesis (Group – 75 Points)

- c. **Write a cohesive report consisting of elements I through IV as described below (Submission Checklist).** The report shall include a cover page in accordance with the Dean's Documentation and Acknowledgment of Academic Work (DAAW) (June 2025), and the report's main body format shall follow the General Donald R Keith Memorial Conference Paper Template. The template can be found on the Course Canvas page, attached to this assignment.

Part II Submission Checklist:

- € Cover page IAW Dean's DAAW (June 2025)
- € Content follows *General Donald R Keith Memorial Conference Paper Template*
- € Content includes the following sections of the report:
 - € I. Introduction **(5 points)**
 - € II. Literature Review **(10 points)**
 - € Include an overview of system dynamics methods and tools
 - € Use at least five (5) scholarly sources for the literature review
 - € III. Problem Articulation **(25 points)**
 - € Includes the following: refined problem statement, at least 10 key variables, reference modes for each variable, time horizon, system boundary, and a justification for each.
 - € Use at least two (2) scholarly sources to describe the problem
 - € IV. Dynamic Hypothesis **(35 Points)**
 - € Formulation of Dynamic Hypothesis (DH)
 - € DH supported by research
 - € CLD
 - € CLD Image (Vensim or handwritten neatly)
 - € Improves on CLD from Part I
 - € CLD is consistent with good modeling practice
 - € Causal links and polarity supported by research
 - € Uses CLD to explain DH to a general audience

Part III: Detailed Formulation: Stock and Flow Diagram (Group – 50 Points)

- a. **Build a functioning stock and flow model and write a cohesive report consisting of elements I through V as described below (Submission Checklist).** The report shall include a cover page in accordance with the Dean's *DAAW (June 2025)* and the report's main body format shall follow the *General Donald R Keith Memorial Conference Paper Template*. It is expected that your model functions, though it may not have all the detail you would like to eventually capture by the final report submission.

b. **Functioning stock and flow model.**

The major component of this deliverable is a stock and flow model of your system. The stock and flow model should capture all of the structure identified in the first part of the project as well as any feedback from your instructor. The model will have more variables than originally identified in the CLDs. The model should function and generate behavior that matches your reference modes. Provide a copy of your complete model with your submission. The Stock and Flow model should adhere to guidelines and format standards in the textbook. Provide output graphs from the model for your key variables. The written description of the model should describe the stock and flow model as well as the dynamic behavior.

Part III Submission Checklist:

- € Cover page IAW Dean's *DAAW (June 2025)*
- € Content follows *General Donald R Keith Memorial Conference Paper Template*
- € Content includes all sections as per *Guidance for Technical Writing in SM484*. A broad overview of the submission requirements is presented here:
 - € I. Introduction **(10 points updates to Section I-IV as needed)**
 - € II. Literature Review as per previous guidance
 - € III. Problem Articulation as per previous guidance
 - € IV. Dynamic Hypothesis (CLD) as per previous guidance
 - € V. Formulation (S&F) of a Simulation Model **(40 points)**
 - € *Post Vensim model with subject "SM484 [project title]"* along with the report document in Canvas.
 - € Includes screenshot of simulation model
 - € Includes table as required defining your variables and units.

Project Presentation (Group – 50 Points)

- a. **Present your work.** Presentation times will be assigned by the instructor. Each group will be given 20 minutes to present their work to the class with at least 5 minutes of questions following the presentation.

b. **Presentation**

The presentation should be comprehensive and cover all five (5) steps of the System Dynamics Modeling Process. All groups must submit their presentation at the same time, IAW timeline at top of assignment and present during lessons 33 and 34.

Presentation Submission Checklist:

- € PowerPoint presentation
- € Content includes all steps in the SD Modeling Process:
 - € I. Introduction **(2 Points)**
 - € II. Literature Review **(3 Points)**
 - € Include an overview of system dynamics methods and tools
 - € Use at least five (5) scholarly sources for literature review
 - € III. Problem Articulation **(5 Points)**
 - € Includes the following: key variables, reference modes, time horizon, system boundary, and a justification for each
 - € Use at least two (2) scholarly sources to describe the problem
 - € IV. Dynamic Hypothesis **(5 Points)**
 - € Formulation of DH
 - € Uses CLD to explain DH to general audience
 - € V. Formulation **(10 points)**
 - € Includes screenshot of simulation model
 - € Includes model outputs for key variables
 - € Includes table as per guidance
 - € VI. Testing **(5 Points)**
 - € Includes, at a minimum, tests for dimensional consistency, extreme conditions, and sensitivity analysis with output graphs where appropriate.
 - € VII. Policy Formation and Evaluation **(5 Points)**
 - € Includes an explanation and depiction of changes to the model as per guidance.
 - € VIII. Conclusion **(3 Points)**
 - € Individual Presentation Evaluation **(12 Points)**

Final Report (Group – 75 Points)

- a. **Write a cohesive report consisting of elements I through VIII as described in *Guidance for Technical Writing in SM484*.** The report shall include a cover page in accordance with the Dean's DAAW (June 2025) and the report's main body format shall follow the *General Donald R Keith Memorial Conference Paper Template*.

b. **Model Testing**

For this deliverable, the team will conduct model testing and report the results of those tests. First, the team will ensure that the model displays dimensional consistency. This includes the check units function in Vensim and all units should be consistent without the use of dummy variables to convert units. The model will also be tested at extreme conditions. The team should test at least two variables at extreme values to ensure the model outputs logical behavior at the extremes. Finally, the team should conduct sensitivity analysis for at least two variables that have some uncertainty. For this the team should present the output and provide a description of any potential impacts to their analysis.

c. **Policy Design and Evaluation**

The purpose of developing system dynamics models is to test policies. The team will develop a policy to correct the problematic behavior in their case study. The team should model the behavior by modifying the structure of the model and the system and generate a new output that shows the corrected behavior. For this, the team should present and describe any changes to the model. The team should also present new output graphs that show the new behavior, this may be at some point in the future when the policy is implemented. The description of the policy should be robust enough to implement.

Submission checklist:

- € Cover page IAW Dean's DAAW (June 2025)
- € Content follows *General Donald R Keith Memorial Conference Paper Template* (Page limit does not apply, just use the same format)
- € Content includes all sections of the report below. This should be a continuation of your work in the first two parts of the project. Include the following elements:
 - € Abstract **(5 Points)**
 - € I. Introduction **(5 Points)**
 - € II. Literature Review as per previous submissions **(5 Points)**
 - € III. Problem Articulation as per previous submissions **(5 Points)**
 - € IV. Dynamic Hypothesis (CLD) as per previous submissions **(5 Points)**

- € V. Formulation (S&F) of a Simulation Model **15 Points)**
 - € Improves upon S&F model from previous submissions
 - € Includes screenshot of updated simulation model
 - € Includes tables as required
- € VI. Model Testing **(15 Points)**
 - € Includes, at a minimum, tests for dimensional consistency, extreme conditions, and sensitivity analysis with output graphs where appropriate. Include the “so-what” of the testing.
- € VII. Policy Design and Evaluation **(15 Points)**
 - € Includes an explanation and depiction of changes to the model that have been made, and recommended future changes based on this research.
- € VIII. Conclusion **(5 Points)**

Outstanding Student Work

A System Dynamics Analysis of Enlisted Retention Policies and Their Impact on Army Readiness

Caden Brungard, Geneva Cesoni, Kavon Pointer, and Kathryn Wegler¹

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Author Note: The authors of this report are cadets at the United States Military Academy completing a system analysis of enlisted retention policies and their impact on Army readiness for a course with the Department of Systems Engineering. Through this system analysis we will be able to better understand and combat retention issues if they rise again in the future. The views expressed herein are those of the authors and do not reflect the position of the United States Military Academy, the Department of the Army, or the Department of Defense.

Abstract: This project develops a system dynamics model to analyze the factors influencing enlisted Soldier retention in the U.S. Army and to assess potential policy interventions. Focusing on the E1–E6 population, the model simulates promotion flows and separation behavior over a 30-year period using 3-month time steps. Model validity is strengthened through sensitivity testing and extreme-conditions analysis, confirming realistic and stable behavior. A policy increasing Off-Duty Hours per Soldier is formulated and tested, but results show minimal impact on long-term retention, indicating that this variable is a low-leverage point. The evaluation suggests that structural factors such as contract length and promotion timing offer greater potential for meaningful change. Overall, the model reveals the feedback loops driving retention outcomes and demonstrates how system dynamics can inform more effective policy design.

Keywords: Retention, Simulation Modeling, Enlisted

1. Introduction

Retention shortfalls among active-duty enlisted infantry Soldiers in ranks E1–E6 pose a significant risk to overall unit readiness. Enlisted Soldiers and junior Non-Commissioned Officers (NCOs) experience different drivers of attrition, stemming from multiple factors such as incentives, family demands, career progression, and civilian employment opportunities. However, current retention efforts often rely on uniform policies that fail to address the unique needs of different groups within the force. The limited effectiveness of these blanket solutions is reflected in rising attrition rates across the enlisted population. If these trends persist, remaining Soldiers will be required to pick up additional workload, leading to fatigue, declining morale, and an increased likelihood of errors during training and operations. These compounding effects ultimately degrade readiness and mission effectiveness.

To address this challenge, this project employs system dynamics modeling to analyze the causal structure of Army retention, with a specific focus on the enlisted cohort. By capturing interrelationships among factors such as compensation, leadership, family stability, and career opportunity, the model seeks to identify leverage points for more targeted and effective retention policies. Understanding these dynamics can help the Army retain its most experienced Soldiers, sustain readiness, and ensure mission success.

2. Literature Review

System dynamics provides an analytical framework for studying and managing complex systems characterized by feedback, delay, and accumulation. Through tools like causal loop diagrams and stock and flow models, system dynamics explain how interconnected variables evolve over time and how policy decisions can produce long term, sometimes unintended, consequences. John Sternman describes the System Dynamics Modeling Process (SDMP) in his book, *Business Dynamics: Thinking and Modeling for a Complex World*. The SDMP consists of five steps: Problem Articulation, Dynamic Hypothesis, Formulation, Testing, and Policy Formulation & Evaluation. Problem articulation is the process of defining the problem to be addressed and establishing its dynamic context. Dynamic hypothesis is a working theory that explains the causes of the problem's behavior over time through feedback structure. Formulation is the translation of the dynamic

hypothesis into a formal stock-and-flow- simulation model. Testing is the process of verifying and validating the model to ensure it is structurally sound and behaviorally credible. Policy Formulation and Evaluation is the stage where alternative policies or strategies are developed, simulated, and compared using the validated model. This process can be used to model real-world systems and problems, test and analyze present and new policies, and create new policies for evaluation.

Among the scholarly works reviewed, the *Dynamic Retention Model* by Asch and Mattock (2023) offers the most comprehensive approach to modeling these dynamics with a focus on compensation. Their model demonstrates how compensation, incentive pay, and retirement benefits influence Soldier behavior and long-term retention. It frames each Soldier's decision to stay or separate as part of a feedback system shaped by pay satisfaction, promotion opportunity, and external job alternatives. For example, when compensation or bonuses increase, retention rates rise; however, the model also highlights secondary effects such as slower career advancement and higher personnel costs, which can reduce satisfaction and organizational efficiency. These insights enable Army organizations to test different compensation designs before implementation to improve both cost-effectiveness and personnel management. Understanding Army retention requires analyzing how multiple variables interact and influence soldiers on staying in or leaving. These characteristics present system dynamics for modeling Army retention, where important factors like pay, leadership, and soldier workload interact dynamically to influence readiness.

Any measurable variables that are most predictive of attrition across service branches presents important factors (Marrone, 2020). In another scholarly article Marrone has a similar mindset by representing a 36-month attrition study used by the DOD and Bureau of Labor Statistics data that track recruits over time. This focuses on economic variables, and demographic factors that include race, gender, marital status, education, and citizenship. More variables tracked are performance scores like the ASVAB and AFQT, medical data, and contract variables that include paygrade, type of occupation, and time of service. Marrone findings of attrition are not shown by just one thing but creates interaction of personal characteristics, and policy incentives. These variables provide valuable inputs for system dynamics models that simulate retention behavior over time.

Further research strengthens the theoretical foundation for applying system dynamics to personnel management. Zanker (2021) emphasizes the value of representing both qualitative and quantitative relationships to capture how organizational systems evolve. Homer (2006) explains how "policy resistance" arises when feedback structures produce unintended consequences, causing organizations to act counter to their goals. In the context of Army retention, this insight helps leaders anticipate how well-intentioned policies—such as bonus incentives—might interact with family or morale issues to create new attrition pressures. Finally, Schoenenberger (2021) expands this logic through *Structural Analysis Methods (SAM)*, which identify leverage points and structural weaknesses within complex systems. When applied to Army retention, SAM can highlight variables such as family support programs, educational opportunities, and deployment frequency that produce the greatest improvements in readiness per policy change.

Collectively, these scholarly sources provide a structured foundation for modeling Army retention. They underscore that retention outcomes are not linear responses to single policies, but the emergent result of interacting feedback loops involving individual, organizational, and policy factors.

3. Problem Articulation

3.1 Problem Statement

The Army struggles to meet recruitment and recruitment goals, creating risks to force readiness, leadership continuity, and mission effectiveness. Enlisted soldiers and officers face different drivers of attrition, ranging from career progression and incentives to family demands and civilian opportunities, but current policies often apply uniform solutions. The project will use system dynamics modeling to analyze the causal structure of Army retention, with emphasis on the enlisted cohort. The model will define key variables, feedback loops, and system boundaries that explain separation decisions, evaluate where existing policies succeed or fail, and test policy interventions to improve force sustainment.

3.2 System Boundary and Time Horizon

For this project we are defining a narrow system boundary, so we can focus on and analyze the effects of specific variables. This model analyzes Army retention dynamics for active-duty infantry soldiers, specifically examining enlisted ranks E1-E6. We chose these ranks because data shows that a majority of voluntary separations occur in the first 6-10 years of service. Once service members reach senior NCO (E7+) retention stabilizes (U.S. Army Talent Management Task Force) (Asch, Heaton, and Martorell). The focus is solely on retention decisions rather than recruitment and on voluntary separations rather than involuntary. Therefore, variables related to accessions and recruiting pipelines are excluded.

This model uses a 30-year time horizon with a time step of 3 months. A 30-year horizon captures multiple career cycles of enlisted Soldiers. This allows us to observe long-term trends in retention and their impact on unit readiness. Using a 3-month time step allows us to see short-term dynamics like retention decisions, promotion gates, and separations. Outcomes are measured through retention rates and changes in enlisted population. Exogenous factors, such as the civilian job market and broader geopolitical events, are outside the system boundary rather than modeled in detail. Out of scope are non-infantry MOSs and Reserve/Guard components.

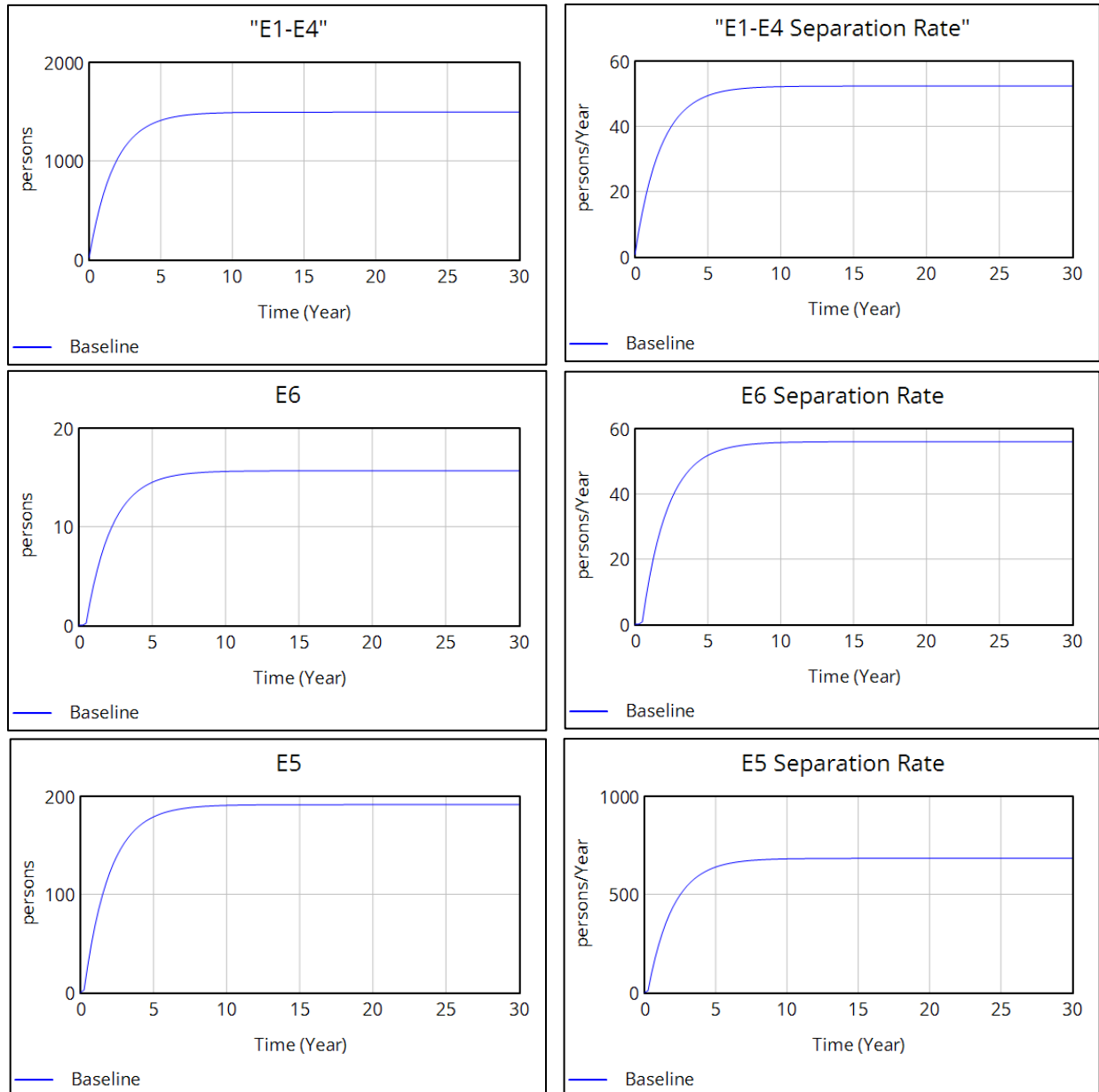
3.3 Key Variables

Table 1. Key Variables and Descriptions

Variable	Variable Description	Units
Enlisted Population	Current number of enlisted soldiers (E1-E6); primary variable that is being influenced by various independent and auxiliary variables.	persons
Enlisted Retention	Number of enlisted soldiers retained within the force each month; reflects the inflow sustaining the overall enlisted population.	Persons/month
Enlisted Retention Rate	Essential to explain % of enlisted personnel who stay in	Persons/month
Workload per Soldier	Demonstrates the operational strain on an individual from their workload; when manpower decreases daily workload will increase	Hours/(persons*week)
Days Off per Soldier	Average number of rest or leave days available to an individual soldier; decreases as workload and operational tempo increase.	Days/(persons*month)
Off Duty Hours per Soldier	Average personal time available to a soldier outside of duty hours; a measure of work-life balance affecting morale and retention.	Hours/(persons*month)
Mission Success Rate	Proportion of missions completed successfully; influenced by readiness, manpower, and soldier fatigue.	Missions/month
Missions Completed	Total number of missions executed by available units within a given time frame; used to measure overall operational productivity.	Missions/month
Squads Available to Complete Training	Total number of trained squads ready to support mission requirements; declines as separations increase or retention falls.	persons
Enlisted Separations	Voluntary/involuntary; we aim to reduce this variable with policies. Essential outflow variable.	Persons/month
Enlisted Separation Rate	Percentage of enlisted soldiers leaving the service; influenced by satisfaction, workload, and external opportunities.	Persons/month
Overtime Hours per Soldier	Additional working hours beyond standard duty time; increases when manpower is low and mission requirements remain high.	Hours/(persons*month)

3.4 Reference Modes

The behaviors of the key variables can be modeled with reference modes. Reference modes are graphs that show how a key variable changes over time, representing the historical trend or expected behavior pattern of the variables in this system. The reference modes for the 8 key variables are depicted below.



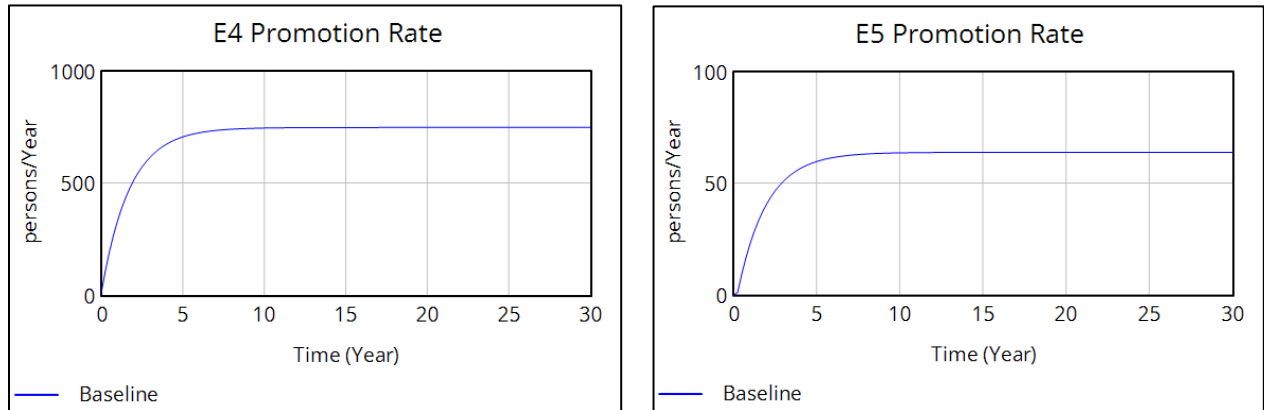


Figure 1. Key Variables Goal-Seeking Reference Modes

The baseline model produces smooth, goal-seeking behavior across all enlisted stocks, separation rates, and promotion flows. Each variable stabilizes over time as inflows and outflows reach equilibrium, demonstrating a structurally consistent and well-behaved system. The E1–E4 population grows rapidly during the first few years of the simulation as accessions accumulate in the lower enlisted ranks. After this early adjustment, the stock approaches a stable equilibrium once promotion and separation flows balance. The corresponding E1–E4 separation rate mirrors this pattern: it rises sharply at the beginning as the population fills, then levels off at a constant rate once the force stabilizes.

The E6 population follows a similar dynamic but at a much smaller scale. After a brief period of growth driven by promotions from the E5 stock, the E6 population converges on a steady-state value. The E6 separation rate increases quickly during the buildup phase and then remains constant once the rank reaches equilibrium. The E5 population exhibits the same characteristic behavior. It increases steeply early in the simulation as soldiers progress from E4, and then it stabilizes as promotions and separations come into balance. The E5 separation rate rises accordingly, reaching a fixed long-term rate once the population settles. Promotion flows also display stable reference modes. The E4 promotion rate increases rapidly at the start as the lower enlisted stock grows, eventually flattening as the system reaches its long-term structure. Similarly, the E5 promotion rate rises quickly during the initial years, then stabilizes once the E5 population reaches equilibrium. Both promotion rates reflect smooth, non-oscillatory behavior consistent with a well-formulated pipeline model.

Overall, the reference modes demonstrate that the baseline system behaves logically and converges to stable equilibria across all ranks and transition flows. This behavior provides confidence in the model’s internal structure and suitability for subsequent sensitivity testing and policy evaluation.

4. Dynamic Hypothesis

Our dynamic hypothesis is that fluctuations in the Army’s enlisted infantry population are primarily driven by internal feedback between workload, rest, and mission success. Enlisted population decreasing means an increase in Soldiers’ workloads, less off-duty time, and more over time hours. This all raises separation rates. The population declines as separation grows, which reinforces this negative cycle. On the other hand, when workload per Soldier decreases, rest and off-duty hours increase and improve retention rates and mission readiness. When readiness is higher and more missions can be completed, morale and retention are boosted, reinforcing growth. Together, these balancing and reinforcing loops explain the long-term behavior of retention and unit readiness and how they impact the enlisted population.

4.1 Causal Loop Diagram (CLD)

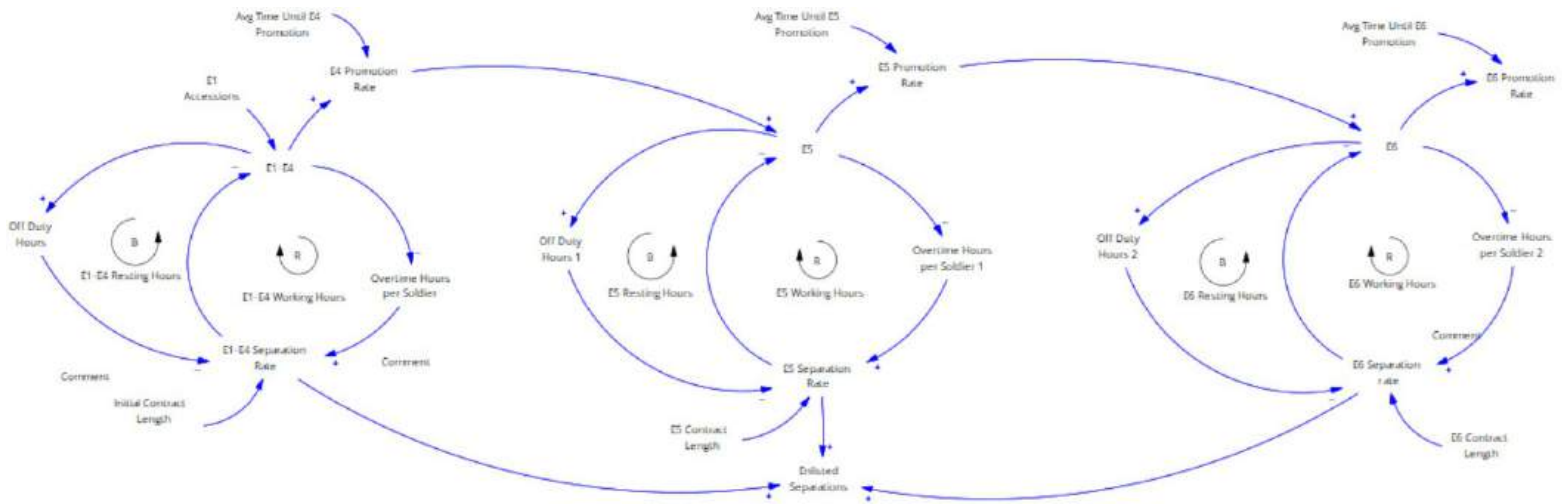


Figure 2. Causal Loop Diagram

4.1.1 Improve on CLD from Part I:

In our original causal loop diagram, several variables were subjective or dimensionless, such as “*Soldier Happiness*,” “*Quality of Life in the Army*,” “*Soldiers’ Work Ethic/Motivation*,” and “*Unit Readiness Level*.” These variables represent important real-world ideas and have strong relationships with each other but were difficult to quantify or assign physical units. This made them unsuitable for a simulation model in Vensim. In our revised CLD, we replaced these abstract concepts with quantifiable, measurable variables. They basically maintain the same or similar causal logic but can now be expressed in real units. We also restructured the loops to ensure that each variable has a clear and physical relationship to the variables it’s connected to, allowing for unit-consistent equations.

Overall, the new CLD is a major improvement because it:

- Converts subjective ideas into quantitative
- Creates unit-based variables suitable for dynamic simulation
- Allows Vensim to pass unit consistency checks
- Clarifies causal pathways by separating retention (reinforcing), separation (balancing), and mission success (reinforcing) dynamics
- Keeps the focus strictly within the defined system boundary

4.1.2 Improve on CLD from Part II:

After Part II of the project, we made some major changes to our system in order to create a functional and effective stock and flow diagram. This means we also have to account for these changes in our CLD and ensure our stock and flow and CLD reflect the same system. The CLD shifts from a single-stock view of the overall enlisted population to a more rank-specific structure. It tracks Soldiers progressing through E1–E4, E5, and E6 instead of just looking at the entire enlisted population. Next, the new CLD will not show retention and separation as broad factors but will separate them by rank and link them to promotion flows, average time in grade, and contract lengths. We also added clearer drivers of separation, such as overtime hours and off-duty hours, and connected them directly to each rank’s separation rate. Overall, the new CLD provides a more specific and realistic depiction of how Soldiers move through the system. How workload-related factors influence separations in a Soldier’s career at different ranks is also highlighted effectively. **The CLD above is updated, complete, and reflects all these changes.**

The table below shows the relationships between variables represented in our causal loop diagram, and the relationships between variables that were introduced to make our stock and flow diagram.

Table 2. Causal Relationships

Independent Variable	Dependent Variable	Polarity	Notes
E1 Accessions	E1-E4	+	Increased accessions expand the size of the E1–E4 population entering the enlisted force.
E1-E4	E4 Promotion Rate	+	Larger E1–E4 cohort increases the pool eligible for promotion to E5.
E4 Promotion Rate	E5	+	Higher promotion rates increase the number of soldiers advancing to E5.
E5	E5 Promotion Rate	+	Greater number of E5s increases the eligible group for promotion to E6.
E5 Promotion Rate	E6	+	Higher E5 promotion rates increase the number advancing to E6.
E6	E6 Promotion Rate	+	Larger E6 population expands the base eligible for further promotion or retention.
E1-E4	E1-E4 Separation Rate	+	A larger E1–E4 population increases the number potentially separating at that level.
E1-E4 Separation Rate	Enlisted Separations	+	Higher separation rates among E1–E4s increase total enlisted separations.
E5	E5 Separation Rate	+	Greater number of E5s increases the potential separations within that rank.
E5 Separation Rate	Enlisted Separations	+	Higher E5 separation rates increase total enlisted separations.
E6	E6 Separation Rate	+	A larger E6 population increases the number separating at that rank.
E6 Separation Rate	Enlisted Separations	+	Higher E6 separation rates increase total enlisted separations.
Overtime Hours per Soldier	E5 Separation Rate	+	More overtime hours raise stress and fatigue, leading to higher separation among E5s.
Overtime Hours per Soldier	E1-E4 Separation Rate	+	Increased workload among junior soldiers contributes to burnout and higher separation.
Off Duty Hours Per Soldier	E1-E4 Separation Rate	-	More off-duty time improves work–life balance, reducing E1–E4 separations.
Off Duty Hours Per Soldier	E5 Separation Rate	-	Increased rest and personal time lower separation likelihood among E5s.
Off Duty Hours Per Soldier	E6 Separation Rate	-	Greater off-duty time improves satisfaction and retention for senior NCOs.
Avg Time until E4 Promotion	E4 Promotion Rate	-	Longer time to reach E4 slows advancement and decreases promotion flow into the E4 population.
Initial Contract Length	E1-E4 Separation Rate	+	Extending the initial contract keeps junior soldiers obligated for more time, reducing early separations.
Avg Time until E5 Promotion	E5 Promotion Rate	-	Delays in reaching E5 can slow promotion flow and increase the chance that soldiers separate before advancing.
E5 Contract Length	E5 Separation Rate	+	Longer contracts for E5s help retain mid-grade NCOs by limiting separation during obligated service.
Avg Time until E6 Promotion	E6 Promotion Rate	-	Increasing the time required to reach E6 reduces advancement rates into the senior NCO tier.
E6 Contract Length	E6 Separation Rate	+	A longer E6 commitment decreases separation likelihood by keeping senior NCOs under obligation.
Overtime Hours per Soldier	E6 Separation Rate	+	Higher overtime demands raise stress and fatigue, making soldiers more likely to separate.

4.2 Describing DH Using CLD

The casual loop diagram is a visual representation of our dynamic hypothesis. Each enlisted population, E1-E4, E5, and E6, all contain a paired set of Resting Hours and Working Hours loops that describe how workload, rest, and separation interact over time. The Resting Hours loop is a balancing loop in each population. It demonstrates the relationship between off-duty hours and separation rate and how their interaction helps stabilize the population. It stabilizes the population because an increase in off-duty hours reduces working hours, which lowers separation rates. The Working Hours loop is a reinforcing loop in each population. It demonstrates the relationship between overtime hours and separation rate and how their interaction decreases the size of the population. It ultimately decreases the population because when higher working hours increase overtime, separation rates rise. These two structures repeat for all three populations we are analyzing for retention. The model captures how similar behavioral mechanism operates at different stages of an enlisted Army career. Together, these reinforcing and balancing loops create a dynamic system in which workload, rest, and separation continuously interact. They shape retention outcomes, which ultimately influences the Army’s overall enlisted population over time.

4.3 Casual Links and Polarities Based on Research

The causal links and polarities in our new CLD reflect findings from Army retention research. In U.S. Army Talent Management Task Force studies by Asch and Mattock (2023), they show how increases in factors such as quality of life, time off, and unit stability improve the fulfillment in soldiers and ultimately retention (+). On the contrary, Marrone (2020) and Homer (2026) demonstrated how higher soldier workload and stress causes higher attrition (-). The positive polarities (+) in our loops represent relationships that have mirrored each other’s behaviors. For example, more off-duty hours lead to more retention, so these variables have a positive polarity because they both have the same behavior of increasing. The negative polarities (-) in our loops represent relationships that have inverse behavior, such as how more workload leads to less rest days. These variables have opposing behavior, making them have negative polarity. All the arrows in our CLD show these relationships. These relationships are between internal factors that drive soldiers’ behavior over time, circling back to our overall analysis of the enlisted population.

5. Formulation

Below is a screenshot of our stock and flow simulation model. It is an aging chain because the stocks (persons) “age” or advance from one stock to the next, governed by inflow and outflow rates that depend on average time spent in each stage. This model simulation captures career progression and models time delays and pipelines effects in enlisted populations and promotion processes.

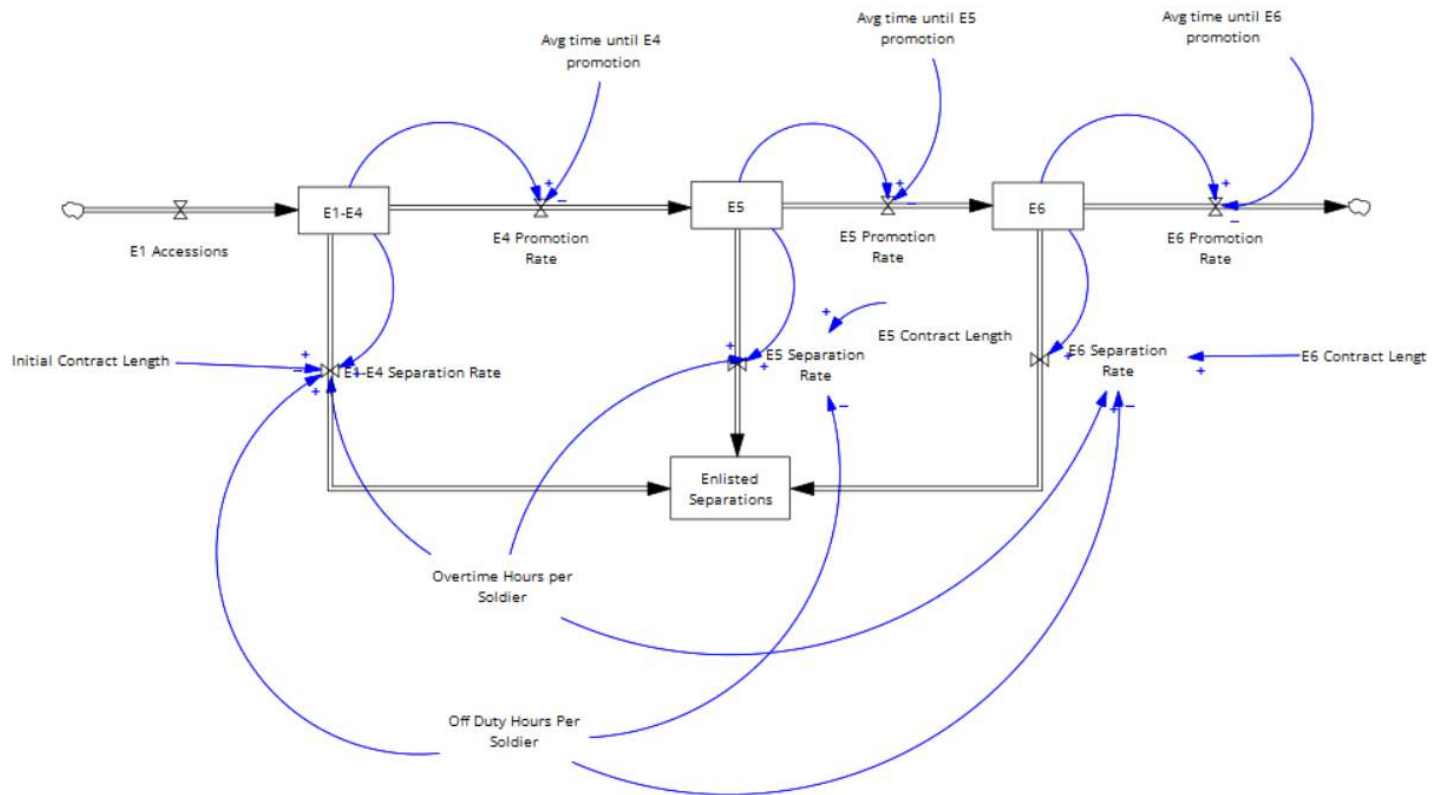


Figure 3. Stock and Flow Diagram

The table below shows key variables and the equations used in the simulation model, as well as the initial value for independent variables.

Table 3. Key Variables and Equations

Variable	Units	Stock or Flow	Initial Value, Equation/Constant Value
E1 Accessions	Persons/Year	Flow	= 11000
E1-E4	Persons	Stock	= 20, INTEG(E1 Accessions-"E1-E4 Separation Rate"-E4 Promotion Rate)
E4 Promotion Rate	Persons/Year	Flow	= "E1-E4"/Avg time until E4 promotion
E1-E4 Separation Rate	Persons/Year	Flow	= ("E1-E4"*(Off Duty Hours Per Soldier/Overtime Hours per Soldier))/Initial Contract Length
Enlisted Separations	Persons	Stock	= INTEG("E1-E4 Separation Rate"+E5 Separation Rate+E6 Separation Rate)
E5	Persons	Stock	= 0, INTEG(E4 Promotion Rate-E5 Promotion Rate-E5 Separation Rate)

E5 Promotion Rate	Persons/Year	Flow	= E5/Avg time until E5 promotion
E5 Separation Rate	Persons/Year	Flow	= (E5*(Off Duty Hours Per Soldier/Overtime Hours per Soldier))/E5 Contract Length
E6	Persons	Stock	= 0, INTEG(E5 Promotion Rate-E6 Promotion Rate-E6 Separation Rate)
E6 Promotion Rate	Persons/Year	Flow	= E6/Avg time until E6 promotion
E6 Separation Rate	Persons/Year	Flow	= (E6*(Off Duty Hours Per Soldier/Overtime Hours per Soldier))/E6 Contract Length
Overtime Hours per Soldier	Hours/(Persons*Month)	Independent Variable	=70
Off Duty Hours per Soldier	Hours/(Persons*Month)	Independent Variable	=500
Initial Contract Length	Year	Independent Variable	=4
Avg Time until E4 Promotion	Year	Independent Variable	=2
E5 Contract Length	Year	Independent Variable	=2
Avg Time until E5 Promotion	Year	Independent Variable	=3
E6 Contract Length	Year	Independent Variable	=2
Avg Time until E6 Promotion	Year	Independent Variable	=2

6. Testing

The following testing was conducted to verify and validate our model:

- Boundary Adequacy: Tests whether the model describes what it was intended for
- Dimensional Consistency: All variables have comparable units so that the model can perform the necessary math to calculate model statistics
- Behavior Reproduction: Shows why a model acts that way it does and whether the behaviors match real world
- Extreme Conditions: Examine each rate equation to determine if the output is feasible at maximum or minimum values
- Sensitivity Analysis: answers how conclusion changes in ways important to your purpose when assumptions are varied over the plausible range of uncertainty

We tested boundary adequacy by referring to our problem articulation and checking that our model addresses the problem statement we outlined and remains within our system boundary. We tested dimensional consistency through our Venism model by using the Units Check function within the software.

6.1 Behavior Reproduction

We tested behavior reproduction by analyzing real army patterns and the behavior the model outputs. The behaviors we analyzed in the table below. Overall, the model successfully reproduced the key qualitative patterns of the real Army enlisted force, demonstrating that it passes the Behavior Reproduction Test.

Table. 4 Behavior Reproduction Testing

Real Army Pattern	Model Behavior	Pass?
Pyramidal rank structure ($E1-E4 > E5 > E6$)	Model stabilizes with correct pyramid	Yes
Sequential Promotions ($E1-E4 \rightarrow E5 \rightarrow E6$)	Aging-chain flows & delays reproduced	Yes
Realistic time to promotion	Delays calibrated to match Army averages	Yes
Higher Junior Separations	$E1-E4 > E5 > E6$ attrition	Yes
Stable long-term force	Stock converges without blow-ups	Yes

6.2 Extreme Conditions

To assess the structural robustness of our model, we conducted extreme conditions test on two key exogenous variables: E1 Accessions and Overtime Hours per Soldier. These tests ensure that the model behaves logically even under unrealistic or boundary values, which is a critical step in model validation.

6.2.1 E1 Accessions Extreme Test

We simulated the model under three accession levels:

- High: 8,000 persons/year
- Baseline: 800 persons/year
- Low: 0 persons/year

The results behaved as expected. Under zero accession, the E1-E4 and E6 populations steadily declined toward zero, demonstrating appropriate model sensitivity to recruitment inflows. Under high accessions, both populations increased sharply and stabilized at a higher equilibrium, confirming that the stock structure and promotion flows respond proportionally to changes in recruitment. The baseline trajectory fell between the two extremes, indicating stable and intuitive model performance. Figures 4 and 5 demonstrate the model outputs for this test.

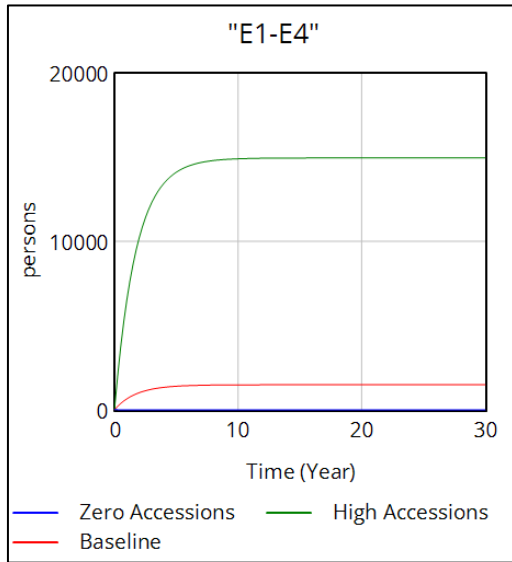


Figure 4. E1-E4 Population Output

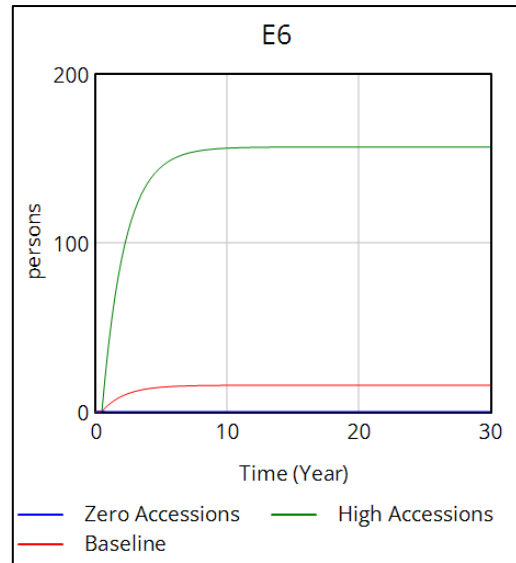


Figure 5. E6 Population Output

6.2.2 Overtime Hours per Soldier Extreme Test

Overtime hours influence soldier fatigue and, in our model, feed into the E1-E4 Separation Rate. We tested three extreme values:

- High: 1,000 hours/month
- Baseline: 70 hours/month
- Low: 0 hours/month

The model responded logically. At extremely high overtime, separation rates spiked dramatically, reflecting unrealistic workload conditions that would cause severe burnout. Under zero overtime, separation dropped substantially, representing idealized work-life balance conditions unlikely to exist in practice. The baseline level sat between these extremes and produced realistic behavior. These results confirm that the separation structure responds sensibly to stress-related workload inputs. Figure 6 demonstrates the model output for this test.

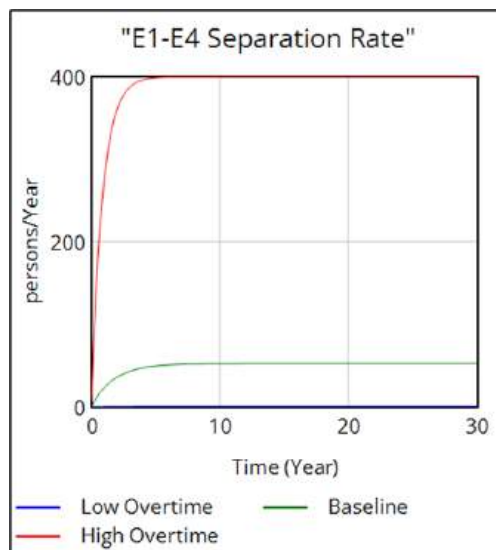


Figure 6. E1-E4 Separation Rate varying Overtime Hours

Across both tests, the model remained stable and produced behavior consistent with real-world expectations. The direction and magnitude of responses were appropriate, and no structural instabilities or implausible behaviors appeared. This increases confidence that the model is sound, internally consistent, and capable of supporting policy experimentation.

6.3 Sensitivity Analysis

Sensitivity analysis was conducted to evaluate how responsive the model is to uncertainty in key parameters and to ensure that small parameter changes produce reasonable and proportional shifts in system behavior. Two variables were tested: Average Time Until E5 Promotion and Enlisted Accessions. For each test, we simulated a high case, baseline case, and low case, then compared the resulting population trajectories for ranks E1–E4, E5, E6, and overall enlisted separations.

6.3.1 Sensitivity Test 1: Average Time Until E5 Promotion

We first tested how changed in the time required for an E4 soldier to be promoted to E5 influenced the population dynamics. The values tested were:

- High: 7 years
- Baseline: 5.5 years
- Low: 4 years

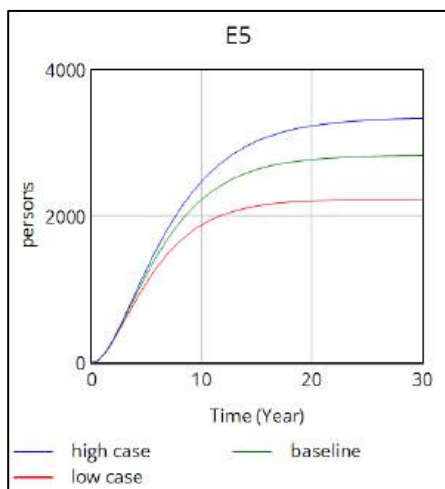


Figure 7. E5 Output

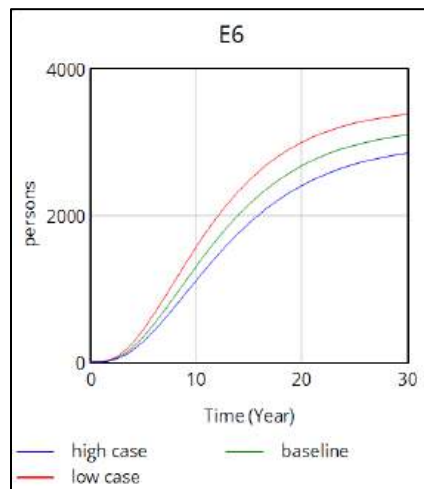


Figure 8. E6 Output

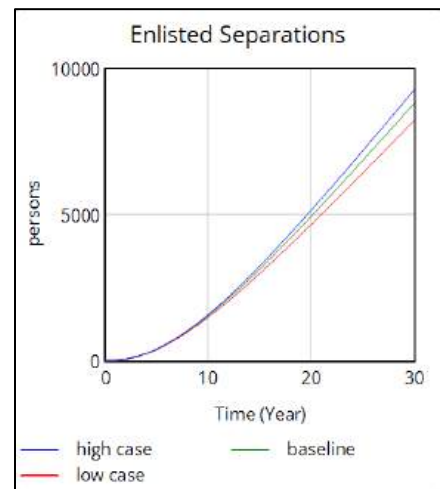


Figure 9. Enlisted Separations Output

The model responded logically and proportionally to these variations. When promotion time increased to seven years, the E5 and E6 populations grew more slowly and reached lower equilibrium levels due to the delayed advancement flow. Shortening promotion time to four years accelerated movement through the system, resulting in higher populations in both ranks. Enlisted separations shifted accordingly, with faster promotion slightly increasing long-term separations because more soldiers remained in the force long enough to reach their separation point. Overall, the model behaved smoothly and logically across all cases.

6.3.2 Sensitivity Test 2: Enlisted Separations

The second sensitivity analysis tested the model's responsiveness to changes in E1 accession rates. The values examined were:

- High: 1,200 persons/year
- Baseline: 800 persons/year
- Low: 500 persons/year

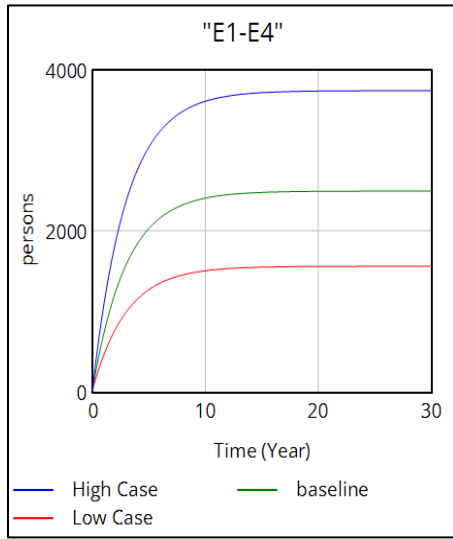


Figure 10. E1-E4 Output

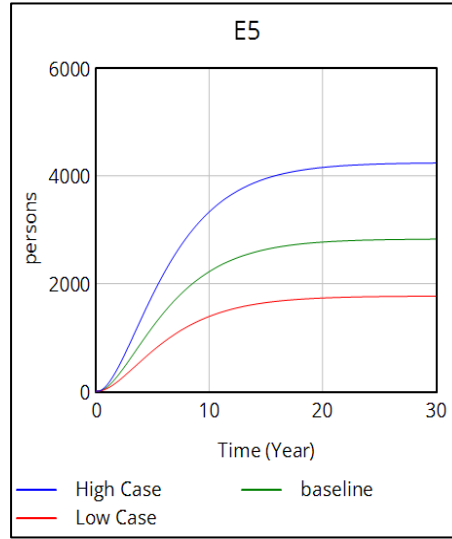


Figure 11. E5 Output

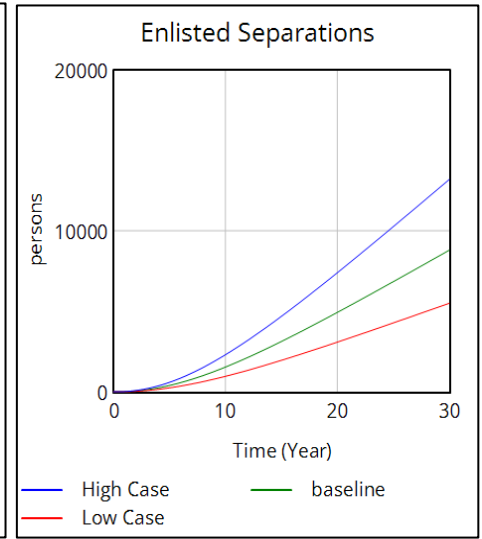


Figure 12. Enlisted Separations Output

Variations in accession levels produced proportional and intuitive changes in system behavior. Higher accession rates led to noticeably larger populations in E1–E4, E5, and E6, while lower accessions resulted in smaller force levels throughout the model. Enlisted separations followed the same pattern, increasing when accessions were high and decreasing when accessions were low. These results confirm that the model is sensitive in appropriate ways to changes in recruitment inputs and maintains stable, realistic behavior across the tested ranges.

7. Policy Formulation & Evaluation

A key purpose of system dynamics modeling is to test potential policies that can alter problematic system behavior, which in this case is declining long-term Army enlisted retention. After looking at our model’s baseline structure, we identified how soldier fatigue and reduced personal time is a contributing factor in soldiers separating. We determined that the exogenous variable “Off-Duty Hours per Soldier” in our model influences this factor and ultimately the fraction of the Soldiers Retained, so we chose it as the focus of our policy intervention. The policy increased “Off-Duty Hours per Soldier” from 500 to 600 hours per month. This was modeled as a direct parameter adjustment. Increasing this variable allows soldiers to rest, recover, and manage personal responsibilities. Conceptually, these factors will increase satisfaction in Soldiers and retention. To implement this policy, we modified the value in our stock and flow model and re-ran the simulation. Below are the updated outputs that show the policy’s effect on each enlisted population:

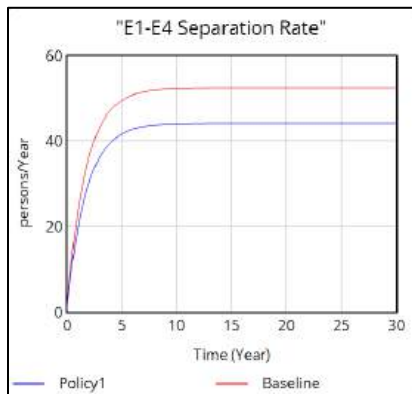


Figure 13. Policy Output 1

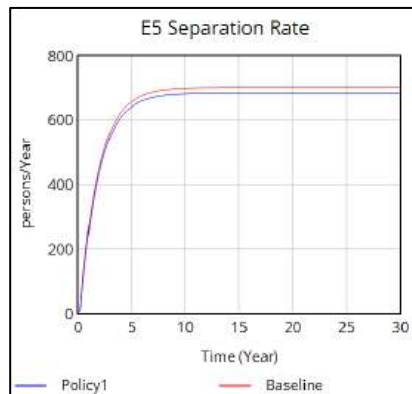


Figure 14. Policy Output 2

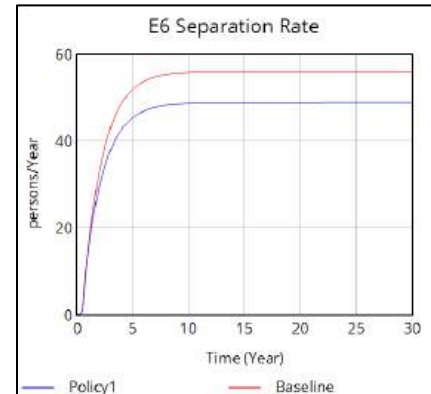


Figure 15. Policy Output 3

It's also important to note that our policy assumes that increasing off-duty hours directly improves retention by making soldiers more satisfied, but this is not universally true. Many soldiers separate for reasons unrelated to workload or personal time, meaning the assumption may not fully capture real-world behavior. The results of the simulation, shown above, after the implementation of our policy showed only minimal change in the system's behavior. This means that increasing the number of hours Soldiers have off did not improve retention enough and its effect was not strong enough to change the long-term pattern behavior in a meaningful way. The trajectories for each population were almost the same as the baseline, and the system returned to an equilibrium very close to its original values. This outcome tells us that while the variable we chose may affect the well-being of Soldiers, it cannot be identified as a high-leverage point in the structure of our model. Overall, adjusting off-duty time did not reshape the feedback loops in our model enough to produce a different retention outcome worthwhile. The dominant loops driving reenlistment and separation are more strongly influenced by other structural factors such as contract cycles, promotion timing, and financial incentives.

Due to our original policy having no meaningful change, we began to look in another, more promising, direction. We decided to look at the contract length variables for each population and determined they may have stronger leverage. The Army uses contracts to structure the timing of re-enlistment decisions and promotion incentives. Contract length is closely related to the following important factors:

- How soon Soldiers become eligible for promotion
- How quickly they gain access to higher pay grades
- Soldier's perception of whether the reward is "close enough" to stay

Longer contracts push financial incentives farther away in time, which reduces motivation in Soldiers and can contribute to separation. On the other hand, shorter contract lengths do the opposite and bring promotions and pay increases sooner, which increases the salience of financial incentives and encourages Soldiers to remain in service. Adjusting contract length would change the timing of outflows in our model as well as modify the re-enlistment decision points. Retention would also be influenced by the more powerful reinforcing loops. These dynamics relate more to the structural foundation of the system, which is why making a new contract length policy is more likely to produce a stronger behavioral change than the off-duty hours policy.

Using the sensitivity analysis from the testing phase, none of the changes in key model assumptions were strong enough to reverse or significantly alter the impact of our proposed policy. In every sensitivity case, whether promotion times were faster or slower, or accessions were higher or lower, the off-duty-hours policy continued to have only a minimal effect on retention. This confirms that the limited impact of our policy is not due to unstable or fragile assumptions but rather because Off-Duty Hours per Soldier is a low-leverage variable in the model. The sensitivity results therefore strengthen the conclusion that a different structural lever, such as contract length, would be more effective for future policy design.

8. Conclusion

We used system dynamics modeling throughout this project to better understand the long-term drivers of Army enlisted retention and how these underlying dynamics influence unit readiness. Our analysis showed that retention challenges do not stem from a single cause but emerge from interacting feedback loops involving workload, promotion timing, separation rates, and other reinforcing and balancing structures. As workload increases, off-duty time and morale decline, causing separation rates to rise and shrinking the enlisted population, ultimately reinforcing a negative cycle. Conversely, improvements in rest, manageable work hours, and stable promotion flows strengthen positive loops that support retention and readiness.

System dynamics tools such as causal loop diagrams, aging chains, reference modes, and stock-and-flow modeling allowed us to identify these patterns far more effectively than traditional analysis. Building and refining the model over time helped highlight true leverage points, such as reducing excessive overtime, improving early-career work-life balance, and shaping promotion timelines to reduce attrition in critical ranks. Sensitivity testing and extreme-condition analysis confirmed that the model behaves reasonably and captures realistic Army dynamics, increasing our confidence in its ability to support policy evaluation. While the model provides a strong foundation, future improvements could enhance realism by incorporating individual soldier behaviors, deployment cycles, transitions between components, and historical retention data. These additions would strengthen the model's ability to evaluate a wider range of policies and generate more precise insights.

Overall, this project demonstrates how system dynamics can reveal the underlying structures driving retention challenges and help leaders design more effective policies that keep cohesive units together and support long-term mission readiness.

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Fair-to-Good Student Work

UNITED STATES MILITARY ACADEMY

FINAL REPORT

SM484: SYSTEM DYNAMICS SIMULATION

SECTION H1

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10 DECEMBER 2025

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Recruit Quality Effect on the Retention of Infantry E5s

Adriana Frias, Landin Jacon-Duffy, Henry Appleton, and Kendrick Kirby

Author Note: The following products presented in this report are the joint ideas and products from the authors listed above through the assistance of the references listed below. The views expressed herein are those of the authors and do not reflect the position of the United States Military Academy, the Department of the Army, or the Department of Defense.

Abstract: In this project we analyzed how the decrease in recruiting standards in recent years has affected the retention of experienced enlisted soldiers (E5's) in the Army. We found that E5's are leaving the Army at a high rate, which causes an increased number of inexperienced soldiers who are not completing their initial contract. For example, by looking at variables like Recruit Quality (ASVAB Scores), AFQT, TAPAS, and Recruiting Goals, these variables gave us a look into the quantitative relationship with retention of soldiers in the Army.

Keywords: Recruit Quality, Recruiting Goals

1. Introduction

Non-Commissioned Officers (Specifically E5s) have been the backbone of the Army since its conception. Retaining experienced, battle-hardened Non-Commissioned Officers (NCOs) is a present problem within our military. NCOs are expected to make important decisions and lead their soldiers during training exercises, combat, or even garrison. This job requires high levels of competence, effort, and the ability to persevere through difficult situations. Unfortunately, to meet recruiting quotas, standards among recruits have dropped, and this has had a negative effect on the NCO Corps. The institutional knowledge that these NCO's have built in their careers is paramount to the Army's lethality. Losing experienced soldiers will negatively impact unit readiness, training effectiveness, and combat capability.

2. Literature Review

When we look at the complex relationship between recruiting standards and E5 retention, we use the Systems Dynamics methodology. Systems Dynamics, developed by Jay Forrester, is a framework for understanding and modeling the behavior of systems over time. This methodology is based on the fact that a system's behavior is mostly driven by its internal feedback structure, which includes accumulations (stocks), rates of change (flows), and delays (Forrester, 1961).

The Army's challenge of recruiting and retention is a very complex system, with multiple variables. George P. Richardson traced the concept of feedback thought, stating that while these modeling tools came from engineering, the core idea of causal loops has always been a concept in social sciences (Richardson, 1991). The modeling approach created by John D. Sterman gives a good approach to building models that help us understand complex systems like the Army. These models serve as "management flight simulators," that allow policy makers to see the long-term, unintended consequences of their decisions (Sterman, 2000).

Donella H. Meadows wrote on the idea of "thinking in systems," which can give us a good technique to not look at the world as full of isolated events, but as a web of interconnections where feedback loops, stocks, and flows change behavior (Meadows, 2008).

The main tool we used in this paper is the causal loop diagram (CLD). A CLD is a visualization of the feedback structure of a system. It shows the key feedback loops in a system, shown as Reinforcing Loops, which increase change and create cycles of growth, and Balancing Loops, which are goal seeking and stabilizing. They counteract change and move towards equilibrium or a goal. In this paper, by applying these tools, we can create a dynamic hypothesis to explain how the decline in E5 retention may be a consequence of the Army's falling recruiting goals, as well as looking at tests like the AFQT and TAPAS (Tailored Army Personality Assessment System) to gauge beforehand the retainment of soldiers.

3. Problem Articulation

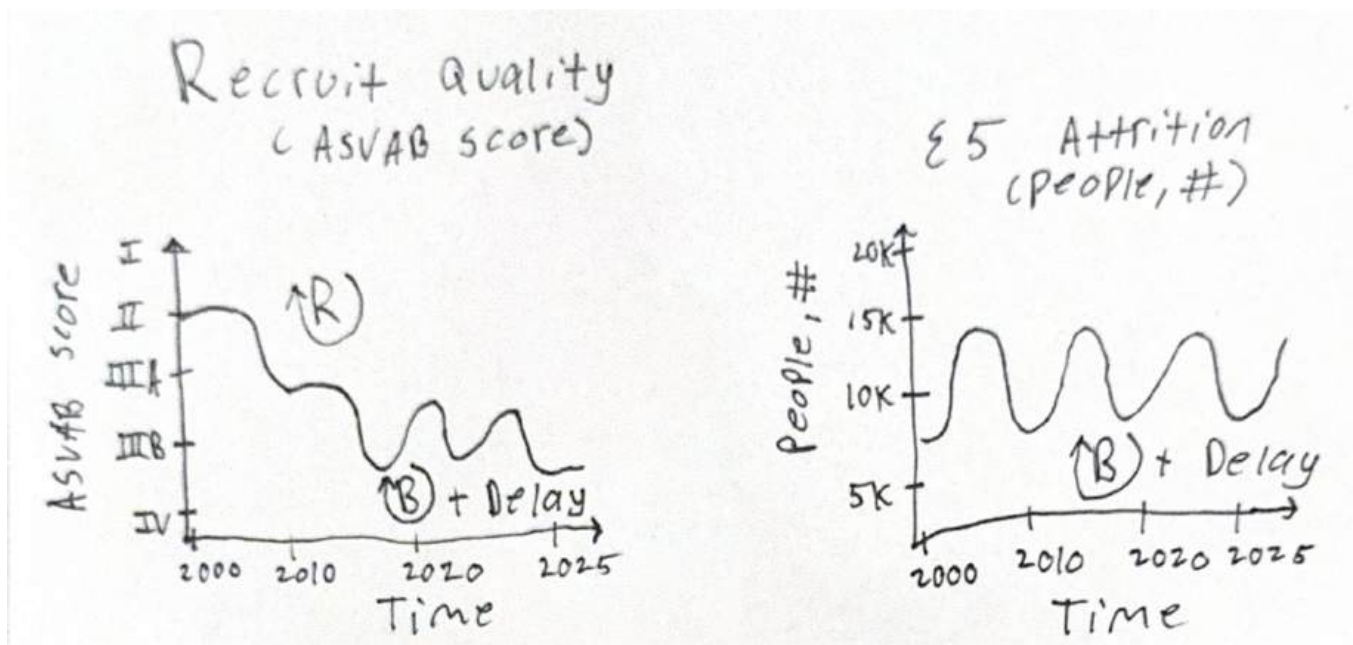
Problem Statement:

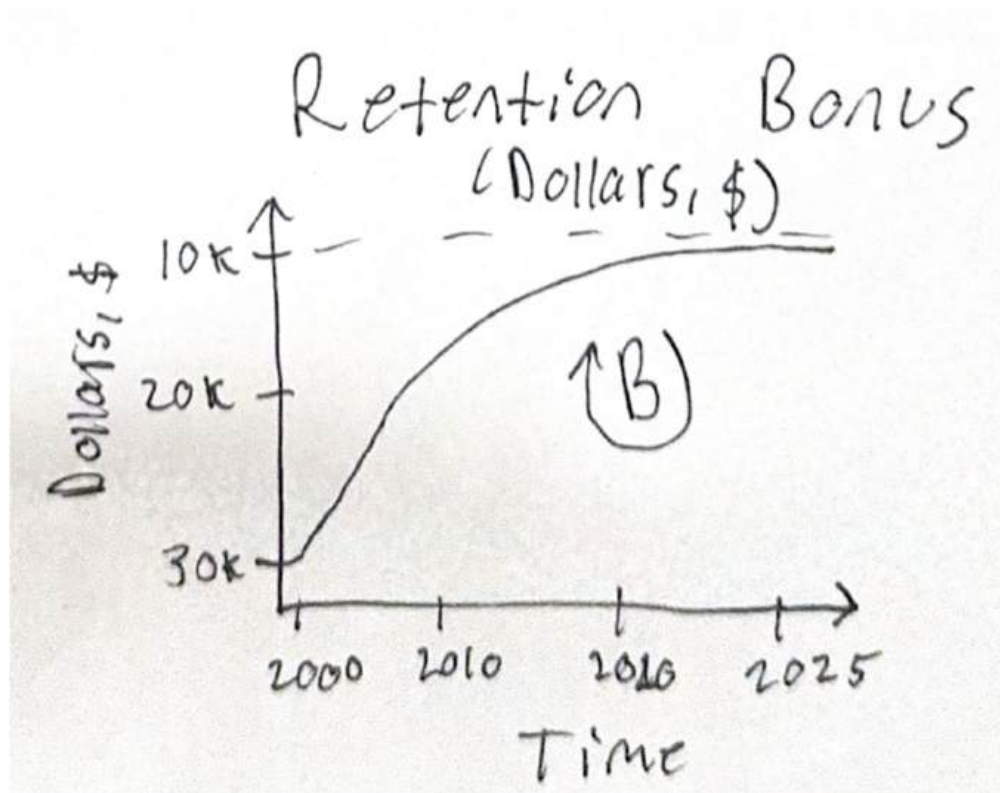
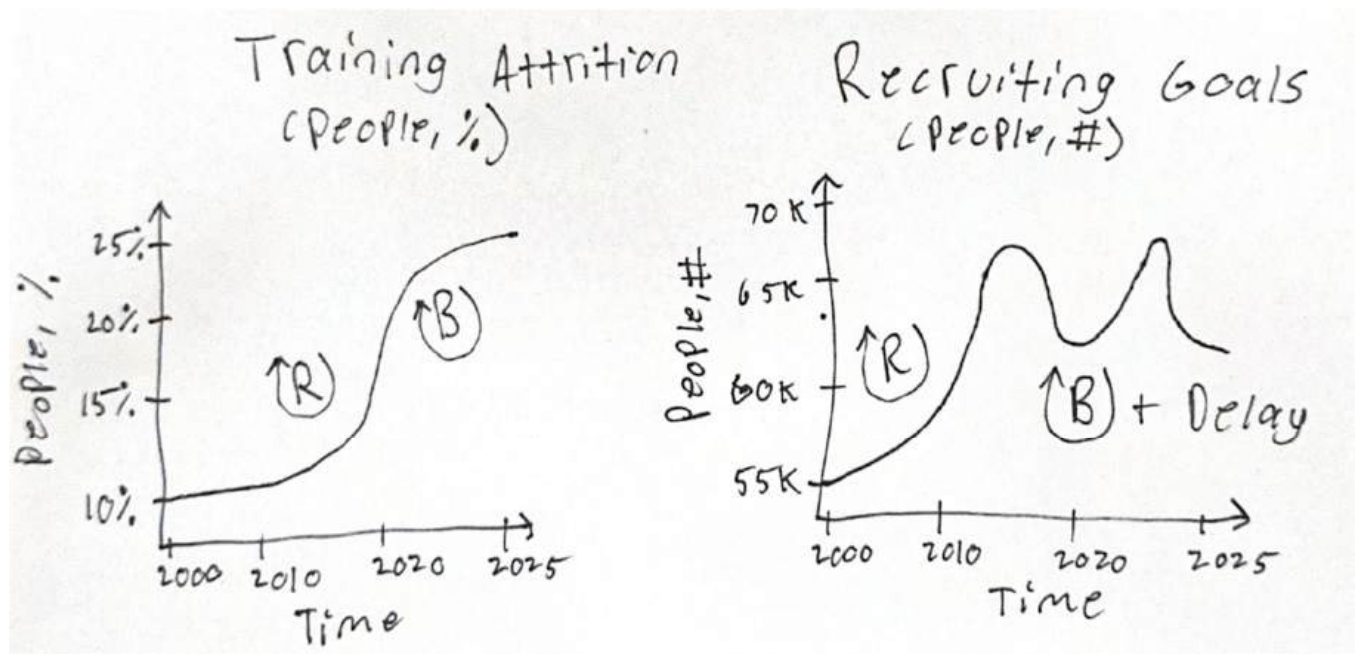
The United States Army often struggles to balance the number and quality of soldiers, as it is an all-volunteer force. Army Recruiters specialize in bringing in recruits, but more Americans each year do not meet the basic intellectual and physical requirements to join the Army. Therefore, recruiting goals in the Army are being met at the expense of decreasing recruiting standards, such as granting waivers for weak ASVAB scores, medical records, and criminal history. This, along with other Army-specific variables, is leading to a decrease in the number of Infantry Sergeants (E5), which means there are fewer experienced NCOs in the Army because soldiers are not completing their initial contracts. If the Army cannot retain its NCOs, the overall quality and lethality of the US Army itself may be jeopardized.

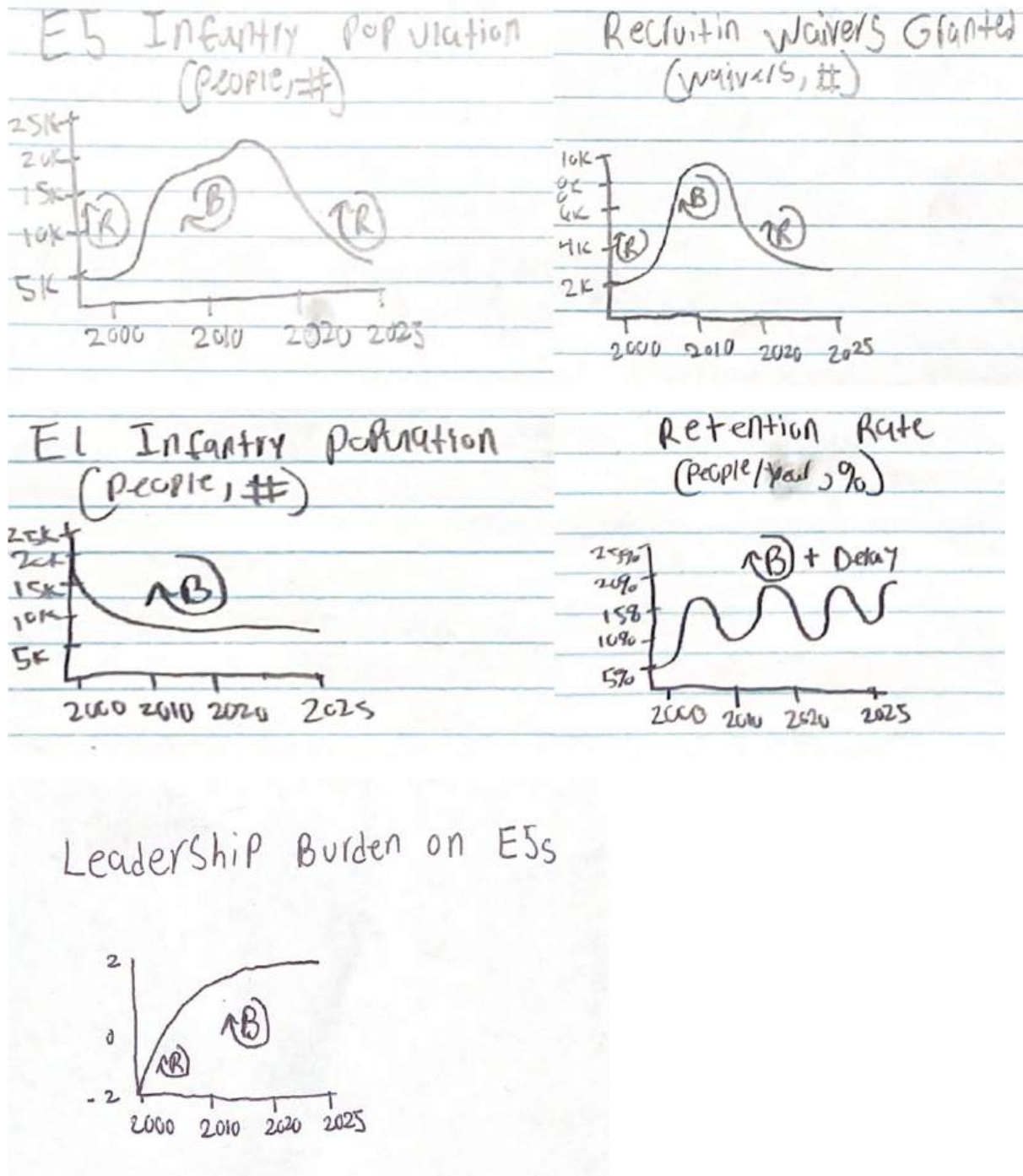
Time Horizon:

25 years (2000-2025). We chose this time horizon because most of the reliable data that is available is focused on the 21st century. Furthermore, we are primarily focused on how the current situation in the army is, and how the current and relevant variables are affecting the number of E5 infantry soldiers in the Army.

Key Variables with Reference Modes:







1. E5 Infantry Population (People, #)

- a. The population of infantry NCO's is decreasing over time as they begin to leave the army. This causes the E5 Infantry population to take an overshoot and collapse shape as the E5 Infantry population increases, but then decreases caused by several variables leading to NCO's leaving the Army.

- b. Endogenous- as the E5 Infantry population increases, public perception increases leading to accessions increasing leading to recruiting goals increasing
- 2. E1 Infantry Population (People, #)
 - a. The E1 Infantry population is decreasing due to lack of proper mentorship and leadership from senior NCOs due to lack of NCO's and increased NCO burden on the NCO's that remain.
 - b. Endogenous – as E1 population increases, the E2 population increases, leading to the E3 population increasing, leading to E4 population increasing, leading to a higher rate of promotion to E5 causing a goal seeking behavior.
- 3. Recruiting Wavers Granted (Waivers, #)
 - a. The increased number of recruiting wavers granted has caused the recruit quality to decline due to lowering the standards for soldiers in order to compensate for a lack of retention.
 - b. Endogenous – as the number of waivers granted increases, the recruit quality decreases, leading to a higher rate of E5 promotions. This leads to an overshoot and collapse shape as the number of waivers granted increases the number of soldiers but leads to a higher attrition due to soldiers with waivers not being fit to stay in the Army.
- 4. Retention Rate (People/Year, %)
 - a. The retention rate of E5's is decreasing as the number of E5's decreases leading to a heavier burden on E5's causing them to not want to stay in the Army. This causes an oscillation behavior.
 - b. Endogenous – as the retention rate decreases, recruiting goals increase, which leads to an increased number of waivers granted which then leads to a decrease in recruit quality.
- 5. Retention Bonus (Dollars, \$)
 - a. 11X soldiers (enlisted infantrymen) are eligible to 0-10k bonuses with 4 years of service (US Army Human Resources Command, 2025). This is one of the key variables to explaining the lack of retention of E5 Infantry soldiers. Infantry soldiers are eligible for one of the lowest reenlistment bonuses compared to other MOSs. The reference mode shows Goal Seeking behavior because the Army is trying to minimize its bonus pay.
 - b. Endogenous – Although the reference node is goal seeking, retention bonuses can be impacted by recruiting goals. If the army is short on infantry soldiers, they may raise the bonus in order to increase the retention rate.
- 6. Recruit Quality (ASVAB Score)
 - a. The infantry has a higher percentage of category IIIB and IV (the two lowest scores on the ASVAB), sergeants (approximately 37 percent of the total) and staff sergeants (approximately 41 percent of the total) than the overall force (Council, 2025). As less young males enlist and more waivers are granted, the quality of the recruit decreases. Likely, as the pressure to meet recruiting goals becomes more throughout the 21st century, the quality of recruits is continuing to decrease forming Growth with Overshoot behavior.
 - b. Endogenous – Recruit quality is caused by higher recruiting goals which lead to more waivers, which in return lead recruits to be lower in quality. This variable has causal relationships within our loop.
- 7. E5 Attrition (People, #)
 - a. The U.S. Army's attrition rate is highest between the enlistment periods of thirty-six months (three years) through seventy-two months (six years), (Osborne, 2021). In 2019 the Army had 19,820 11B Infantrymen

on active duty (15-17% of the total force), (Elie, 2019). The reference node shows oscillating behavior due to the factors of time.

- b. Endogenous – One of the underlying variables affecting E5 attrition is the leadership burden as an E5. However, other sources affecting E5 attrition may be reenlistment bonus's and moving out of the barracks. Regardless, E5 attrition is within our loop and is affected by other variables and has a key causal relationship with the E5 population.

8. Recruiting Goals (People, #)

- a. The recruiting goals for regular army soldiers show a Growth with Overshoot behavior throughout the past five years, varying from roughly 50,000 to 65,000 (US Army Recruiting Command, 2024). Some years the army needs more soldiers due to not meeting last year's goal, so the goal increases. If the goal is met, then the goal will be less the following year.
- b. Endogenous – A lower retention rate will increase the recruiting goals, which has a causal relationship with waivers granted and recruit quality.

9. Training Attrition (People, %)

- a. Research shows that nearly one-quarter of Army recruits are failing out of training and not finishing the first two years of their contract (Beynon, 2025). We modeled this behavior with S-Shaped Growth because in more recent years the number of recruits failing training has increased.
- b. Endogenous – Training attrition is directly affected by the recruit quality, which is caused by higher pressure from recruiting goals which leads to more waivers granted.

10. Leadership Burden on E5s (Dmnl)

- a. As soldiers increase in rank, so does their responsibility and leadership burden. Some soldiers thrive in leadership roles and enjoy it, but others with more responsibilities outside of work such as family or education may find the leadership role to be a burden. Ultimately, with an Army at peace, the leadership burden on E5s can be a less rewarding position and cause a decrease in the E5 population.
- b. Endogenous – The leadership burden on E5s is related to the retention rate of E5s because the increased work responsibility and hours can deter soldiers from reenlisting, effecting E5 attrition.

System Boundary:

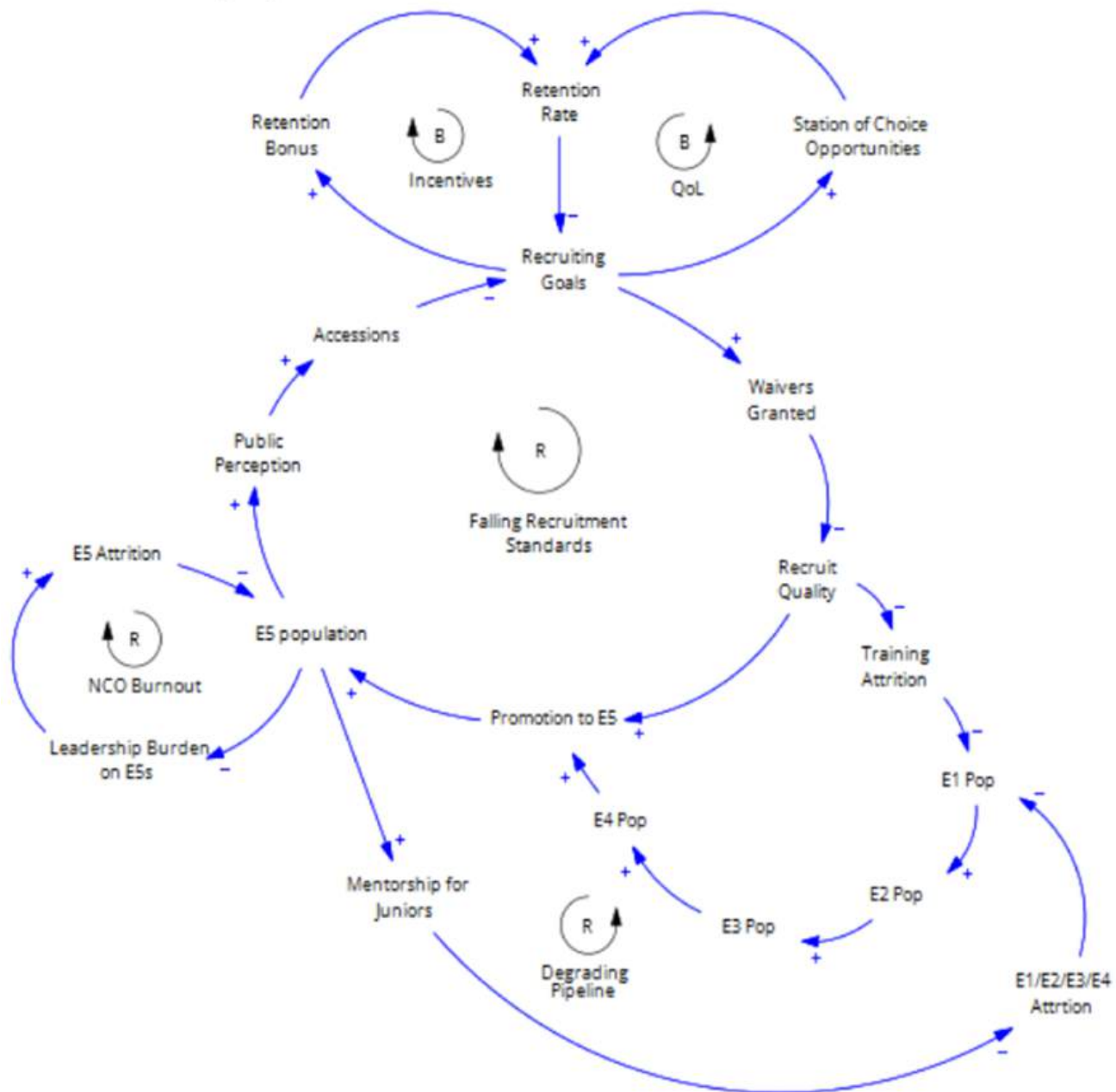
The focus of this project is to determine the key variables that may affect a decrease in the number of E5 Infantry soldiers in the active-duty US Army. The system boundaries in relation to the key variables identified above exclude civilian political or economic influences. We also exclude race, gender, and hometown as it relates to our key variables. Furthermore, we are excluding how these trends may vary between peace vs war. Our system boundary is focused more directly on the Army's resources, decisions, and trends as it relates to recruits and enlisted soldiers. The boundary is scoped to organizational decisions that the Army can directly influence over a 25-year time horizon from 2000 to 2025 to explain why higher recruiting goals and lowering recruitment standards have decreased the E5 population.

4. Dynamic Hypothesis

The observed pattern of attempting to meet recruiting goals but having a persistent decline in Infantry E5 population and experience arises endogenously from the interaction of goal pressure to increased waivers to quality-driven accessions to higher E5 burden to E5 attrition, with incentives only providing temporary fixes. When recruiting goals rise faster than the applicant pool, the Army attempts to reach its target by increasing waivers and lowering entry standards, a practice RAND identifies as occurring during difficult recruiting periods (Knapp et al, 2018). Decreasing recruit quality subsequently increases E1/E2/E3/E4 attrition, a relationship studied in Army research showing scores on the cognitive quality measure,

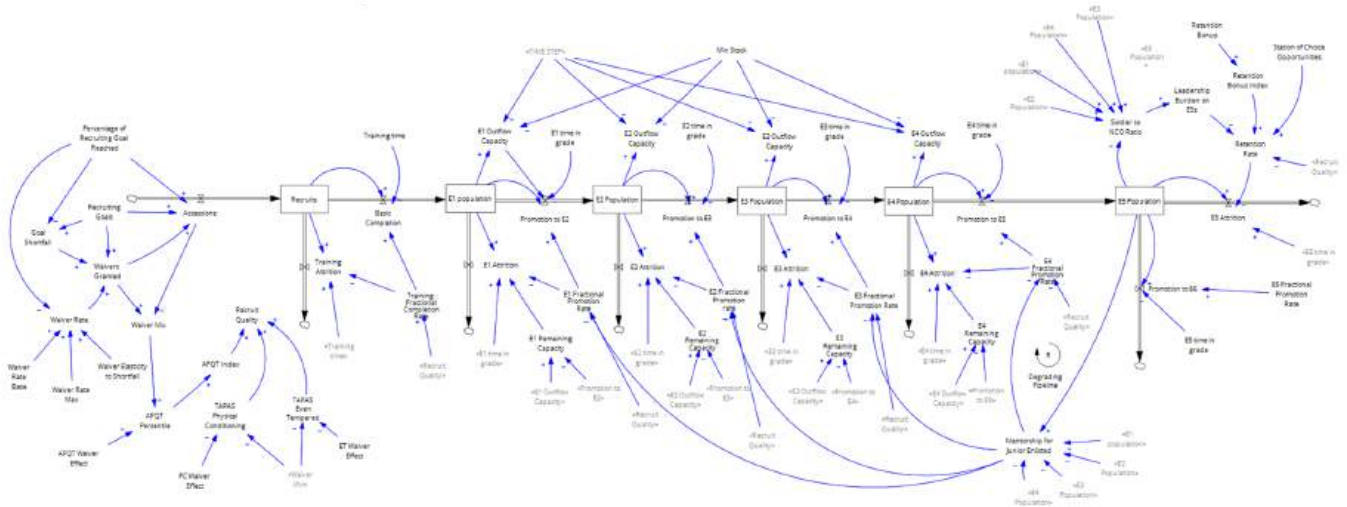
Armed Forces Qualification Test (AFQT), and the non-cognitive quality measure, Tailored Adaptive Personality Assessment System (TAPAS), as predictors for training graduation and attrition later on (Hughes et. al, 2020). This depletes downstream cohorts of Infantry enlisted soldiers so the promotion of E5s falls shrinking the E5 population. Fewer E5s means more leadership burden on those that remain, and less mentorship for the junior enlisted. Weaker mentorship amplifies junior attrition and a decline in promotions, reinforcing the degrading pipeline to E5. In the short run, incentives like retention bonus and Station-of-Choice opportunities raise the retention rate and combat the Army's need for new recruits, but they do not improve the quality composition of the force (Asch, 2021). Meanwhile, the consequences of falling recruitment standards and declining NCO experience create an image of reduced competence which lowers trust in the Army and therefore lowers public perception of the Army (Allen and Braun, 2012). This lowers accessions and increases pressure to meet recruitment goals, triggering additional waivers and closing the reinforcing loop.

a) Causal loop diagrams (CLD)



5. Formulation

a) Stock and Flow Diagram



The model illustrates the process by which individuals progress from accession through training from the junior ranks and into the E5 grade. This process starts with cohort quality which can be accessed by the AFQT, TAPAS traits, and the rate of waivers. When cohort quality is lower, the losses from training are higher, and fewer recruits promote from basic training. This generates smaller populations of new Soldiers in the grade of E1–E4 and lowers the rate of promotions into the E5 grade. As the population of E5’s shrinks, the Soldier-to-NCO ratio increases, which transfers more workload to E5 leaders, further shrinking their population. The retention issues for E5s also produces a loss of mentorship for junior enlisted and repeats the cycle of lower Soldier retention and promotion rates. There are levers to balance retention, like incentives and station of choice, which positively impact retention outcomes, however, the observable pattern in the past can be characterized as a self-reinforcing spiral in which low cohort quality leads to low retention rates for E5s unless those incentives increase entry quality or reduce the workload on NCOs. Overall, the diagram shows that low cohort quality leads to low retention rates for E5s, and that pattern is likely to continue unless policy levers increase entry quality, reduce the burden on NCOs, or increase their incentives to reenlist.

b) Assumptions: A key assumption we made is that 15% of the Army is Infantry. Other assumptions we made with the help of ChatGPT are listed in the explanation for each variable in the variable table.

c) Variable Table

Variable	Initial value	Equation (Vensim)	Units	Explanation
Accessions	—	$\text{MIN}(\text{Recruiting Goals}, \text{Percentage of Recruiting Goal Reached} * \text{Recruiting Goals} + \text{Waivers Granted})$	Persons/Year	Annual accessions entering BCT. Caps realized accessions at the recruiting goal; shortfalls can be backfilled by waivers.
AFQT Index	—	$\text{AFQT Percentile}/100$	Dmnl	Scales AFQT percentile to 0–1 so it can be blended with TAPAS indices (Hughes et al., 2020).
AFQT Percentile	—	$40 - \text{AFQT Waiver Effect} * ((\text{Waiver Mix} - 0.1)/0.1)$	Dmnl	Cohort mean AFQT percentile. Baseline 40 (infantry cohorts below 50th pct; Hughes et al., 2020). Higher waiver mix pulls the mean down by the effect slope.

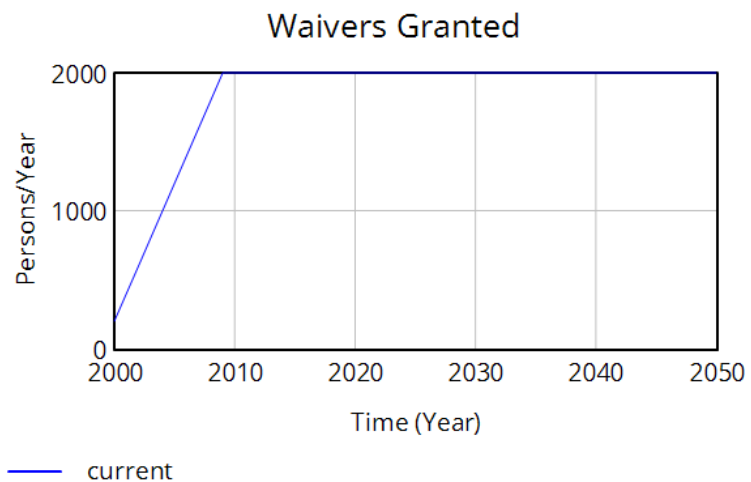
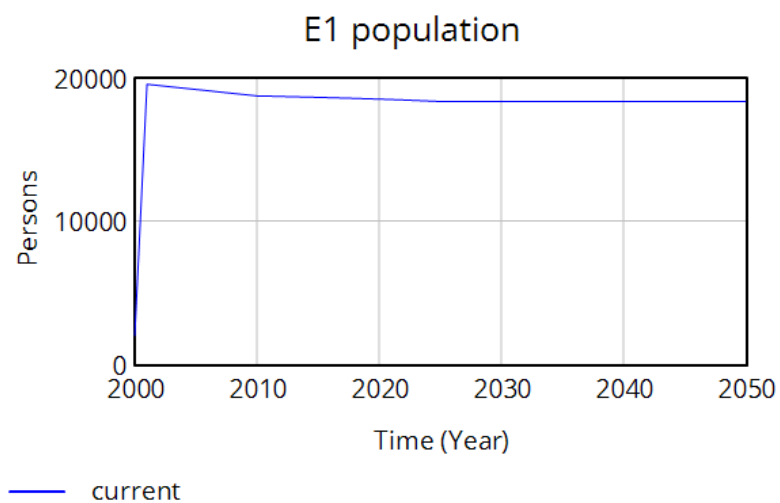
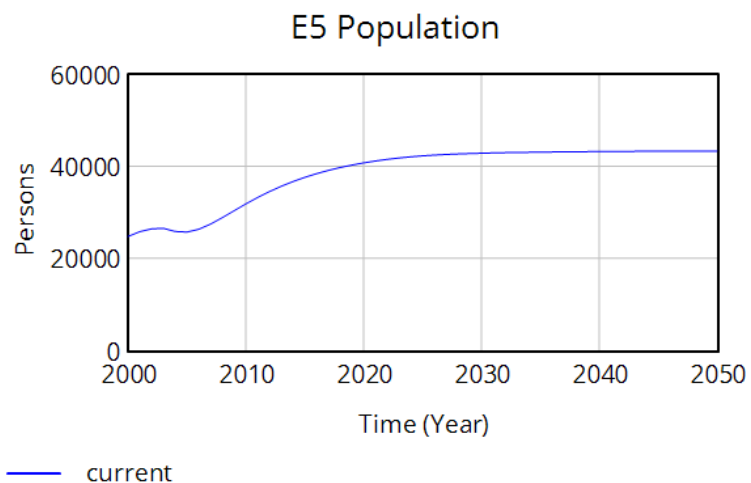
Variable	Initial value	Equation (Vensim)	Units	Explanation
AFQT Waiver Effect	4	—	Dmnl	Sensitivity: each +0.10 above baseline waiver mix reduces mean AFQT by ~4 percentile points (assumption, consistent with mixed-credential pools).
Basic Completion	—	$(\text{Recruits} * \text{Training Fractional Completion Rate}) / \text{Training time}$	Persons/Year	Flow of trainees completing all phases and joining E1; completion fraction apportioned over average training time.
E1 Attrition	—	$\text{MIN}(((\text{E1 population} * (1 - \text{E1 Fractional Promotion Rate})) / \text{E1 time in grade}), \text{E1 Remaining Capacity})$	Persons/Year	Losses from E1 not promoted within time-in-grade; capped so the stock can't drop below the minimum stock rule.
E1 Fractional Promotion Rate	—	$\text{MIN}(0.99, \text{MAX}(0, 0.95 * (1 + 0.3 * (\text{Mentorship for Junior Enlisted} - 1)) * (1 + 0.5 * (\text{Recruit Quality} - 0.5))))$	Dmnl	Share of E1 advancing to E2 per TIG. Base 0.95 (ChatGPT, 2025), lifted by mentoring and recruit quality sensitivities.
E1 Outflow Capacity	—	$\text{MAX}(0, (\text{E1 population} - \text{Min Stock}) / \text{TIME STEP})$	Persons/Year	Mechanical guardrail ensuring outflows (promotions + attrition) can't overshoot the stock floor.
E1 population	2000	$\text{INTEG}(\text{Basic Completion} - \text{E1 Attrition} - \text{Promotion to E2}, 2000)$	Persons	E1 stock level; initial value per assumption (ChatGPT, 2025).
E1 Remaining Capacity	—	$\text{MAX}(0, \text{E1 Outflow Capacity} - \text{Promotion to E2})$	Persons/Year	Residual outflow room available for attrition after promotions, honoring the stock floor.
E1 time in grade	0.203	—	Year	Avg TIG for E1 (~2.4 months; assumption documented as ChatGPT, 2025).
E2 Attrition	—	$\text{MIN}(((\text{E2 Population} * (1 - \text{E2 Fractional Promotion rate})) / \text{E2 time in grade}), \text{E2 Remaining Capacity})$	Persons/Year	Losses from E2 not promoted within TIG; respects stock-floor capacity.
E2 Fractional Promotion rate	—	$\text{MIN}(0.99, \text{MAX}(0, 0.85 * (1 + 0.25 * (\text{Mentorship for Junior Enlisted} - 1)) * (1 + 0.4 * (\text{Recruit Quality} - 0.5))))$	Dmnl	Share of E2 advancing to E3. Base 0.85 (ChatGPT, 2025); improved by mentoring and quality.
E2 Outflow Capacity	—	$\text{MAX}(0, (\text{E2 Population} - \text{Min Stock}) / \text{TIME STEP})$	Persons/Year	Outflow ceiling for E2, enforcing non-negative stocks.
E2 Population	8360	$\text{INTEG}(\text{Promotion to E2} - \text{E2 Attrition} - \text{Promotion to E3}, 8360)$	Persons	E2 stock level; initial value by assumption (ChatGPT, 2025).
E2 Remaining Capacity	—	$\text{MAX}(0, \text{E2 Outflow Capacity} - \text{Promotion to E3})$	Persons/Year	Capacity left for attrition once promotions are accounted for.
E2 time in grade	0.8	—	Year	Avg TIG for E2 (~9.6 months; assumption, ChatGPT, 2025).
E3 Attrition	—	$\text{MIN}(((\text{E3 Population} * (1 - \text{E3 Fractional Promotion Rate})) / \text{E3 time in grade}), \text{E3 Remaining Capacity})$	Persons/Year	Losses from E3 not promoted; constrained by floor.
E3 Fractional Promotion Rate	—	$\text{MIN}(0.99, \text{MAX}(0, 0.82 * (1 + 0.2 * (\text{Mentorship for Junior Enlisted} - 1)) * (1 + 0.3 * (\text{Recruit Quality} - 0.5))))$	Dmnl	Share of E3 advancing to E4. Base 0.82; mentoring and quality effects taper with seniority.

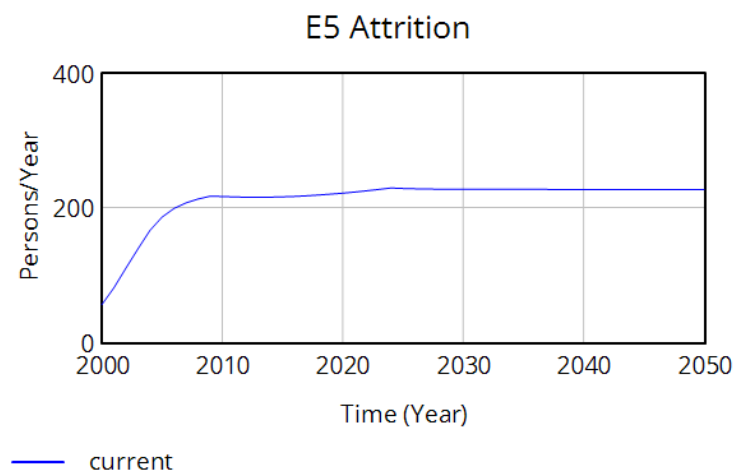
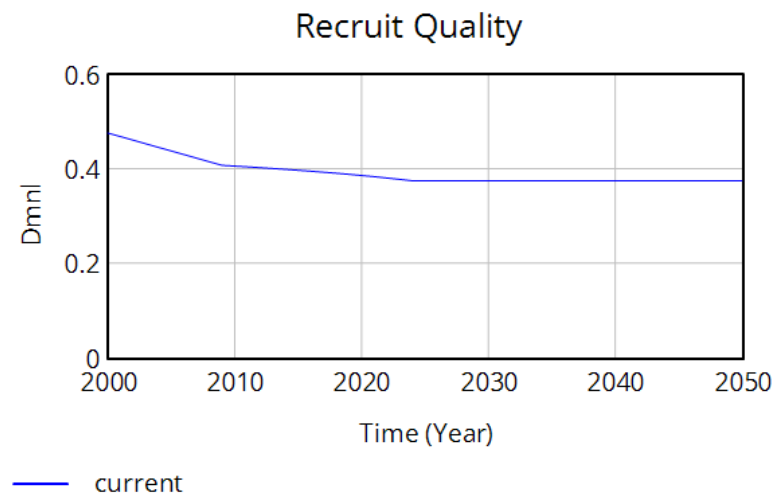
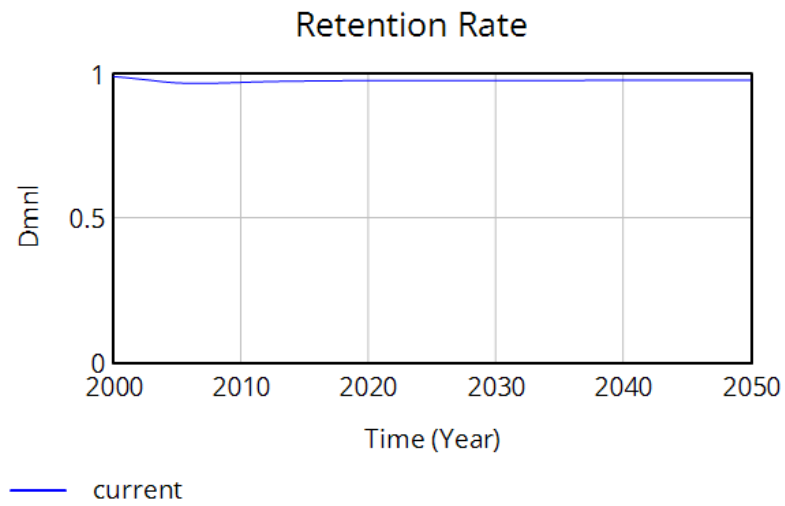
Variable	Initial value	Equation (Vensim)	Units	Explanation
E3 Outflow Capacity	—	$\text{MAX}(0, (\text{E3 Population} - \text{Min Stock})/\text{TIME STEP})$	Persons/Year	Outflow ceiling for E3.
E3 Population	16380	$\text{INTEG}(\text{Promotion to E3} - \text{E3 Attrition} - \text{Promotion to E4}, 16380)$	Persons	E3 stock level; initial value by assumption.
E3 Remaining Capacity	—	$\text{MAX}(0, \text{E3 Outflow Capacity} - \text{Promotion to E4})$	Persons/Year	Residual outflow capacity for E3 attrition.
E3 time in grade	1.8	—	Year	Avg TIG for E3 (~21.6 months; assumption).
E4 Attrition	—	$\text{MIN}(((\text{E4 Population} * (1 - \text{E4 Fractional Promotion Rate}))/\text{E4 time in grade}), \text{E4 Remaining Capacity})$	Persons/Year	Losses from E4 not promoted; floor-aware.
E4 Fractional Promotion Rate	—	$\text{MIN}(0.99, \text{MAX}(0, 0.7 * (1 + 0.15 * (\text{Mentorship for Junior Enlisted} - 1)) * (1 + 0.2 * (\text{Recruit Quality} - 0.5))))$	Dmnl	Share of E4 advancing to E5. Base 0.7; weaker sensitivity to mentoring/quality at higher grades.
E4 Outflow Capacity	—	$\text{MAX}(0, (\text{E4 Population} - \text{Min Stock})/\text{TIME STEP})$	Persons/Year	Outflow ceiling for E4.
E4 Population	23100	$\text{INTEG}(\text{Promotion to E4} - \text{E4 Attrition} - \text{Promotion to E5}, 23100)$	Persons	E4 stock level; initial value by assumption.
E4 Remaining Capacity	—	$\text{MAX}(0, \text{E4 Outflow Capacity} - \text{Promotion to E5})$	Persons/Year	Residual capacity for E4 attrition.
E4 time in grade	3	—	Year	Avg TIG for E4 (~36 months; assumption).
E5 Attrition	—	$(\text{E5 Population} * (1 - \text{Retention Rate}))/\text{E5 time in grade}$	Persons/Year	Losses from E5, inversely proportional to retention and spread over TIG.
E5 Fractional Promotion Rate	0.7	—	Dmnl	Share of E5 advancing to E6 (assumed baseline, ChatGPT, 2025). Promotion out does not feed earlier-grade attrition.
E5 Population	24750	$\text{INTEG}(\text{Promotion to E5} - \text{E5 Attrition} - \text{Promotion to E6}, 24750)$	Persons	E5 stock level; initial value by assumption.
E5 time in grade	4.5	—	Year	Avg TIG for E5 (~54 months; assumption).
ET Waiver Effect	0.02	—	Dmnl	Sensitivity: each +0.10 waiver-mix above baseline reduces TAPAS Even-Tempered by ~0.02 on 0–1 scale (assumption consistent with non-cognitive selection drift; Hughes et al., 2020).
FINAL TIME	2100	—	Year	Simulation horizon end.
Goal Shortfall	—	$\text{Recruiting Goals} * (1 - \text{Percentage of Recruiting Goal Reached})$	Persons/Year	The gap between target and realized accessions before waivers.
INITIAL TIME	2000	—	Year	Simulation start.
Leadership Burden on E5s	—	Soldier to NCO Ratio/8	Dmnl	Workload index per E5 vs. target 8:1 soldier:NCO (approx. infantry squad design; doctrinally ~9:1).

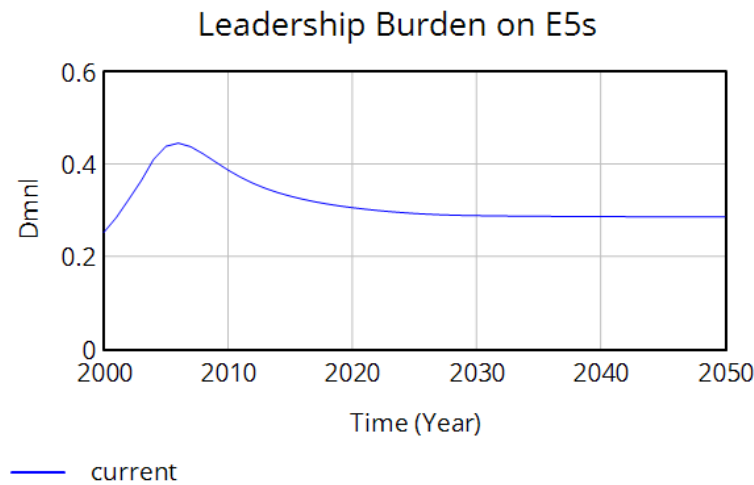
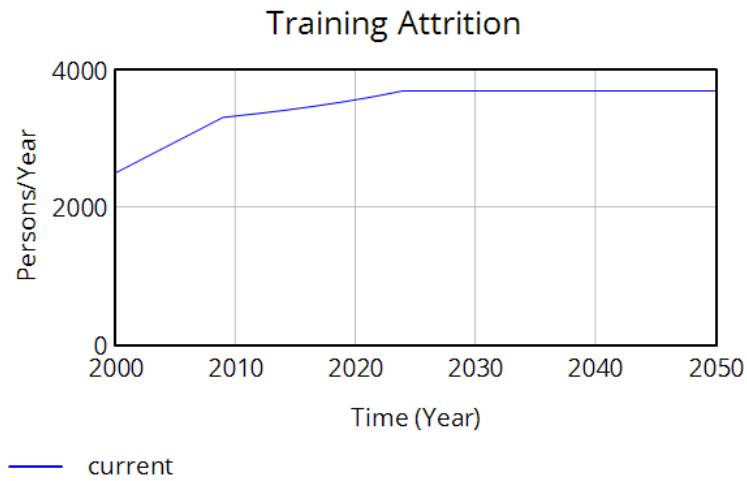
Variable	Initial value	Equation (Vensim)	Units	Explanation
Mentorship for Junior Enlisted	—	$1 * (E5 \text{ Population} / (E1 \text{ population} + E2 \text{ Population} + E3 \text{ Population} + E4 \text{ Population}))$	Dmnl	Mentoring capacity proxy: more E5s per junior increases mentoring (bounded implicitly around ~0–1 scale).
Min Stock	50	—	Persons	Stock-floor guard so levels never collapse below 50.
PC Waiver Effect	0.04	—	Dmnl	Sensitivity: each +0.10 waiver-mix above baseline reduces TAPAS Physical Conditioning by ~0.04 (assumption; aligns with higher physical risk in waiver cohorts).
Percentage of Recruiting Goal Reached	—	$\text{MAX}(0.5, 0.98 - \text{RAMP}(0.02, 2000, 2050))$	Dmnl	Realism device: gradual decline from 0.98 toward 0.5 over 2000–2050, floor at 0.5 (USAREC 2024 trend-style).
Promotion to E2	—	$\text{MIN}((E1 \text{ population} * E1 \text{ Fractional Promotion Rate}) / E1 \text{ time in grade}, E1 \text{ Outflow Capacity})$	Persons/Year	Flow from E1 to E2; limited by outflow capacity to keep stocks non-negative.
Promotion to E3	—	$(E2 \text{ Population} * E2 \text{ Fractional Promotion rate}) / E2 \text{ time in grade}$	Persons/Year	Flow from E2 to E3; proportional to stock, promotion rate, and TIG.
Promotion to E4	—	$(E3 \text{ Population} * E3 \text{ Fractional Promotion Rate}) / E3 \text{ time in grade}$	Persons/Year	Flow from E3 to E4.
Promotion to E5	—	$(E4 \text{ Population} * E4 \text{ Fractional Promotion Rate}) / E4 \text{ time in grade}$	Persons/Year	Flow from E4 to E5.
Promotion to E6	—	$(E5 \text{ Population} * E5 \text{ Fractional Promotion Rate}) / E5 \text{ time in grade}$	Persons/Year	Flow from E5 to E6.
Recruit Quality	—	$0.5 + 0.7 * (\text{TAPAS Physical Conditioning} - 0.5) + 0.1 * (\text{TAPAS Even Tempered} - 0.5) + 0.2 * (\text{AFQT Index} - 0.5)$	Dmnl	Composite 0–1 index weighting PC most (medical/physical attrition risk), ET modestly (misconduct risk), and AFQT moderately (learning demands) (Hughes et al., 2020).
Recruiting Goals	10000	—	Persons/Year	Annual target accessions; set near infantry scale (USAREC 2024; rounded).
Recruits	11000	$\text{INTEG}(\text{MAX}(0, \text{Accessions} - \text{Basic Completion} - \text{Training Attrition}), 11000)$	Persons	Training-pipeline stock; uses MAX(0, ...) to avoid negative stock.
Retention Bonus	10000	—	Dmnl	Average cash incentive level used to compute an index; value \$10k assumed (ChatGPT, 2025).
Retention Bonus Index	—	$\text{Retention Bonus} / 8000$	Dmnl	Normalizes current bonus to an \$8k baseline for sensitivity scaling.
Retention Rate	—	$\text{MIN}(0.99, \text{MAX}(0.5, 0.9 + 0.05 * (\text{Retention Bonus Index} - 1) + 0.05 * (\text{Station of Choice Opportunities} - 0.3) - 0.1 * (\text{Leadership Burden on E5s} - 1) + 0.1 * (\text{Recruit Quality} - 0.5)))$	Dmnl	E5 stay rate with baseline 0.90, modest positive effects of bonus and station choice, negative burden effect, and small positive recruit-quality spillover (USAREC 2024 for incentive plausibility; assumptions).
SAVEPER	—	TIME STEP	Year	Output write frequency.

Variable	Initial value	Equation (Vensim)	Units	Explanation
Soldier to NCO Ratio	—	$(E1 \text{ population} + E2 \text{ Population} + E3 \text{ Population} + E4 \text{ Population}) / E5 \text{ Population}$	Dmnl	Average junior-soldier load per E5.
Station of Choice Opportunities	0.4	—	Dmnl	Fraction of reenlisting E5s receiving preferred station (assumption within policy ranges; USAREC 2024).
TAPAS Even Tempered	—	$0.5 - ET \text{ Waiver Effect} * ((\text{Waiver Mix} - 0.1) / 0.1)$	Dmnl	Cohort mean ET on 0–1 scale; lower with higher waiver share (Hughes et al., 2020).
TAPAS Physical Conditioning	—	$0.45 - PC \text{ Waiver Effect} * ((\text{Waiver Mix} - 0.1) / 0.1)$	Dmnl	Cohort mean PC on 0–1 scale; infantry baseline slightly below average (Hughes et al., 2020); declines with waiver share.
TIME STEP	1	—	Year	Numerical integration step.
Training Attrition	—	$(\text{Recruits} * (1 - \text{Training Fractional Completion Rate})) / \text{Training time}$	Persons/Year	Loss flow from training implied by the complement of completion, spread over training time.
Training Fractional Completion Rate	—	$\text{MIN}(0.99, \text{MAX}(0.5, 0.9 * (1 + 0.6 * (\text{Recruit Quality} - 0.5))))$	Dmnl	Completion fraction anchored at 0.9, responsive to recruit quality (biggest driver of training success; Hughes et al., 2020).
Training time	0.5	—	Year	Mean elapsed time from BCT start to E1 at unit (~6 months).
Waiver Elasticity to Shortfall	0.5	—	Dmnl	Policy knob: each 10% goal shortfall lifts waiver rate by 0.05 (assumption).
Waiver Mix	—	$\text{IF THEN ELSE}(\text{Accessions} > 0, \text{Waivers Granted} / \text{Accessions}, 0)$	Dmnl	Share of accessions arriving with waivers.
Waiver Rate	—	$\text{MIN}(\text{Waiver Rate Max}, \text{Waiver Rate Base} + \text{Waiver Elasticity to Shortfall} * (1 - \text{Percentage of Recruiting Goal Reached}))$	Dmnl	Annual waiver share policy as a function of shortfall; bounded by a maximum.
Waiver Rate Base	0.1	—	Dmnl	Typical steady-state waiver share ≈10% (Knapp et al., 2018).
Waiver Rate Max	0.2	—	Dmnl	Practical upper bound on waiver approvals (~20%+ in tough years; Knapp et al., 2018).
Waivers Granted	—	$\text{MIN}(\text{Goal Shortfall}, \text{Waiver Rate} * \text{Recruiting Goals})$	Persons/Year	Number of waivers actually used; cannot exceed the shortfall or the policy cap.

d) Key Variable Model Outputs



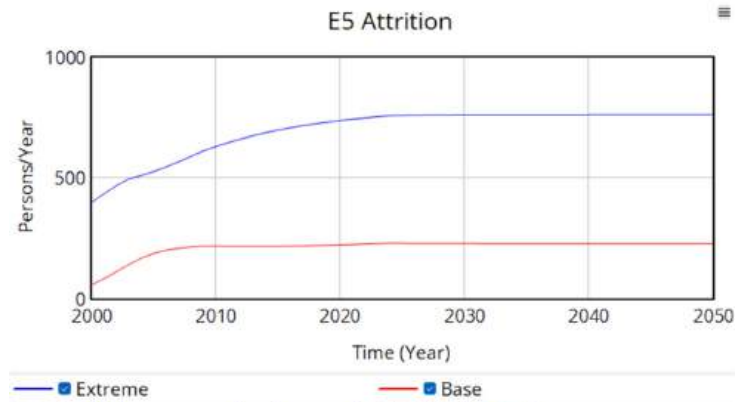




6. Model Testing

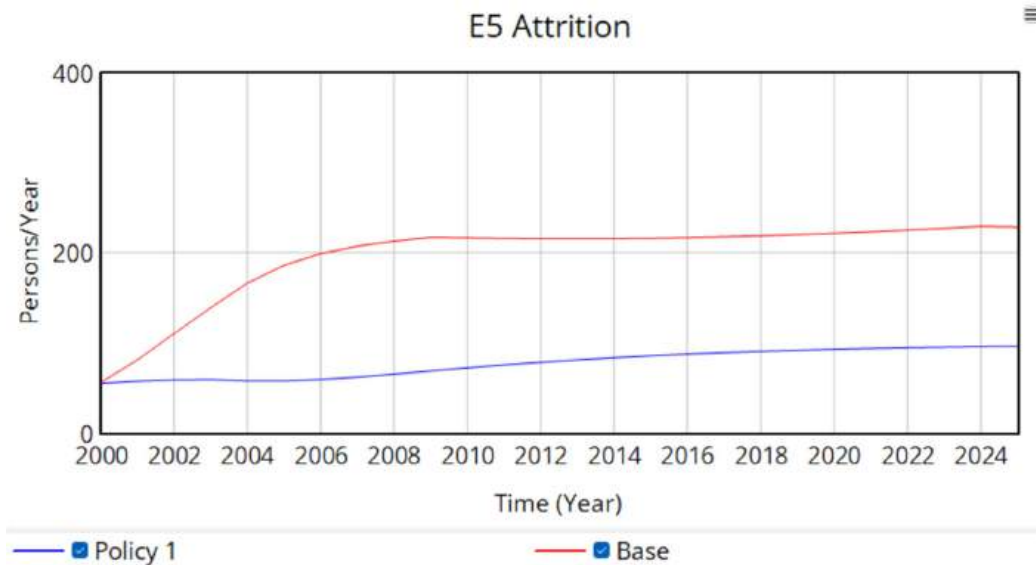
Through the testing of our model by using Extreme Conditions, Dimensional Consistency, and Sensitivity analysis, we were able to see that our model was reasonable and made sense with our current data. For Extreme Conditions, we tested the model by using the variable of retention bonus and pushing it outside of the normal or expected range, and we were able to see our model still behaved within reason. For Dimensional Consistency, we tested our model through checking to see if our equations and variables were dimensionally consistent. Because our model ran with no errors, we were able to see that our model was Dimensionally Consistent. To test our model using sensitivity analysis, we had to identify what the most influential parameter was, which was E5 attrition in our model. Through identifying this parameter, we were able to see that this parameter changed the model the most. Through testing our model using these techniques, we were able to see our model is reasonable and make sense for what we are trying to display.





7. Policy Design and Evaluation

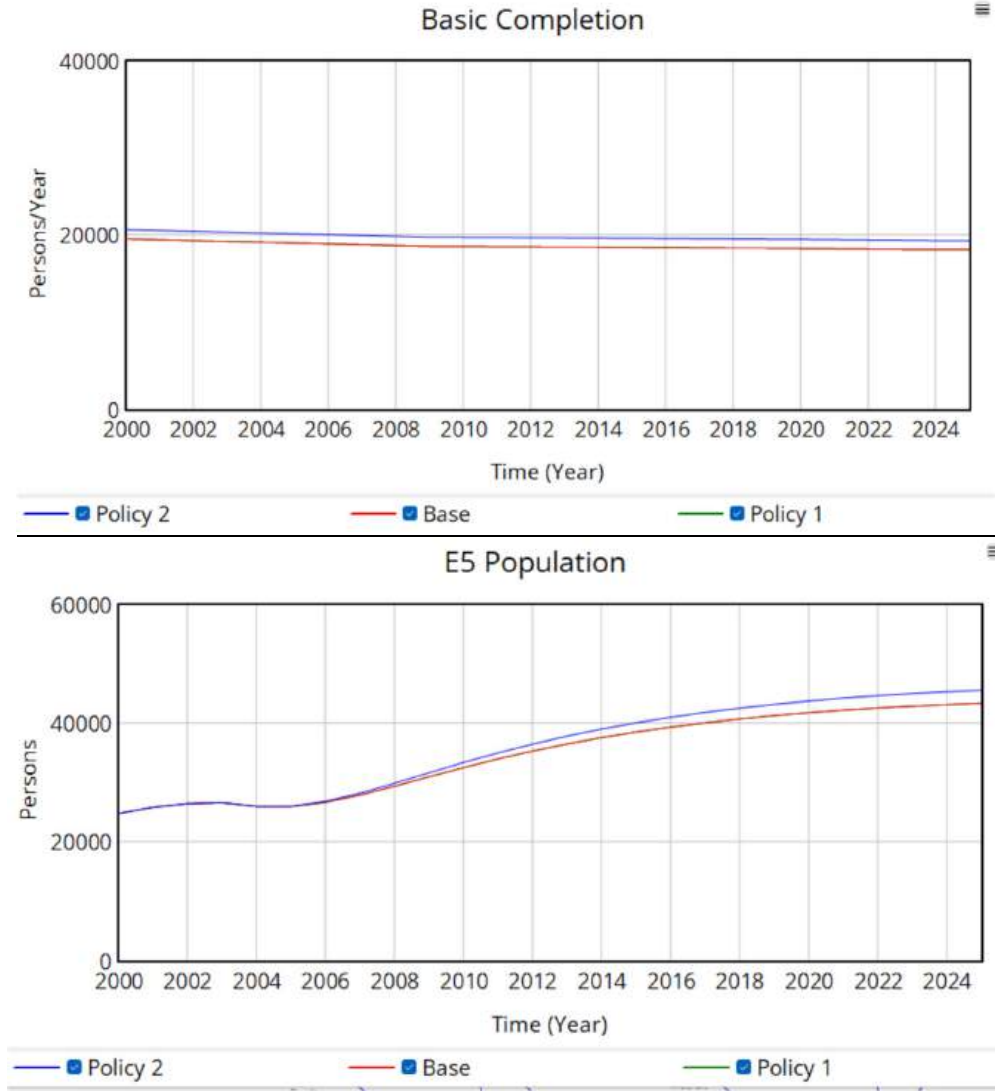
Policy Recommendation 1: Increase Retention Bonus for E5s from \$10,000 to \$15,000



Evaluation of Policy 1:

Money is a driving factor for any accepting and staying with any job, and soldiers are no different. Policy 1 shows that increasing the monetary bonus for E5s leads to a significant decrease in the number of E5s who are leaving the Army. The primary focus of this project is to validate our thesis that Infantry E5s are leaving the Army at a higher rate, and the base model confirms this. Retention Bonus is a variable that directly affects the Retention Rate and ultimately the E5 Attrition Flow. Although the impact of this policy is significantly positive, the Army runs on a strict budget, which makes implementing this policy difficult. Furthermore, the source of the lack of Infantry E5s is the decreasing quality of recruits. This policy will likely help solve the problem short-term but will not be a long-term solution due to the changing budget of the Army and the continued lack of quality recruits.

Policy Recommendation 2: Reduce Recruiting Goals & Waivers Granted to Increase Recruit Quality



Evaluation of Policy 2 (Optimal Policy):

The purpose of Policy 2 is to reduce the recruiting goals of the Army, which will lead to fewer waivers being granted for recruits with criminal records, physically unfit recruits, medical exemptions, and weak ASVAB scores. Refining the selection process will ensure that there are higher-quality recruits entering the Army and completing basic training. In our research, we found that 25% of recruits did not complete the first 2 years of their contract, giving the Army fewer opportunities to retain potential NCOs. Furthermore, this does not require an increase in military spending or technological advancements. The downside to this policy is that there will likely be a significant decrease in the number of enlisted soldiers since more recruits will become ineligible. However, recruiting men and women who are intellectually and physically competent can increase the E5 population long-term. The output graphs show minimal change due to the variety and complexity of our model, which simulates and represents the real-world challenges to this problem. We recommend this policy because we believe that these changes can increase the long-term E5 Infantry population and increase the quality of soldiers in the Army.

8. Conclusion

Based on our results and research findings, we found that our model shows a decrease in E4 and E5 populations over time. The Army is retaining fewer of these soldiers due to poor recruit quality (Recruiting Goals, TAPAS, AFQT) and weak retention bonuses that discourage soldiers from staying in. Through more testing, we found that our second policy, which reduces recruiting goals and the number of waivers given to recruits will increase recruitment quality the most within the infantry. There are, however, limitations with this policy. We had to make a lot of assumptions when testing. These assumptions include the total number of Infantry E5s, retention rates for E1s through E5's, and the fraction of E5s offered their preferred duty station. For future data gathering, if we widened our scope to include every E5 in the Army, not just infantry, we would be able to make less assumptions given the fact that most data given by the government is on the Army as a whole.

In conclusion, NCOs run the Army, specifically E5s. Without a steady flow of high quality, capable recruits, as well as effective re-enlistment bonuses, the experienced NCOs are put under an increased load and are leaving the Army because of this. This decreases the Army's lethality, making it vulnerable to any upcoming peer to peer conflicts. Prioritizing the retainment of NCOs should be a major focus for the Army in the Coming decade.

9. References

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Acknowledgment

ChatGPT. Assistance given to the author, AI. I used ChatGPT to create reasonable assumptions for initial stocks of recruits through E5s for Infantry soldiers in 2000, the average time in grade as a recruit from E1-E5, retention rate for E1-E5, fractional completion rate of infantry basic training, fractional promotion rate from recruit-E5, average and baseline reenlistment bonuses for E5s, and the fraction of reenlisting E5s who are offered their preferred duty station. I also used ChatGPT to convert my Vensim produced list of variables with their corresponding variable name, equation, initial value, units, and explanation into an organized table. OpenAI, (<https://chatgpt.com/share/69153191-eb10-8008-8013-234f839a0462>). West Point, NY, 12NOV2025.

Poor Student Work

UNITED STATES MILITARY ACADEMY

FINAL REPORT

SM484: SYSTEM DYNAMICS SIMULATION

SECTION J1

MAJ JACOB LUEDER

BY

CADET LUCAS SORENSEN '27, CO F2
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WEST POINT, NEW YORK

10 DECEMBER 2025

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Modeling Retention of Soldiers in the U.S. Army

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Author Note: “The views expressed herein are those of the authors and do not reflect the position of the United States Military Academy, the Department of the Army, or the Department of Defense.” For any questions, please contact the authors at the email addresses listed above.

1. Introduction

Each year, the United States Army has a goal of a certain number of soldiers it wants to recruit to maintain its operations. As service members retire, complete their contractual agreements, or are unable to fulfill their contractual agreements and process out of the Army, there is a demand to fill these roles which requires an inflow of new soldiers. To model the relationship of how the Army recruits, retains, and loses its service members is crucial to understanding if there can be any improvements to the current system. This report will deliver models reflecting the correlation and causation of the Army's system to service member retention and a dynamic hypothesis to provide guidance to key decision makers.

2. Literature Review

Systems dynamics allows for the rigorous modeling of real-world problems and provides a variety of tools to do this (Sterman, 2000). Causal loop diagrams (CLDs) allow the quick capturing of hypotheses, illustrating important feedback which may cause problems, and representing the feedback structure of systems (Sterman, 2000). While useful, CLDs fail to accurately capture accumulation in a system. To solve this, systems dynamics uses stocks and flows diagrams, allowing for the separation of stocks to provide inertia and delays for a system, as well as flows to represent rates of change (Sterman, 2000). Additional variables familiar to CLDs can be added to these stocks and flows diagrams to provide additional information and accuracy in modelling real-world scenarios.

The Army has the highest attrition rate of any branch of the military (Marrone, 2020). For each soldier attritted, the Army loses an estimated \$70,000 on their training (Hughes et al., 2020). From 2001-2013, the Army averaged a 15% and 29.7% attrition rate over 12 months and 36 months respectively. Using probit regression revealed that over half of attrition in the first term of a contract can be predicted by a trainee's Armed Forces Qualification Test (AFQT) score, with a negative causal link between a trainee's AFQT score and their likelihood to attrit (Hughes et al., 2020). Analysis shows that this is no fault of the service, and these individuals are often unable to perform at the level required of their occupation (Marrone, 2020). Looking at other individual variables, being married is associated with a higher attrition rate, with the Army having the highest proportion of married recruits at 18.6% (Marrone, 2020). Specifically in the Army, females have a 13% higher attrition rate between months 7 to 36, around 10% higher than every other branch (Marrone, 2020). Using variable coefficients between services yields much higher attrition rates, inferring that each service can properly recruit individuals who are suited to their specific service (Marrone, 2020). Baseline attrition can be attributed to the specific military lifestyle or skillset not being a good match for the recruit, and with the Army evenly distributing recruits across occupational specialties, compared to other services keeping their recruits unassigned, the specialization in AIT may account for a great amount of Army attrition, with trainees not being a good fit for their military occupational specialty despite being a good fit for the Army (Marrone, 2020). A key variable in the results of attrition rates is how recruiting and enlistment occurs from the start. The findings in studies are indicative of the motivation to enlist, and recruitment advertising based on whether military service is compulsory or mandated. A common pattern found within the U.S. Army is that soldiers claim to enlist due to the opportunities to travel, earn money, and pay for a higher education, with the intention of not serving a full career in the military. (Fraser, 2019)

Papers looking at other branches of the armed forces, other nations, and other points in a contract where an individual can attrit and yield similar findings. A 2023 paper looking at Israeli Defense Forces found 1 in 6 men and 1 in 12 women were unable to complete their initial contracts, with the main cause of attrition being a positive causal link to what the authors call mental difficulties (Ben-Ari et al., 2023). The paper additionally states a positive causal link from being in a rear unit, as well

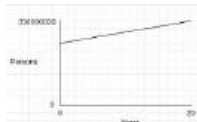
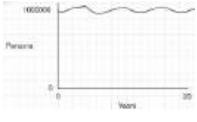
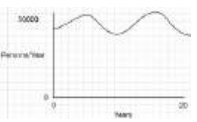
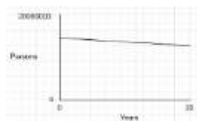
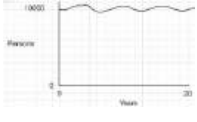
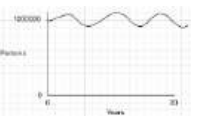
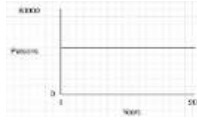
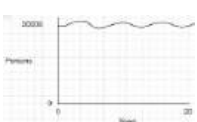
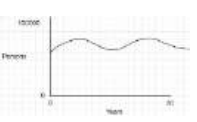
as belonging to a family facing welfare problems, to an early discharge (Ben-Ari et al., 2023). Additionally looking at attrition in a servicemember's first unit shows age and gender to affect the likelihood an individual will complete their first contract. Soldiers aged 21-25 have a .85 probability of attrition, and soldiers ages 26-50 have a .80 probability, relative to the base rate of attrition found in soldiers ages 17-20 (Marrone et al., 2021). Females additionally have a 2.11 probability of not completing their first contract when using men as the base rate (Marrone et al., 2021). Finally, relative to White enlistees, Black enlistees are 10% less likely, Hispanics 31% less likely, and Asians 38% less likely to attrit (Marrone et al., 2021). identify that attrition rates have been relatively steady over the past two decades, with a noticeable drop between 2003-2006, which they attribute to the lead up to the surge in Iraq, which would have put additional pressure on the Army to reduce attrition and improve recruitment. The paper additionally notes a positive causal link between a soldier's level of education and their likelihood of attrition (Marrone et al., 2021). Receiving a bonus decreases odds of attrition by around 7%, while entry pay-grade has a negative causal link, with odds of attrition dropping around 20% for each additional pay grade a soldier receives on entry (Marrone et al., 2021).

There have been several studies looking at the attrition rates of each branch which have attempted to identify causes of attrition. While several variables are commonly accepted as factors that could result in a higher attrition rate, many sources agree that pre-screening applicants is a poor idea because it would also remove potential non-attritors from the applicant pool (Marrone, 2020). There is not enough data on how changes to Basic Training and AIT would affect attrition. This paper aims to fill this gap in research.

3. Problem Articulation

Almost 25% of soldiers fail to make it through their first enlistment. Most of this dropout rate is a result of low-quality enlistees, who are unable to meet the standard necessary to join the army but are sent to basic or AIT before failing there, usually being discharged before reaching their unit. As a result, despite meeting recruiting goals, the number of soldiers enlisting into the army is much lower than the recruiting numbers reported to Congress (Hughes et al., 2020). Decreasing the Army's attrition rate while maintaining current standards would result in a larger number of soldiers in the Army, as well as greater flexibility in recruiting (Marrone, 2020). In defining this problem, we considered a time horizon of twenty years to be an appropriate time horizon. This is the time horizon often used to identify changes between administrations while ensuring numbers aren't skewed by population growth (Marrone et al., 2021). Key variables and reference modes are depicted below.

Table 1. Key Variables and Reference Modes

Key Variable	Reference Mode	Key Variable	Reference Mode	Key Variable	Reference Mode
Non-Recruitable Population		Desired Size of Army		Enlistment	
Recruitable Population		Dropouts		Active Duty	
Basic Training and AIT		Graduates		Recruitment Goal	

Stock variables such as Recruitable Population, Non-Recruitable Population, Basic Training and AIT, and Active Duty is the US population, minus every other stock. Active Duty is all soldiers currently serving in an active-duty role. Recruitable Population is the number of people not currently serving but who meet the requirements to join the Army. Basic Training and

AIT consists of soldiers who have joined the armed forces but have not yet made it to their unit. Enlistment is the rate at which people join the Army, moving them from the Recruitable Population to the Basic Training and AIT stock. Desired Size of the Army is defined as 5% of the recruitable population, a number that can realistically be achieved while retaining a large standing Army. Recruitment Goal is a derivative of this, informing recruiters of how many additional people they must recruit to bridge the gap between the actual size of the Army and its desired size. Dropouts are defined as those within AIT and Basic training who are unable to complete their training due primarily to holistic health and fitness (H2F) complications. Graduates are the remainder who complete their training and are then moved to the Active-Duty pool. To help model these numbers, a dropout rate was modeled to create the number of dropouts that mimic reality. Standards are an arbitrary variable to affect the desired size of the Army. As the Army seeks to shrink its size, it tightens standards, forcing people out. As the Army aims to grow, it relaxes standards to allow greater enlistment. Waivers are exceptions to standards, allowing in those who would otherwise be unable to enlist. Like standards, this is unitless and is meant to show the ‘proportion’ of standards that can be ignored.

4. Dynamic Hypothesis

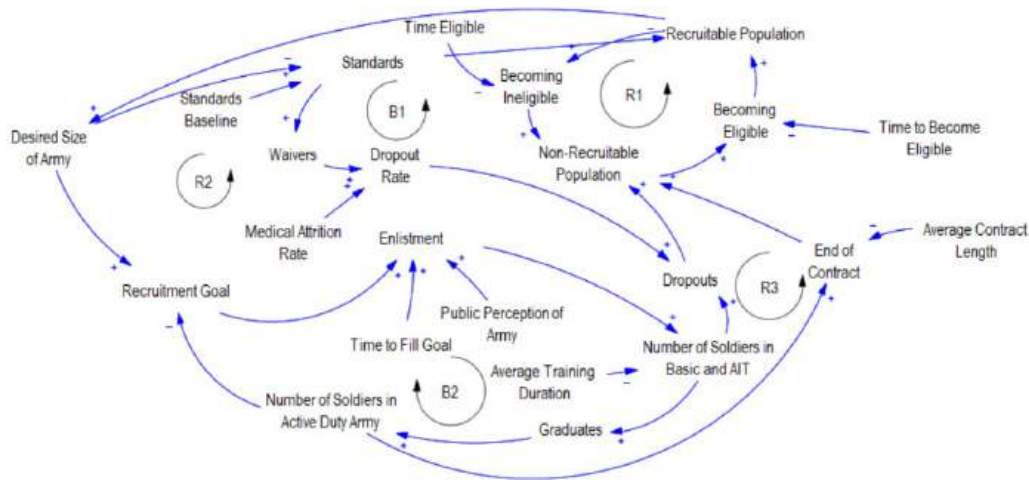


Figure 1. Causal Loop Diagram of Attrition and Retention in the Army

The causal loop diagram above models Army accessions in various loops. This includes three reinforcing loops. R1 shows the cycle of the US population. As individuals approach 18, get in better shape, or exogenous variables have some outsized effect, they enter the recruitable population pool. Likewise, as they get too old, have H2F complications, or as another exogenous variable takes hold, they are moved to the non-recruitable population pool. R2 indicates how the desired size of the pool is a function of the population the US can draw on, affecting the recruitment goal, the number of people enlisting, going to basic training and AIT, and before those soldiers drop out, ending up at R1 where they are eventually recycled back into the recruitable population. R3 follows an almost identical path except instead of dropping out, soldiers go into active duty and complete their first contract. The model additionally has 2 balancing loops. B1 shows that as the desired size of the Army increases, new standards create slack, which increases the recruitable population. As this number increases the desired size of the Army, standards tighten up. B2 shows that as the size of the Army increases, recruiting goals decrease as there is less of a gap to fill. This decreases enlistment, eventually decreasing the number of soldiers in the Army, which in turn raises the recruitment goal as the gap becomes more apparent.

In the model, standards are in place to limit eligibility to certain people the army is looking for, decreasing the Recruitable Population. Marrone’s data shows applicants without a high school diploma, or with weaker pre-service records, were more likely to attrit early, which is why the Army tightens standards to narrow eligibility to those more qualified (Marrone, 2020). This supports the idea that stricter standards reduce the recruitable population and increase the non-recruitable pool, increasing the Non-Recruitable Population. With this, waivers are put into effect by the army to allow certain people through those standards to the recruitable population pool. Marrone explains that when the Army faces recruitment shortfalls, it sometimes grants waivers for educational, moral, or medical disqualifications to expand the eligible pool. Horton (2024) similarly notes that the Army “adjusts entry requirements” to meet manpower goals, which aligns with the negative polarity

Public Perception of the Army increases Enlistment. Fransen (2019) found that Army recruitment campaigns highlighting education, career opportunities, and patriotism are directly linked to higher enlistment motivation, showing that positive marketing and outreach contribute to recruitment success. Horton (2024) adds that negative public opinion, family pressures, and retention issues can discourage potential recruits, while improving the Army's image and communication strategies helps boost enlistment, supporting the positive link from Public Perception to Enlistment. Once individuals enlist, the process continues through the Enlistment of Soldiers in Basic/AIT to Soldiers in the Active-Duty Army, with attrition during training and early service moderating this flow. Marrone's 2020 report shows characteristics like education level and work history at the time of recruitment to predict early and 36-month attrition, reinforcing why eligibility and attrition are important balancing factors in the system (Marrone, 2020). Feedback loops also capture policy adjustments: when Active-Duty Strength drops below the Recruitment Goal, the Army may respond by loosening standards or issuing more waivers to Standards); conversely, when the force meets targets, standards can tighten. This behavior, reflected in Marrone (2020) and Horton (2024), demonstrates how recruitment policies adapt to maintain force levels while balancing quality and quantity. Overall, these connections are supported by evidence showing that outreach and perception drive enlistment (Fransen 2019 and Horton 2024), while eligibility and attrition determine how recruits progress into the active-duty force (Marrone, 2020). Given these conditions, we hypothesize that at present, the recruitable population pool will continue to decrease, resulting in a smaller Army. Changing variables such as waivers, standards, and outsized influence caused by human decisions such as executive orders or Department of the Army memos may change this trend, allowing the Army to draw from a larger pool of applicants. This will in turn influence the number of individuals who drop out of Basic or AIT or are unable to complete their first contract.

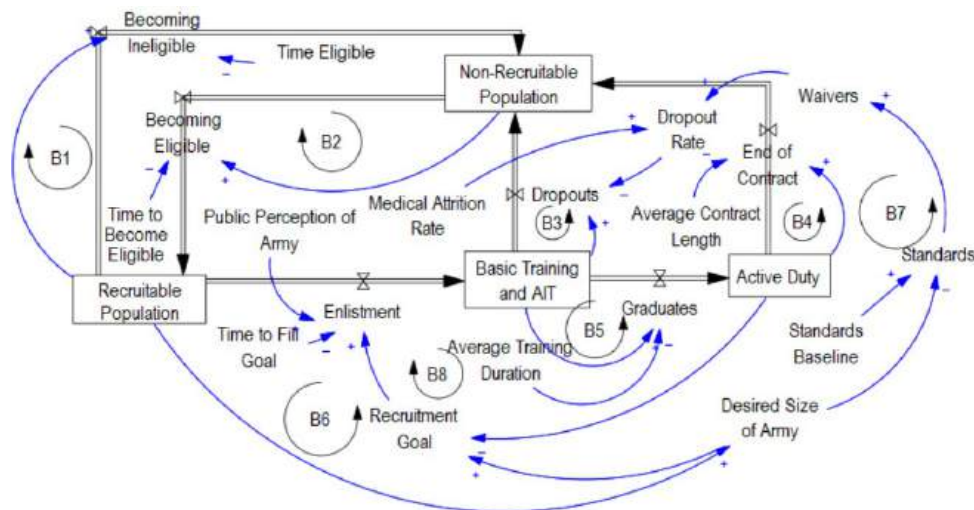


Table 2. Key Variables and Associated Units within Stock and Flow Model					
Variable	Units	Variable	Units	Variable	Units
Becoming Ineligible	Persons/month	Time eligible	Month	Recruitable population	Persons

Becoming eligible	Persons/month	Time to become eligible	Month	Non-recruitable population	Persons
Public perception of army	Dmnl	Time to fill goal	Month	Enlistment	Persons/month
Recruitment goal	Persons	Medical attrition rate	1/month	Dropout rate	1/month
Average contract length	Month	Waivers	Dmnl	Standards	Dmnl
Standards baseline	Persons	Desired size of army	Persons	Average training duration	Month
Dropouts	Persons/month	Graduates	Persons/month	Active duty	Persons
End of contract	Persons/month	Basic Training and AIT	Persons		

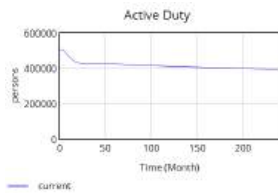


Figure 3. Simulation Output of Active Duty

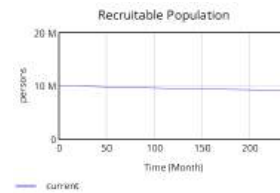


Figure 4. Simulation Output of Recruitable Population



Figure 5. Simulation Output of Basic Training and AIT

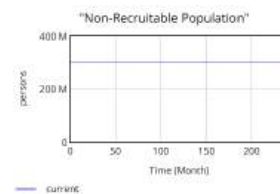


Figure 6. Simulation Output of Non-Recruitable Population

These graphs show as time goes on, the number of active-duty soldiers and the recruitable population will slightly decrease, while the non-recruitable population stays constant. The Basic and AIT population needs a period of calibration before likewise remaining constant.

6. Model Testing

The final simulation stock and flow model were able to effectively and accurately capture what we believed would happen. Our model was able to maintain realistic outcomes in extreme conditions and was dimensionally consistent throughout, shown in Figure 7, Figure 8, and Figure 9. These images confirm the validity of our model, allowing us to draw conclusions from its output.

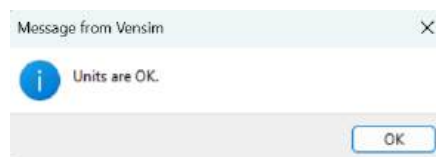


Figure 7. Dimensional Consistency Testing

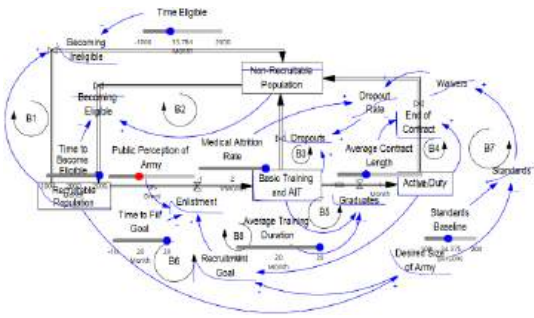


Figure 8. Extreme Conditions and Sensitivity Testing

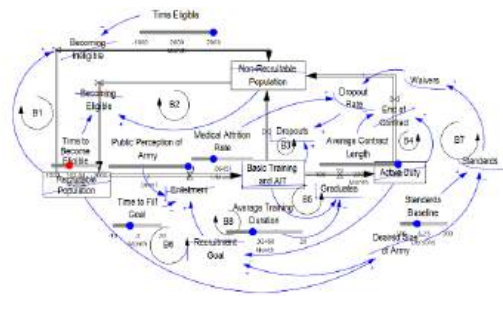


Figure 9. Additional Extreme Conditions and Sensitivity Testing

7. Policy Design and Evaluation

The significant inertia contained in the Recruitable Population and Non-Recruitable Population stocks made it extremely difficult to affect significant change. This outcome is validated in relevant literature by Marrone et al. With the number of individuals in Basic and AIT being relatively consistent year by year, our policy focuses primarily on increasing the number of active duty soldiers rather than decreasing the number of dropouts, as the Army has made no significant effort to change its attrition rate, using it as a way to weed out soldiers for whom the Army may not be a good fit. To increase recruitment and retention, the average contract length could be increased. This would deter individuals who are not fully committed from joining the Army but would increase retention every year. Additionally, increasing the public perception of the Army, primarily with advertising campaigns, could increase recruitment by drawing from the recruitable population. Finally, reducing the medical attrition rate could decrease the number of individuals who drop out within the first year of their contract. This would be very hard to significantly changes to the medical attrition rate would likely affect the entirety of the American population. Such a change would additionally change the number of individuals in the Non-Recruitable Population pool, moving a significant number of them to the Recruitable Population pool. Our final model recognized the conclusions drawn within the literature review and recognized that Army attrition will stay relatively consistent unless the Army deems it necessary to keep soldiers in for an extended period, as was shown in the lead up to the surge between 2003-2006.

Future research should focus on analyzing the specific effects medical attrition has on retention, and how the Army can screen out these individuals prior to them enlisting. Additionally, research should focus on identifying the desired size of the Army, and how the Army determines what size it would like to year. Being able to predict this number years in advance would provide better data on the model's recruitment goal as well as provide data for additional variables that could be added, showing what number of people the Army is comfortable to let attrit.

8. Conclusion

Attrition in the U.S. Army is reflective of many variables, but it remains consistent despite policy changes. The issue of making a significant change in policy to affect attrition are the barriers required to change, including ethical considerations that would allow more service members to join. These ethical considerations include lowering the enlistment age and giving more leniency to medical exemptions. Additionally, with the size of the population the Army can draw on, paired with the current geopolitical situation of the United States, the Army is not hard-pressed to find a solution to reduce attrition. Rather, the Army has identified attrition as a cost it is willing to bear to ensure soldiers entering units are committed to the values of the Army, using attrition to weed out those it was unable to effectively screen prior to enlistment. Were the geopolitical situation to change, the Army would most likely make drastic change to this, significantly reducing the number of soldiers it is willing to cut from Basic Training and AIT and using a variety of tools to significantly increase the number of recruits. However, for the foreseeable future the Army is faced with the challenge of drawing on a recruitable population that decreases as a total percentage of the US population. Rather than making a concerted effort to expand that pool, the Army has contented itself to maximize recruitment within that pool, with lifestyle changes to increase the recruitable population being hard for the Army to implement and force upon the American public.

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Acknowledgement of Assistance

Gemini. Assistance given to the authors, AI. We used the following prompt in Gemini: “summarize this paper” in reference to the paper by Marrone. Gemini in turn gave us a list of key points from the paper. After determining which points were relevant, we asked which page number it found those points on to verify the information was true. This allowed us to read the relevant passages ourselves to generate in-text citations. Gemini, (<https://gemini.google.com/share/2cdedb5b5b09>). West Point, NY, 20OCT2025.

Gemini. Assistance given to the authors, AI. We used the following prompt in Gemini: “summarize this paper” Gemini in turn gave us a list of key points from the Hughes paper. After determining which points were relevant, we asked which page number it found those points on to verify the information was true. This allowed us to read the relevant passages ourselves to generate in-text citations. Gemini, (<https://gemini.google.com/share/3fa078d02a85>). West Point, NY, 20OCT2025.

Gemini. Assistance given to the authors, AI. We used the following prompt in Gemini: “why do i get a floating point error in my vensim and how do i fix it?” referencing one of our draft stock and flow models. In turn, Gemini provided sources causing the error, as well as ways to fix it. The AI primarily looked at ways to stop our recruitable population from immediately falling to zero. Further prompting helped resolve unit errors and simplify the model, resulting in the turned in product. Additionally, Gemini assisted in identifying loops within the CLD and S&F models, helping point out an additional two loops after initially only spotting six. Gemini, (<https://gemini.google.com/share/505dc82ab3bf>). West Point, NY, 12NOV2025



UNITED STATES MILITARY ACADEMY
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UNITED STATES MILITARY ACADEMY
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SECTION III

ASSIGNMENTS



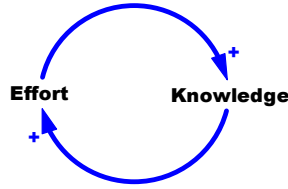
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Homework



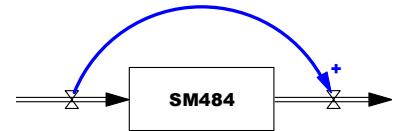
Homework 1

Instructor Solution



System Dynamics Simulation
SM 484 – AT 26-1
MAJ Jake Lueders
MAJ Pat Lupfer

SOLUTION



Homework 1: Problem Definition and Feedback Structure

Assigned: 20AUG25; Due: 10SEP25 at 2359
75 Points

This is an individual assignment. Your submission must be created and submitted IAW your instructor's directions and the assignment coversheet. As always, submit your work with a coversheet IAW the Dean's DAAW.

The design and evaluation of high-leverage policies demand a long-term, dynamic perspective. The analyst or manager must be able to characterize the strategic problem, including its current symptoms and history. The time horizon should be explicit and must be long enough to include possible side effects, delayed responses to policies, and other feedback. This assignment develops your ability to develop a *reference mode* that characterizes graphically patterns of problem behavior over time in various situations. The assignment also develops your skill in constructing *causal loop diagrams* that capture the feedback structure of the system, and your ability to relate the structure in a causal map to the problem behavior in the reference mode.

Problem definition involves both textual and graphical statements of problematic behavior. Conceptualization entails identifying feedback loops that are hypothesized to underlie observed patterns of system behavior. Model formulation is the process of moving from a theory of underlying structure to a fully specified mathematical model to test the theory. In this assignment, the skills involved in problem definition and model conceptualization are separate. Later assignments will bring these skills together with those of formulation and analysis to focus on a variety of strategic and operational problems.

Your CLD should be hand-drawn legibly for this assignment. Your graphs and diagrams should be clear and legible. Ensure you use your time to think about the issues and develop your ideas so that you can convey the structure effectively.

A. Read the Case Study

Read about the global Issues identified by the United Nations and **select one** of the following case studies for the assignment: Ageing, Artificial Intelligence, Outer Space, or Water.

<https://www.un.org/en/global-issues>

B. Identifying the Behavior of the System

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Homework 1: Problem Definition and Feedback Structure

The case studies from the UN examine issues of vital importance to the future of our world, and are problems uniquely suited to analysis using System Dynamics. **Select one** of the four case studies (Ageing, Artificial Intelligence, Outer Space, or Water) that is interesting to you and identify the critical variables, behavior, and feedback structure that creates the dynamic behavior of the individual case. Within each specified issue, there exists a summary of the issue, as well as links to supplemental articles. While the summary provides some background on each and is sufficient to create a basic causal loop diagram, additional research is recommended to better understand the system. You **MUST** reference at least **2x additional sources**, other than the summary.

Additional Source 1: -2 each missing source (acceptable if in works cited only)

Additional Source 2: _____

- ☐ B1 (5 Points). **Select one** of the case studies (Ageing, Artificial Intelligence, Outer Space, or Water) and provide a short paragraph describing what your focus area will be.

Case Study: 3pts selecting study, 2pts short description

- ☐ B2 (10 Points). There are several important variables required to understand the cases. Make a list of the most important variables or concepts characterizing the problem described in the case study. Your list should be comprehensive but as short as possible (8 to 12). You may aggregate similar concepts when possible.

Variable	Units
Credit for 8+ variables and correctness (i.e. if a unit doesn't make sense could take away .25 to .5pt, if they only include 7 variables then take away 1pt) Generally they should do well here as long as they put in an effort and follow stock and flow rules	

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Homework 1: Problem Definition and Feedback Structure

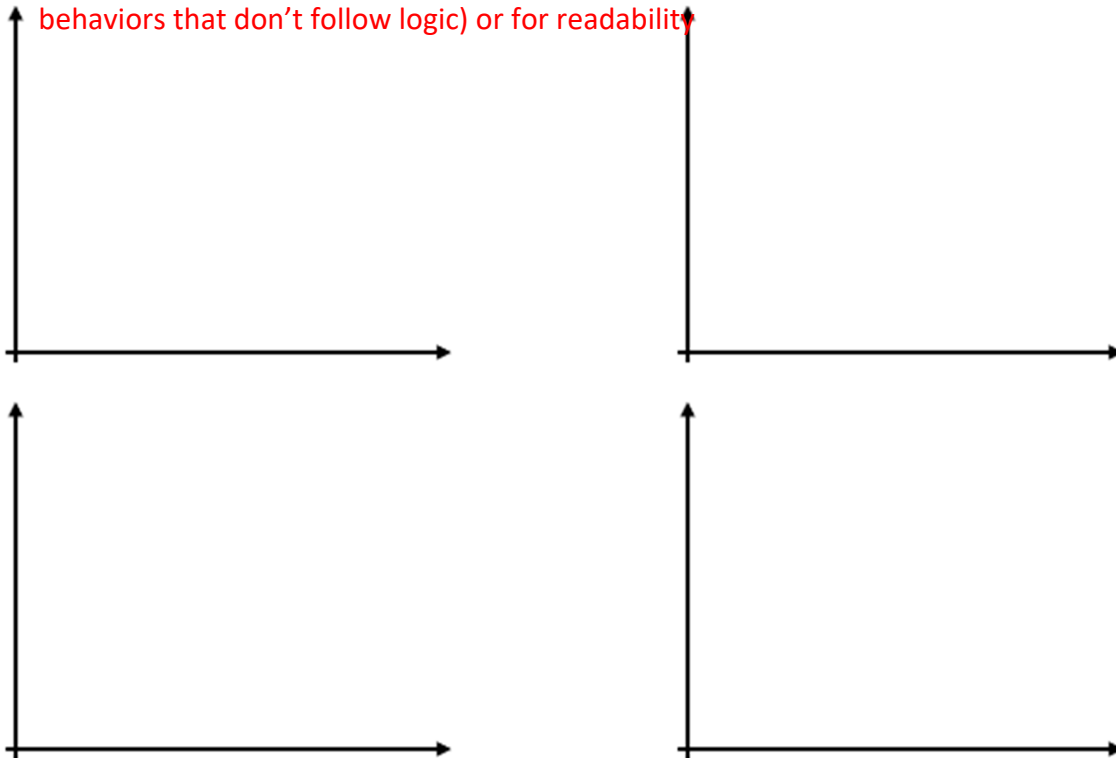
- ☐ B3 (5 Points). Use the description and analysis in the case study to depict the reference mode – the set of dynamics that characterize the problematic behavior. First, identify the time horizon over which the dynamics unfold. Do not be constrained by the publication date of the article or the range of data given in the case study. Select a time period long enough to capture the dynamics of the critical variables, including their past development and enough of the future for the dynamics and impact of policies to play out.

5pts for time horizon that makes sense, adjust points for time horizons that have issues (unless they provide short description for why they chose it)

Time Horizon: _____
Why Chosen? _____

- ☐ B4 (20 Points). Next, sketch a graph showing your best estimate for the behavior of each variable over the time horizon. You do not need to gather actual data for the variables you identify. In the early stages of problem definition, recognizing the fundamental modes of behavior are more important than numerical precision. For the purposes of this assignment, it is sufficient to use your best judgment to estimate the behavior of these variables. Be sure to provide appropriate scales and units of measure for your estimates. If two or more variables have the same units of measure, plot them on the same scales (for example, in a model of a business one would show revenue and cost on the same graph so that the difference between them (profit) can be easily seen). You should develop a reference mode for every variable you identified in Part B1.

Full points if they draw for all their variables, can make minor deductions for major errors (i.e. behaviors that don't follow logic) or for readability



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Homework 1: Problem Definition and Feedback Structure



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Homework 1: Problem Definition and Feedback Structure

C. Identifying the Structure of the System

Drawing on the case study and your analysis above, create a causal loop diagram that captures the important feedback loops underlying the dynamics you described above. Your diagram should include the loops you believe are essential to understanding and explaining the dynamics but should be simple enough to understand. Remember that your client cannot understand a diagram that includes everything. ***Be sure to identify the polarity of each link and loop.***

- C1 (25 Points). Draw a causal loop diagram by hand that captures the feedback structure of the system. Use your hypothesized reference mode and associated structure as a basis for your causal loop diagram. Your causal loop diagram should include all your variables identified in part B. You may add additional variables as needed to ensure you create unambiguous loops. (you should have no more than ~15 Variables, and 2-4 Loops)

Focus on correctness of links and polarities, and loop identification. Again this should start at full points, and make deductions for errors that are inconsistent with Vensim syntax or for glaring errors with SD principles. As instructors don't have the same background as cadets in each topic area, can make general minor deductions such as "several causal links don't appear causal", but not major deductions since they may have found research that implies causation.

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Homework 1: Problem Definition and Feedback Structure

- C2 (10 Points). Briefly describe how the loops you identify in your causal loop diagram create the behavior you describe in your reference mode. A good explanation captures the causal relationships (feedback) that generate the behaviors in the reference mode. You should explain the thought behind each of the feedback loops. Additionally, your explanation should show how the loops interact to create the reference mode.

5pts identifying and discussing all loops they have labeled, 5pts for correctness of their explanation on how their structure drives behavior in the reference modes

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Homework 1: Problem Definition and Feedback Structure

Outstanding Student Work

UNITED STATES MILITARY ACADEMY

HOMEWORK 1: PROBLEM DEFINITION AND FEEDBACK STRUCTURE

SM484: SYSTEM DYNAMICS SIMULATION

SECTION H1

MAJ PATRICK LUPFER

BY

CDT ADRIANA FRIAS '26, CO I2

WEST POINT, NEW YORK

10 SEPTEMBER 2025

AF I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

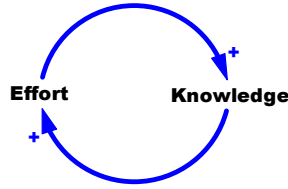
 I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

AF I ACKNOWLEDGE THAT I USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

 I ACKNOWLEDGE THAT I DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: _____ DATE: 10 SEPTEMBER 2025

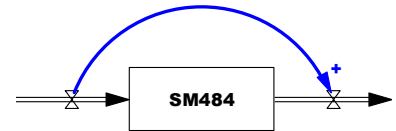


System Dynamics Simulation

SM 484 – AT 26-1

MAJ Jake Lueders

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Homework 1: Problem Definition and Feedback Structure

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Homework 1: Problem Definition and Feedback Structure

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Additional Source 1: Federal Bank of Atlanta: We're getting older and the Labor Market Shows It

Additional Source 2: SHRM: US Labor Shortage Looms Who Will Do the Work?

- ☐ B1 (5 Points). **Select one** of the case studies (Ageing, Artificial Intelligence, Outer Space, or Water) and provide a short paragraph describing what your focus area will be. Case Study: Ageing

Elderly population is growing in number and proportion of the total population for virtually every country. My focus area will be the shrinking workforce caused by an increasing elderly population, specifically labor shortage.

- ☐ B2 (10 Points). There are several important variables required to understand the cases. Make a list of the most important variables or concepts characterizing the problem described in the case study. Your list should be comprehensive but as short as possible (8 to 12). You may aggregate similar concepts when possible.

Variable	Units
Working age population (16-64)	People
Elderly Population (65+)	People
Fertility Rate	Births/woman
Labor Force Participation Rate	People/Total Pop
Immigration	People/Year
Deaths	People/Year
Births	People/Year
Job Opening to Applicant Ratio	Opening/People
Automation and Tech Substitution	Robots per 10,000 workers
Trained Caretakers Population	People
Untrained (Family) Caretakers Population	People
Elderly in Need of Caretakers Population	People

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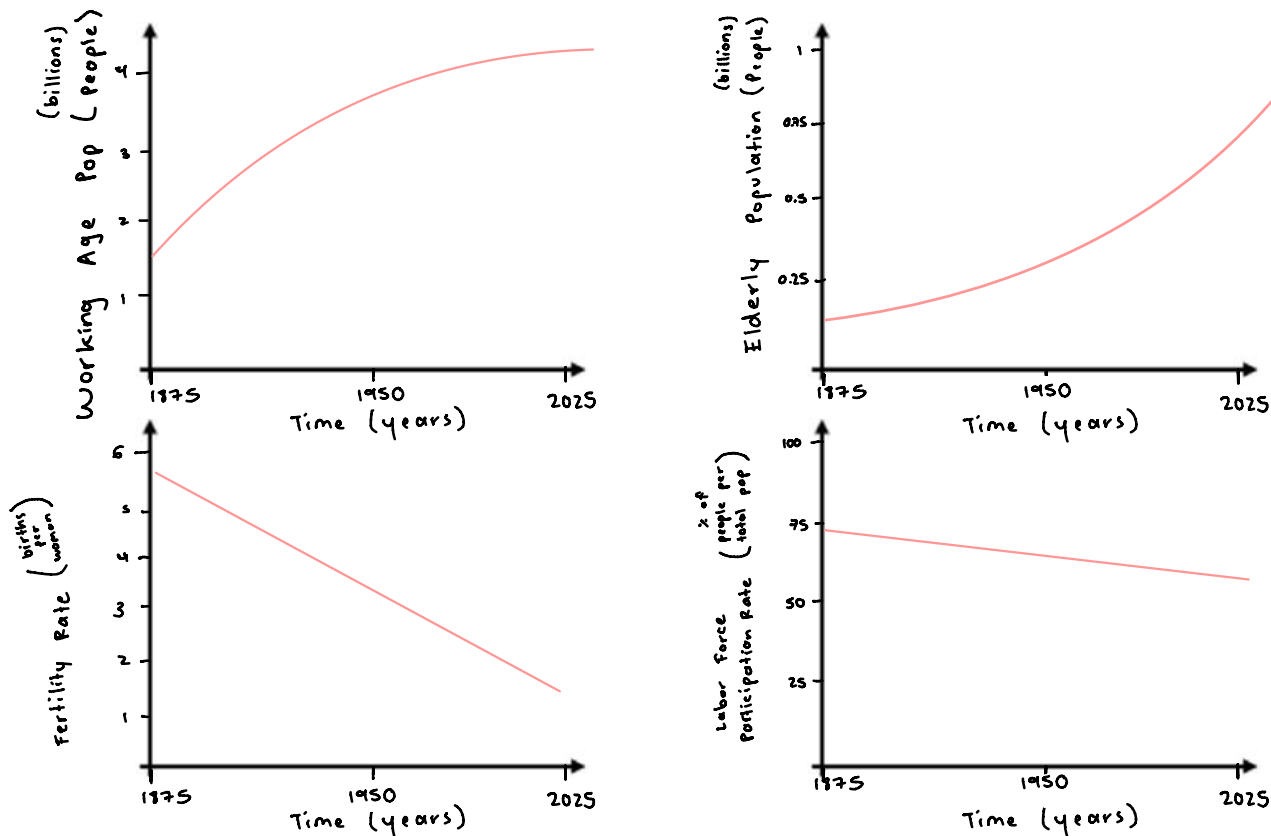
Homework 1: Problem Definition and Feedback Structure

- B3 (5 Points). Use the description and analysis in the case study to depict the reference mode – the set of dynamics that characterize the problematic behavior. First, identify the time horizon over which the dynamics unfold. Do not be constrained by the publication date of the article or the range of data given in the case study. Select a time period long enough to capture the dynamics of the critical variables, including their past development and enough of the future for the dynamics and impact of policies to play out.

Time Horizon: 150 years

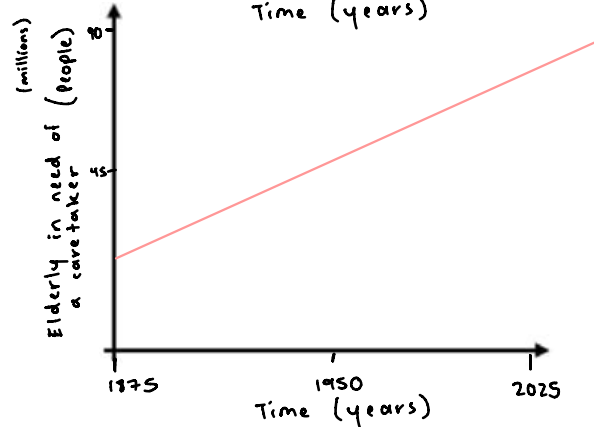
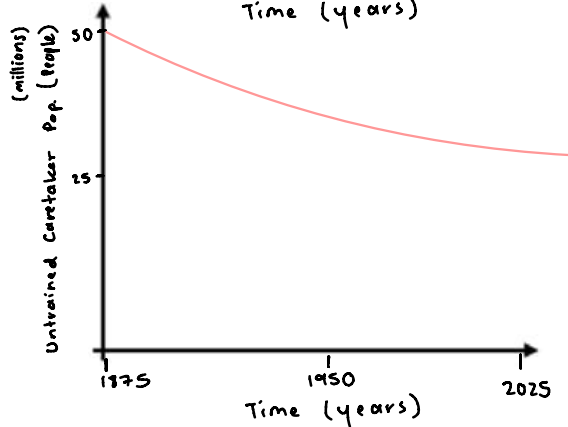
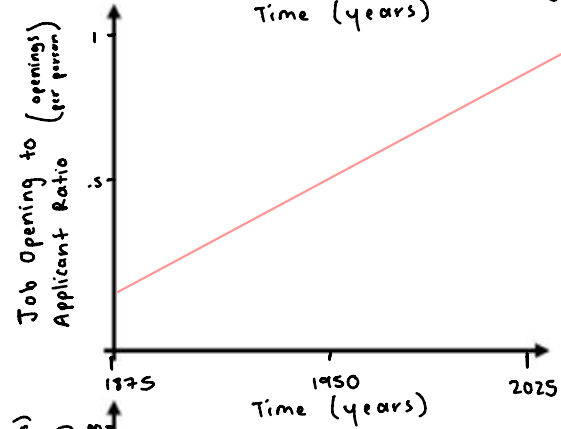
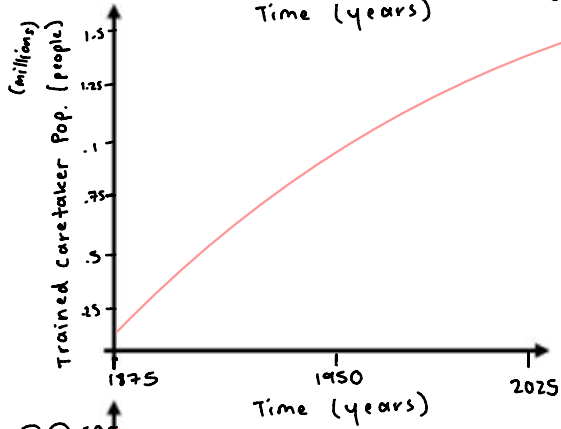
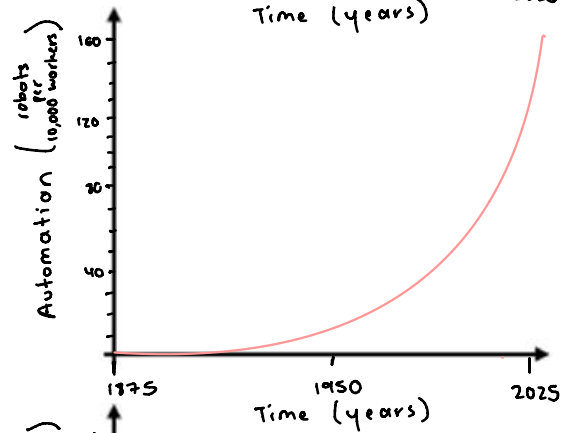
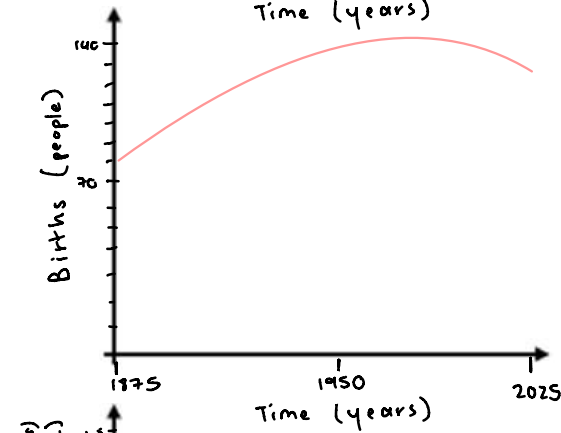
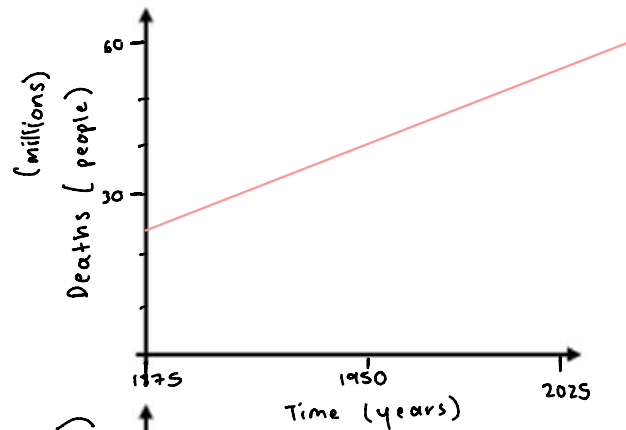
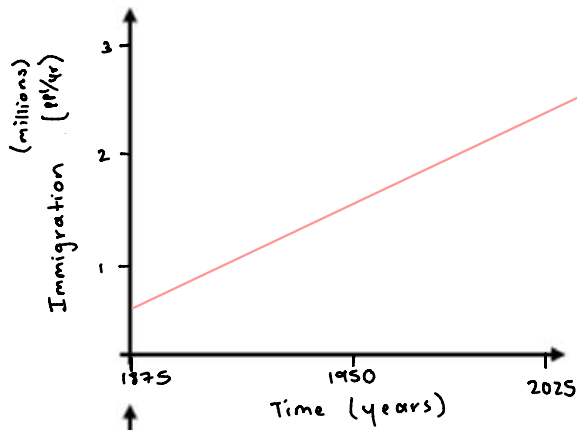
Why Chosen? Its a long enough time period to see norms before and how they have changed in recent times causing the problem observed.

B4 (20 Points). Next, sketch a graph showing your best estimate for the behavior of each variable over the time horizon. You do not need to gather actual data for the variables you identify. In the early stages of problem definition, recognizing the fundamental modes of behavior are more important than numerical precision. For the purposes of this assignment, it is sufficient to use your best judgment to estimate the behavior of these variables. Be sure to provide appropriate scales and units of measure for your estimates. If two or more variables have the same units of measure, plot them on the same scales (for example, in a model of a business one would show revenue and cost on the same graph so that the difference between them (profit) can be easily seen). You should develop a reference mode for every variable you identified in Part B1.



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Homework 1: Problem Definition and Feedback Structure



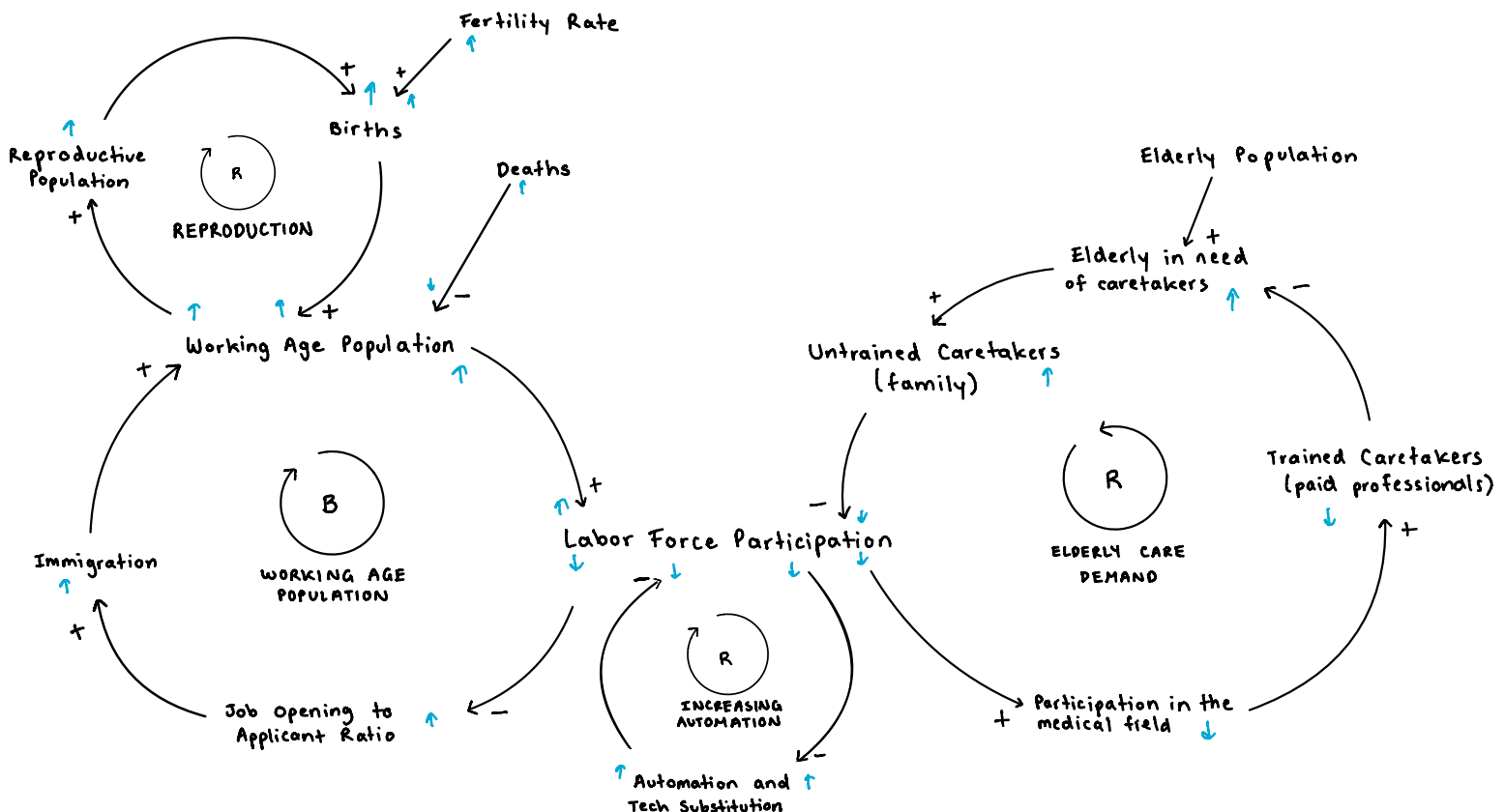
SM 484: System Dynamics Simulation – AT 26-1

Homework 1: Problem Definition and Feedback Structure

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Homework 1: Problem Definition and Feedback Structure

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The Elderly Care Demand loop shows how as labor force participation goes down, participation in the medical field goes down, which results in a lowered amount of trained caretakers, which results in an increased number of elderly people in need of a caretaker, leading to family members of working age having to step in as caretakers, ultimately created an even lower labor force participation. This is a reinforcing loop that shows how the labor force is depleted over time as the elderly population rises and requires family to care for them instead of work. This reflects the behavior of the trained and untrained caretaker reference modes, as they have both started flattening out and changing directions in recent years due to the labor shortage and increasing elderly population in need of medical care.

The Working Age Population loop shows how decreased labor force participations leads to a higher job opening to applicant ratio, causing more immigration for job opportunities which adds to the working age population and increases labor force participation. This reflects the behavior of the immigration reference mode increasing at a similar linear rate as the job opening to applicant ratio. the promise of economic opportunity attracts people to immigrate. However, the immigration rate is not increasing as rapidly as the working age population is decreasing, which results in the decreasing behavior of the labor force participation reference mode.

The Reproduction loop shows how fertility rate affects births per year, which contributes to the size of the working population, who in turn contain the reproductive population that influences the number of births. This is a reinforcing loop that shows how the rate and scale at which people are reproducing influences the size of the working population. This reflects the behavior of the births and working age population reference modes. Because the Fertility Rate decreases rapidly over time, the number of births per year starts to decrease significantly towards the end, and so influences the decreasing behavior of the working age population reference mode.

The Increasing Automation loop shows how as labor force participations decreases, automation and the creation of technological substitutions increases, which in turns decreases the rate of labor force participation. This is a reinforcing loop that shows how automation and labor force participation cyclically affect each other. This reflects the decreasing behavior of the labor force participation reference mode. As technological advancements have boomed in recent years, the ability for machines to substitute people has only worsened the shrinkage of the labor force.

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The Reproduction loop and Immigration Need loop interact to influence the decreasing behavior of the working age population reference mode. In recent years the fertility rate has gone down, which as shown in the Reproduction loop reduces the Working Age population, and so is resulting in the decrease in labor force participation. This behavior is further explained by the Elderly Care Demand loop which shows how elderly need for caretakers is pulling family members, or untrained working aged people, from participating in the labor force and how this effect snowballs. This behavior is also influenced by the Increasing Automation loop which shows the inverse relationship of technological automation and labor force participation. As the workforce shrinks, more technology is produced to automate those tasks. However, as more technology to automate labor is produced, less people need to participate in the labor force, and so the effect snowballs resulting in the decreasing labor force participation rate seen in the reference mode and ultimately the decreasing population of working age people also seen in the reference mode.

Works Cited

- Davidson, Charles. "Population Aging Is the Single Biggest Reason for the US Worker Shortage, Fueling Inflation. This Article Reviews Research That Shows Almost All of the Increase in the 16-And-Older Population Has Been among Seniors." *Www.atlantafed.org*, 12 Jan. 2023,
www.atlantafed.org/economy-matters/economic-research/2023/01/12/we-re-getting-older-and-the-labor-market-shows-it.
- Maurer, Roy. "U.S. Labor Shortage Looms: Who Will Do the Work?" *Shrm.org*, 14 Jan. 2025,
www.shrm.org/topics-tools/news/hr-quarterly/us-labor-shortage-looms-who-will-do-work.
- United Nations. "Global Issues: Ageing." *United Nations*, 2024,
www.un.org/en/global-issues/ageing.

Acknowledgment of Assistance

ChatGPT. *Assistance given to the author, AI. I used the following prompts in ChatGPT: “Cite the following pdfs in accordance with the history department style and formatting guide attached (do not cite the style and formatting guide)” and “What are units measuring participation in the medical field?” The information obtained from the query was used to generate units for my variables.* OpenAI, (<https://chatgpt.com/c/67e2bb24-d290-800a-b46d-b154faa095d3>). West Point, NY, 25MAR2025.

Aycock, Walker CDT F-4 '26. Assistance given to the author, verbal and written discussion. After drafting my causal loop diagram, I asked CDT Walker to review my diagram for logic. He two revisions he had for me were to make a handwritten copy as I had originally made it digital, and to change where I had fertility rate because the flow did not make sense. With this knowledge, I changed my diagram to a handwritten version and asked my instructor for help revising the fertility rate flow. West Point, NY, 10OCT2025.



UNITED STATES MILITARY ACADEMY
WEST POINT.

Homework 2

Instructor Solution

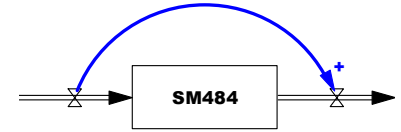
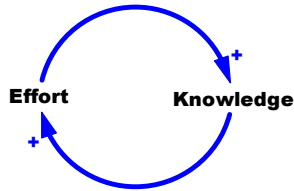


System Dynamics Simulation

SM 484 – AT 26-1

MAJ Jake Lueders

MAJ Pat Lupfer



Homework 2: Modeling a Pandemic

Assigned: 05SEP25; Due: 19SEP25 at 2359

75 Points; *This is an individual assignment*

In this assignment, you will use Vensim software to explore how an infectious disease propagates through a population of susceptible individuals. Infectious diseases are critical management and public policy challenges (consider SARS, COVID-19, HIV, or H1N1) and provide a useful setting to explore *feedback*—how the states of a system (the *levels* or *stocks*) influence (or “feedback” to) the flows that alter those states. This assignment also introduces you to the structure and behavior of fundamental feedback systems.

These systems are the building blocks from which systems that are more complex are composed. In particular, the feedback structure governing the spread of a contagious disease is also important in the growth of new products, the diffusion of innovations, the spread of rumor and riot, and other contexts of social contagion.

SUBMISSION: *Paste or type your answers into this PDF, add your DAAW coversheet and citations, and submit. You should have more than enough space. If you find that you need more, please make a note under that question and add pages to the end of this document. **No need to submit your model** if you follow the instructions and answer the questions COMPLETELY.*

A. Build the Base Model

On Canvas, this assignment also contains a VensimPLE tutorial called “SM484 Vensim Tutorial AY26-1”. Make sure you also download the file: Italy_Covid_Cases.vdf, which has the Italy COVID data embedded in it. Put the Italy COVID file in the Vensim directory. The tutorial will walk you through the development of the pandemic model.

- ☐ A1 (25 points) Follow the tutorial to build the model it describes. Be sure to document your model by adding enough commentary and explanation so that the meaning of each variable, the reasoning behind each formulation, and the data sources for each parameter value are clear to your audience. Do not wait until you are done to fill in the documentation. Document each equation as you create it. Doing so saves time in the long run. Forcing yourself to describe, in writing, the rationale for each formulation and sources for each parameter tests your understanding: if you cannot write a concise, clear description of a formulation, you probably do not understand it yourself. Documenting as you go avoids costly delays in the discovery of errors and rework.

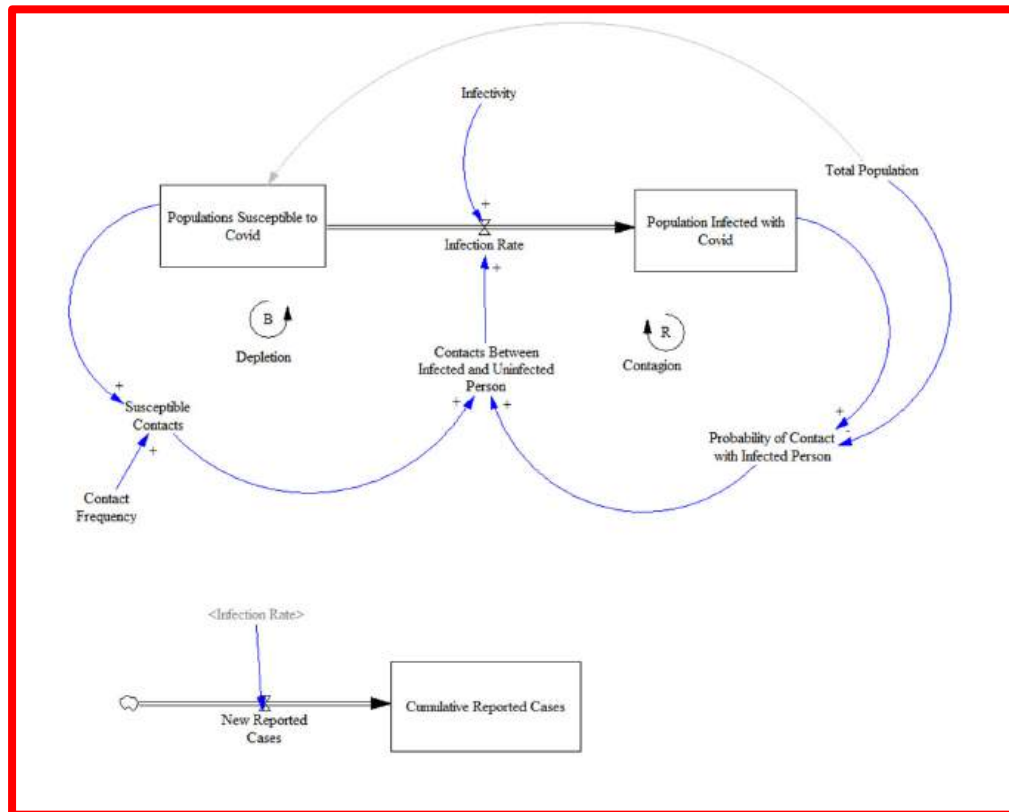
* Updated by MAJ Jake Lueders for VensimPLE v 10.2, from material developed by Professors Hazhir Rahmandad and John Sterman.

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Homework 1 - Modeling a Pandemic

Always provide the units of measure for every variable, and check that every equation is dimensionally consistent. Vensim can test your model for dimensional consistency (see the tutorial). Models that fail the dimensional consistency test are meaningless. Often dimensional errors are symptoms of more serious conceptual difficulties.

- ☐ a. Submit a clear screenshot of the stock and flow diagram of your model. If you click the “Lock Sketch” button, you can take a screenshot with all the handles removed from the causal links.



- ☐ b. Provide the equation list from your model using the document all function, ensure you have units and a comment for each variable you’ve created.

(01) Contact Frequency= 2.5, but numbers that are logical and produce a valid model are acceptable

Units: Persons/Persons/Day

(02) Contacts Between Infected and Uninfected People =
Susceptible Contacts*"Probability of Contact w/ Infected Person"

Units: People/Day

(03) Cumulative Reported Cases= INTEG (
New Reported Cases), Initial Value = 0

Units: Persons

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Homework 1 - Modeling a Pandemic

- (05) Infection Rate=
Contacts Between Infected and Uninfected Persons*Infectivity
Units: Persons/Day
- (06) Infectivity=
0.05 (or other logical number)
Units: Dmnl
- (08) New Reported Cases=
Infection Rate
Units: Persons/Day
This assumes that new cases are reported immediately
- (11) Probability of Contact w/ Infected Person=
Population Infected with COVID / Total Population
Units: Dmnl
- (13) Susceptible Contacts=
Contact Frequency * Populations Susceptible to COVID
Units: People/Day
Susceptible contacts is the total number of contacts that susceptible people have per day calculated by multiplying the susceptible population by their number of contacts per day.
- (15) Total Population=
240,500
Units: Persons
- (04) FINAL TIME = 150
Units: Day
- (07) INITIAL TIME = 0
Units: Day
The initial time for the simulation.
- (09) Population Infected with COVID = INTEG (
Infection Rate)
Units: Persons
- (10) Population Susceptible to COVID= INTEG
-Infection Rate. Initial Value: Total Population
Units: Persons

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Homework 1 - Modeling a Pandemic

(12) $\text{SAVEPER} =$
 TIME STEP
Units: Day [0,?]
The frequency with which output is stored.

(14) $\text{TIME STEP} = 0.25$
Units: Day [0,?]
The time step for the simulation.

☐ A2 (15 points) Answer the following questions and provide output graphs for each question. You can plot multiple simulation runs on the same graph to answer the questions.

☐ a. What happens when you initialize the stock of infected people at zero? Briefly explain why. How do the dynamics change if you assume that one or more members of the population in question are already infected?

(-1 if no output graph) When you initialize the population at zero, the model 'breaks'—it is no longer able to examine the effects of the virus. This is because no one in the population is susceptible, therefore, the disease cannot penetrate the population. Said another way, the probability of coming into contact with an infected person goes to zero.

If you assume that one or more members of the population is already infected, the probability of contact with an infected person becomes positive, driving the contagion reinforcing feedback loop

☐ b. How do the dynamics change as the contact frequency increases? Does changing the contact frequency influence the total number of people who get COVID? Explain why or why not with reference to the structure of the model.

(-1 if no output graph) As contact frequency increases, the number of susceptible contacts increases, which drives the depletion feedback loop. The dynamics of this loop are governed by the total number of people in the population. Increasing the contact frequency does not change the cumulative number of people who can get the disease; however, it does increase the number of people who contract the disease sooner. On the other hand, a lower contact frequency could result in a lower number of cases, but that is determined by the time horizon. Eventually, everyone in this model becomes infected, given a long enough time horizon.

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Homework 1 - Modeling a Pandemic

- c. How does the time horizon (the length of the simulation) change the apparent shape of the curves? Why? What could you adjust to make the time horizon more impactful?
- To answer the question on time horizon, you need to examine your simulation over different periods of time. You can run the model once with a long time horizon (e.g., 1000 days) and then change the time range viewed in graphs and tables. Select Time Axis from the Control Panel dialog box to change the time frame displayed in graphs and tables (see “The Control Panel” in the tutorial or User’s Guide). Vensim allows you to load multiple runs (or “datasets”) and then view the output of different runs in the same graphs and tables. See tutorial and the section on “The Control Panel” in the User’s Guide for details on selecting datasets.

(-1 if no output graph) Given that there is one balancing and one reinforcing feedback loop, we continue to see s-shaped growth for the infected population and cumulative cases. Changing the exogenous variables of the model results in steeper/flatter curves and/or delays in the total population becoming infected. This change occurs because the virus has more time to move throughout the population. When we zoom in to a small time horizon, we may not see the entire change based on the exogenous variable values. If we zoom far out (i.e. 1000 days), we can change exogenous variables that pushes the time far to the right when all people contract the disease.

B. Improving Your Model

All models are wrong; however, some models are useful. In your critique of the model, you may have identified a wide range of unrealistic assumptions. In this section, you will relax one such assumption. Make one change at a time and test it thoroughly, rather than making multiple changes at once. With multiple changes, it is impossible to determine which altered assumption is responsible for any changes in behavior.

- B1 (20 points) To this point, we’ve assumed people remain infectious indefinitely. In epidemiology, this is known as the Susceptible-Infectious (SI) model. The SI model is appropriate for chronic infections that the body cannot clear and for which there is no cure. For most infectious diseases, however, including COVID, smallpox, chicken pox, measles, and influenza, patients either recover (or die).

Reformulate your model to include the removal of infectious people from the stock of infectious individuals. In epidemiology, the revised model is known as the Susceptible-Infectious-Removed (SIR) model.

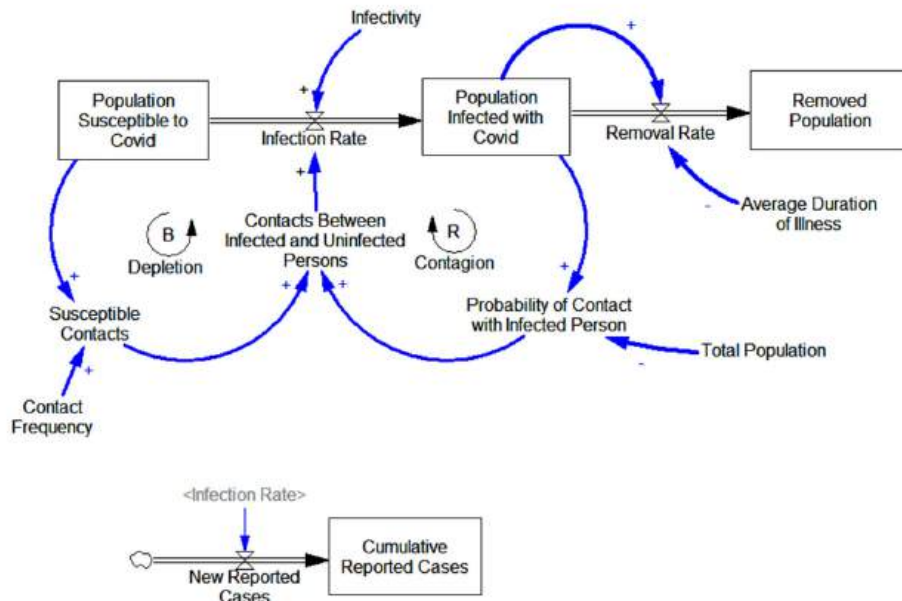
To modify your model, first create a new stock, the “Removed” population, placing it to the right of the population infected with COVID. Next create a new flow, called the “Removal Rate”, that moves people from the Infectious to the Removed states (see the

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tutorial for creating stock and flow variables). Note: For simplicity, we will not distinguish mortality from removal, so the removal flow includes those recovering and those dying.

Next, add a variable representing the average duration of infectivity (that is, how long people remain infectious, on average). Now you need to capture the feedback structure determining the rate at which people are removed. The removal rate must depend on the population infected with COVID and the average duration of their illness.



There are several formulations for recovery, but the simplest (and most widely used) is:

$$\text{Removal Rate} = \text{Population Infected with COVID} / \text{Average Duration of Illness}$$

To implement this formulation, you need to add causal links connecting the infected stock and the average duration of infectivity to the removal flow. Next, you must change the equation for the infected population, so that the removal flow is now included as an outflow that drains the infected stock. Alter the equation for the Population Infected with COVID to read:

$$\text{Population Infected with Covid} = \text{INTEG}(\text{Infection Rate} - \text{Removal Rate})$$

The equation states that the infectious population increases as people become infectious and decreases as people are removed from the population susceptible to COVID; the population infected with COVID accumulates the difference between infection and

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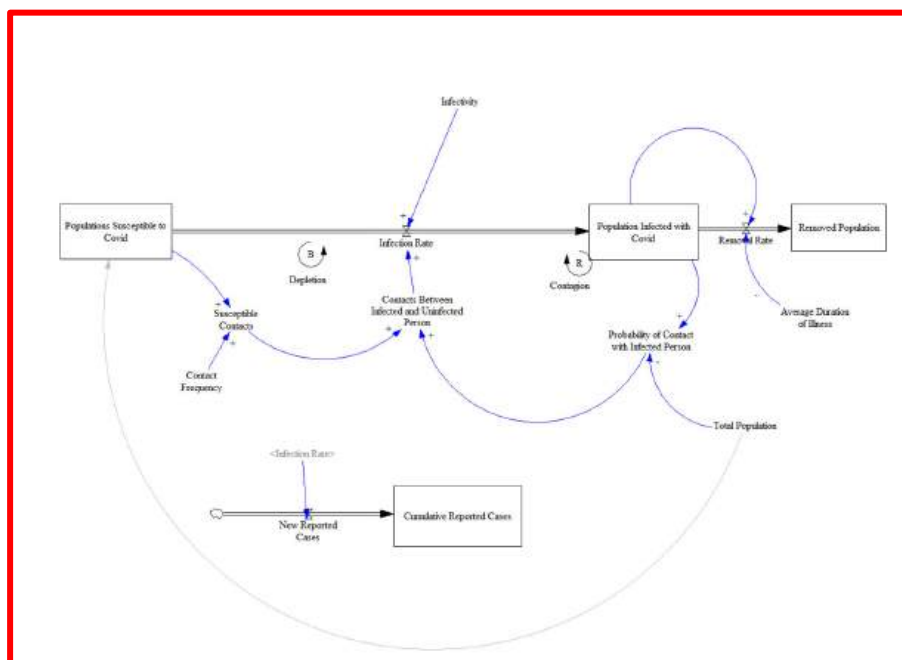
Homework 1 - Modeling a Pandemic

removal. Note that the infectious population is no longer the same as the cumulative number of cases.

Finally, you must estimate a value for the average duration of infectivity. Epidemiologists believe the infectious stage of COVID lasts about 1 to 3 weeks (10 - 20 days). Use your judgment to select a base case value. Be sure to make any notes in the comments section of the equation editor about your assumed duration and include them in your report.

Make sure your revised model is documented and dimensionally consistent (Check Units).

- ☐ a. Submit a clear screenshot of the stock and flow diagram of your new model.



- ☐ b. Provide the additional equations you added to your model using the document all function, ensure you have units and a comment for each variable you've created.

(16) Removal Rate=

Population Infected with COVID/Average Duration of Illness

Units: Persons/Day

The removal rate is equal to the people who are infected with COVID (initialized with the total population) divided by the Average Duration of Illness in days.

(17) Average Duration of Illness=14

Units: Day

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Homework 1 - Modeling a Pandemic

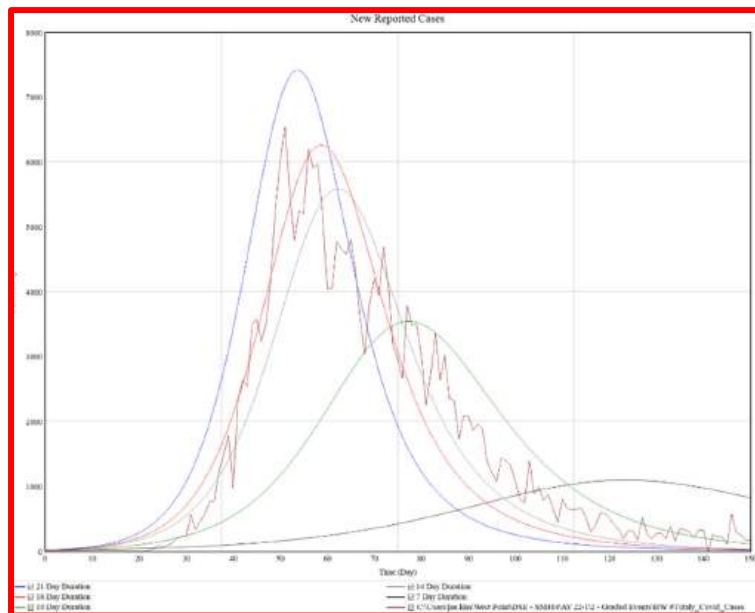
(18) $\text{Removed Population} = \text{integ}(\text{Removal Rate})$
Units: Persons

- ☐ B2 (15 points) Analyze your model to determine the effect of adding removal, provide output graphs when appropriate to show the different dynamics of the models.

- ☐ a. What kind of feedback loop have you added to your model? How do you know?

Balancing. As the population infected with COVID increases, so does the removal rate; however, as the removal rate increases, the population infected with COVID goes down.

- ☐ b. How do the dynamics change as the duration of infectivity is increased or decreased? Vary the key parameters over a wide range (try 4+ different numbers for duration) to be sure you understand the possible dynamics of the system. Provide an output graph with the various durations indicated as separate simulation runs.



- ☐ c. How do the dynamics differ from those of the SI model in which there is no removal? Explain with reference to the structure of your model.

In the SIR model, in contrast to the SI model, people once infected can be removed from the population within the system. The model now captures immunity and recovery and no longer assumes that once you get the disease, it is permanent/chronic. Additionally, an increase in the average duration of infection

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Homework 1 - Modeling a Pandemic

drives the contagion loop, increasing the magnitude of the peaks for probability of contact w/ an infected person and contacts between infected and uninfected people.

Outstanding Student Work

UNITED STATES MILITARY ACADEMY

HOMEWORK 2

SM484: SYSTEMS DYNAMIC SIMULATION

SECTION H1

MAJ PATRICK LUPFER

BY

CDT GENEVA CESONI '27, CO F3

WEST POINT, NEW YORK

19 SEPTEMBER 2025

GC I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

 I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

 I ACKNOWLEDGE THAT I USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

GC I ACKNOWLEDGE THAT I DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: *Geneva Cesoni* DATE: 19 SEPTEMBER 2025

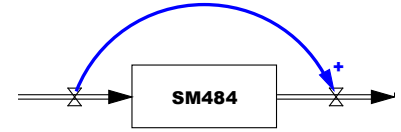
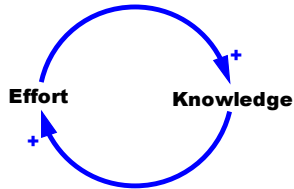


System Dynamics Simulation

SM 484 – AT 26-1

MAJ Jake Lueders

MAJ Pat Lupfer



Homework 1: Modeling a Pandemic

Assigned: 05SEP25; Due: 19SEP25 at 2359

75 Points; *This is an individual assignment*

In this assignment, you will use Vensim software to explore how an infectious disease propagates through a population of susceptible individuals. Infectious diseases are critical management and public policy challenges (consider SARS, COVID-19, HIV, or H1N1) and provide a useful setting to explore *feedback*—how the states of a system (the *levels* or *stocks*) influence (or “feedback” to) the flows that alter those states. This assignment also introduces you to the structure and behavior of fundamental feedback systems.

These systems are the building blocks from which systems that are more complex are composed. In particular, the feedback structure governing the spread of a contagious disease is also important in the growth of new products, the diffusion of innovations, the spread of rumor and riot, and other contexts of social contagion.

SUBMISSION: *Paste or type your answers into this PDF, add your DAAW coversheet and citations, and submit. You should have more than enough space. If you find that you need more, please make a note under that question and add pages to the end of this document. **No need to submit your model** if you follow the instructions and answer the questions COMPLETELY.*

A. Build the Base Model

On Canvas, this assignment also contains a VensimPLE tutorial called “SM484 Vensim Tutorial AY26-1”. Make sure you also download the file: Italy_Covid_Cases.vdf, which has the Italy COVID data embedded in it. Put the Italy COVID file in the Vensim directory. The tutorial will walk you through the development of the pandemic model.

- ☐ A1 (25 points) Follow the tutorial to build the model it describes. Be sure to document your model by adding enough commentary and explanation so that the meaning of each variable, the reasoning behind each formulation, and the data sources for each parameter value are clear to your audience. Do not wait until you are done to fill in the documentation. Document each equation as you create it. Doing so saves time in the long run. Forcing yourself to describe, in writing, the rationale for each formulation and sources for each parameter tests your understanding: if you cannot write a concise, clear description of a formulation, you probably do not understand it yourself. Documenting as you go avoids costly delays in the discovery of errors and rework.

* Updated by MAJ Jake Lueders for VensimPLE v 10.2, from material developed by Professors Hazhir Rahmandad and John Sterman.

Homework 2 - Modeling a Pandemic

☐ a. Submit a clear screenshot of the stock and flow diagram of your model. If you click the “Lock Sketch” button, you can take a screenshot with all the handles removed from the causal links.



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Homework 2 - Modeling a Pandemic

- ☐ b. Provide the equation list from your model using the document all function, ensuring you have units and a comment for each variable you've created.

(01) Contact Frequency=

6

Units: Persons/Persons/Day

Contact Frequency is a constant.

(02) Contacts Between Infected and Uninfected Persons=

Probability of Contact with Infected Person*Susceptible Contacts

Units: Persons/Day

Contacts Between Infected and Uninfected Persons is equal to the product of Probability of Contact with Infected Person and Susceptible Contacts.

(03) Cumulative Reported Cases= INTEG (

New Reported Cases,

0)

Units: Persons

Cumulative Reported Cases is equal to New Reported Cases.

(04) FINAL TIME = 150

Units: Day

The final time for the simulation.

(05) Infection Rate=

Contacts Between Infected and Uninfected Persons*Infectivity

Units: Persons/Day

The infection rate is determined by the total number of contacts between infected and uninfected persons each day (Contacts Between Infected and Uninfected Persons), and the probability that each such contact results in transmission from the infected to uninfected person (Infectivity).

(06) Infectivity=

0.02

Units: Dmnl

Infectivity is a constant.

(07) INITIAL TIME = 0

Units: Day

The initial time for the simulation.

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Homework 2 - Modeling a Pandemic

- (08) New Reported Cases=
Infection Rate
Units: Persons/Day
Assuming new cases are reported immediately, New Reported Cases is equal to Infection Rate.
- (09) Population Infected with Covid= INTEG (Infection Rate,
2)
Units: Persons
The Population Infected with Covid is equal to the Infection Rate.
- (10) Population Susceptible to Covid= INTEG (-Infection Rate,
Total Population)
Units: Persons
The Population Susceptible to Covid is equal to the population susceptible prior to the onset of the disease less all of those that have contracted it. It is initialized to the Total Population, which assumes that all individuals are initially susceptible (no prior natural or vaccine- conferred immunity).
- (11) Probability of Contact with Infected Person=
Population Infected with Covid/Total Population
Units: Dimensionless
Probability of Contact with Infected Person is equal to Population Infected with Covid divided by Total Population.
- (12) SAVEPER =
TIME STEP
Units: Day [0,?]
The frequency with which output is stored.
- (13) Susceptible Contacts=
Contact Frequency*Population Susceptible to Covid
Units: Persons/Day
Susceptible Contacts is equal to the product of Contact Frequency and Population Susceptible to Covid.

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Homework 2 - Modeling a Pandemic

(14) TIME STEP = 0.25

Units: Day [0,?]

The time step for the simulation.

(15) Total Population=
240500

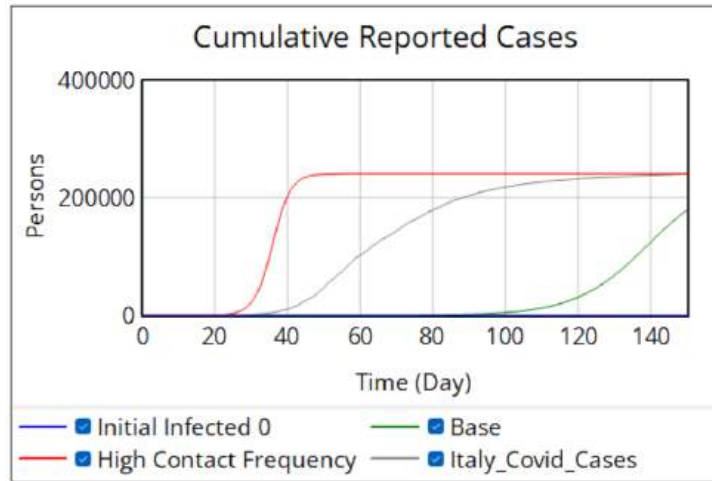
Units: Persons

Total Population is a constant.

SM 484: System Dynamics Simulation – AT 26-1

Homework 2 - Modeling a Pandemic

- ☐ A2 (15 points) Answer the following questions and provide output graphs for each question. You can plot multiple simulation runs on the same graph to answer the questions.
- ☐ a. What happens when you initialize the stock of infected people at zero? Briefly explain why. How do the dynamics change if you assume that one or more members of the population in question are already infected?



When I initialized the stock of infected people at zero, the chart shows that when it starts at zero, it stays at zero (the horizontal blue line staying at 0). This occurs due to the variable equations that lead up to the Cumulative number of Reported Cases and what happens to them when infected people is set at zero. It causes a chain reaction:

1. Probability of Contact with Infected Person= **Population Infected with Covid (0) /Total Population**
2. Contacts Between Infected and Uninfected Persons=
Probability of Contact with Infected Person (0)*Susceptible Contacts
3. Infection Rate= **Contacts Between Infected and Uninfected Persons (0)*Infectivity**
4. New Reported Cases= **Infection Rate (0)**
5. Cumulative Reported Cases= **New Reported Cases (0)**

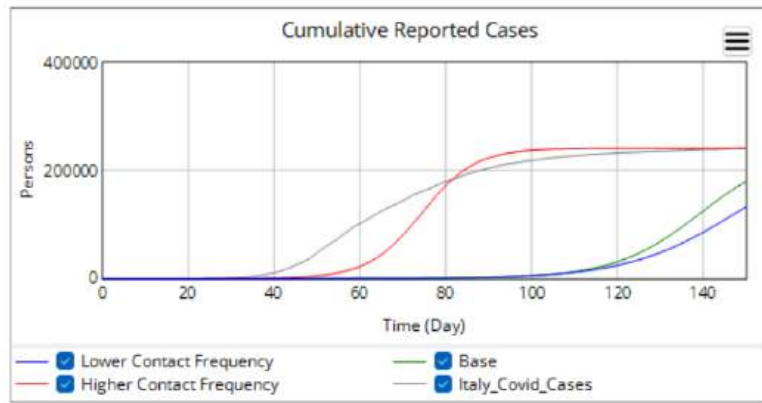
All because I **changed the numerator of Population Infected to zero in equation 1**, it caused all of the above variables to also be equal to zero, eventually leading to Cumulative Reported Cases being zero, as shown in the graph.

When I make the initial amount of people infected one or more, this causes an increase in the cumulative number of reported cases. The higher the increase in initial people infected, the faster the cumulative number of reported cases will increase as well.

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Homework 2 - Modeling a Pandemic

- ☐ b. How do the dynamics change as the contact frequency increases? Does changing the contact frequency influence the total number of people who get COVID? Explain why or why not with reference to the structure of the model.



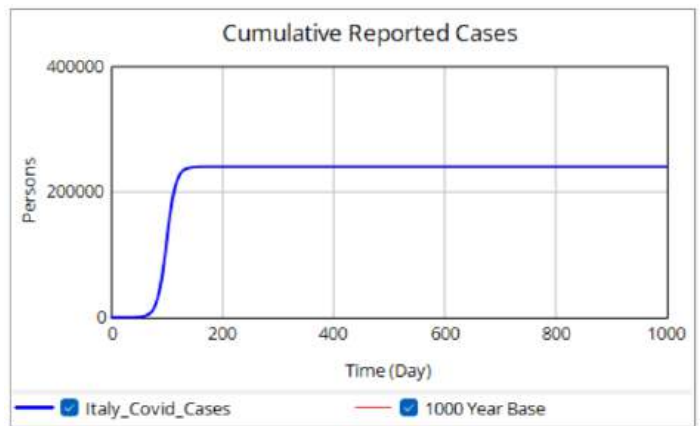
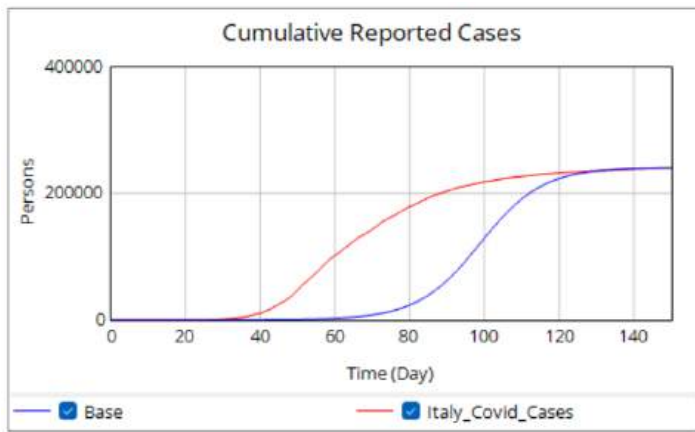
As contact frequency increases, Cumulative Reported Cases increases exponentially and then into goal seeking behavior to form a s-shaped growth. This means that it increases rapidly and eventually plateaus with a higher contact frequency.

Yes, Changing the contact frequency influences the total number of people who get COVID by increasing it as well. This is because an increase in Contact Frequency increases Susceptible Contacts, which then increases Contacts Between Infected and Uninfected Persons, which then increases Infection Rate. Infection Rate is an inflow to the Population Infected with Covid, so an increase in it leads to an increase in the total number of people who get COVID.

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- ☐ c. How does the time horizon (the length of the simulation) change the apparent shape of the curves? Why? What could you adjust to make the time horizon more impactful?
- To answer the question on time horizon, you need to examine your simulation over different periods of time. You can run the model once with a long time horizon (e.g., 1000 days) and then change the time range viewed in graphs and tables. Select Time Axis from the Control Panel dialog box to change the time frame displayed in graphs and tables (see “The Control Panel” in the tutorial or User’s Guide). Vensim allows you to load multiple runs (or “datasets”) and then view the output of different runs in the same graphs and tables. See the tutorial and the section on “The Control Panel” in the online User’s Guide for details on selecting datasets.



The Time Horizon changes the apparent shape of the curves because over a longer period of time, less detail can be seen. The actual shapes of the curves are not changing, but what can be seen is. The difference in behavior of the curves for the Base and for Italy_Covid_Cases can clearly be seen in the 150 day time horizon. However, since 1000 days is so zoomed out, we can only see the major change in slope behavior for each line, which is the exact same. We cannot see the differences of the two lines in the 1000 day time horizon. For the time horizon to be most impactful, the time horizon should stay at 150, since it shows a clearer demonstration of the two curves differences.

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Homework 2 - Modeling a Pandemic

B. Improving Your Model

All models are wrong; however, some models are useful. In your critique of the model, you may have identified a wide range of unrealistic assumptions. In this section, you will relax one such assumption. Make one change at a time and test it thoroughly, rather than making multiple changes at once. With multiple changes, it is impossible to determine which altered assumption is responsible for any changes in behavior.

- B1 (20 points) To this point, we've assumed people remain infectious indefinitely. In epidemiology, this is known as the Susceptible-Infectious (SI) model. The SI model is appropriate for chronic infections that the body cannot clear and for which there is no cure. For most infectious diseases, however, including COVID, smallpox, chicken pox, measles, and influenza, patients either recover (or die).

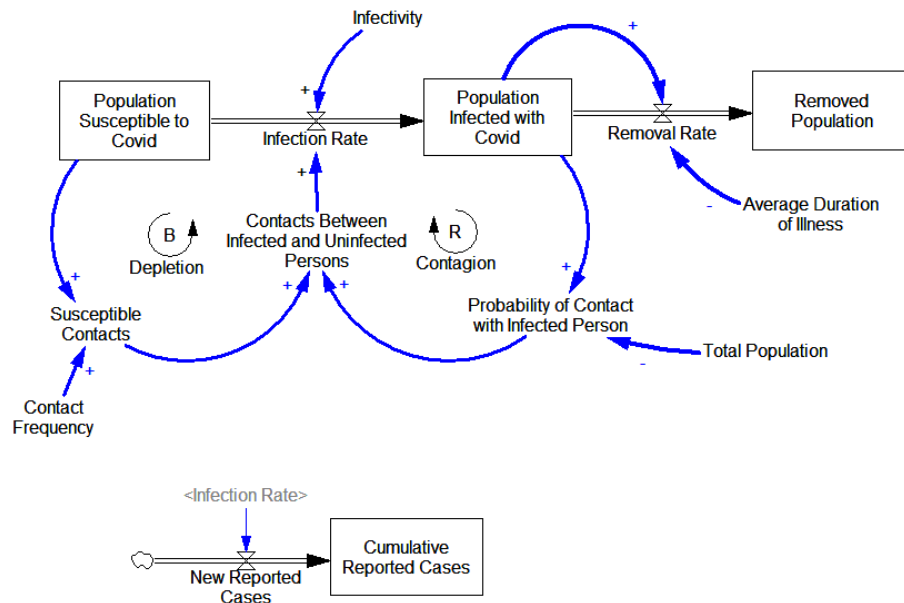
Reformulate your model to include the removal of infectious people from the stock of infectious individuals. In epidemiology, the revised model is known as the Susceptible-Infectious-Removed (SIR) model.

To modify your model, first create a new stock, the "Removed" population, placing it to the right of the population infected with COVID. Next create a new flow, called the "Removal Rate", that moves people from the Infectious to the Removed states (see the tutorial for creating stock and flow variables). Note: For simplicity, we will not distinguish mortality from removal, so the removal flow includes those recovering and those dying.

Next, add a variable representing the average duration of infectivity (that is, how long people remain infectious, on average). Now you need to capture the feedback structure, determining the rate at which people are removed. The removal rate must depend on the population infected with COVID and the average duration of their illness.

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There are several formulations for recovery, but the simplest (and most widely used) is:

$$\text{Removal Rate} = \text{Population Infected with COVID} / \text{Average Duration of Illness}$$

To implement this formulation, you need to add causal links connecting the infected stock and the average duration of infectivity to the removal flow. Next, you must change the equation for the infected population, so that the removal flow is now included as an outflow that drains the infected stock. Alter the equation for the Population Infected with COVID to read:

$$\text{Population Infected with Covid} = \text{INTEG}(\text{Infection Rate} - \text{Removal Rate})$$

The equation states that the infectious population increases as people become infectious and decreases as people are removed from the population susceptible to COVID; the population infected with COVID accumulates the difference between infection and removal. Note that the infectious population is no longer the same as the cumulative number of cases.

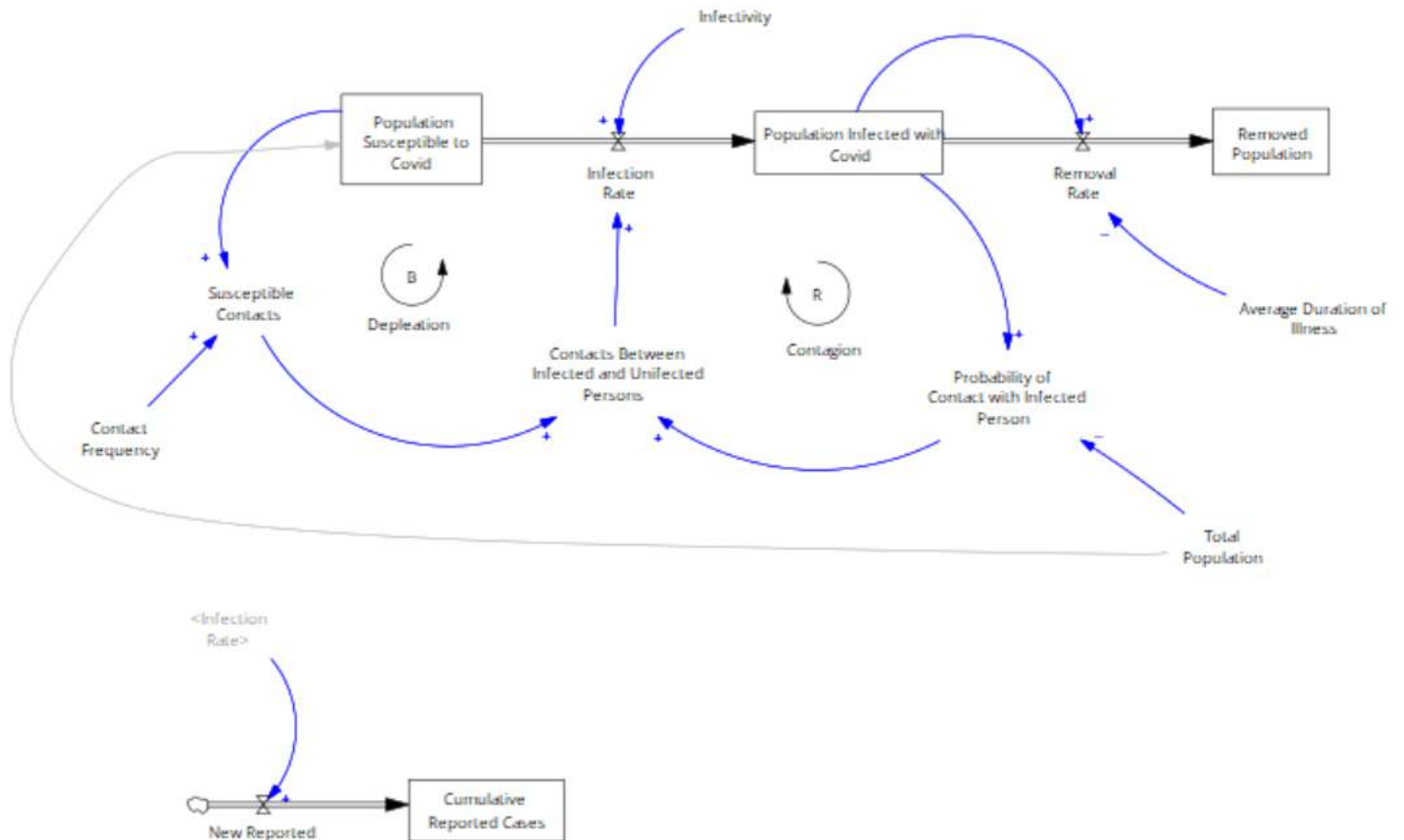
Finally, you must estimate a value for the average duration of infectivity. Epidemiologists believe the infectious stage of COVID lasts about 1 to 3 weeks (10 - 20 days). Use your judgment to select a base case value. Be sure to make any notes in the comments section of the equation editor about your assumed duration and include them in your report.

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Make sure your revised model is documented and dimensionally consistent (Check Units).

- ☐ a. Submit a clear screenshot of the stock and flow diagram of your new model.



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Homework 2 - Modeling a Pandemic

- ☐ b. Provide the additional equations you added or adjusted to your model using the document all function, ensure you have units and a comment for each variable you've created.

(01) Average Duration of Illness=
15

Units: Day

Going off of how long Epidemiologists believe the infectious stage of COVID lasts, the Average Duration of Illness is a constant between 10-20 days.

(10) Population Infected with Covid= INTEG (
Infection Rate-Removal Rate,
2)

Units: Persons

The Population Infected with Covid is equal to the Integral of the Infection Rate minus the Removal Rate.

(13) Removal Rate=
Population Infected with Covid/Average Duration of Illness

Units: Persons/Day

Removal Rate is equal to the Population Infected with COVID divided by the Average Duration of Illness.

(14) Removed Population= INTEG (
Removal Rate,
0)

Units: Persons

The Removed Population is equal to Removed Rate.

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- ☐ B2 (15 points) Analyze your model to determine the effect of adding removal, provide output graphs when appropriate to show the different dynamics of the models.
- ☐ a. What kind of feedback loop have you added to your model? How do you know?

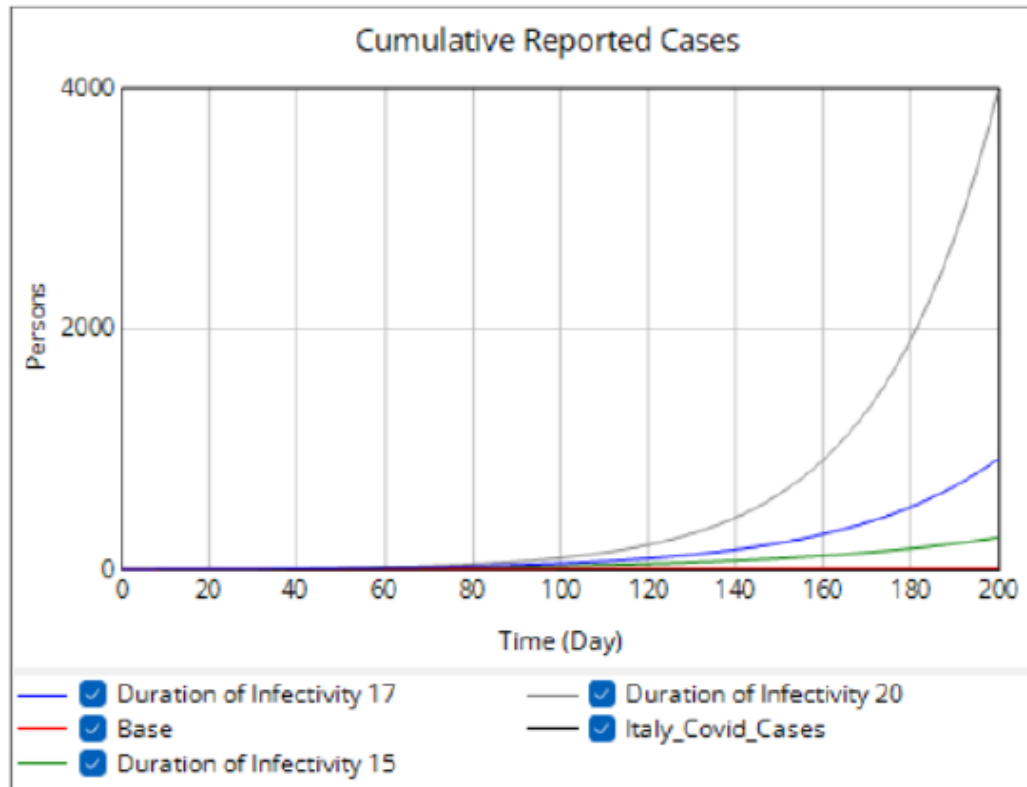
The feedback loop that I added to my model consists of one positive causal link from Population Infected with Covid to the a new flow variable, Removal Rate. Removal Rate flows from Population Infected with Covid to Removed Population, which is a new stock. Finally, a new auxiliary variable, Average Duration of Illness, has been added with a negative causal link to Removal Rate.

Adding these elements may form a balancing feedback loop. This is due to the fact that when the amount of infected people increases, the removal rate increases, and the infected population is reduced. The causal link to Removal Rate from Average Duration of Illness being negative allows this process to be regulated because it adjusts how quickly people move out of the infected stock. I know this can form a balancing loop because growth is counteracting itself and a self-correcting effect is shown.

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Homework 2 - Modeling a Pandemic

- b. How do the dynamics of the system change as the duration of infectivity is increased or decreased? Vary the key parameters over a wide range (try 4+ different numbers for duration) to be sure you understand the possible dynamics of the system. Provide an output graph of the New Reported Cases with the various durations indicated as separate simulation runs.

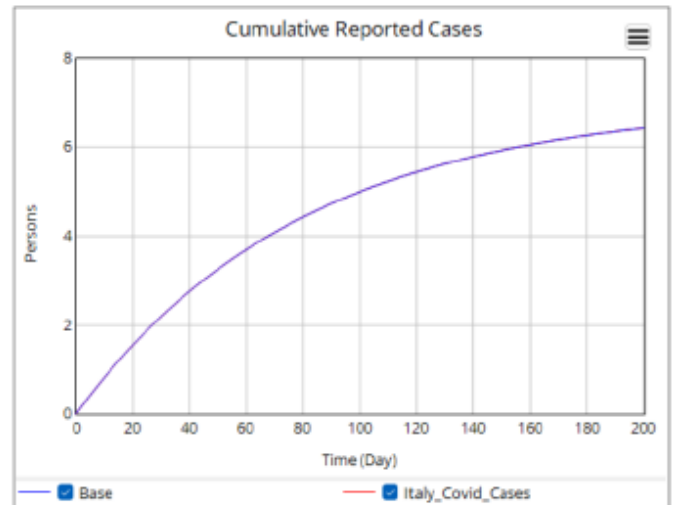
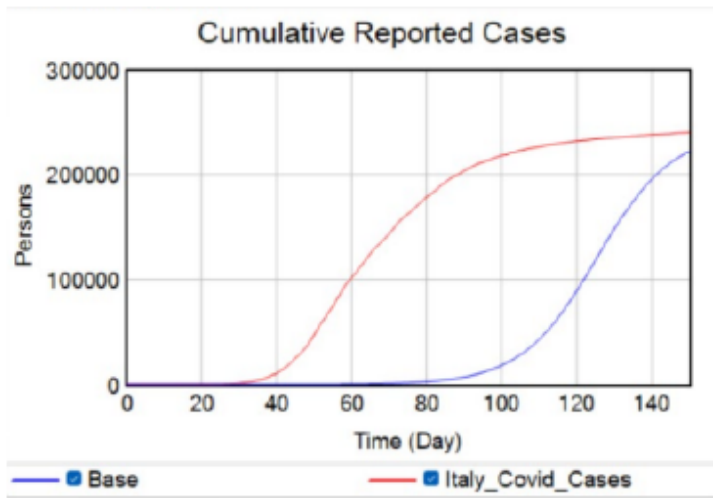


In my understanding of the system's dynamics and output graph, people will stay sick and contagious for more days when the duration of infectivity is longer. This means the number of new cases will increase quicker and reach a peak until it will fall sharply since many people become infected at once. However, when the opposite occurs, and the duration is shorter, people will recover faster. This means the outbreak will spread slower and peak later in time and at a lower value. The disease can possibly fade out before it can spread quickly when the duration is shorter.

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Homework 2 - Modeling a Pandemic

- ☐ c. How do the dynamics differ from those of the SI model in which there is no removal? Explain with reference to the structure of your model.



The system behaves differently from the basic SI model when removal is added. Infected individuals stay in the infected group in the SI model. This means the number of cases continues to increase rapidly until the population is entirely infected. This forms a steep S-shaped curve.

When removal is added, people can't spread the disease anymore after a set time, because they recover. This slows the growth of new cases, and means a lower and slower increase in cumulative cases occurs unlike the SI model. This happens because the infected population keeps decreasing instead of growing with no limitations.

WORKS CITED

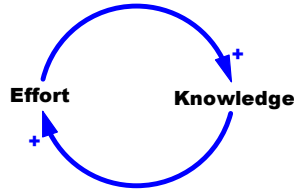
Wegler, CDT Kathryn '26 G-2. Assistance given to the author. CDT Wegler explained to me how to make the graphs for the last two questions. West Point, NY. September 19, 2025.



UNITED STATES MILITARY ACADEMY
WEST POINT.

Homework 3

Instructor Solution

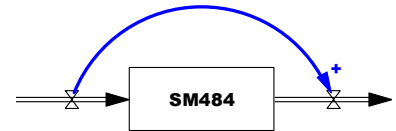


1 System Dynamics Simulation

SM 484 – AT 26-1

MAJ Jacob Lueders

MAJ Patrick Lupfer



Homework 3: Stocks and Flows **SOLUTION**

Assigned: 29 September 2025; Due: 27 October 2025 at 2359
75 Points Total

This is an individual assignment. Follow these submission instructions exactly. You will submit a single PDF containing 11 pages. The DAAW and a space for works cited have been included, as well as a blank extra work page (if needed); simply add your name, the date, and circle your section and instructor. DO NOT convert to word and back to PDF, either type directly into the PDF, or print and handwrite then SCAN it back in using a scanner. DO NOT TAKE A PICTURE AND CONVERT TO PDF UNLESS YOU KNOW HOW TO CONVERT IT BACK TO 8.5"x11". There are scanners in the DSE Cadet Study/Capstone Area and the library, or you can stop by and ask an instructor for help scanning.

This homework gives you practice with the structure and dynamics of stocks and flows. Stocks and flows are the building blocks from which every more complex system is composed. The ability to identify, map, and understand the dynamics of the networks of stocks and flows in a system is essential to understanding the processes of interest in any modeling effort.

A. Identifying Stock and Flow Variables

The distinction between stocks and flows is crucial for understanding the source of dynamics. In physical systems it is usually obvious which variables are stocks and which flows. In human and social systems, often characterized by intangible, "soft" variables, identification is more difficult. An example is provided below.

Name	Type	Units
Example: Inventory of soda	Stock	Cases
Example: Soda order rate	Flow	Cases/week

SM 484: System Dynamics Simulation – AT 26-1**Homework 3 – Stocks and Flows**

- A1 (5 Points) For each of the following variables, state whether it is a stock or a flow, and give units of measure for each. Additionally, identify two stocks and two flows with supporting units from your project case study.

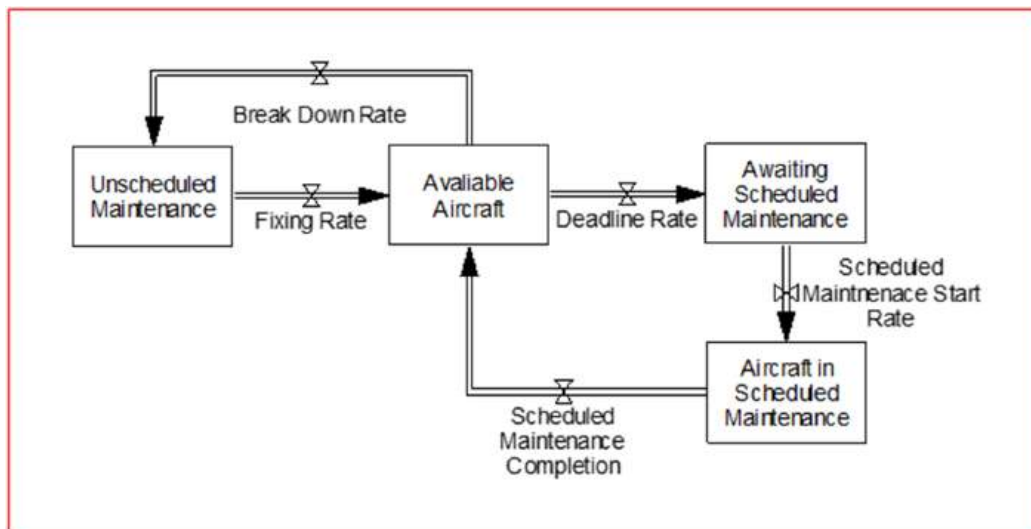
Name	Type	Units
Company Balance Sheet	Stock	\$
Customer service calls on hold	Stock	Customers or Calls
GDP (Gross Domestic Product)	Flow	\$/Year
US federal budget deficit	Flow or Stock	\$/Year or \$
Products under development	Stock	Products
Corporate accounts payable	Stock	\$, could be # of accounts too
Cash flow	Flow	\$/Time
Book value of inventory	Stock	\$
Promotion of Captains to Majors	Flow	Captains/Month
Incidence of attacks on DoD web sites	Flow	Attacks/Minute
Greenhouse gas emissions of the US	Stock Flow	Tons CO2 Tons CO2/Year
Yen/dollar exchange rate	Stock	Yen/Dollar
Employee morale	Stock	Happiness or some other dmnl unit
Interest Rate on 30-year US Treasury Bond	Stock	Percent or %/year
	Stock	
	Stock	
	Flow	
	Flow	

B. Representing Stocks and Flows

Systems are composed of interconnected networks of stocks and flows. Modelers must be able to represent the stock and flow networks of people, material, goods, money, energy, etc. from which systems are built.

For each of the following cases, construct a stock and flow diagram that properly connects the variables.

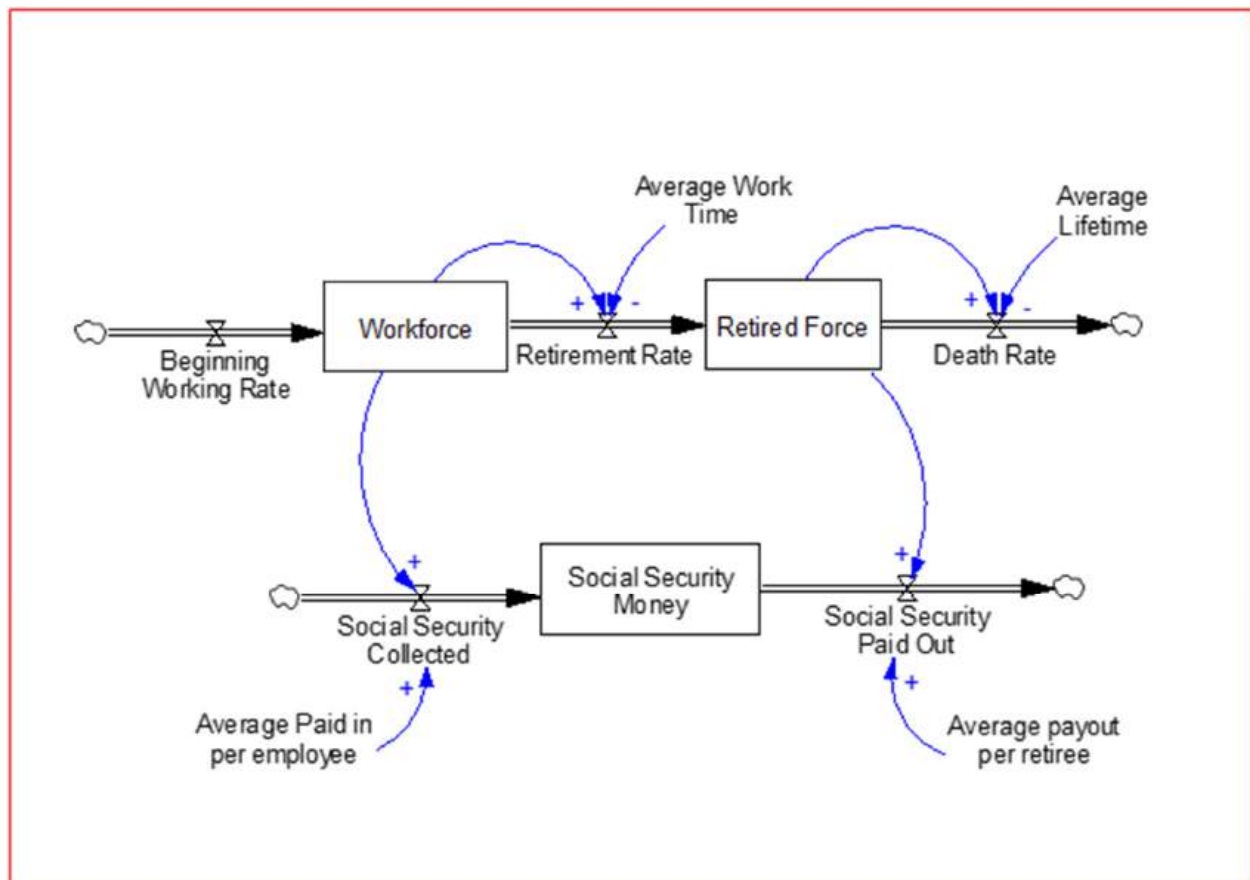
- Not all the variables need to be connected by physical flows; they may be linked by information flows.
 - You may need to add additional stocks or flows beyond those specified to complete your diagram (but keep it simple). Be sure to consider the boundary of your stock and flow map. That is, what are the sources and sinks for the stock and flow networks? Are you tracking sources and sinks far enough upstream and downstream? This process of deciding how far to extend the stock and flow network is called “challenging the clouds” because you challenge the assumption that the clouds are in fact unlimited sources or sinks.
 - Consider the units of measure for your variables and make sure they are consistent within each stock and flow chain.
- B1 (20 Points) A Combat Aviation Brigade has hundreds of aircraft that require scheduled preventative maintenance. The goal of maintenance is to maximize aircraft availability while keeping maintenance costs low. Consider only the UH-60s assigned to a lift battalion for this exercise. These aircraft can exist in four states, they are either *available*, *awaiting scheduled maintenance*, *in scheduled maintenance*, or *undergoing unscheduled maintenance* (i.e. when a part breaks).



SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

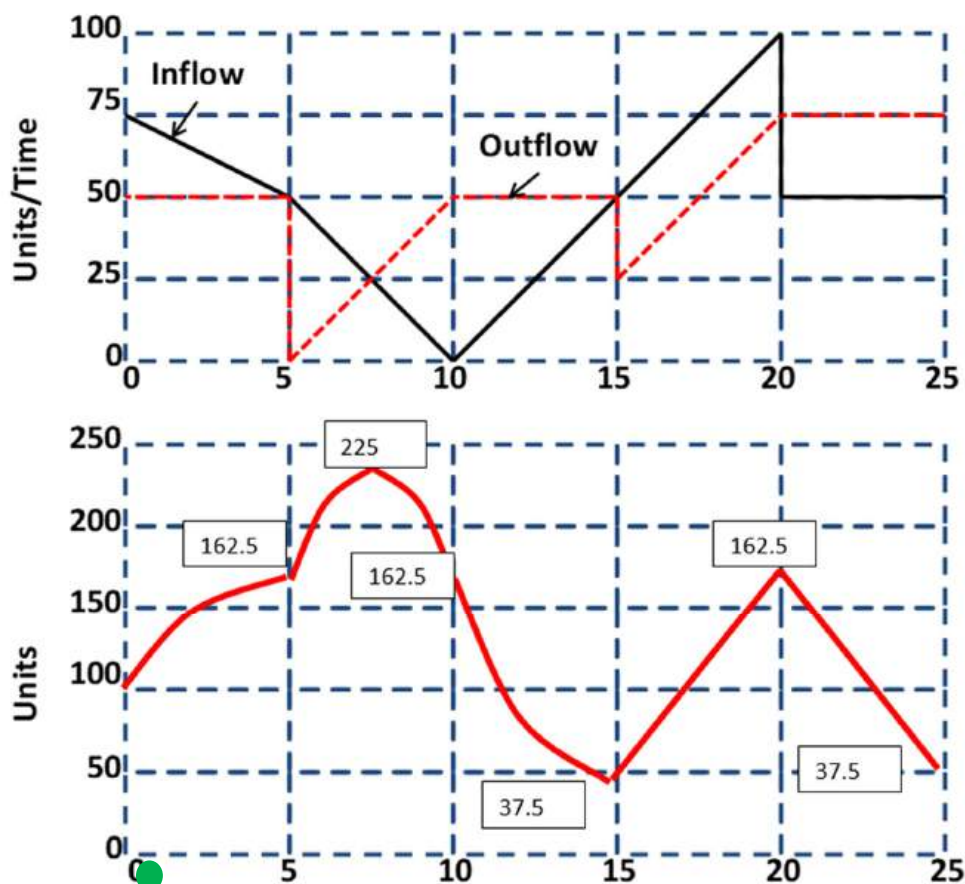
- B2 (20 Points) A topic frequently in the news is the health (or lack thereof) of the US Social Security system. As in many nations, the US Social Security system is “pay as you go”: revenue received today from taxes paid by workers and their employers is used to pay benefits to those over the age of sixty-five. For the purpose of this question, assume that people start working at age twenty and begin receiving social security benefits at age sixty-five. Map the stock and flow structure of the US population as it relates to the social security system. Also map the stock and flow structure of money as it goes through the system. You should have two separate stock flow chains in this answer, one for people, one for money. Make sure they are appropriately linked with information arrows.



C. Accumulation and Graphical Integration and Differentiation

Stocks are accumulations. The difference between the inflows and outflows of a stock accumulates, altering the level of the stock variable. The process of accumulation gives stocks inertia and memory and creates delays. Since realistic models are far too complex to solve with formal analysis, it is important to understand the relationship between flows and the behavior of stocks intuitively. Consider the following system:

- C1 (10 Points) On the graph provided, draw the trajectory of the stock given the inflow and outflow rates shown. Indicate the numerical values for any maxima or minima, and for the maximum or minimum values of the slope for the stock. Assume the initial quantity in the stock is 100 units (marked with green circle below).

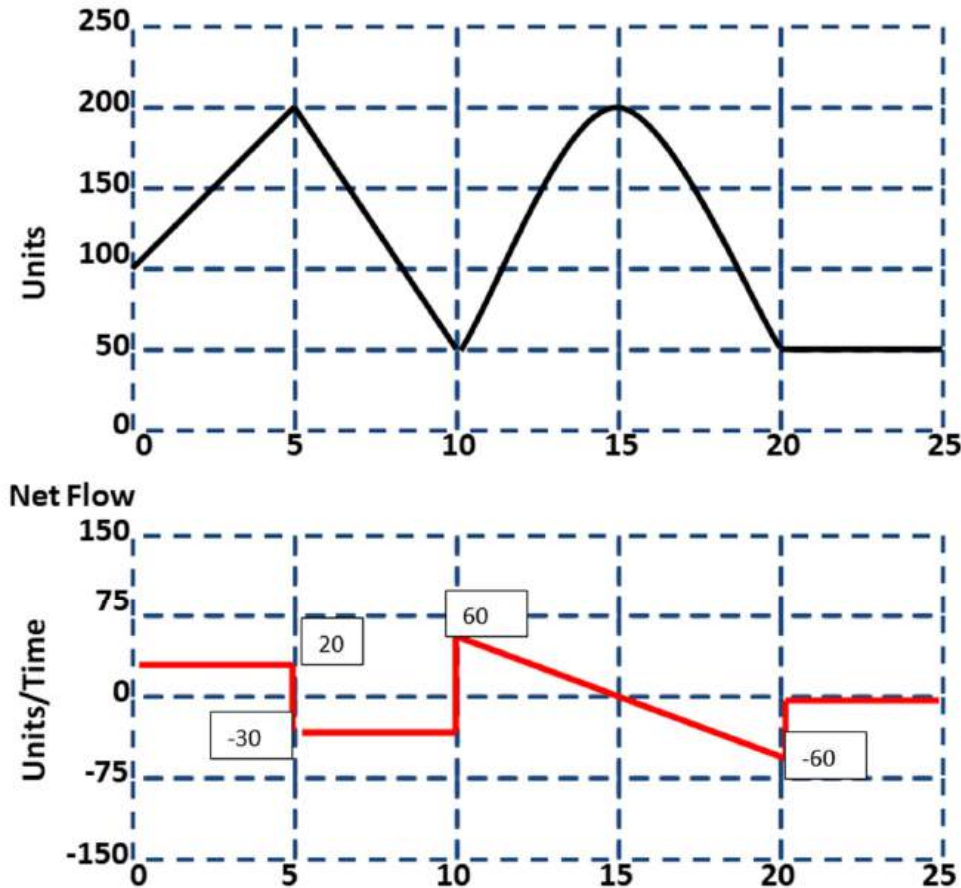


- C2 (10 Points) This exercise asks you to graphically differentiate a stock to understand the behavior of the net rate of flow to the stock. By estimating the slope of the stock at each point, create a graph of the net rate of change in the stock. That is, in the graph below you are given the behavior of the stock over time and must infer the behavior of its net

SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

inflow. (This is the inverse operation of graphical integration, the previous exercise.) Graphically differentiate the stock and provide numerical values for key points in the net flow graph (maxima and minima).



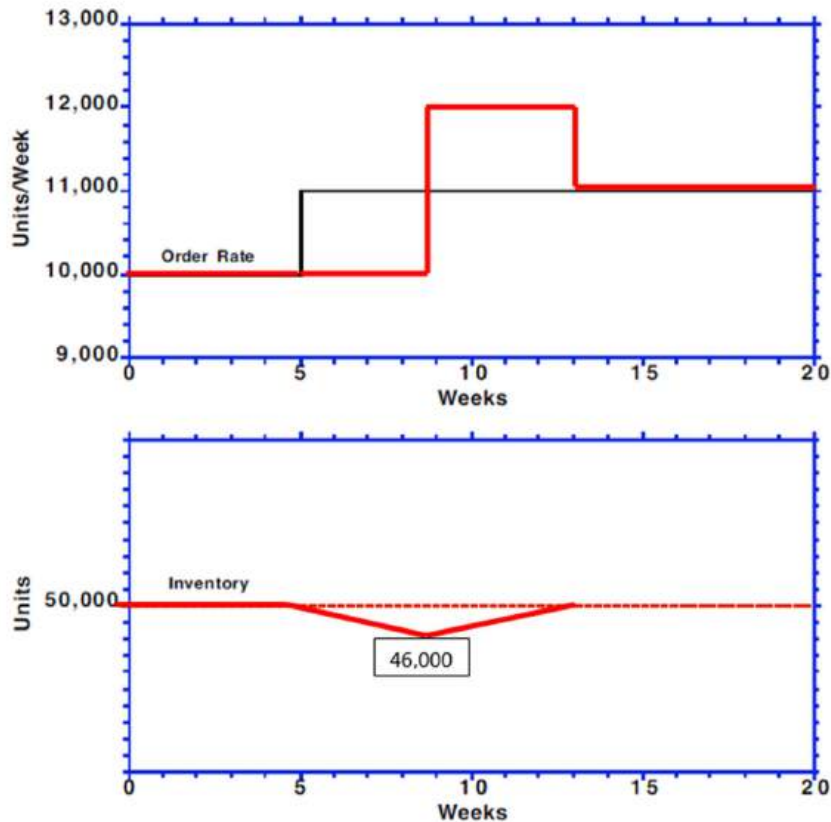
- C3 (10 Points) Consider a manufacturing firm. The firm maintains an inventory of finished product. The firm uses this inventory to fill customer orders as they come in. Historically, orders have averaged 10,000 units per week. The firm strives to maintain an inventory of 50,000 units to provide excellent customer service (that is, to be able to fill essentially 100% of every order). Although the firm has ample capacity to handle variations in demand, it takes time to adjust the production schedule, and to make the product—a *total lag (delay) of four weeks*.

Now imagine that the order rate for the firm's products suddenly and unexpectedly rises by 10%, and remains at the new, higher rate, as shown in the flow graph below. Before the change in demand, production was equal to orders at 10,000 units/week, and inventory was equal to the desired level of 50,000 units.

SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

Sketch the likely path of *production* (inflow) on the top graph in relation to the order rate (outflow). Sketch the likely path of the *actual inventory* stock on the lower graph, in relation to the *goal inventory*. Provide an appropriate scale for the graph of inventory.



Outstanding Student Work

Homework 3 Gradescope

● Graded

Student

Lucas Sorensen

Total Points

74.5 / 75 pts

Question 1

Q1-Fill in Table

4.5 / 5 pts

– 0 pts Correct

✓ – 0.5 pts 1 to 5 incorrect

– 1 pt 6 to 10 incorrect

Question 2

Q2-S&F for CAB

20 / 20 pts

✓ – 0 pts Correct

– 3 pts missing one stock

– 1 pt one incorrect/missing flows

– 2 pts two incorrect/missing flows

– 3 pts three incorrect/missing flows

– 2 pts stocks not connected via flows back/from available correctly

– 2 pts stocks not mutually exclusive (i.e. can exist in multiple stocks at the same time)

– 1 pt errors not in line with good modeling practice such as causal links into stocks, causal links into sources/sinks, flows into flows

– 6 pts missing two stocks

– 20 pts no submission

– 2 pts stocks switched

Question 3

Q3-S&F for Social Security

20 / 20 pts

✓ - 0 pts Correct

- 3 pts flows connecting people stocks to money stocks
- 2 pts multiple stocks not influencing flows
- 1 pt one stock not influencing flows
- 2 pts missing multiple appropriate sources and sinks
- 1 pt missing one source/sink
- 1 pt 1-2 incorrect/missing causal links
- 2 pts missing/incorrect multiple causal links
- 1 pt missing one flow
- 2 pts missing multiple flows
- 1 pt errors not in line with good modeling practice. wrong direction loops, causal links to stocks, took screenshot without lock sketch, not mutually exclusive stocks
- 20 pts no answer
- 2 pts image quality too poor to read

Question 4

Q4-Draw Stock Given Inflow/Outflow

10 / 10 pts

✓ - 0 pts Correct

- 1 pt 1 SEGMENT PARTIALLY INCORRECT
- 2 pts ONE SEGMENT INCORRECT
- 1 pt SECOND SEGMENT PARTIALLY INCORRECT
- 2 pts SECOND SEGMENT INCORRECT
- 1 pt THIRD SEGMENT PARTIALLY INCORRECT
- 2 pts THIRD SEGMENT INCORRECT
- 1 pt FOURTH SEGMENT PARTIALLY INCORRECT
- 10 pts NO ANSWER

Question 5

Q5-Draw Net Flow Given Stock

10 / 10 pts

✓ - 0 pts Correct

- 1 pt ONE SEGMENT PARTIALLY INCORRECT
- 2 pts ONE SEGMENT INCORRECT
- 1 pt SECOND SEGMENT PARTIALLY INCORRECT
- 2 pts SECOND SEGMENT INCORRECT
- 1 pt THIRD SEGMENT PARTIALLY INCORRECT
- 2 pts THIRD SEGMENT INCORRECT
- 10 pts NO ANSWER

Question 6

Q6-Estimating Production and Inventory

10 / 10 pts

✓ - 0 pts Correct

- 1 pt FLOW PARTIALLY INCORRECT
- 3 pts FLOW INCORRECT
- 1 pt STOCK PARTIALLY INCORRECT
- 3 pts STOCK INCORRECT
- 0 pts STOCK INCORRECT BUT FOLLOWS ERROR IN FLOW
- 10 pts NO ANSWER
- 5 pts NO ANSWER ON FLOW

Question 7

Extra Work

0 / 0 pts

✓ - 0 pts Correct

Question 8

Works Cited

0 / 0 pts

✓ - 0 pts Correct

Question 9

Cover Page

0 / 0 pts

✓ - 0 pts Correct

- 5 pts Failed to follow submission instructions (don't add pages, don't convert to word then back to pdf, etc)

UNITED STATES MILITARY ACADEMY

HOMEWORK #3

SM484: SYSTEM DYNAMICS SIMULATION

SECTION H1/I1/J1

MAJ LUEDERS/LUPFER

BY

CADET LUCAS SORENSEN
Type FIRST LAST name from Canvas

WEST POINT, NEW YORK
27 OCTOBER 2025

 I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

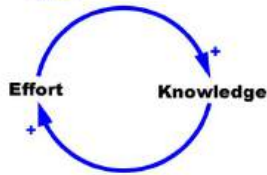
LS I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

 I ACKNOWLEDGE THAT I USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

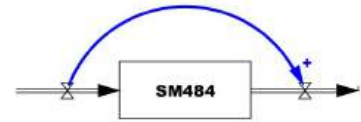
LS I ACKNOWLEDGE THAT I DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: SORENSEN.LUCAS.SPENCER.162 3696908 Digitally signed by
SORENSEN.LUCAS.SPENCER.1623696908
Date: 2025.10.27 20:40:49 -04'00' DATE: 27OCTOBER2025



System Dynamics Simulation SM 484 – AT 26-1

MAJ Jake Lueders
MAJ Pat Lupfer



Homework 3: Stocks and Flows

Assigned: 29 September 2025; Due: 27 October 2025 at 2359
75 Points Total

This is an individual assignment. Follow these submission instructions exactly. You will submit a single PDF containing 11 pages. The DAAW and a space for works cited have been included, as well as a blank extra work page (if needed); simply add your name, the date, and circle your section and instructor. DO NOT convert to word and back to PDF, either type directly into the PDF, or print and handwrite then SCAN it back in using a scanner. DO NOT TAKE A PICTURE AND CONVERT TO PDF UNLESS YOU KNOW HOW TO CONVERT IT BACK TO 8.5"x11". There are scanners in the DSE Cadet Study/Capstone Area and the library, or you can stop by and ask an instructor for help scanning.

This homework gives you practice with the structure and dynamics of stocks and flows. Stocks and flows are the building blocks from which every more complex system is composed. The ability to identify, map, and understand the dynamics of the networks of stocks and flows in a system is essential to understanding the processes of interest in any modeling effort.

A. Identifying Stock and Flow Variables

The distinction between stocks and flows is crucial for understanding the source of dynamics. In physical systems it is usually obvious which variables are stocks and which flows. In human and social systems, often characterized by intangible, "soft" variables, identification is more difficult. An example is provided below.

Name	Type	Units
Example: Inventory of soda	Stock	Cases
Example: Soda order rate	Flow	Cases/week

SM 484: System Dynamics Simulation – AT 26-1**Homework 3 – Stocks and Flows**

- **Q1** (5 Points) For each of the following variables, state whether it is a stock or a flow, and give units of measure for each. Additionally, identify two stocks and two flows with supporting units from your project case study.

Name	Type	Units
Company Balance Sheet	Stock	Dollars
Customer service calls on hold	Stock	# of Calls
GDP (Gross Domestic Product)	Stock	Dollars/Year
US federal budget deficit	Stock	Dollars
Products under development	Stock	# of Products
Corporate accounts payable	Stock	Dollars
Cash flow	Flow	Dollars
Book value of inventory	Stock	Dollars
Promotion of Captains to Majors	Flow	# of Promotions/Year
Incidence of attacks on DoD web sites	Flow	# of Attacks/Year
Greenhouse gas emissions of the US	Flow	Tons of CO2/Year
Yen/dollar exchange rate	Stock	Yen/Dollar
Employee morale	Stock	Happiness Points
Interest Rate on 30-year US Treasury Bond	Stock	Percent/Year
US Population	Stock	Persons
AIT Graduates	Stock	Persons
AIT Graduation Rate	Flow	Persons/Month
Basic Graduation Rate	Flow	Persons/Month

SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

B. Representing Stocks and Flows

Systems are composed of interconnected networks of stocks and flows. Modelers must be able to represent the stock and flow networks of people, material, goods, money, energy, etc. from which systems are built.

For each of the following cases, construct a stock and flow diagram that properly connects the variables.

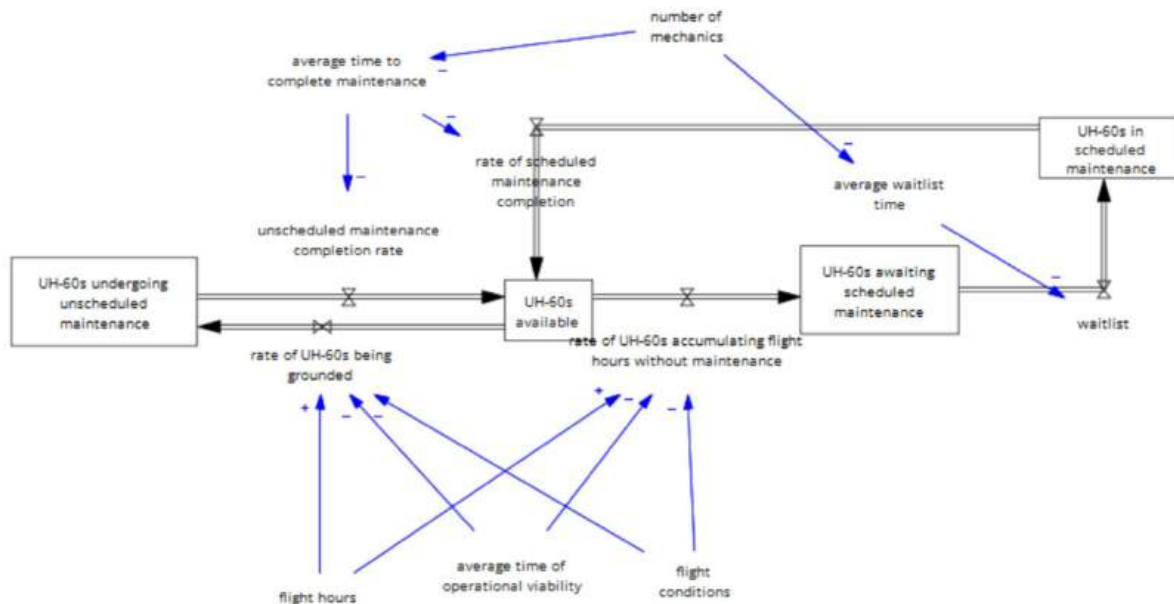
- Not all the variables need to be connected by physical flows; they may be linked by information flows.
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- Consider the units of measure for your variables and make sure they are consistent within each stock and flow chain.

SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

- **Q2 (20 Points)** A Combat Aviation Brigade has hundreds of aircraft that require scheduled preventative maintenance. The goal of maintenance is to maximize aircraft availability while keeping maintenance costs low. Consider only the UH-60s assigned to a lift battalion for this exercise. These aircraft can exist in four states, they are either *available*, *awaiting scheduled maintenance*, *in scheduled maintenance*, or *undergoing unscheduled maintenance* (i.e. when a part breaks).

In Vensim, map the stock and flow structure relating these various states, and include a legible screenshot below taken with the “lock sketch” button selected.

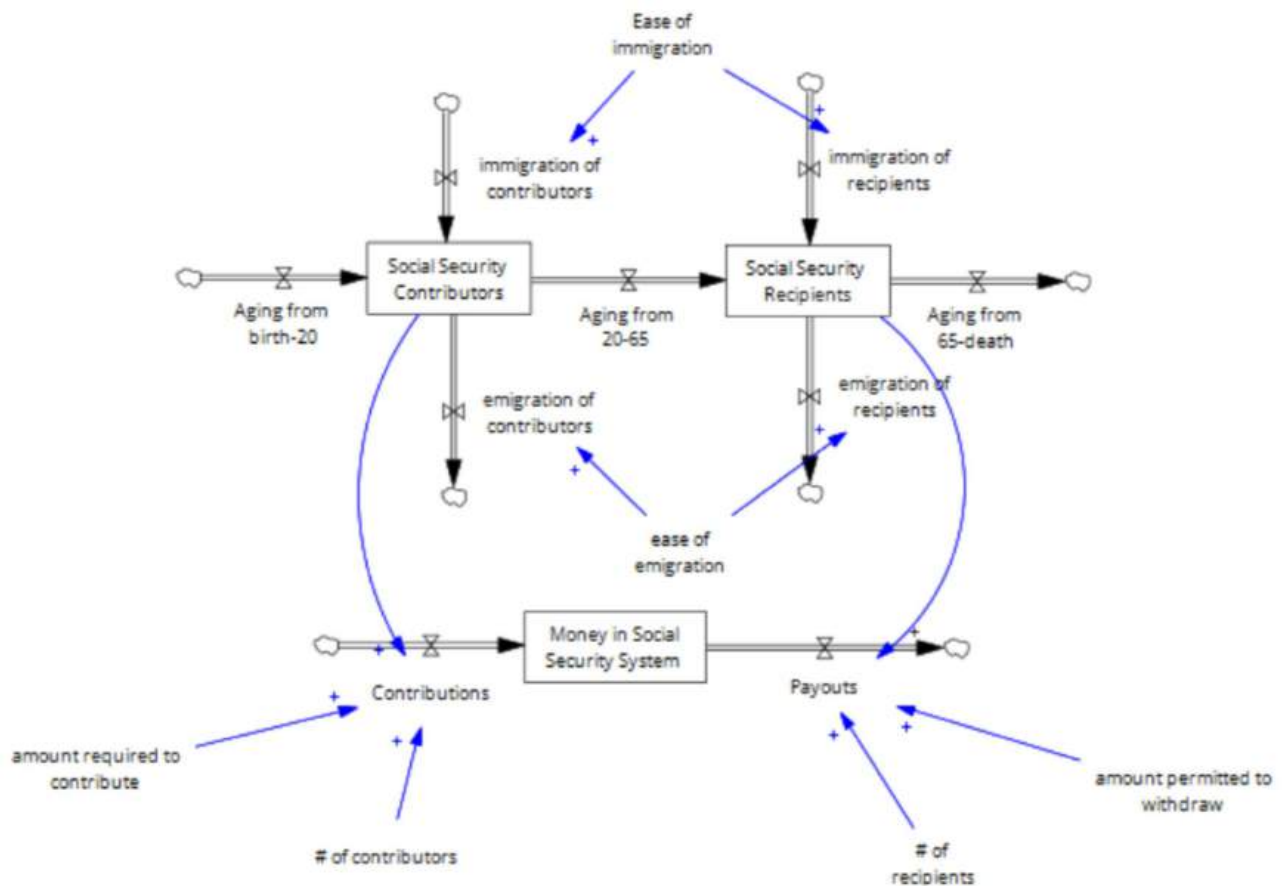


SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

- **Q3 (20 Points)** A topic frequently in the news is the health (or lack thereof) of the US Social Security system. As in many nations, the US Social Security system is “pay as you go”: revenue received today from taxes paid by workers and their employers is used to pay benefits to those over the age of sixty-five. For the purpose of this question, assume that people start working at age twenty and begin receiving social security benefits at age sixty-five. Map the stock and flow structure of the US population as it relates to the social security system. Also map the stock and flow structure of money as it goes through the system. You should have two separate stock flow chains in this answer, one for people, one for money. Make sure they are appropriately linked with causal arrows.

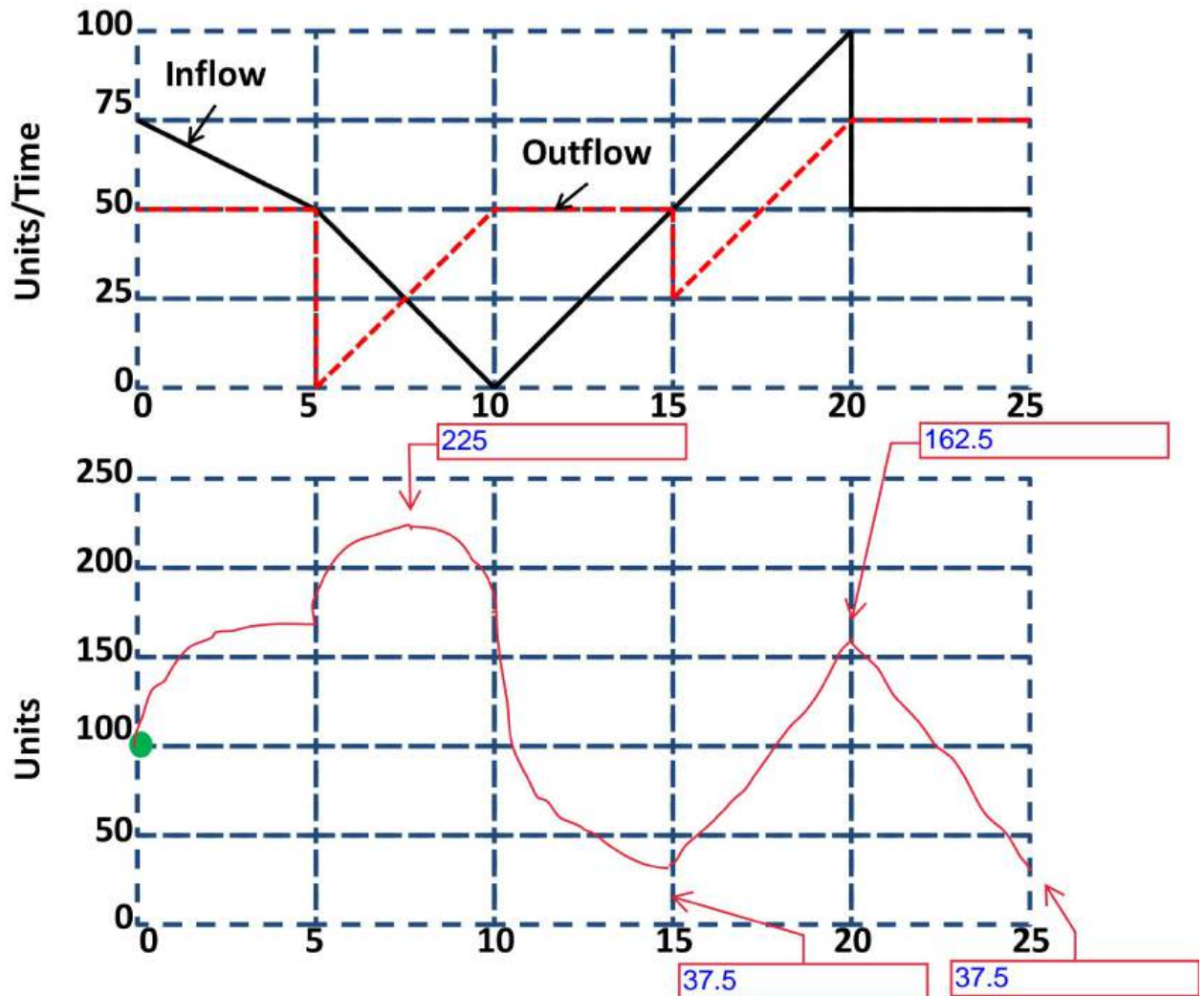
In Vensim, map the stock and flow structure of the US population as it relates to the social security system, and include a legible screenshot below taken with the “lock sketch” button selected.



C. Accumulation and Graphical Integration and Differentiation

Stocks are accumulations. The difference between the inflows and outflows of a stock accumulates, altering the level of the stock variable. The process of accumulation gives stocks inertia and memory and creates delays. Since realistic models are far too complex to solve with formal analysis, it is important to understand the relationship between flows and the behavior of stocks intuitively. Consider the following system:

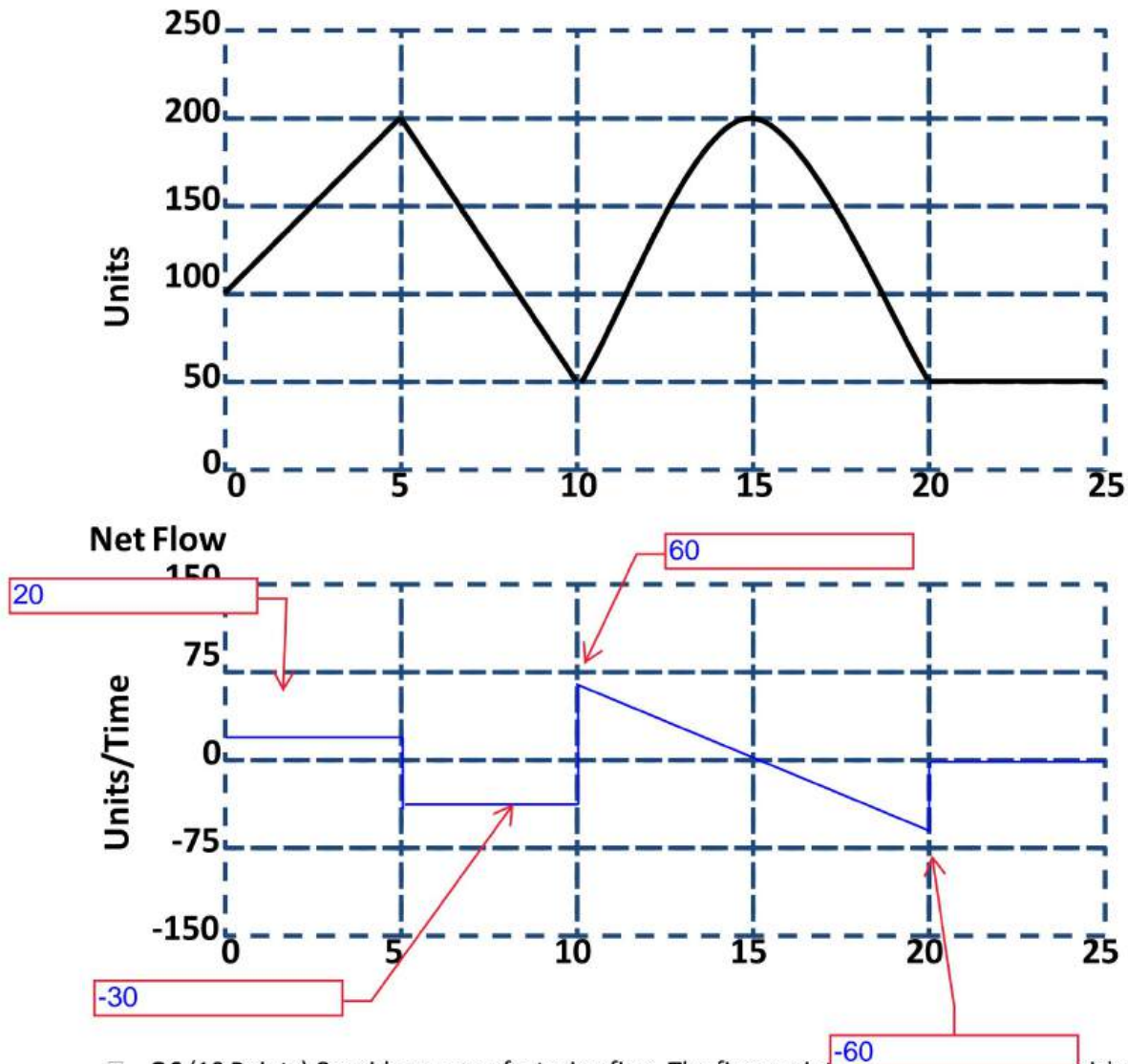
- **Q4** (10 Points) On the graph provided, draw the trajectory of the stock given the inflow and outflow rates shown. Indicate the numerical values for any maxima or minima, and for the maximum or minimum values of the slope for the stock. Assume the initial quantity in the stock is 100 units (marked with green circle below).



SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

- **Q5 (10 Points)** This exercise asks you to graphically differentiate a stock to understand the behavior of the net rate of flow to the stock. By estimating the slope of the stock at each point, create a graph of the net rate of change in the stock. That is, in the graph below you are given the behavior of the stock over time and must infer the behavior of its net inflow. (This is the inverse operation of graphical integration, the previous exercise.) Graphically differentiate the stock and provide numerical values for key points in the net flow graph (maxima and minima).



- **Q6 (10 Points)** Consider a manufacturing firm. The firm maintains an inventory of finished product. The firm uses this inventory to fill customer orders as they come in. Historically, orders have averaged 10,000 units per week. The firm strives to maintain an inventory of

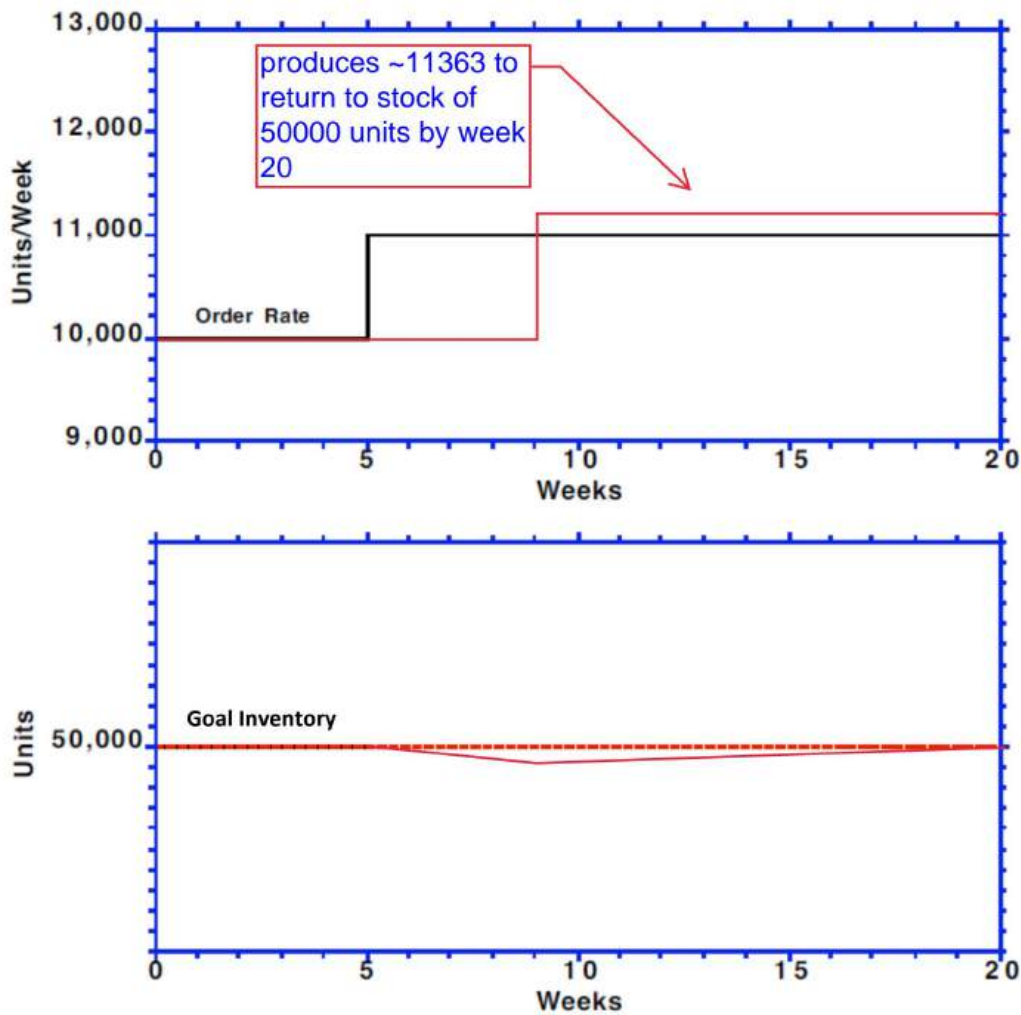
SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

50,000 units to provide excellent customer service (that is, to be able to fill essentially 100% of every order). Although the firm has ample capacity to handle variations in demand, it takes time to adjust the production schedule, and to make the product—a *total lag (delay) of four weeks*.

Now imagine that the order rate for the firm's products suddenly and unexpectedly rises by 10%, and remains at the new, higher rate, as shown in the flow graph below. Before the change in demand, production was equal to orders at 10,000 units/week, and inventory was equal to the desired level of 50,000 units.

Sketch the likely path of *production* (inflow) on the top graph in relation to the order rate (outflow). Sketch the likely path of the *actual inventory* stock on the lower graph, in relation to the *goal inventory*. Provide an appropriate scale for the graph of inventory.



SM 484: System Dynamics Simulation – AT 26-1

Homework 3 – Stocks and Flows

EXTRA WORK (Do not delete, just leave blank if not needed)

SM 484: System Dynamics Simulation – AT 26-1

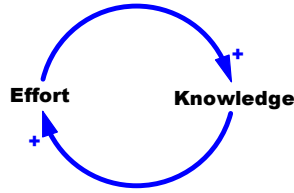
Homework 3 – Stocks and Flows

Works Cited: Include your works cited below, do not delete even if you didn't use additional sources.



Homework 4

Instructor Solution

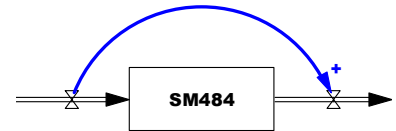


System Dynamics Simulation

SM 484 – AY 26-1

MAJ Jacob Lueders

MAJ Pat Lupfer



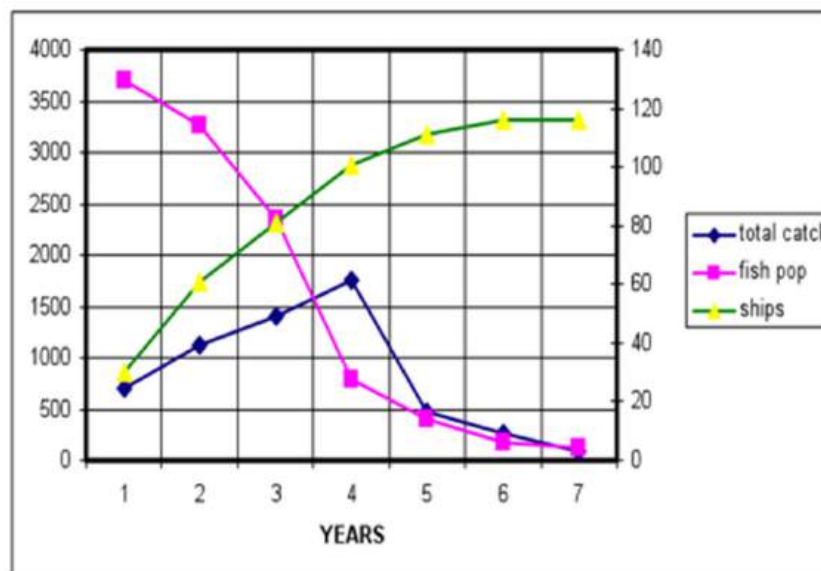
Homework 4: Fishbanks Policy Analysis

Assigned: 29 October 2025; Due: 19 November 2025

75 Points

This is an individual assignment. Follow these submission instructions exactly. You will submit a single PDF containing 14 pages. The DAAW and a space for works cited have been included, as well as a blank extra work page (if needed); simply add your name, the date, and circle your section and instructor. DO NOT convert to word and back to PDF, either type directly into the PDF, or print and handwrite then SCAN it back in using a scanner. DO NOT TAKE A PICTURE AND CONVERT TO PDF UNLESS YOU KNOW HOW TO CONVERT IT BACK TO 8.5"x11". There are scanners in the DSE Cadet Study/Capstone Area and the library, or you can stop by and ask an instructor for help scanning.

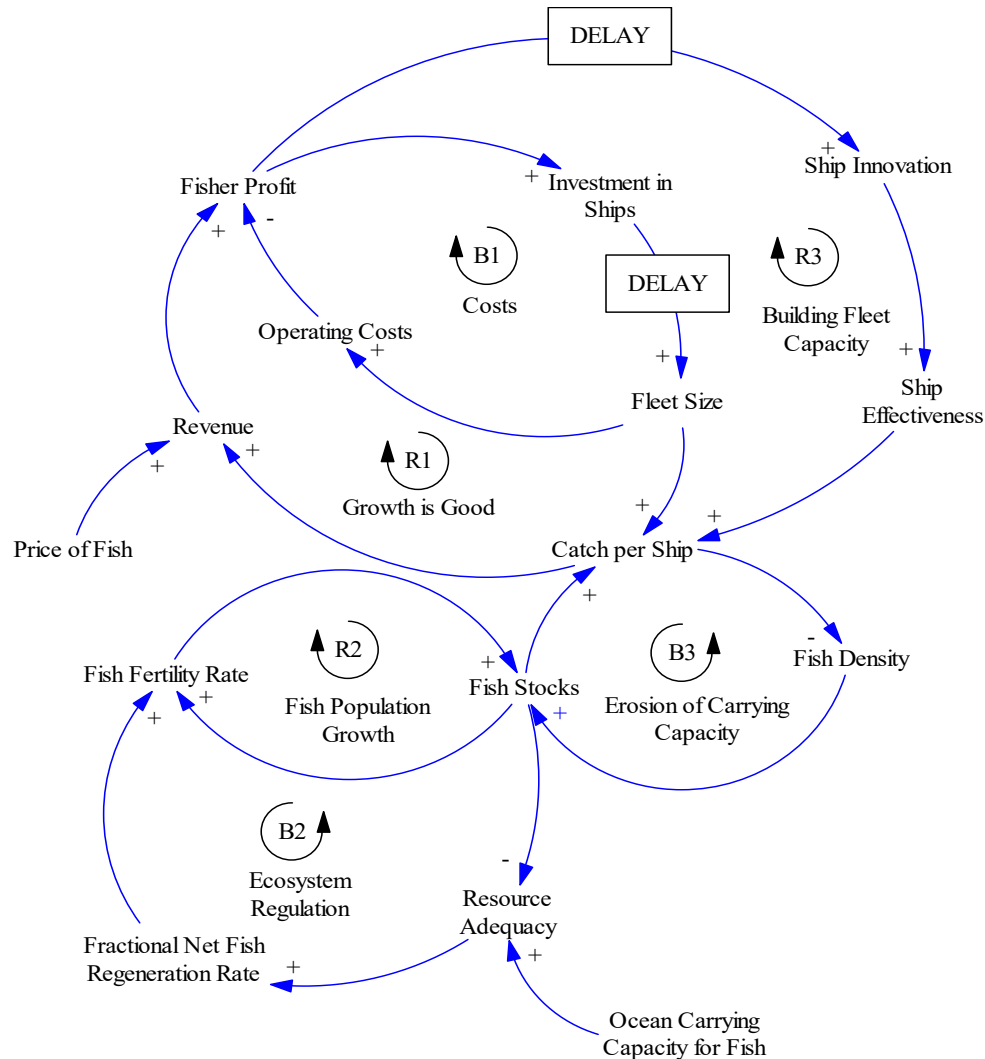
The year is 1970. You are a manager in the national fish and wildlife service. You are charged with the responsibility of making policy suggestions regarding the management of a fishery surrounding your national borders. Fishing is an important source of economic income for your country so your decisions must ensure the healthy maintenance of the fish population and fishing industry.



Your assessment of the fishing in the public waters is that the size of the fishing fleet increases while the fish population is decimated and the fishing industry collapses. Indeed, you speculate that the existing dynamics operating among these three variables results in the outcomes depicted in the graph.

Homework 4: Fishbanks Policy Analysis

The image of the commons fishery is that the fish population is assumed to be unlimited. Fishing companies need merely invest in a fleet of ships, which, over time, will generate profits that exceed operating expenses. As evident in the CLD below, this mental model characterizes a business as usual approach (R1) of unlimited growth. Fish stocks are assumed to grow at a constant rate, which produces plentiful harvests (R2). Companies reinvest profits into their companies, which, in the aggregate, grows fleet size (R3).



This image presents a mental model of unlimited growth. And for a while, the model behaves in accordance with the mental model. Exponential growth occurs in the growth of fleet size, fish harvest, and assets. But then the business as usual approach goes terribly wrong. Costs rise as ships must spend longer hours in search of fish further removed from the coastal waters (B1). The fish stocks, already constrained by physiological limits of density based on availability of food

Homework 4: Fishbanks Policy Analysis

and air, level off as they reach these natural limits (B2), then fall quickly as their numbers decline faster via fishery extraction than can be sustained via natural reproduction (B3).

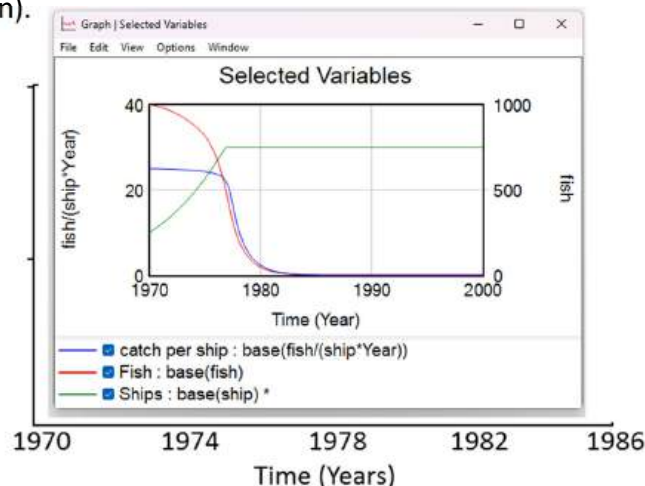
This scenario can be explained by the latent underlying structure of these feedback mechanisms. Essentially, the fishing industry is growing so fast that it is eroding the carrying capacity (i.e., fish stock) of their own industry. The resulting behavior is regulated by a latent structure that permits growth until a tipping point is reached, which is defined as the point at which continued depletion of the resource occurs faster than the regeneration of the resource.

Your challenge is to devise a regulatory policy or complimentary set of policies capable of sustaining the fishery. Beginning with the standard model given to you (*SM484Fishbanks.mdl*), answer the following questions.

1. **(10 points)** The standard model given to you, which includes minimal structure to represent the system, is currently set up to simulate the regulatory situation present within the national fishing industry as of 1970 (no regulation). Observe the behavior of the number of ships, catch per year, fish population, and total accumulated profits.

Clearly, leaving the present system undisturbed is an unacceptable situation. The fish population is decimated, the fishing industry dies, and you lose your job. One regulatory policy, which might help to avoid this situation is to levy a large tax on new ships (say \$200 per ship), thereby raising the price and curtailing purchases.

- a. Sketch on the axes below the effect you believe that this policy will have on the number of ships, total catch per year, and fish population (do this **without** running a simulation).



4pts showing all three with rational modes of behavior. If only showing 1, use partial credit, example below:

For Fish Population Only shown: 3 points overshoot and collapse, 2 Points Goal Seeking, 1 Points some rational mode of behavior.

Homework 4: Fishbanks Policy Analysis

- b. Try running this policy by changing the ship cost from \$300 to \$500. Once you have changed the ship cost, simulate the system, and generate a graph with both the base case and policy to show the results for ships, total catch per year, and fish population. Was this policy successful? How does the behavior generated in this simulation compare with the previous behavior? Explain the differences in behavior.

1 Point – Was the policy successful (no)

1 Point – How does the behavior change (lower total number of ships)

1 Point – Explain the differences in behavior (same behavior, just delayed)

-1 Point-no graph

- c. Vary the fish price and re-simulate to observe the effects on behavior. What behavior is exhibited if the price is too high?

3 Points – Steeper S Shaped- Decline for Fish Population as price increases/Quicker Ship Building overshoot

2 Points – Only mention one variable

1 Points – Limited explanation

2. **(20 points)** Another possible policy alternative is to keep building ships until there is a noticeable problem with the fish population. Since the only feedback signal from the fish population that fishermen receive is from the fish catch, the policy could make building ships illegal once the catch per ship decreases to an unacceptable rate.

To implement this policy, first change the ship cost back to 300 dollars per ship.

Next insert the feedback from Catch per Ship to Ship Building Rate. During good fishing, the average ship will catch 25 fish per year. See what happens when the ship building is limited after the catch drops below 22 fish per year.

IF THEN ELSE (catch per ship > 22, yearly profits * fraction invested / ship cost, 0)

This equation says that if the catch per ship is greater than 22 fish per year then purchase the normal number of ships (Profits * Fraction Invested / Ship Cost). However, if the catch per ship falls below 22, then build zero ships.

- a. Run the model and produce a graph of the behavior. How does the behavior generated based on the new regulation differ from a system with no regulation? Explain any differences or lack of differences.

5 Points – Fish Population still exhibits the same dynamics (S-Shaped Decline), but Ships stop building more quickly

4 Points – Ships stop building – no mention of additional variables

3 Points – No Change, but explains why they think it happens

2

Homework 4: Fishbanks Policy Analysis

- b. Was this policy successful? If not, try increasing the threshold on Catch per Ship from 22 to a higher number. At what number does the system produce a stable fish population? What factors not present within the model might affect the catch per ship?

5 Points – 25 is the limit, adds additional factors that they think might be relevant

4 Points – Wrong limit, but close and talks additional factors

3 Points – Wrong limit, talks factors but not very well

2 Points – 25 Limit, no explanation

- c. How easy do you think it will be for the government to collect data on the catch per ship? What problems might be encountered?

5 Points – Thorough explanation with more than one example of difficulties

4 Points – Explains one challenge well

3 Points – Limited explanation

- d. Do you get the impression that fishing systems tend to give a great deal of warning before a catastrophe or do you feel these catastrophes come on more suddenly? Is the catch per ship an accurate indication of the fish population?

5 Points – Very little warning, catch per ship not a great indicator with some explanation

4 Points – No explanation

3 Points – 1 of 2 concepts

3. (5 points) It is clear that a tax on ships seems to do nothing more than delay the destruction of the fish population. In fact, all of the challenges listed in the text box limit the options for effective policy implementation.

Finish the quote using your course text. *Business Dynamics* Chapter 3, section 3.5.5 states

- Fish population is never really known, it can only be estimated from the past behavior of the catch.
- It is not possible to place an overseer on each boat to make sure it adheres to a specified catch. Hence regulating the catch is not possible. However, you might be able to randomly inspect catches from ships that return to the docks to unload the fish they caught, fining or confiscating ships that exceed regulatory limits.
- It is not possible to curb investment into boats in the face of expectations of catch and availability of financial resources, especially when owning assets is a basis for winning (success in real life).
- It is not possible to restrict people from building new boats. However, you could limit the number of boats allowed to fish on any given day by licensing the total number of ships or the number authorized to fish during any given time interval.

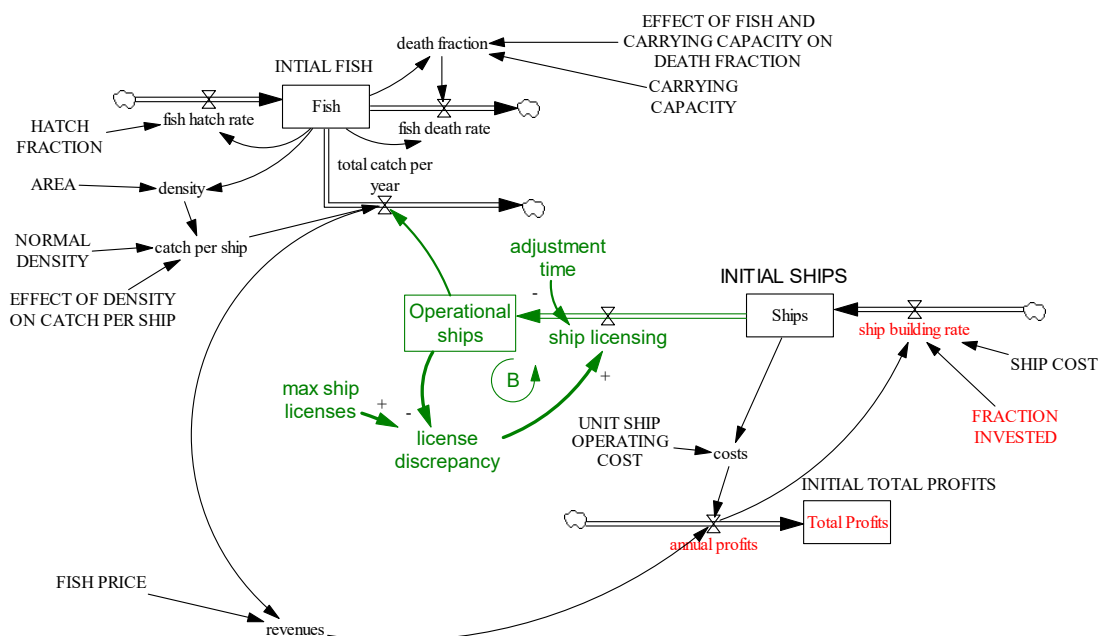
“Policy design is much more than...

than changing the values of parameters such as a tax rate or markup ratio. Policy design includes the creation of entirely new strategies, structures, and decision rules. Since the feedback structure of a system determines its dynamics, most of the time high leverage policies will involve changing the dominant feedback loops by redesigning the stock and flow structure, eliminating time delays, changing

Homework 4: Fishbanks Policy Analysis

the flow and quality of information available at key decision points, or fundamentally reinventing the decision processes of the actors in the system. (Any part or whole of this is fine)

4. **(40 points)** A policy is depicted in the stock and flow diagram below, with new/modified structure in green. Modify the standard model provided to you, incorporating this structure with fully formed equations. Use realistic values to calibrate initial and exogenous variable values.



- a. Explain the purpose of this policy.

6 Points

Homework 4: Fishbanks Policy Analysis

- b. Provide all data for the variables in the table below (if initial value is not applicable, write “NA”): 4 Points/Variable (24 Points Total)-reasonable description and initial values

Variable	Description	Equation	Initial Value	Units
Operational Ships		= INTEG (Ship Lic)	0	Ship
Ship Licensing		Lic Discrepancy/AT		Ship/Year
Max Ship Licenses		Any reasonable value		Ship
License Discrepancy		Max Ship Lic- Operational Ships		Ship
Adjustment Time		Any reasonable value		Year
Total Catch Per Year		Operational Ships*catch per ship		Fish/Year

- c. Simulate the model and discuss the performance of your policy on the sustainability of the fishery. Use output graphs to support your answer. Discuss lessons learned from your experimentation, even if the policy you suggest is unable to create a balance between the human and natural systems.

5 Points-Output graphs & relevant discussion

- d. Based on research, develop an additional policy recommendation. Explain how you would update the model to capture the policy recommendation.

5 Points-new policy & how to update the model discussed

Outstanding Student Work

Homework 4 Gradescope

● Graded

Student

Brent Keefer

Total Points

73 / 75 pts

Question 1

1a

4 / 4 pts

✓ - 0 pts Graphed all three with some rational behavior changed from the initial given plot

- 0.5 pts No Labels

- 1 pt Just copied initial graph, think through how the policy would change this

- 3 pts only included one plot

- 1 pt all three plotted, but not rational behavior (linear change, instantaneous change, etc)

Question 2

1b

3 / 3 pts

✓ - 0 pts Correct. Included graph, policy unsuccessful (or temporarily successful), explained why (population still crashes), explained differences (delayed crash)

- 0.5 pts Correct analysis, but incorrectly stated policy was successful when the fish population still collapses

- 0.5 pts Only base case or higher cost shown, not both

- 0.5 pts missing some detail in the answer

- 1 pt No/incorrect Graph

Question 3

1c

3 / 3 pts

✓ - 0 pts Correct, discussion of at least two 1) steeper decline for fish population, 2) quicker ship building resulting from revenue increase, and 3) catch per ship peaking and collapsing sooner

- 1 pt Only discussed one variable/limited discussion

- 2 pts incorrect or severely lacking description of behavior, possibly due to model errors.

- 3 pts No Answer

Question 4

2a

5 / 5 pts

- ✓ - 0 pts Correct, fish population exhibits same dynamics (s shaped decline) but ships stop building more quickly as the catch per ship drops. Additional explanation why this happens. Multiple graphs shown with base case and additional policy highlighted.
- 1 pt only references a single variable (ship building, fish population, total ships, etc) and why, but no other variables discussed. Reference multiple variables with minimal explanation for some.
- 2 pts states minimal change to some variables, but lacks explanation why.
- 1 pt no image included/screenshot of slider changes only

Question 5

2b

5 / 5 pts

- ✓ - 0 pts Correct, identifies it wasn't successful and that 25 is the point at which a stable population emerges. Adds additional factor discussion.
- 2 pts Explanation present, but no limit or incorrect limit discussed
- 3 pts No/incorrect explanation and wrong or no limit discussed
- 2 pts Correct limit, no explanation
- 1 pt correct explanation and limit but incorrectly says the policy was successful at 22 catch per ship.

Question 6

2c

5 / 5 pts

- ✓ - 0 pts Correct, thorough explanation with more than one example of problems that could be encountered.
- 1 pt Explains one challenge well, or partial explanation of multiple
- 2 pts limited explanation
- 5 pts didn't answer the question

Question 7

2d

5 / 5 pts

- ✓ - 0 pts Correct, very little warning, catch per ship not a great indicator with some additional detailed explanation on how by the time catch per ship decreases the population is experiencing significant decline
- 1 pt lacking some detail in the explanation
- 2 pts only answered one of the questions
- 1 pt incorrect claim either that catch per ship is accurate, or that there is significant warning before a population dropoff
- 2 pts Incorrect claim that both catch per ship is accurate and that there is significant warning, but includes good supporting detail

Question 8

3

5 / 5 pts

✓ - 0 pts Correct

- 1 pt Incomplete

Question 9

4a

6 / 6 pts

✓ - 0 pts Correct

- 2 pts Partial

Question 10

4b

23 / 24 pts

- 0 pts Correct

✓ - 1 pt Click here to replace this description.

- 2 pts Click here to replace this description.

- 3 pts Click here to replace this description.

- 4 pts Click here to replace this description.

- 5 pts Click here to replace this description.

Question 11

4c

4 / 5 pts

- 0 pts Correct

✓ - 1 pt No optimal level identified

- 2 pts Lacks detail

- 2 pts No graph

Question 12

4d

5 / 5 pts

✓ - 0 pts Correct

- 1 pt Lacks detail

Question 13

EXTRA WORK 1

0 / 0 pts

✓ - 0 pts Correct

Question 14

EXTRA WORK 2

0 / 0 pts

✓ - 0 pts Correct

Question 15

WORKS CITED

0 / 0 pts

✓ - 0 pts Correct

- 1 pt incorrect submission-added pages or moved around the pages that answers were found on

- 1 pt Click here to replace this description.

UNITED STATES MILITARY ACADEMY

HOMEWORK #4

SM484: SYSTEM DYNAMICS SIMULATION

SECTION H1/I1/J1

MAJ LUEDERS/LUPFER

BY

CADET Brent Keefer
Type FIRST LAST name from Canvas

WEST POINT, NEW YORK

19 novemeber
2025

 I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

BK I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

AND:

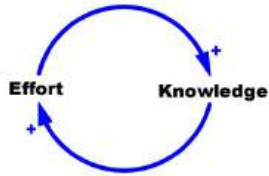
 I ACKNOWLEDGE THAT I USED GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

BK I ACKNOWLEDGE THAT I DID NOT USE GENERATIVE AI TO COMPLETE THIS ASSIGNMENT.

SIGNATURE: brent keefer DATE: 11/16
2025

SM 484: System Dynamics Simulation – AT 26-1

Homework 4 – Fishbanks Policy Analysis

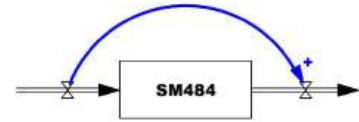


System Dynamics Simulation

SM 484 – AT 26-1

MAJ Jake Lueders

MAJ Pat Lupfer

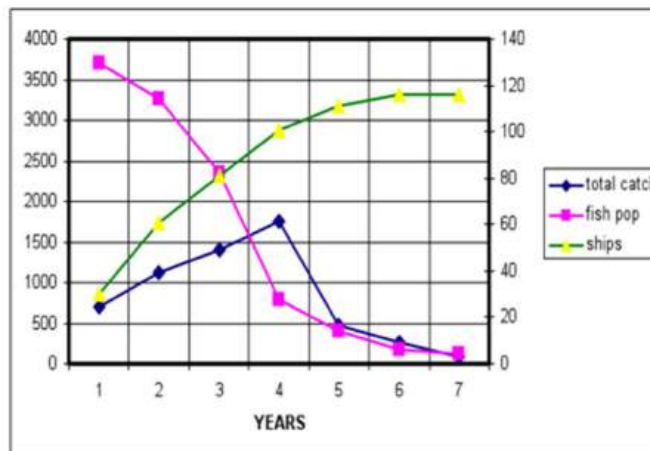


Homework 4: Fishbanks Policy Analysis

Assigned: 29 October 2025; Due: 19 November 2025 at 2359
75 Points

*This is an individual assignment. Follow these submission instructions exactly. You will submit a single PDF containing 14 pages. The DAAW and a space for works cited have been included, as well as a blank extra work page (if needed); simply add your name, the date, and circle your section and instructor. **DO NOT convert to word and back to PDF**, either type directly into the PDF, or print and handwrite then SCAN it back in using a scanner. **DO NOT TAKE A PICTURE AND CONVERT TO PDF UNLESS YOU KNOW HOW TO CONVERT IT BACK TO 8.5"x11"**. There are scanners in the DSE Cadet Study/Capstone Area and the library, or you can stop by and ask an instructor for help scanning.*

The year is 1970. You are a manager in the national fish and wildlife service. You are charged with the responsibility of making policy suggestions regarding the management of a fishery surrounding your national borders. Fishing is an important source of economic income for your country so your decisions must ensure the healthy maintenance of the fish population and fishing industry.



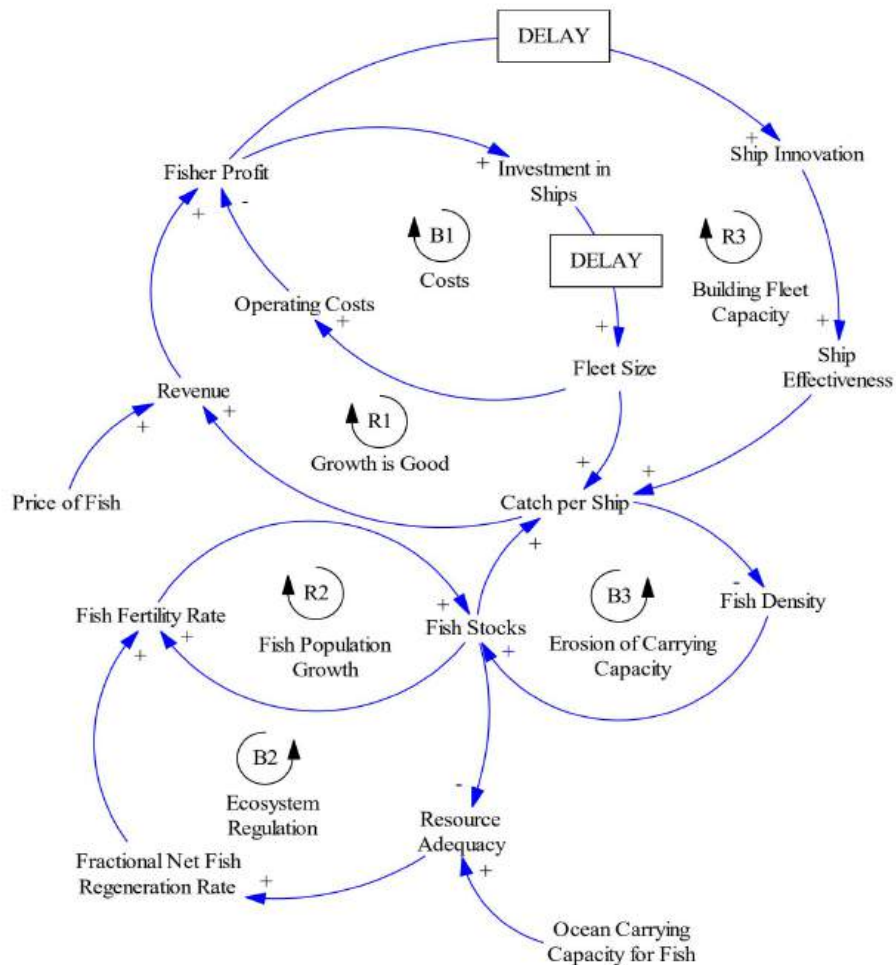
Your assessment of the fishing in the public waters is that the size of the fishing fleet increases while the fish population is decimated and the fishing industry collapses. Indeed, you speculate

SM 484: System Dynamics Simulation – AT 26-1

Homework 4 – Fishbanks Policy Analysis

that the existing dynamics operating among these three variables results in the outcomes depicted in the graph.

The common perception is that the fish population is assumed to be unlimited. Fishing companies need merely invest in a fleet of ships, which, over time, will generate profits that exceed operating expenses. As evident in the CLD below, this mental model characterizes a business as usual approach (R1) of unlimited growth. Fish stocks are assumed to grow at a constant rate, which produces plentiful harvests (R2). Companies reinvest profits into their companies, which, in the aggregate, grows fleet size (R3).



This image presents a mental model of unlimited growth. And for a while, the model behaves in accordance with the mental model. Exponential growth occurs in the growth of fleet size, fish harvest, and assets. But then the business as usual approach goes terribly wrong. Costs rise as ships must spend longer hours in search of fish further removed from the coastal waters (B1).

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The fish stocks, already constrained by physiological limits of density based on availability of food and air, level off as they reach these natural limits (B2), then fall quickly as their numbers decline faster via fishery extraction than can be sustained via natural reproduction (B3).

This scenario can be explained by the latent underlying structure of these feedback mechanisms. Essentially, the fishing industry is growing so fast that it is eroding the carrying capacity (i.e., fish stock) of their own industry. The resulting behavior is regulated by a latent structure that permits growth until a tipping point is reached, which is defined as the point at which continued depletion of the resource occurs faster than the regeneration of the resource.

Your challenge is to devise a regulatory policy or complimentary set of policies capable of sustaining the fishery. Beginning with the standard model given to you (*SM484Fishbanks.mdl*), answer the following questions.

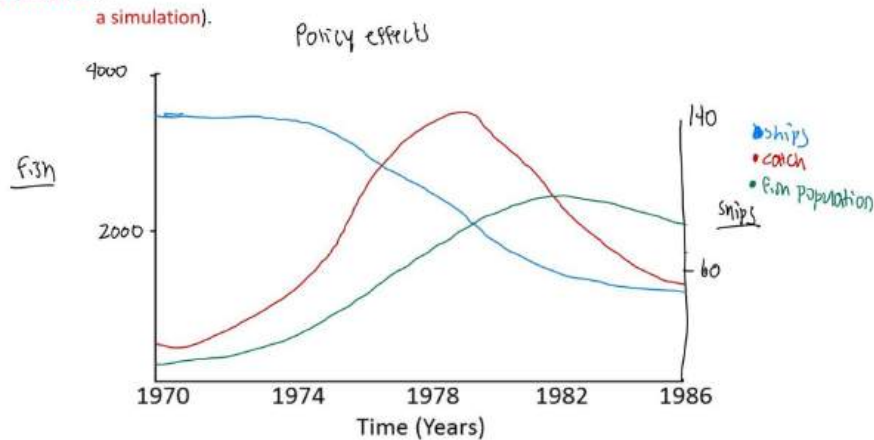
SM 484: System Dynamics Simulation – AT 26-1

Homework 4 – Fishbanks Policy Analysis

1. **(10 points)** The standard model given to you, which includes minimal structure to represent the system, and is currently set up to simulate the regulatory situation present within the national fishing industry as of 1970 (no regulation). Observe the behavior of the number of ships, catch per year, fish population, and total accumulated profits.

Clearly, leaving the present system undisturbed is an unacceptable situation. The fish population is decimated, the fishing industry dies, and you lose your job. One regulatory policy, which might help to avoid this situation is to levy a large tax on new ships (say \$200 per ship), thereby raising the price and curtailing purchases.

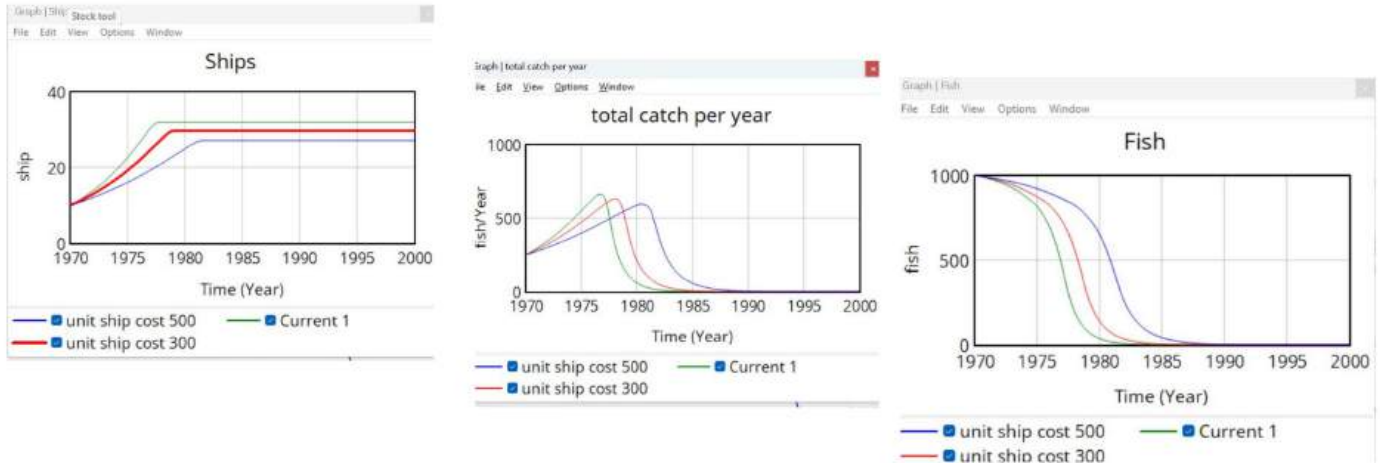
- a. Sketch on the axes below *the effect you believe* that this policy will have on the number of ships, total catch per year, and fish population (do this **without running a simulation**).



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Homework 4 – Fishbanks Policy Analysis

- b. Try running this policy by changing the ship cost from \$300 to \$500. Once you have changed the ship cost, simulate the system, and generate a graph (insert screenshot below) with both the base case and policy to show the results for ships, total catch per year, and fish population. Was this policy successful? How does the behavior generated in this simulation compare with the previous behavior? Explain the differences in behavior.



The policy of increasing ship costs proved only moderately effective. Although it postponed the collapse of the fish population by around four years, it ultimately failed to prevent the depletion of the fishery. The central objective maintaining a sustainable fish population and ensuring long-term stability was not achieved, showing that this measure alone was inadequate. Relative to the baseline scenario, the higher costs reduced fleet size and slowed the rate of depletion, offering a brief period of stability in total catch. Yet this stabilization was temporary, as the fundamental causes of overfishing persisted. The resulting system behavior reflected a recurring pattern: rapid fleet expansion followed by resource exhaustion. This outcome underscores that cost-based interventions alone are insufficient. For true sustainability, such a policy would need to be paired with broader measures, such as stricter fleet size limits or regulated catch quotas.

- c. Vary the fish price and re-simulate to observe the effects on behavior. What

When the fish price is set too high, the system shows a pattern of overexploitation and rapid resource depletion. Higher prices make fishing more profitable, which drives a sharp expansion of the fleet. As shown in vensim, this causes a steep increase in the number of ships and a rapid surge in total catch particularly evident in the \$100 fish price scenario. However, this fast-paced growth leads to the early collapse of the fish population compared to cases with lower fish prices which is shown in vensim

Conversely, a lower fish price (such as \$10) results in slower fleet growth and a more stable catch rate, allowing the fish population to remain sustainable for a longer period. This comparison demonstrates how strongly fish price influences system behavior and reinforces the importance of balancing economic profitability with ecological sustainability to avoid overexploitation.

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2. **(20 points)** Another possible policy alternative is to keep building ships until there is a noticeable problem with the fish population. Since the only feedback signal from the fish population that fishermen receive is from the fish catch, the policy could make building ships illegal once the catch per ship decreases to an unacceptable rate.

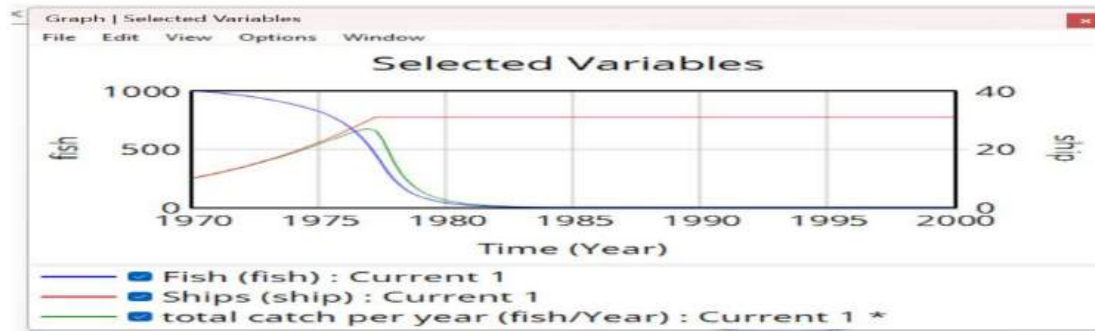
To implement this policy, first change the ship cost back to 300 dollars per ship.

Next insert the feedback from Catch per Ship to Ship Building Rate. During good fishing, the average ship will catch 25 fish per year. See what happens when the ship building is limited after the catch drops below 22 fish per year, using the following equation:

IF THEN ELSE (catch per ship > 22, yearly profits * fraction invested / ship cost, 0)

This equation says that if the catch per ship is greater than 22 fish per year then purchase the normal number of ships (Profits * Fraction Invested / Ship Cost). However, if the catch per ship falls below 22, then build zero ships.

- a. Run the model and produce a graph of the behavior (insert screenshot). How does the behavior generated based on the new regulation differ from a system with no regulation? Explain any differences or lack of differences.



The graph shows that the new regulation produces only a minor effect on the system's overall behavior when compared to the baseline scenario without regulation. In both cases, the fish population still experiences a rapid decline, with the regulation merely delaying depletion slightly. This indicates that while the Catch per Ship threshold slightly affects the rate of shipbuilding, it is not strong enough to meaningfully alter the system's dynamics.

The minimal difference between the two outcomes emphasizes the regulation's limited ability to address the root cause of overfishing. Even with the policy in place, the fishing fleet continues to expand quickly in the early years, ultimately leading to the collapse of the fish population. This outcome highlights the need for more substantial regulatory changes such as stricter fleet size limits, catch quotas, or stronger enforcement to effectively complement the Catch per Ship threshold. Without such measures, the regulation does little to shift the system's long-term trajectory.

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- b. Was this policy successful? If not, try increasing the threshold on Catch per Ship from 22 to a higher number. At what number does the system produce a stable fish population? What factors not present within the model might affect the catch per ship?

The policy using an initial Catch per Ship threshold of 22 failed to create a stable fish population. While this threshold slightly postponed the population's collapse compared to the baseline scenario, the fish stock ultimately became fully depleted over time. Raising the threshold to a higher level around 25 or above would slow the rate of new ship construction, giving the fish population a better chance to recover and sustain itself more effectively. Several external factors not represented in the model could also influence the Catch per Ship rate and the overall effectiveness of this policy. Environmental factors such as habitat loss, pollution, and climate change may reduce fish reproduction or alter catch rates. Technological advancements in fishing could increase efficiency, raising catch rates even under stricter thresholds. Market forces and policy changes like fluctuating fish prices or tighter fishing regulations might also affect fishing intensity. Furthermore, shifts in fish behavior, such as migration or availability, could further impact catch rates. Incorporating these elements into the model would offer a more complete understanding of the system's dynamics and policy outcomes.

- c. How easy do you think it will be for the government to collect data on the catch per

Gathering accurate data on Catch per Ship poses several challenges for government agencies. Although calculating it by dividing total catch by the number of active ships seems simple in theory, numerous practical and logistical obstacles can complicate the process. For example, some vessels may under report or falsify their catches to avoid taxes, quotas, or regulatory scrutiny, resulting in unreliable data. In addition, inconsistencies in reporting standards and record-keeping across different regions and fleets can further reduce data accuracy. Illegal, unreported, and unregulated (IUU) fishing adds another layer of difficulty, as these activities are inherently hard to track. Seasonal fluctuations in fish populations, regional differences in fishing intensity, and variations in technology or ship capacity can also distort averages if not properly accounted for. Finally, the administrative demands of monitoring and verifying reports from large fleets particularly in international waters can strain government resources. These challenges indicate that while collecting Catch per Ship data is feasible, it requires strong systems for monitoring, verification, and enforcement to ensure reliability.

- d. Do you get the impression that fishing systems tend to give a great deal of warning before a catastrophe or do you feel these catastrophes come on more suddenly? Is the catch per ship an accurate indication of the fish population?

Fishing systems often experience abrupt collapses rather than providing clear early warning signs. This pattern is reflected in the model's behavior, where the fish population drops sharply once it passes a critical threshold. The system's internal dynamics particularly the delay between the decline in fish numbers and the observable effects on catch rates allow overfishing to continue until recovery becomes nearly impossible. Such delayed feedback makes it difficult to detect and respond to warning signs before the system reaches collapse.

Moreover, catch per ship is not a fully reliable or timely indicator of the fish population's health. Although it measures fish availability relative to fishing effort, it fails to capture the underlying regenerative capacity of the population. By the time catch per ship begins to decrease, the stock may already be too depleted to recover. Technological improvements and more efficient fishing methods can further obscure these signals, masking the true state of the resource and reducing the effectiveness of catch per ship as an early warning measure.

Homework 4 – Fishbanks Policy Analysis

Finish the quote using your course text. *Business Dynamics* Chapter 3, section 3.5.5 states

- “Policy design is much more than the values of parameters such as a tax rate or markup ratio. Policy design includes the creation of entirely new strategies, structures, and decision rules.”

The diagram illustrates the dynamics of a fishery management system, showing the interaction between fish stock and ship fleet size.

Stocks:

- Fish:** The primary stock, influenced by hatch and death rates.
- Ships:** The fleet size, influenced by licensing and building rates.

Flows:

- fish hatch rate:** Influenced by Hatch Fraction and Fish stock.
- fish death rate:** Influenced by Fish stock and Carrying Capacity.
- total catch per year:** Determined by Fish stock, density, and catch per ship.
- ship licensing:** Influenced by Fish stock, Operational ships, and adjustment time.
- ship building rate:** Influenced by Initial Ships, Ship Cost, and Fraction Invested.
- costs:** Determined by Ship stock and Unit Ship Operating Cost.
- annual profits:** Determined by Ship stock, Fish Price, and Revenues.

Feedback Loops:

- Density-Dependent Catch:** Fish stock increases density, which decreases catch per ship, leading to a decrease in total catch and fish stock.
- Profit-Driven Ship Building:** Annual profits lead to investment, which increases the ship building rate, leading to an increase in the ship fleet.
- Operational Ships:** The difference between ship licensing and ship building rate determines the change in the ship stock.

a. Explain the purpose of this policy.

This policy implements a licensing system designed to regulate the number of ships operating within the fishery. By establishing a maximum number of fishing licenses, it restricts how many vessels can be active at any given time. The mechanism functions by assessing the difference between the current fleet size and the license cap preferred to as the license discrepancy. Adjustments to the fleet occur gradually based on this discrepancy, allowing ships to enter or leave operation over a defined period. The primary objective of this policy is to curb overfishing and promote long-term sustainability of fish populations. By controlling fleet size, it helps prevent rapid depletion of fish stocks while still supporting the economic viability of the fishing industry. The policy also incorporates profitability considerations by linking fleet growth to total profits and reinvestment in new vessels. Ultimately, this approach seeks to balance economic incentives with ecological preservation.

b. Provide all data for the variables in the table below (if initial value is not applicable, write "NA"):

Variable	Description	Equation	Initial Value	Units
Operational Ships	<i>Number of ships currently operating in the fishery</i>	<i>INTEG(Ship Licensing)</i>	<i>50 ships</i>	<i>ships</i>
Ship Licensing	<i>The mechanism controlling how many ships are allowed to operate based on licensing</i>	<i>License Discrepancy / Adjustment Time</i>	<i>N/A</i>	<i>Ship/Year</i>
Max Ship Licenses	<i>The maximum number of ships allowed to operate in the system</i>	<i>50</i>	<i>N/A</i>	<i>Ship</i>
License Discrepancy	<i>The difference between the actual number of operational ships and the maximum licenses allowed</i>	<i>Max Ship Licenses – Operational Ship</i>	<i>N/A</i>	<i>Ship</i>
Adjustment Time	<i>The time it takes for the system to adjust the number of operational ships to meet licensing limits</i>	<i>N/A</i>	<i>1</i>	<i>YEAR</i>
Total Catch Per Year	<i>The amount of fish caught by the operational ships in a year</i>	<i>Operational Ships * Catch Per Ship</i>	<i>N/A</i>	<i>Fish/Year</i>

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Homework 4 – Fishbanks Policy Analysis

- c. Simulate the model and discuss the performance of your policy on the sustainability of the fishery. **Use output graphs to support your answer.** Discuss lessons learned from your experimentation, even if the policy you suggest is unable to create a balance between the human and natural systems.



The sustainability policy implemented in the fishery model reveals several important effects through its performance outputs. As illustrated in the graphs, catch per ship declines rapidly as the fish population decreases, demonstrating that the policy is unsuccessful in maintaining the fishery over time. The total fish stock becomes completely depleted within a few decades, even under the policy, indicating a collapse of the natural system. Although the number of operational ships initially falls in response to the policy and later stabilizes, this adjustment ultimately proves insufficient to preserve the fish population.

This simulation underscores several important insights. Even with regulation in place, the balance between human activity and natural resources remains unstable when overfishing and population pressures are not effectively managed. The continued decline in fish stocks reveals the policy's main weakness its lack of adaptability to the ongoing depletion of the resource. This emphasizes the need for stronger, more flexible measures such as adaptive catch limits or stricter enforcement of fleet reduction strategies to better align human behavior with ecological sustainability. Although the policy represents progress, it also highlights the complexity of achieving long-term sustainability and the necessity of addressing the interconnected feedback loops that drive system behavior.

- d. Based on research, develop an additional policy recommendation. Explain how you

While I was doing research for this assignment I decided that an effective policy would be to implement adaptive catch limits that respond to real-time assessments of fish populations. This approach would adjust the allowable catch per ship each year based on the current estimated fish stock, ensuring that harvest levels remain aligned with ecosystem capacity. In addition, a dynamic licensing system could be introduced to regulate the number of active ships according to population trends reducing licenses during periods of low stock and gradually increasing them as fish populations recover.

The adaptive catch limit would help reduce overfishing by changing how much fish can be caught based on how many fish are available. This helps prevent the population from dropping too quickly. In the model, this can be added to the total catch per year variable by making it the smaller value between the actual catch and the set catch limit. The dynamic licensing part would work with this by adjusting the number of ships allowed to fish, depending on how healthy the fish population is. This could be shown as the difference between the current fish population and the target population over time, which would then decide how many licenses are given out. Together, these policies would help balance fishing activity with the fish population, promoting long-term sustainability. Testing and adjusting these settings could make the policy more effective in real situations.

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EXTRA WORK (Do not delete, just leave blank if not needed)

References

Papaioannou, E. A. et al. Not all those who wander are lost—responses of fishers' communities to shifts in the distribution and abundance of fish. Front. Marine Sci. (2021).

Whelan, J. G. (1994). Building the fish banks model and renewable resource depletion. System Dynamics Education Project, Sloan School of Management, Massachusetts Institute of Technology. https://ocw.mit.edu/courses/15-988-system-dynamics-self-study-fall-1998-spring-1999/cc6338ec4f6f8e53b2f56f406b6a4908_fish.pdf

SM 484: System Dynamics Simulation – AT 26-1

Homework 4 – Fishbanks Policy Analysis

EXTRA WORK (Do not delete, just leave blank if not needed)

SM 484: System Dynamics Simulation – AT 26-1

Homework 4 – Fishbanks Policy Analysis

Works Cited: Include your works cited below, do not delete even if you didn't use additional sources.



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WEST POINT.

TAB

Quizzes



Quiz 1

Instructor Solution

This quiz was given in Canvas, and a PDF export or print function does not work. Words below shown in brackets [] appear as blanks in the Quiz.

Question 1

System Dynamics Basics 13 pts

System Dynamics is a methodology for studying and managing complex [systemtype1] systems over time utilizing feedback loops, time delays, and [tool1] and [tool2] diagrams.

Show Answers for systemtype1

→ feedback

→ Feedback

→ dynamic

System Dynamics Basics 13 pts

System Dynamics is a methodology for studying and managing complex [systemtype1] systems over time utilizing feedback loops, time delays, and [tool1] and [tool2] diagrams.

Show Answers for tool1

→ stocks

→ stock

→ stock and flow

System Dynamics Basics 13 pts

System Dynamics is a methodology for studying and managing complex [systemtype1] systems over time utilizing feedback loops, time delays, and [tool1] and [tool2] diagrams.

Show Answers for tool2

→ flows

→ flow

→ causal loop

Question 2

⋮

System Dynamics Basics 2

2 pts

If you understand the basic [structure] of a system you will understand the [behavior] it can produce.

Show Answers for

→ structure

⋮

System Dynamics Basics 2

2 pts

If you understand the basic [structure] of a system you will understand the [behavior] it can produce.

Show Answers for

→ behavior

Question 3

⋮

System Dynamics Modeling Process

5 pts

List the steps in the System Dynamics Modeling Process in Order:

1. [step1]
2. [step2]
3. [step3]
4. [step4]
5. [step5]

Show Answers for

→ Problem Articulation

→ problem articulation

System Dynamics Modeling Process

5 pts

List the steps in the System Dynamics Modeling Process in Order:

1. [step1]
2. [step2]
3. [step3]
4. [step4]
5. [step5]

Show Answers for

- Dynamic Hypothesis
- dynamic hypothesis

System Dynamics Modeling Process

5 pts

List the steps in the System Dynamics Modeling Process in Order:

1. [step1]
2. [step2]
3. [step3]
4. [step4]
5. [step5]

Show Answers for

- Formulation
- formulation

System Dynamics Modeling Process

5 pts

List the steps in the System Dynamics Modeling Process in Order:

1. [step1]
2. [step2]
3. [step3]
4. [step4]
5. [step5]

Show Answers for

- Testing
- testing

System Dynamics Modeling Process

5 pts

List the steps in the System Dynamics Modeling Process in Order:

1. [step1]
2. [step2]
3. [step3]
4. [step4]
5. [step5]

Show Answers for

- ☒ Policy Formulation & Evaluation
- ☐ policy formulation & evaluation
- ☐ Policy Formulation and Evaluation
- ☐ policy formulation and evaluation

Question 4

Feedback Structure 1

3 pts

Positive feedback

- ☒ indicates that the effect is positively related to the cause.
- ☐ indicates that the accumulation of items is increasing.
- ☐ indicates that the effect is always increasing in value.
- ☐ indicates that the effect is larger than the cause.

Question 5

Feedback Structures 2

3 pts

Negative feedback

- ☐ indicates that the effect is always decreasing.
- ☐ indicates that the effect has a bad impact
- ☒ is self correcting or counteracts change
- ☐ indicates that the effect is less than the cause

Question 6

Link Polarity 1	2 pts
Identify the link polarity for the following variables:	
 <pre>graph LR; Population --> ResourceAdequacy[Resource Adequacy]</pre>	
☐ Positive	
☒ Negative	
☐ Cannot be determined	

Question 7

Link Polarity 2	2 pts
Identify the link polarity for the following variables:	
 <pre>graph LR; Production --> Inventory</pre>	
☒ Positive	
☐ Negative	
☐ Cannot be determined	

Question 8

Loop Polarity
3 pts

Identify the loop polarity for the following causal loop diagram:

☐ Reinforcing
☒ Balancing
☐ Cannot be determined

Question 9

Loop Polarity 2
3 pts

Identify the loop polarity for the following causal loop diagram:

☐ reinforcing
☐ balancing
☐ cannot be determined
☒ not a loop

Question 10

Stocks and Flow

5 pts

Label the following elements on the diagram:

The diagram illustrates a system with a central stock, 'Bank Balance', represented by a blue box. Two flows connect it to external clouds. A green flow, labeled 'Deposit', originates from 'Cloud 1' (a cloud icon) and points into the 'Bank Balance' box. A red flow, labeled 'Withdrawal', originates from the 'Bank Balance' box and points to 'Cloud 2' (another cloud icon). Both flows have a valve symbol in the middle. The labels 'Cloud 1' and 'Cloud 2' are in purple, 'Deposit' is in green, and 'Withdrawal' is in red.

Cloud1[Cloud1]
Deposit[Deposit]
Bank Balance[BankBalance]
Withdrawal[Withdrawal]
Cloud2[Cloud2]

Show Answers for

→ Source

→ source

Stocks and Flow

5 pts

Label the following elements on the diagram:

The diagram is identical to the one above, showing a central 'Bank Balance' stock with 'Deposit' inflow from 'Cloud 1' and 'Withdrawal' outflow to 'Cloud 2'.

Cloud1[Cloud1]
Deposit[Deposit]
Bank Balance[BankBalance]
Withdrawal[Withdrawal]
Cloud2[Cloud2]

Show Answers for

→ Inflow

→ inflow

Stocks and Flow

5 pts

Label the following elements on the diagram:



Cloud1[Cloud1]

Deposit[Deposit]

Bank Balance[BankBalance]

Withdrawal[Withdrawal]

Cloud2[Cloud2]

Show Answers for

BankBalance

→ Stock

→ stock

Stocks and Flow

5 pts

Label the following elements on the diagram:



Cloud1[Cloud1]

Deposit[Deposit]

Bank Balance[BankBalance]

Withdrawal[Withdrawal]

Cloud2[Cloud2]

Show Answers for

Withdrawal

→ Outflow

→ outflow

Stocks and Flow

5 pts

Label the following elements on the diagram:



Cloud1[Cloud1]

Deposit[Deposit]

Bank Balance[BankBalance]

Withdrawal[Withdrawal]

Cloud2[Cloud2]

Show Answers for

Cloud2

→ Sink

→ sink

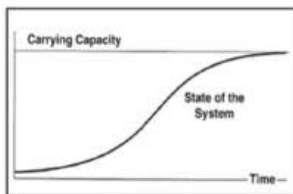
Question 11

Behavior 1

4 pts



Which structure generates the following behavior? Select all that apply



→ ☐ reinforcing loop

→ ☐ balancing loop

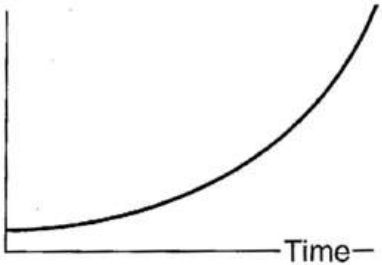
→ ☐ exogenous goal

☐ endogenous goal

Question 12

Behavior 2

3 pts



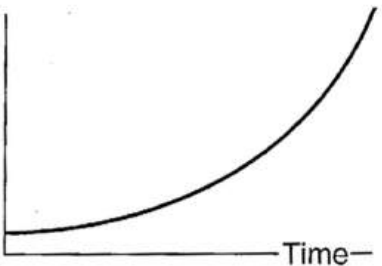
The following behavior is called [name] and is generated by the structure containing a [structure] loop.

Show Answers for

☒ exponential growth

Behavior 2

3 pts



The following behavior is called [name] and is generated by the structure containing a [structure] loop.

Show Answers for

☒ reinforcing

☒ positive

Question 13

Struggles

2 pts

What areas of system dynamics are you struggling to understand?



Quiz 2

Instructor Solution

1

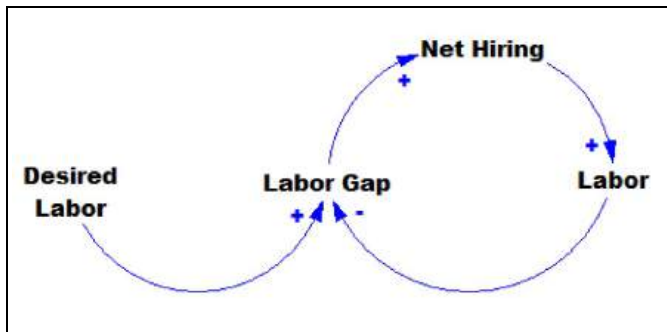
Multiple Choice 3 points Dynamics of Simple Structures 1

Based on the following causal loop diagram, what type of dynamic behavior would the system demonstrate?



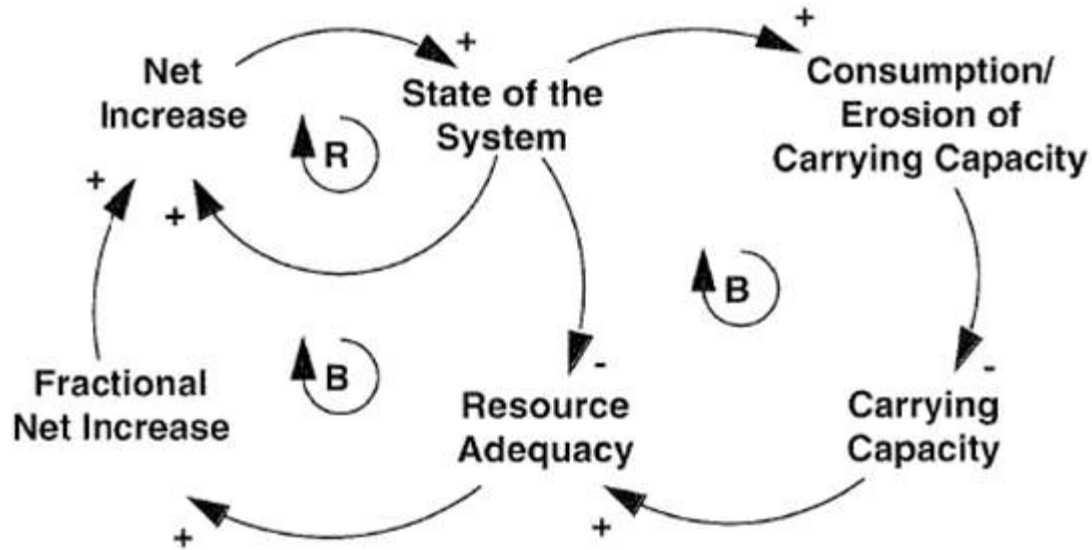
- ☐ S-Shaped Growth
- ☒ Exponential Growth
- ☐ Goal-Seeking
- ☐ Overshoot and Collapse

Based on the following causal loop diagram, what type of dynamic behavior would the system demonstrate?



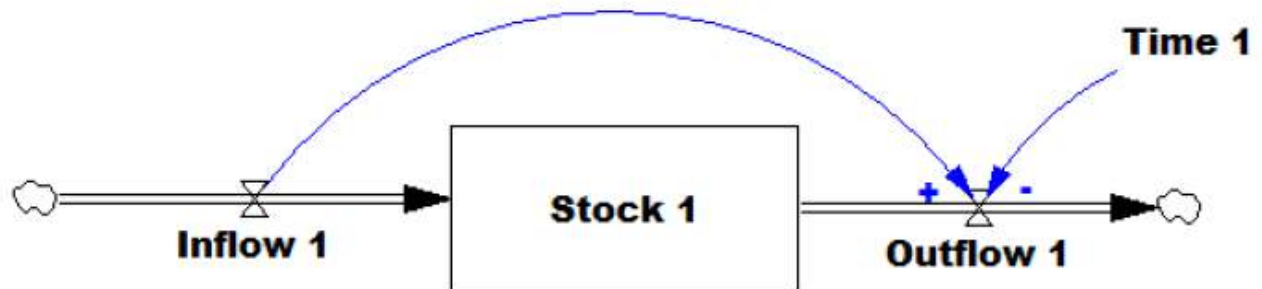
- ☐ S-Shaped Growth
- ☐ Exponential Growth
- ☒ Goal-Seeking
- ☐ Overshoot and Collapse

Based on the following causal loop diagram, what type of dynamic behavior would the system demonstrate?



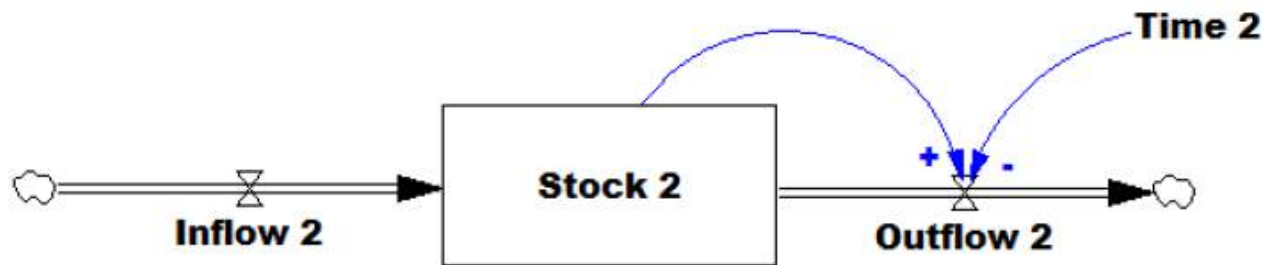
- ☐ S-Shaped Growth
- ☐ S-Shaped with Overshoot
- ☐ Goal-Seeking
- ☒ Overshoot and Collapse

The following stock and flow diagram indicates what type of delay?



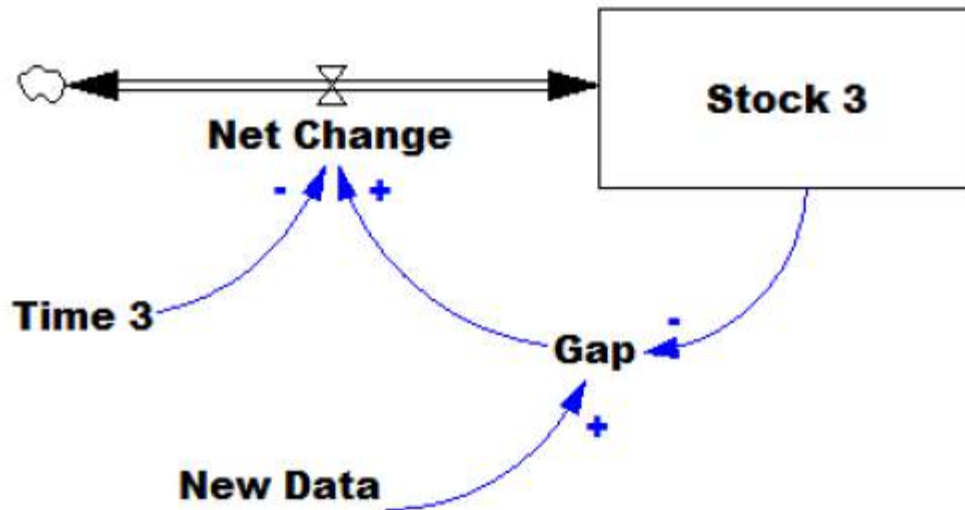
- ☐ 1st Order Material Delay
- ☐ 1st Order Information Delay
- ☒ Pipeline Delay
- ☐ 2nd Order Material Delay

The following stock and flow diagram indicates what type of delay?



- ☒ 1st Order Material Delay
- ☐ 1st Order Information Delay
- ☐ Pipeline Delay
- ☐ 2nd Order Material Delay

The following stock and flow diagram indicates what type of delay?

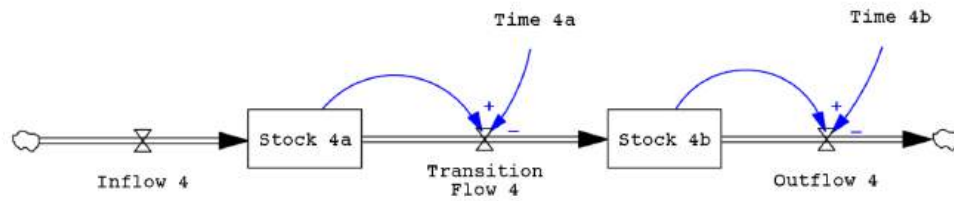


- ☐ 1st Order Material Delay
- ☒ 1st Order Information Delay
- ☐ Pipeline Delay
- ☐ 2nd Order Material Delay

7

Multiple Choice 4 points Delays 4

The following stock and flow diagram indicates what type of delay?



- ☐ 1st Order Material Delay
- ☐ 1st Order Information Delay
- ☐ 3rd Order Material Delay
- ☒ 2nd Order Material Delay

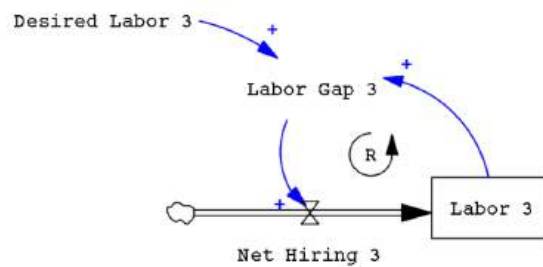
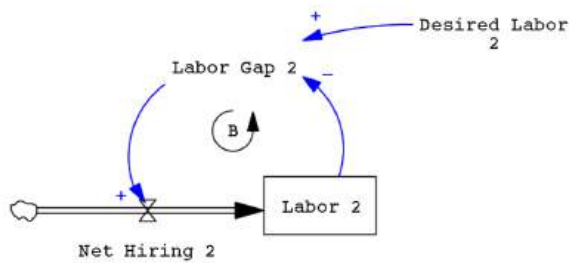
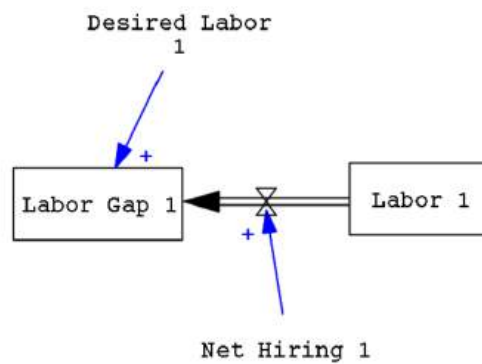
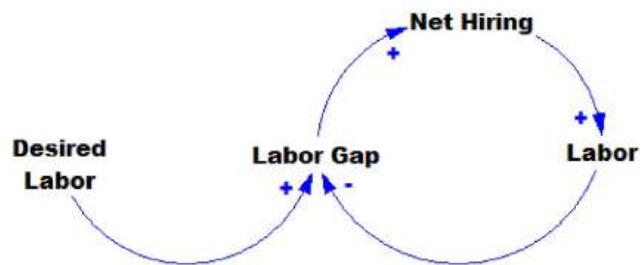
8

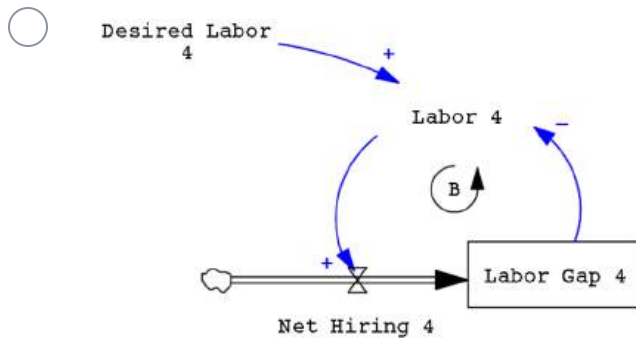
Multiple Answer 2 points Delays 5

Select all that apply for describing information delays.

- ☒ Adaptive Expectations
- ☒ Exponential Smoothing
- ☐ Balancing the Output
- ☐ Separate Inputs and Outputs

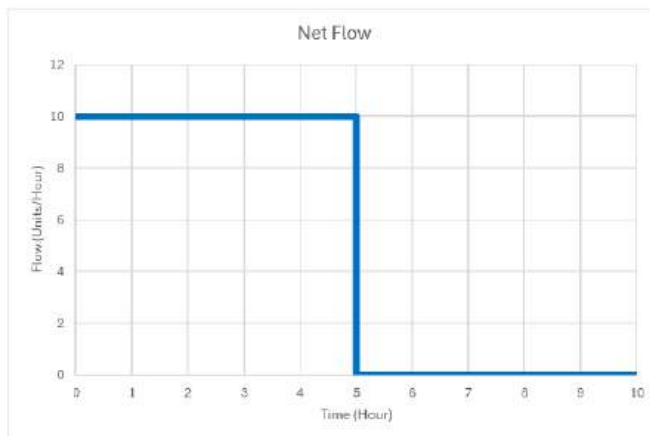
Select the correct stock and flow for the following CLD.





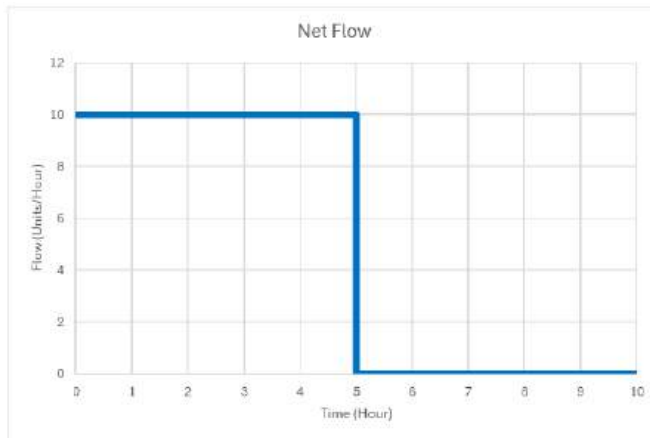
10 Numeric 3 points Graphical Integration 1

Based on the following graph of net flow, calculate the level of the stock at time 3, if the initial level is 50 units. (enter number only, the units are assumed to be "units")



80

You can determine the inflow and outflow from this net flow graph when the initial stock is given as 50 units.

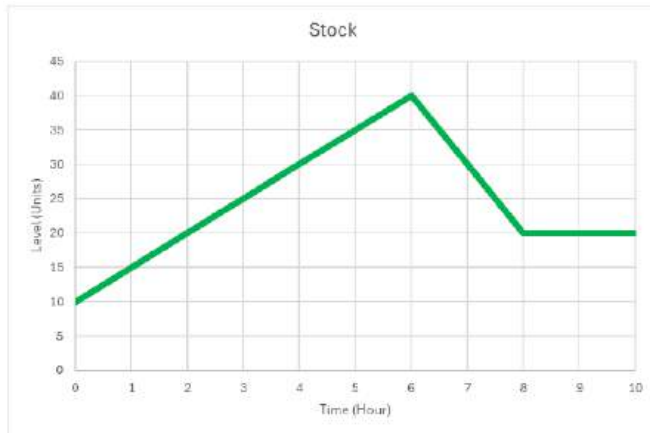


☐ True

☒ False

12 Numeric 3 points Graphical Differentiation 1

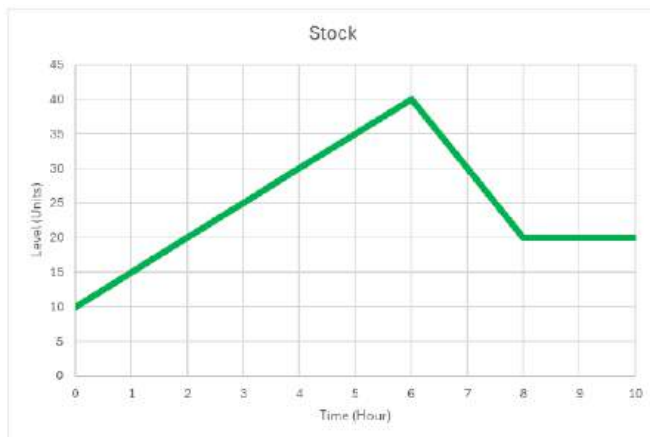
Based on the following graph of the stock, calculate the net flow at time 4. (enter number only, the units are assumed as "units/hour")



5

13 True or False 2 points Graphical Differentiation 2

Given the below stock graph, you can determine the initial value of the stock.



☒ True

☐ False